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AMAR**DELHI UNIVERSITY SERIES**

B.Sc. (Honours) II Year

Semester - III

MATHEMATICS

Core Course (Honours in Mathematics)

C 5 : Theory of Real Functions

C 6 : Group Theory - I

C 7 : Multivariate Calculus

Revised Syllabus

(with effect from Academic session 2020-21)

₹ 280.00

AMAR GUPTA & SONS

SYLLABUS

SEMESTER-III

REVISED SYLLABUS (With effect from Academic year 2019-20)

B.Sc. (Hons.) Mathematics II Year

BMATH 305 : Theory of Real Functions

BMATH 306 : Group Theory-I

BMATH 307 : Multivariate Calculus

BMATH 305 : THEORY OF REAL FUNCTIONS

Total Marks: 100 (Theory: 75, Internal Assessment: 25)

Workload: 5 Lectures, 1 Tutorial (per week) Credits: 6 (5+1)

Duration: 14 Weeks (70 Hrs.) Examination: 3 Hrs.

Course Objectives: It is a basic course on the study of real valued functions that would develop an analytical ability to have a more matured perspective of the key concepts of calculus, namely, limits, continuity, differentiability and their applications.

Course Learning Outcomes: This course will enable the students to:

- Have a rigorous understanding of the concept of limit of a function.
- Learn about continuity and uniform continuity of functions defined on intervals.
- Understand geometrical properties of continuous functions on closed and bounded intervals.
- Learn extensively about the concept of differentiability using limits, leading to a better understanding for applications.
- Know about applications of mean value theorems and Taylor's theorem.

Unit 1 : Limits of Functions : Limits of functions (ϵ - δ approach), Sequential criterion for limits, Divergence criteria, Limit theorems, One-sided limits, Infinite limits and limits at infinity.

Unit 2 : Continuous Functions and their Properties : Continuous functions, Sequential criterion for continuity and discontinuity, Algebra of continuous functions, Properties of continuous functions on closed and bounded intervals; Uniform continuity, Non-uniform continuity criteria, Uniform continuity theorem.

Unit 3 : Derivability and its Applications : Differentiability of a function, Algebra of differentiable functions, Carathéodory's theorem, Chain rule; Relative extrema, Interior extremum theorem, Rolle's theorem, Mean-value theorem and applications, Intermediate value property of derivatives, Darboux's theorem.

Unit 4 : Taylor's Theorem and its Applications : Taylor polynomial, Taylor's theorem with Lagrange form of remainder, Application of Taylor's theorem in error estimation; Relative extrema, and to establish a criterion for convexity; Taylor's series expansions of e^x , $\sin x$ and $\cos x$.

Keywords: Continuity, Convexity, Differentiability, Limit, Relative extrema, Rolle's theorem, Taylor's theorem, Uniform continuity.

BMATH 306 : GROUP THEORY-I

Total Marks : 100 (Theory: 75, Internal Assessment: 25)

Workload : 5 Lectures, 1 Tutorial (per week) Credits: 6 (5+1)

Duration : 14 Weeks (70 Hrs.) Examination: 3 Hrs.

Course Objectives: The objective of the course is to introduce the fundamental theory of groups and their homomorphisms. Symmetric groups and group of symmetries are also studied in detail. Fermat's Little Theorem as a consequence of the Lagrange's theorem on finite groups.

Course Learning Outcomes: The course will enable the students to:

- Recognize the mathematical objects that are groups, and classify them as abelian, cyclic and permutation groups, etc.
- Link the fundamental concepts of groups and symmetrical figures.
- Analyze the subgroups of cyclic groups and classify subgroups of cyclic groups.
- Explain the significance of the notion of cosets, normal subgroups and factor groups.

(ii)

SYLLABUS

(v) Learn about Lagrange's theorem and Fermat's Little theorem.
 (vi) Know about group homomorphisms and group isomorphisms.

Unit 1 : Groups and its Elementary Properties : Symmetries of a square, Dihedral groups, Definition and examples of groups including permutation groups and quaternion groups (illustration through matrices), Elementary properties of groups.

Unit 2 : Subgroups and Cyclic Groups : Subgroups and examples of subgroups, Centralizer, Normalizer, Center of a group, Product of two subgroups; Properties of cyclic groups, Classification of subgroups of cyclic groups.

Unit 3 : Permutation Groups and Lagrange's Theorem : Cycle notation for permutations, Properties of permutations, Even and odd permutations, Alternating groups; Properties of cosets, Lagrange's theorem and consequences including Fermat's Little theorem, Normal subgroups, Factor groups, Cauchy's theorem for finite abelian groups.

Unit 4 : Group Homomorphisms : Group homomorphisms, Properties of homomorphisms, Group isomorphisms, Cayley's theorem, Properties of isomorphisms, First, Second and Third isomorphism theorems for groups.

Keywords : Cauchy's theorem for finite Abelian groups, Cayley's theorem, Centralizer, Cyclic group, Dihedral group, Group homomorphism, Lagrange's theorem, Normalizer, Permutations.

BMATH 307 : MULTIVARIATE CALCULUS

Total Marks : 150 (Theory: 75, Internal Assessment: 25 and Practical: 50)

Workload : 4 Lectures, 4 Practicals (per week) Credits: 6 (4+2)

Duration : 14 Weeks (56 Hrs. Theory + 56 Hrs. Practical) Examination: 3 Hrs.

Course Objectives : To understand the extension of the studies of single variable differential and integral calculus to functions of two or more independent variables. Also, the emphasis will be on the use of Computer Algebra Systems by which these concepts may be analyzed and visualized to have a better understanding. This course will facilitate to become aware of applications of multivariable calculus tools in physics, economics, optimization, and understanding the architecture of curves and surfaces in plane and space etc.

Course Learning Outcomes: This course will enable the students to:

(i) Learn the conceptual variations when advancing in calculus from one variable to multivariable discussion.

(ii) Understand the maximization and minimization of multivariable functions subject to the given constraints on variables.

(iii) Learn about inter-relationship amongst the line integral, double and triple integral formulations.

(iv) Familiarize with Green's, Stokes' and Gauss divergence theorems.

Unit 1 : Calculus of Functions of Several Variables : Functions of several variables, Level curves and surfaces, Limits and continuity, Partial differentiation, Higher order partial derivative, Tangent planes, Total differential and differentiability, Chain rule, Directional derivatives, The gradient, Maximal and normal property of the gradient, Tangent planes and normal lines.

Unit 2 : Extrema of Functions of Two Variables and Properties of Vector Field : Extrema of functions of two variables, Method of Lagrange multipliers, Constrained optimization problems; Definition of vector field, Divergence and curl.

Unit 3 : Double and Triple Integrals : Double integration over rectangular and nonrectangular regions, Double integrals in polar coordinates, Triple integration over a parallelepiped and solid regions, Volume by triple integrals, Triple integration in cylindrical and spherical coordinates, Change of variables in double and triple integrals.

Unit 4 : Green's, Stokes' and Gauss Divergence Theorem : Line integrals, Applications of line integrals: Mass and Work, Fundamental theorem for line integrals, Conservative vector fields, Green's theorem, Area as a line integral, Surface integrals, Stokes' theorem, Gauss divergence theorem.

Keywords : Directional derivatives, Double integral, Gauss divergence theorem, Green's theorem, Lagrange's multipliers, Level curves, Stokes' theorem, Volume integral, Vector field.

AGS-S

Name of the Paper : Paper : C-5 : Theory of Real Functions
 Name of Course : B.Sc (Hons.) Mathematics II Year (CBCS)
 Semester : III
 Duration : 3 Hours
 Maximum Marks : 75

MODEL TEST PAPER-I

Q. 1. (a) Prove that $\lim_{x \rightarrow a} f(x)$ exists and is equal to a number l iff

both left limit $\lim_{x \rightarrow a^-} f(x)$ and right limit $\lim_{x \rightarrow a^+} f(x)$ exist and are equal to l .

Sol. Condition is necessary :

Let $\lim_{x \rightarrow a} f(x) = l$, then for any $\epsilon > 0$, there exists some $\delta > 0$ such that

$$|f(x) - l| < \epsilon, \text{ when } 0 < |x - a| < \delta$$

$$|f(x) - l| < \epsilon, \text{ when } |a - \delta < x < a + \delta, x \neq a$$

Which gives :

$$|f(x) - l| < \epsilon, \text{ where } a - \delta < x < a \quad \dots (1)$$

and

$$|f(x) - l| < \epsilon, \text{ when } a < x < a + \delta \quad \dots (2)$$

From (1) and (2), we see that :

$$\lim_{x \rightarrow a^-} f(x) \text{ and } \lim_{x \rightarrow a^+} f(x) \text{ both exist and are equal to } l$$

Condition is sufficient :

$$\text{Let } \lim_{x \rightarrow a^-} f(x) = l = \lim_{x \rightarrow a^+} f(x)$$

then for any $\epsilon > 0$, $\exists$ some $\delta_1 > 0$ and $\delta_2 > 0$.

Such that :

$$|f(x) - l| < \epsilon, \text{ when } a < x < a + \delta_1 \quad \dots (3)$$

and

$$|f(x) - l| < \epsilon, \text{ when } a - \delta_2 < x < a \quad \dots (4)$$

Let

$$\delta = \min(\delta_1, \delta_2), \text{ then } \delta < \delta_1 \text{ and } \delta < \delta_2$$

$\Rightarrow$

$$a + \delta < a + \delta_1 \text{ and } a - \delta > a - \delta_2 \quad \dots (5)$$

From (3) and (5):

$$|f(x) - l| < \epsilon, \text{ where } a < x < a + \delta$$

and

$$|f(x) - l| < \epsilon, \text{ where } a - \delta < x < a$$

$\therefore$

$$|f(x) - l| < \epsilon, \text{ where } a - \delta < x < a + \delta, x \neq a$$

$\Rightarrow$

$$|f(x) - l| < \epsilon, \text{ where } 0 < |x - a| < \delta$$

Which gives

$$\lim_{x \rightarrow a} f(x) = l$$

Which proves the given theorem

: Proved.

(3)

(b) Using $\epsilon - \delta$ method, prove that

$$\lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a} = 2a$$

 Sol. We know that $\lim_{x \rightarrow a} f(x) = l$, iff gives $\epsilon > 0$, there exists a positive real number δ of depending upon ϵ s.t.

$$|f(x) - l| < \epsilon \text{ for } 0 < |x - a| < \delta$$

$$\begin{aligned} \text{Let: } f(x) &= \frac{x^2 - a^2}{x - a} \\ |f(x) - 2a| &= \left| \frac{x^2 - a^2}{x - a} - 2a \right| \\ &= \left| \frac{x^2 - a^2 - 2a(x - a)}{x - a} \right| \\ &= \left| \frac{x^2 - a^2 - 2ax + 2a^2}{x - a} \right| \\ &= \left| \frac{x^2 - 2ax + a^2}{x - a} \right| \\ &= \left| \frac{(x - a)^2}{x - a} \right| \\ &= |x - a| \end{aligned}$$

$$|f(x) - 2a| < \epsilon, \text{ whenever } 0 < |x - a| < \epsilon$$

 Choosing $\delta = \epsilon$,

$$|f(x) - 2a| < \epsilon, \text{ whenever } 0 < |x - a| < \delta$$

Hence,

$$\lim_{x \rightarrow a} f(x) = 2a$$

$$\lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a} = 2a$$

Proved.

 (c) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, and $m \neq 0$, then prove that

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l}{m}$$

Sol. Given,

$$\lim_{x \rightarrow a} f(x) = l \quad \dots(1)$$

$$\lim_{x \rightarrow a} g(x) = m \quad \dots(2)$$

$$\begin{aligned} \left| \frac{f(x)}{g(x)} - \frac{l}{m} \right| &= \left| \frac{mf(x) - lg(x)}{g(x)m} \right| \\ &= \frac{|mf(x) - lg(x)|}{|g(x)m|} \\ &= \frac{|mf(x) - lm + lm - lg(x)|}{|g(x)||m|} \\ &= \frac{|m(f(x) - l) + l(m - g(x))|}{|g(x)||m|} \\ &\leq \frac{|m(f(x) - l)| + |l(m - g(x))|}{|g(x)||m|} \\ &= \frac{|m||f(x) - l| + |l||m - g(x)|}{|g(x)||m|} \\ &= \frac{|m||f(x) - l| + |l||g(x) - m|}{|g(x)||m|} \quad \dots(3) \end{aligned}$$

Since,

$$\lim_{x \rightarrow a} g(x) = m \neq 0, \text{ so there exists } \delta_1 > 0 \text{ s.t. :}$$

$$|g(x)| > \frac{1}{2} |m|, \text{ when } 0 < |x - a| < \delta_1$$

$$\frac{1}{|g(x)|} < \frac{2}{|m|}, \text{ when } 0 < |x - a| < \delta_1 \quad \dots(4.1)$$

Since,

$$\lim_{x \rightarrow a} f(x) = l, \text{ there exists some } \delta_2 > 0 \text{ s.t. :}$$

$$|f(x) - l| < \frac{1}{4} \epsilon |m|, \text{ when } 0 < |x - a| < \delta_2 \quad \dots(4.2)$$

Since,

$$\lim_{x \rightarrow a} g(x) = m, \text{ there exists some } \delta_2 > 0$$

$$|g(x) - m| < \frac{1}{3} \in \frac{|m|^2}{|l|}, \text{ where } 0 < |x - a| < \delta_2 \quad \dots(4.3)$$

Let
We get:

$\delta = \min \{\delta_1, \delta_2, \delta_3\}$, then, from (3) (4.1), (4.2), (4.3),

$$\begin{aligned} \left| \frac{f(x)}{g(x)} - \frac{l}{m} \right| &\leq \frac{|m| |f(x) - l| + |l| |g(x) - m|}{|m| |g(x)|} \\ &= \frac{|m| \left[\frac{\epsilon |m|}{4} + |l| \frac{1}{4} \epsilon \frac{|m|^2}{|l|} \right]}{|m|} \cdot \frac{2}{|m|} \\ &= \frac{2}{|m|^2} \left[\frac{1}{4} \epsilon |m|^2 + \frac{1}{4} \epsilon |m|^2 \right] \\ &= \frac{2}{|m|^2} \left[\frac{1}{2} \epsilon |m|^2 \right] \\ &= \epsilon \end{aligned}$$

$$\left| \frac{f(x)}{g(x)} - \frac{l}{m} \right| < \epsilon, \text{ when } 0 < |x - a| < \delta$$

Which gives :

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l}{m}, m \neq 0 \quad \text{Proved.}$$

Q. 2. State and prove intermediate value theorem.

Sol. Intermediate Value Theorem :

If a function $f(x)$ is continuous in $[a, b]$ and $f(a) \neq f(b)$, then $f(x)$ assumes every value between $f(a)$ and $f(b)$.

Proof of theorem :

let us suppose that $f(a) < f(b)$ and let M be any number lying between $f(a)$ and $f(b)$ i.e., $f(a) < M < f(b)$ (1)

We have to show that there exists a point $c \in [a, b]$ s. t. $f(c) = M$

Let us define a function ϕ on $[a, b]$ as follows:

$$\phi(x) = f(x) - M \quad \forall x \in [a, b] \quad \dots(2)$$

Since, $f(x)$ is given to be continuous in $[a, b]$ and M being a constant is always continuous, so ϕ is continuous in $[a, b]$.

From (2)

$$\phi(a) = f(a) - M$$

and

$$f(b) = f(b) - M \quad \dots(2)$$

From (1),

$$f(a) < M < f(b)$$

$$f(a) < M$$

and

$$M < f(b)$$

$$f(a) - M < 0$$

and

$$0 < f(b) - M$$

$$f(a) - M < 0$$

and

$$f(b) - M > 0$$

$$\phi(a) < 0$$

and

$$\phi(b) > 0$$

Which shows that $\phi(a)$ and $\phi(b)$ are of opposite signs.

Hence, by theorem, there exists some point $c \in [a, b]$ such that,

$$\phi(c) = 0$$

$\Rightarrow$

$$f(c) - M = 0$$

$\frac{1}{\Rightarrow}$

$$f(c) = M$$

$\Rightarrow$

$$f(c) = M, c \in [a, b]$$

Hence, we proved intermediate value theorem.

(b) Using ϵ, δ definition, prove that

$f(x) = 3x + 1$ is continuous at $x = 2$.

Sol.

$$f(x) = 3x + 1$$

$$f(x) = 3(2) + 1 = 6 + 1 = 7$$

Let $\epsilon > 0$ be given,

$$|f(x) - f(2)| = |(3x+1) - 7|$$

$$= |3x + 1 - 7|$$

$$= |3x - 6|$$

$$= |3(x - 2)|$$

$$= 3|x - 2| < \epsilon$$

Whenever

$$= |x - 2| < \frac{\epsilon}{3}$$

$$= |x - 2| < \frac{\epsilon}{3}$$

$\therefore$ If we choose $\delta = \frac{\epsilon}{3}$, then

$$|f(x) - f(2)| < \epsilon, \text{ whenever } |x - 2| < \delta$$

$\Rightarrow f(x)$ is continuous at $x = 2$

$\Rightarrow f(x) = 3x + 1$ is continuous at $x = 2$

Proved.

(c) Examine the continuity of the following function at the indicated point. Also, point out the type of discontinuity, if any.

$$f(x) = \frac{e^{1/x} - 1}{e^{1/x} + 1} ; x \neq 0$$

$$= 0 \quad ; x = 0$$

at $x = 0$.

Sol.

$$f(x) = \frac{e^{1/x} - 1}{e^{1/x} + 1} ; x \neq 0$$

$$f(0) = 0$$

$$\text{r.h.l.} = f(0+0)$$

$$= \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{x \rightarrow 0^+} \frac{e^{1/x} - 1}{e^{1/x} + 1}$$

$$= \lim_{h \rightarrow 0^+} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1}$$

$$= \lim_{h \rightarrow 0^+} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1}$$

$$= \lim_{h \rightarrow 0^+} \frac{e^{\frac{1}{h}} (1 - e^{-\frac{1}{h}})}{e^{\frac{1}{h}} (1 + e^{-\frac{1}{h}})}$$

$$= \lim_{h \rightarrow 0^+} \frac{1 - e^{-\frac{1}{h}}}{1 + e^{-\frac{1}{h}}}$$

$$= \frac{1 - e^{-\infty}}{1 + e^{-\infty}}$$

$$= \frac{1 - 0}{1 + 0}$$

$$= 1$$

$$\text{l.h.l.} = f(0-0)$$

$$= \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1}$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1}$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{-\frac{1}{h}} - 1}{e^{-\frac{1}{h}} + 1}$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{-\infty} - 1}{e^{-\infty} + 1}$$

$$= \frac{0 - 1}{0 + 1} = -1$$

Since $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$ exist, but are not equal

$\Rightarrow \lim_{x \rightarrow 0} f(x)$ does not exist

Type of discontinuity

$$\lim_{x \rightarrow 0^+} f(x) = 1 \neq f(0)$$

$$\lim_{x \rightarrow 0^-} f(x) = -1 \neq f(0)$$

$\Rightarrow f(x)$ has a discontinuity of the first kind at $x = 0$.

Ans.

Q. 3. (a) Define the relationship between continuity and uniform continuity. Give examples to illustrate the same.

Sol. Uniform continuity : A function f defined on an interval I is said to be uniformly continuous in the interval I , if for each $\epsilon > 0$, there exists some $\delta > 0$ such that

$$|f(x_2) - f(x_1)| < \epsilon, \text{ where } |x_2 - x_1| < \delta \text{ and for all } x_1, x_2 \in I$$

Continuity :

A function $f(x)$ is said to be continuous at $x = a$, if for $\epsilon > 0$, there exist a positive number δ (depending upon ϵ) such that

$$|f(x) - f(a)| < \epsilon \text{ if } 0 < |x - a| < \delta$$

Hence,

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Relationship between continuity and Uniform continuity :

(1) The continuity of a function is defined at a point, whereas uniform continuity of a function is defined in an interval.

(2) In the case of continuity of a function at point c , the choice of $\delta > 0$ depend upon $\epsilon > 0$ and the point c . But in case of uniform continuity of a function in an interval the choice of $\delta > 0$ depend only on $\epsilon > 0$ and not on pair of points of the given interval.

(3) Every uniform continuous function on an interval is continuous on that interval but the converse is not true.

(4) If a function f is continuous on a closed and bounded interval $[a, b]$, then it is uniformly continuous on $[a, b]$.

Examples :

(1) The function $f(x) = x^2 \forall x \in [1, 2]$ is continuous in the closed interval $[1, 2]$. Hence $f(x)$ is uniformly continuous in $[1, 2]$.

(2) The function $f(x) = \frac{1}{x}$ is not uniformly continuous in $(0, 1)$. But $f(x) = \frac{1}{x}$ is continuous in $(0, 1)$ as $x \neq 0 \in (0, 1)$ is continuous in $(0, 1)$.

(b) Define Uniform continuity of a function on the interval I . Show that $f(x) = \frac{1}{x^2}$ is uniformly continuous on $[a, \infty]$, $a > 0$.

Sol. Uniform Continuity :

A function f defined on an interval I is said to be uniformly continuous in the interval I , if for each $\epsilon > 0$ there exists some $\delta > 0$ such that

$$|f(x_2) - f(x_1)| < \epsilon, \text{ when } |x_2 - x_1| < \delta \text{ and for all } x_1, x_2 \in I$$

Solution of the problem :

$$\text{Let } f(x) = \frac{1}{x^2}$$

For $x, y \geq a > 0$, we have

$$\begin{aligned} |f(x) - f(y)| &= \left| \frac{1}{x^2} - \frac{1}{y^2} \right| \\ &= \left| \left(\frac{1}{x} + \frac{1}{y} \right) \left(\frac{1}{x} - \frac{1}{y} \right) \right| \\ &= \left| \frac{1}{x} + \frac{1}{y} \right| \left| \frac{1}{x} - \frac{1}{y} \right| \\ &= \left| \frac{1}{x} + \frac{1}{y} \right| \left| \frac{y-x}{xy} \right| \\ &= \left| \frac{1}{x} + \frac{1}{y} \right| \frac{|y-x|}{|y||x|} \end{aligned}$$

$$\Rightarrow |f(x) - f(y)| \leq \frac{2|x-y|}{a|x||y|} \text{ as } x, y \geq a > 0$$

$$\text{Since } x \geq a, y \geq a \Rightarrow xy \geq a^2$$

$$\frac{1}{xy} \leq \frac{1}{a^2}$$

$$\Rightarrow |f(x) - f(y)| \leq \frac{2}{a^2} |x - y|$$

Let $\epsilon > 0$ be given.

$$\text{Let } \delta = \frac{\epsilon a^2}{2}$$

$$\text{Then } |f(x) - f(y)| < \epsilon, \text{ when } |x - y| < \delta \forall x, y \geq a$$

Hence, f is uniformly continuous on $[a, \infty]$.

(c) Show that $\sin x$ is uniformly continuous on $[0, \infty]$.

Sol. Let $f(x) = \sin x$

Let $\epsilon > 0$ be given and x, y be any two points in $[0, \infty]$

$$\begin{aligned} \text{Then } |f(x) - f(y)| &= |\sin x - \sin y| \\ &= \left| 2 \sin \frac{x-y}{2} \cos \frac{x+y}{2} \right| \\ &= 2 \left| \sin \frac{x-y}{2} \right| \left| \cos \frac{x+y}{2} \right| \\ &\leq 2 \left| \frac{x-y}{2} \right| \end{aligned}$$

$$\text{as } |\sin \theta| \leq |\theta|$$

$$|\cos \theta| \leq 1$$

$$\Rightarrow |f(x) - f(y)| \leq |x - y|$$

$$\Rightarrow |f(x) - f(y)| < \epsilon, \text{ when } |x - y| < \delta, (\delta = \epsilon), x, y \in [0, \infty]$$

Hence $f(x) = \sin x$ is uniformly continuous on $[0, \infty]$. Proved

Q. 4. State and prove Lagrange's mean value theorem and give its geometrical interpretation.

Sol. Lagrange's Mean Value Theorem:

Let f be a function defined on $[a, b]$ such that

(i) f is continuous in $[a, b]$

(ii) f is derivable in $]a, b[$

then, there exist some $c \in]a, b[$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof

Let $f(x)$ be a function defined on $[a, b]$, $f(x)$ is continuous in $[a, b]$ and $f(x)$ is derivable in $]a, b[$.

Let us define a function $\phi(x)$ on $[a, b]$ as follows:

$$\phi(x) = f(x) + Ax \quad \dots(1)$$

Where A is a constant to be chosen as follows:

$$\phi(a) = \phi(b) \text{ i.e.,}$$

$$f(a) + Aa = f(b) + Ab$$

$$-Ab + Aa = f(b) - f(a)$$

$$A(b-a) = f(b) - f(a)$$

$$A = \frac{f(b) - f(a)}{b - a} \quad \dots(2)$$

Now $\phi(x)$ satisfies the following conditions:

(i) $\phi(x)$ is continuous in $[a, b]$, since $f(x)$ and Ax are both continuous in $[a, b]$ i.e., $f(x) + Ax$ is continuous in $[a, b]$.

(ii) $\phi(x)$ is derivable in $]a, b[$ since $f(x)$ and Ax are both derivable in $]a, b[$ i.e., $f(x) + Ax$ is derivable in $]a, b[$.

(iii) $\phi(a) = \phi(b)$

Since all the conditions of Rolle's theorem are satisfied, there exists some $\xi \in]a, b[$ s.t.

$$\phi'(\xi) = 0$$

From (2)

$$\phi'(x) = f'(x) + A$$

$$\text{Theorem Eq. (3)} \quad \phi'(\xi) = 0$$

$$f'(\xi) + A = 0$$

$$f'(\xi) = -A = -\frac{f(b) - f(a)}{b - a}$$

$$f'(\xi) = \frac{f(b) - f(a)}{b - a} \quad \dots(5)$$

Which prove Lagrange's mean value theorem.
Geometrical Interpretative of Lagrange's Mean Value Theorem:

If $A(a, f(a))$ and $B(b, f(b))$ are any two points on a continuous curve $y = f(x)$, then the slope of the chord AB is:

$$m = \frac{f(b) - f(a)}{b - a}$$

Hence, geometrically point of view Lagrange's mean value theorem states that there exists a point $x = \xi$ on the curve $y = f(x)$ lying of between A and B at which the tangent to the curve is parallel to the chord AB .

(b) If a function f is derivable at c and if $f'(c) \neq 0$, then $\frac{1}{f'}$ is also derivable at c and

$$\left(\frac{1}{f'}\right)'(c) = \frac{-f''(c)}{[f'(c)]^2}$$

Sol. Since f is derivable at c , so f is continuous at c .
Thus

$$\lim_{x \rightarrow c} f(x) = f(c)$$

and

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

$$\lim_{x \rightarrow c} \frac{\frac{1}{f(x)} - \frac{1}{f(c)}}{x - c} = \lim_{x \rightarrow c} \frac{\frac{f(c) - f(x)}{f(x)f(c)}}{x - c}$$

$$= \lim_{x \rightarrow c} \frac{-\{f(x) - f(c)\}}{(x - c)f(x)f(c)}$$

$$= -\frac{1}{f(c)} \lim_{x \rightarrow c} \frac{\{f(x) - f(c)\}}{(x - c)f(x)}$$

$$= -\frac{1}{f(c)} \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \cdot \lim_{x \rightarrow c} \frac{1}{f(x)}$$

$$= -\frac{1}{f(c)} f'(c) \cdot \frac{1}{f(c)}$$

$$= -\frac{f'(c)}{[f(c)]^2}$$

(c) A function $f(x)$ is defined as follows,

$$f(x) = \begin{cases} x^p \cos\left(\frac{1}{x}\right) & ; \text{ when } x \neq 0 \\ 0 & ; \text{ when } x = 0 \end{cases}$$

What conditions should be imposed on p so that f may be derivable at $x = 0$.

Sol. The given function $f(x)$ is defined as :

Proved.

$$f(x) = \begin{cases} x^p \cos\left(\frac{1}{x}\right) & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases} \quad \dots(1)$$

at $x=0$

$$\begin{aligned} \text{R.H.D} = R_f'(0) &= \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0^+} \frac{x^p \cos\left(\frac{1}{x}\right) - 0}{x - 0} \\ &= \lim_{x \rightarrow 0^+} x^{p-1} \cos\left(\frac{1}{x}\right) \\ &= \lim_{x \rightarrow 0^+} (0+h)^{p-1} \cos\left(\frac{1}{0+h}\right) \\ &= \lim_{x \rightarrow 0^+} h^{p-1} \cos\left(\frac{1}{h}\right) \end{aligned} \quad \dots(2)$$

$$\begin{aligned} \text{LHD} = L_f'(0) &= \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0^-} \frac{x^p \cos\left(\frac{1}{x}\right) - 0}{x - 0} \\ &= \lim_{x \rightarrow 0^-} x^{p-1} \cos\left(\frac{1}{x}\right) \\ &= \lim_{x \rightarrow 0^-} (0-h)^{p-1} \cos\left(\frac{1}{-h}\right) \\ &= \lim_{x \rightarrow 0^-} (-1)^{p-1} (h)^{p-1} \cos\left(\frac{1}{h}\right) \end{aligned} \quad \dots(3)$$

In order that f may be differentiable at $x=0$, the limits (2) and (3) both must be equal. Thus is possible only when $p > 1$ for in that case,

$$L_f'(0) = 0 = R_f'(0)$$

Hence, the required condition for differentiability of f at $x=0$ is $p > 1$.

Q. 5. (a) State and prove Taylor's theorem with Lagrange form of remainder after n terms.

Sol. Taylor's Theorem with Lagrange form of Remainder :

- If a function $f(x)$ is such that
- $f(x), f'(x), f''(x), \dots, f^{(n-1)}(x)$ are continuous in $[a, a+h]$
 - $f^{(n)}(x)$ exists in $]a, a+h[$
- Then $\exists$ at least one value θ between 0 and 1 s.t.

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{h^n}{n!} f^{(n)}(a+\theta h)$$

Where, $\frac{h^n}{n!} f^{(n)}(a+\theta h)$ is called Lagrange's form of remainder after n terms i.e.

$$R_n = \frac{h^n}{n!} f^{(n)}(a+\theta h)$$

Proof :

consider the function defined by :

$$\begin{aligned} F(x) &= f(x) + (a+h-x) f'(x) + \frac{(a+h-x)^2}{2!} f''(x) + \dots \\ &\quad + \frac{(a+h-x)^{n-1}}{(n-1)!} f^{(n-1)}(x) + \frac{(a+h-x)^n}{n!} A \end{aligned} \quad \dots(1)$$

where A is a constant which is to be determined s.t.

$$F(a) = F(a+h)$$

$$\begin{aligned} F(a) &= f(a) + (a+h-a) f'(a) + \frac{(a+h-a)^2}{2!} f''(a) + \dots \\ &\quad + \frac{(a+h-a)^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{(a+h-a)^n}{n!} A \\ &= f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) \\ &\quad + \frac{h^n}{n!} A \end{aligned} \quad \dots(2)$$

and

$$F(a+h) = f(a+h)$$

Hence,

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{h^n}{n!} A \dots (3)$$

As we are given that $f(x), f'(x), f''(x), \dots, f^{(n-1)}(x)$ are continuous in $[a, a+h]$ and $f'(x), f''(x), \dots, f^{(n-1)}(x)$ all exists in $]a, a+h[$.

$$A(x) = (a+h-x)f'(x) + \dots + \frac{(a+h-x)^{n-1}}{(n-1)!} f^{(n-1)}(x)$$

are continuous in $[a, a+h]$ and also differentiable in $]a, a+h[$.

$$\text{Thus } f(x) = (a+h-x)f'(x) + \frac{(a+h-x)^2}{2} f''(x) + \dots + \frac{(a+h-x)^{n-1}}{(n-1)!} f^{(n-1)}(x)$$

$\frac{(a+h-x)^n}{n!} A$ is also continuous in $[a, a+h]$ and differentiable in $]a, a+h[$.

$\therefore F(x)$ is continuous in $[a, a+h]$ and differentiable in $]a, a+h[$ and $F(a) = F(b)$.

Thus all the conditions of Rolle's theorem has been satisfied. Therefore $\exists$ at least one value θ between $0 < \theta < 1$ s.t.:

$$F'(a+\theta h) = 0$$

$$F'(x) = F'(x) - F'(x) + (a+h-x)F''(x) - (a+h-x)$$

$$F''(x) = \frac{(a+h-x)^{n-1}}{(n-1)!} f''(x) - \frac{(a+h-x)^{n-1}}{(n-1)!} A$$

$$F'(x) = \frac{(a+h-x)^{n-1}}{(n-1)!} f''(x) - \frac{A(a+h-x)^{n-1}}{(n-1)!}$$

$$F'(x) = [f''(x) - A] \frac{(a+h-x)^{n-1}}{(n-1)!}$$

$$F'(x) = [f''(x) - A] \frac{h^{n-1}(1-\theta)^{n-1}}{(n-1)!}$$

$$F'(a+\theta h) = [f''(a+\theta h) - A] \frac{h^{n-1}(1-\theta)^{n-1}}{(n-1)!}$$

$$F'(a+\theta h) = 0$$

Since,

$$0 = [f''(a+\theta h) - A] \frac{h^{n-1}(1-\theta)^{n-1}}{(n-1)!}$$

Since,

$$0 < \theta < 1$$

$$1-\theta \neq 0 \text{ moreover } h \neq 0$$

$$f''(a+\theta h) = 0$$

$$A = f''(a+\theta h)$$

On putting above value of A in Eq. (3)

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{h^n}{n!} f''(a+\theta h) \dots (4)$$

Which proves Taylor's theorem with Lagrange form of remainder.

(b) Find the Lagrange's and Cauchy's remainder after n times in the expansion of e^x .

Sol. Let

$$f(x) = e^x$$

$$f^n(x) = e^x$$

(i) Lagrange's Remainder is calculated as-

$$R_n = \frac{x^n}{n!} f^n(\theta x)$$

$$= \frac{x^n}{n!} (e^{\theta x})$$

$$= \frac{x^n}{n!} e^{\theta x}$$

Ans.

(ii) Cauchy's Remainder is calculated as-

$$R_n = \frac{x^n}{(n-1)!} (1-\theta)^{n-1} f^n(\theta x)$$

$$= \frac{x^n}{(n-1)!} (1-\theta)^{n-1} (e^{\theta x})$$

$$= \frac{x^n}{(n-1)!} (1-\theta)^{n-1} (e^{\theta x})$$

Ans.

(c) Expand e^{ax} using Maclaurin's series.

Sol. Let

$$y = e^{ax}$$

$$y_n = a^n e^{ax}$$

At $x=0$

...(1)

...(2)

$$(y)_0 = e^0 = 1$$

$$(y_n)_0 = a^n e^0 = a^n (1) = a^n \quad \dots(3)$$

On puttey

$$n = 1, 2, 3, 4, \dots \text{ in Eq. (4)} \quad \dots(4)$$

$$n = 1 \Rightarrow (y_1)_0 = a$$

$$n = 2 \Rightarrow (y_2)_0 = a^2$$

$$n = 3 \Rightarrow (y_3)_0 = a^3$$

$$n = 4 \Rightarrow (y_4)_0 = a^4$$

Using Maclaurin's series, we have

$$y = (y)_0 + x(y_1)_0 + \frac{x^2}{2!}(y_2)_0 + \dots + \frac{x^n}{n!}(y_n)_0$$

$$\Rightarrow e^{ax} = 1 + x(a) + \frac{x^2}{2!}(a^2) + \frac{x^3}{3!}(a^3) + \dots + \frac{x^n}{n!}(a^n)$$

$$\Rightarrow e^{ax} = 1 + ax + \frac{a^2 x^2}{2!} + \frac{a^3 x^3}{3!} + \dots + \frac{a^n x^n}{n!} + \dots \quad \text{Ans.}$$

Q. 6. (a) Derive expresive for Taylor series.

Sol. Taylors Series:

Let a function f posses continuous derivative of every order in $[a, a+h]$, then for all $n \in \mathbb{N}$ we have by Taylor's theorem:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a)$$

$$+ \frac{h^n}{n!}f^{(n)}(a+\theta h) \quad \dots(1)$$

where

$$0 < \theta < 1$$

Let

$$S_n = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a)$$

$$R_n = \frac{h^n}{n!}f^{(n)}(a+\theta h) \quad \dots(2)$$

Here, R_n is the taylor's reminder after n times using (2), Eq. (1) becomes—

$$f(a+h) = S_n + R_n \quad \dots(3)$$

For $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} (R_n) = \lim_{n \rightarrow \infty} \frac{h^n}{n!}f^{(n)}(a+\theta h) = 0$$

Eq. (3) gives—

$$f(a+h) = \lim_{n \rightarrow \infty} S_n + \lim_{n \rightarrow \infty} R_n$$

$$f(a+h) = \lim_{n \rightarrow \infty} S_n + 0$$

$$\lim_{n \rightarrow \infty} S_n = f(a+h)$$

Thus, the infinte series $f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + \dots$ converges to $f(a+h)$.

This if

(i) f passesses continuous derivatives of every order in $[a, a+h]$

(ii) The taylor's reminder after n terms i.e. $R_n = 0$ as $n \rightarrow \infty$

Then,

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^n}{n!}f^{(n)}(a) + \dots$$

Which is called as taylor's infinte series.

(b) Expand e^x in ascending power of $(x-1)$.

Sol. Let

$$f(x) = e^x$$

$$f(x) = f(1+x-1)$$

$$= f(1+(x-1))$$

$$= f(1+h)$$

Where,

$$h = x-1$$

$$f(x) = e^x$$

$$f'(x) = e^x$$

$$f''(x) = e^x$$

$$f'''(x) = e^x$$

$$f^{(4)}(x) = e^x$$

At $x=1$

$$f(1) = e$$

$$f'(1) = e$$

$$f''(1) = e$$

$$f'''(1) = e$$

$$f^{(4)}(1) = e$$

Using taylor's series we get:

$$f(1+h) = f(1) + hf'(1) + \frac{h^2}{2!}f''(1) + \dots$$

$$= e^x = \frac{h}{1!} (x) + \frac{h^2}{2!} (x^2) + \dots$$

$$= \left(1 + \frac{h}{1!} + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots \right)$$

On putting

h = 1-x, we get:

$$f(x) = e^x \left[1 + \frac{(1-x)}{1!} + \frac{(1-x)^2}{2!} + \frac{(1-x)^3}{3!} + \dots \right]$$

$$= e^x \left[1 + \frac{(1-x)}{1!} + \frac{(1-x)^2}{2!} + \frac{(1-x)^3}{3!} + \dots \right] \quad \text{Ans.}$$

(c) If $y = e^m \tan^{-1} x$, then prove that

$$y = 1 + mx + \frac{m^2}{2} x^2 + \frac{m(m^2-2)}{3} x^3 + \dots$$

Sol.

$$y = e^m \tan^{-1} x$$

d.w.r.t.x

$$y_1 = \tan^{-1} x \cdot \frac{m}{1+x^2}$$

$$y_1 = y \cdot \frac{m}{1+x^2}$$

$$(1+x^2) y_1 = m y$$

d.w.r.t.x

$$(1+x^2) y_2 + (2x) y_1 = m y_1$$

$$(1+x^2) y_2 + (2x-m) y_1 = 0$$

Using Leibnitz theorem diff. Eq. (3) n terms-

$$D^n [(1+x^2) y_2 + (2x-m) y_1] = 0$$

$$= [y_{n+2} (1+x^2) + n C_1 y_{n+1} (2x) + n C_2 y_n (2)]$$

$$+ [y_{n+1} (2x-m) + n C_1 y_n (2)] = 0$$

$$\Rightarrow y_{n+2} (1+x^2) + n y_{n+1} (2x) + \frac{n(n-1)}{2} y_n (2) + y_{n+1} (2x-m) + n y_n (2) = 0$$

$$\Rightarrow (1+x^2) y_{n+2} + y_{n+1} [2nx + 2x - m] + y_n [n^2 - n + 2n] = 0$$

$$\Rightarrow (1+x^2) y_{n+2} + y_{n+1} [2nx + 2x - m] + y_n [n^2 - n + 2n] = 0$$

$$\Rightarrow (1+x^2) y_{n+2} + [(n+1) 2x - m] y_{n+1} + n(n+1) y_n = 0$$

At $x = 0$

$$(y)_0 = e^0 = 1$$

$$(y_1)_0 = m$$

$$(y_2)_0 = m^2$$

$$(y_{n+2})_0 = m(y_{n+1})_0 - n(n+1)(y_n)_0$$

$$n = 1, 2, 3, \dots \text{in Eq. (5)}$$

$$(y_3)_0 = m(y_2)_0 - 1 \cdot 2 (y_1)_0$$

$$= m(m^2) - 1 \cdot 2 (m)$$

$$= m^3 - 2m = m(m^2 - 2)$$

$$(y_4)_0 = m(y_3)_0 - 2 \cdot 3 (y_2)_0$$

$$= m(m^3 - 2m) - 6 m^2$$

$$= m^4 - 2m^2 - 6m^2$$

$$= m^4 - 8m^2$$

$$= m^2 (m^2 - 8)$$

Using Maclaurien series:

$$y = (y)_0 + x (y_1)_0 + \frac{x^2}{2!} (y_2)_0 + \frac{x^3}{3!} (y_3)_0 + \dots$$

$$y = (1) + x(m) + \frac{x^2}{2!} (m^2)_0 + \frac{x^3}{3!} [m(m^2 - 2)]$$

$$+ \frac{x^4}{4!} [m^2(m^2 - 8)] + \dots$$

$$= 1 + mx + \frac{x^2}{2!} m^2 + \frac{m(m^2 - 2)}{3!} x^3 + \dots \quad \text{Proved.}$$

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Name of the Paper : Paper : C-5 : Theory of Real Functions
 Name of Course : B.Sc. (Hons.) Mathematics II Year (CBCS)
 Semester : III
 Duration : 3 Hours
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MODEL TEST PAPER-II

Q. 1. (a) Prove that the limit of a function at a point, when it exists, is unique.

Sol. Proof:

Suppose $\lim_{x \rightarrow a} f(x)$ exists and is not unique.

Let $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} f(x) = l'$

Where $l \neq l'$

Now, $l = l' \Rightarrow |l - l'| > 0$... (1)

If we take $\epsilon = \frac{1}{2} |l - l'| > 0$, then

$\lim_{x \rightarrow a} f(x) = l \Rightarrow$ given $\epsilon > 0, \exists \delta_1 > 0$ s.t.

$|f(x) - l| < \epsilon$, whenever $0 < |x - a| < \delta_1$... (2)

and

$\lim_{x \rightarrow a} f(x) = l' \Rightarrow$ given $\epsilon > 0, \exists \delta_2 > 0$ s.t.

$|f(x) - l'| < \epsilon$, whenever $0 < |x - a| < \delta_2$... (3)

Let $\delta = \min(\delta_1, \delta_2)$, then from (2) and (3),

We have:

$|f(x) - l| < \epsilon$ and $|f(x) - l'| < \epsilon$

Whenever

$0 < |x - a| < \delta$

$$|l - l'| = |l - f(x) + f(x) - l'|$$

$$|l - l'| \leq |l - f(x)| + |f(x) - l'|$$

$$|l - l'| \leq |f(x) - l| + |f(x) - l'|$$

$$|l - l'| < \epsilon + \epsilon$$

$$|l - l'| < 2\epsilon, \text{ whenever } 0 < |x - a| < \delta$$

$$|l - l'| < |l - l'| \text{ whenever } 0 < |x - a| < \delta$$

$\Rightarrow$ Which is absurd therefore our supposition is wrong. Hence $l = l'$, which

process that $\lim_{x \rightarrow a} f(x)$, if it exists is unique

(b) Using $f(x) = x \sin\left(\frac{1}{x}\right)$ approach, prove that

$$\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$$

(22)

Sol. We show that $\lim_{x \rightarrow 0} f(x) = l$, iff given $\epsilon > 0$, there exists a positive real number δ depending upon ϵ s.t.

$$|f(x) - l| < \epsilon \text{ for } 0 < |x - a| < \delta$$

Let

$$f(x) = x \sin\left(\frac{1}{x}\right)$$

$$|f(x) - 0| = \left| x \sin\left(\frac{1}{x}\right) - 0 \right|$$

$$= \left| x \sin\left(\frac{1}{x}\right) \right|$$

$$= |x| \left| \sin\left(\frac{1}{x}\right) \right|$$

$$\leq |x| \text{ as } \left| \sin\left(\frac{1}{x}\right) \right| \leq \left| \sin\left(\frac{1}{x}\right) \right| \leq 1$$

$$|f(x) - 0| < \epsilon, \text{ whenever } 0 < |x| < \epsilon$$

We choose

$$\delta = \epsilon, \text{ then:}$$

$$|f(x) - 0| < \epsilon, \text{ whenever } 0 < |x| < \delta$$

$$|f(x) - 0| < \epsilon, \text{ whenever } 0 < |x - 0| < \delta$$

Hence,

$$\lim_{x \rightarrow 0} f(x) = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$$

Proved.

(c) IF $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$ then prove that

$$\lim_{x \rightarrow a} [f(x) g(x)] = lm$$

Sol.

$$\lim_{x \rightarrow a} f(x) = l$$

$$\lim_{x \rightarrow a} g(x) = m$$

$$|f(x) g(x) - lm| = |f(x) g(x) - l g(x) + l g(x) - lm|$$

$$= |g(x) (f(x) - l) + l (g(x) - m)|$$

$$\leq |g(x) (f(x) - l)| + |l (g(x) - m)|$$

$$= |g(x)| |f(x) - l| + |l| |g(x) - m|$$

Since $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$ for $0 < \epsilon, < 1$, there exists some $\delta < 0$

such that $|f(x) - l| < \epsilon$ and $|g(x) - m| < \epsilon$, $0 < |x - a| < \delta$... (3)

$$|g(x)| = |m + g(x) - m|$$

$$|g(x)| \leq |m| + |g(x) - m|$$

$$|g(x)| < |m| + \epsilon, \text{ when } 0 < |x - a| < \delta \quad \dots (4)$$

From (2), (3) and (4), we have:

$$|f(x)g(x) - lm| \leq |g(x)||f(x) - l| + |l||g(x) - m|$$

$$|f(x)g(x) - lm| < (|m| + \epsilon) + |l|\epsilon, \text{ whenever } 0 < |x - a| < \delta$$

$$|f(x)g(x) - lm| < (|m| + |l| + 1)\epsilon, \text{ whenever } 0 < |x - a| < \delta$$

$$(\because \epsilon' < 1)$$

Let us choose ϵ' such that

$$\epsilon' < \frac{\epsilon}{(|m| + |l| + 1)}, \epsilon > 0$$

$$|f(x)g(x) - lm| < \epsilon, \text{ when } 0 < |x - a| < \delta$$

Hence,

$$\lim_{x \rightarrow a} f(x)g(x) = lm$$

Proved.

Q. 2. (a) If a function f is continuous in $[a, b]$ and $f(a)$ and $f(b)$ are of opposite sign, then there exists some point $c \in]a, b[$ such that $f(c) = 0$.

Sol. Since, $f(a)$ and $f(b)$ are of opposite sign. We may take,

$$f(a) > 0$$

$$f(b) < 0$$

and

Let S be a subset of $[a, b]$ defined as follows,

$$S = \{x : a \leq x \leq b \text{ and } f(x) > 0\} \quad \dots (1)$$

Then S is non empty and S is bounded also with b as its upper bound.

By order completeness property of $\mathbb{R}$, S has the super mum.

Let $c = \sup S, c \in [a, b]$

Now, we shall prove that $c \neq a$ and $c \neq b$ that $c \in]a, b[$.

Since, f is continuous at $x = a$ and $f(a) > 0, \exists \alpha, \delta_1 > 0$ such that:

$$f(x) > 0 \forall x \in [a, a + \delta_1]$$

$$[a, a + \delta_1] \subset S$$

If $c = a$, then $\sup S = a \Rightarrow$ every element of $S \leq a$

$$a + \delta_1 \leq a$$

This is impossible and so $c \neq a$

Now, we show that $c \neq b$

Since,

$$f(b) < 0, \exists \alpha, \delta_2 > 0 \text{ s.t.}$$

$$f(x) < 0 \forall x \in]b - \delta_2, b]$$

$\Rightarrow b - \delta_2$ is an upper bound of S for otherwise, there exists some $y \in S$ s.t.

$$b - \delta_2 < y \leq b$$

$$\Rightarrow f(y) < 0$$

As $y \in S, f(y) > 0$ by (1). This contradicts (3).

Since $c = \sup S$, so $c \leq b - \delta_2 \Rightarrow c \neq b$

Thus, $c \neq a$ and $c \neq b \Rightarrow c \in]a, b[$

Finally, we show that $f(c) = 0$

Case I : Let $f(c) > 0$

Since, f is continuous at c and $f(c) > 0$, so $\exists \alpha, \delta_3 > 0$ s.t.

$$f(x) > 0 \forall x \in]c - \delta_3, c + \delta_3[$$

Let a be a number s.t. $c < a < c + \delta_3$

$$\therefore a \in]c, c + \delta_3[$$

$$\Rightarrow f(a) = 0 \text{ by (4)}$$

$$\Rightarrow a \in S \text{ by (1).}$$

$$a \leq c$$

$$c \geq a$$

Which contradicts Eq. (2).

Thus, $f(c) \neq 0$

Case II : Let $f(c) < 0$

Since, f is continuous at $x = c$ and $f(c) < 0$, so $\exists \alpha, \delta_4 > 0$ s.t.

$$f(x) < 0 \forall x \in]c - \delta_4, c + \delta_4[$$

Now,

$$c = \sup S \Rightarrow \exists \text{ at least one } \beta \in S \text{ s.t.}$$

$$c - \delta_4 < \beta \leq c$$

$$\Rightarrow \beta \in]c - \delta_4, c[$$

$$\Rightarrow f(\beta) < 0 \text{ using (6)}$$

$$\text{But, } \beta \in S \Rightarrow f(\beta) > 0$$

Which is a contradiction

$$f(c) \neq 0$$

Since, $f(c) \neq 0$ and also $f(c) \neq 0$

Hence, $f(c) = 0$ for some $c \in]a, b[$.

Which proves given theorem.

(b) Using $\epsilon - \delta$ definition, prove that

$$f(x) = \cos x \text{ is continuous.}$$

Sol. Let

$$f(x) = \cos x$$

Let α be any real number, so that $f(\alpha) = \cos \alpha$

Let $\epsilon < 0$ be given

$$|f(x) - f(\alpha)| = |\cos x - \cos \alpha|$$

$$= \left| -2 \sin \left(\frac{x + \alpha}{2} \right) \sin \left(\frac{x - \alpha}{2} \right) \right|$$

$$\begin{aligned}
 &= 2 \left| \sin \left(\frac{x+\alpha}{2} \right) \right| \left| \sin \left(\frac{x-\alpha}{2} \right) \right| \\
 &\leq 2 \left| \sin \left(\frac{x-\alpha}{2} \right) \right| \text{ because } \left| \sin \frac{x+\alpha}{2} \right| \leq 1 \\
 &< 2 \left| \frac{x-\alpha}{2} \right| \\
 &= |x-\alpha|
 \end{aligned}$$

$$= |x-\alpha| < \epsilon, \text{ whenever } |x-\alpha| < 1$$

$\therefore$ If we choose

$$\delta = \epsilon, \text{ then,}$$

$$|f(x) - f(\alpha)| < \epsilon, \text{ whenever } |x - \alpha| < \delta$$

$\Rightarrow f$ is continuous at $x = \alpha$

$\Rightarrow f$ is continuous at every $\alpha \in \mathbb{R}$

$\Rightarrow f$ is continuous.

(c) Examine the continuity of the following function at the indicated point. Also, point out the type of discontinuity, if any.

$$\begin{aligned}
 f(x) &= \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}; x \neq 0 \\
 &= 0; x = 0
 \end{aligned}$$

at $x = 0$.

Sol.

$$\begin{aligned}
 f(x) &= \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}; x \neq 0 \\
 &= 1; x = 0
 \end{aligned}$$

at $x = 0$

r. h. l.

$$\begin{aligned}
 f(0) &= 1 \\
 &= f(0+0) \\
 &= \lim_{x \rightarrow 0^+} f(x)
 \end{aligned}$$

$$= \lim_{x \rightarrow 0^+} \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}$$

$$= \lim_{h \rightarrow 0^+} \frac{e^{\frac{1}{0+h}} - e^{-\frac{1}{0+h}}}{e^{\frac{1}{0+h}} + e^{-\frac{1}{0+h}}}$$

$$= \lim_{h \rightarrow 0^+} \frac{e^{\frac{1}{h}} - e^{-\frac{1}{h}}}{e^{\frac{1}{h}} + e^{-\frac{1}{h}}}$$

$$= \lim_{h \rightarrow 0^+} \frac{e^{\frac{1}{h}} - (1 - e^{-\frac{1}{h}})}{e^{\frac{1}{h}} + (1 + e^{-\frac{1}{h}})}$$

$$= \lim_{h \rightarrow 0^+} \frac{(1 - e^{-\frac{1}{h}})}{(1 + e^{-\frac{1}{h}})}$$

$$= \frac{1 - e^{-\infty}}{1 + e^{-\infty}}$$

$$= \frac{1 - 0}{1 + 0}$$

$$= 1$$

$$\text{l.h.l.} = f(0-0)$$

$$= \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{\frac{1}{h}} - e^{-\frac{1}{h}}}{e^{\frac{1}{h}} + e^{-\frac{1}{h}}}$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{\frac{1}{h}} - e^{-\frac{1}{h}}}{e^{\frac{1}{h}} + e^{-\frac{1}{h}}}$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{\frac{1}{h}} - e^{-\frac{1}{h}}}{e^{\frac{1}{h}} + e^{-\frac{1}{h}}}$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{\frac{1}{h}} (e^{-\frac{1}{h}} - 1)}{e^{\frac{1}{h}} (e^{-\frac{1}{h}} + 1)}$$

$$= \lim_{h \rightarrow 0^-} \frac{e^{-\frac{1}{h}} - 1}{e^{-\frac{1}{h}} + 1}$$

$$= \frac{1 - 1}{1 + 1}$$

$$= \frac{0-1}{0+1} \\ = -1$$

$\lim_{x \rightarrow 0^-} f(x)$ exist and equal to -1

and $\lim_{x \rightarrow 0^+} f(x)$ exist and equal to 1

$$\Rightarrow \lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$$

$\lim_{x \rightarrow 0} f(x)$ does not exist

$\Rightarrow f(x)$ is not continuous at $x=0$

Type of discontinuity :

Since, $\lim_{x \rightarrow 0^-} f(x)$ exists and $\lim_{x \rightarrow 0^+} f(x) = 1$

and $\lim_{x \rightarrow 0^-} f(x)$ exists and $\lim_{x \rightarrow 0^+} f(x) = -1$

Hence, $f(x)$ has a discontinuity of first kind at $x=0$.

Q. 3. (a) Define Uniform continuity of a function on the interval I. Show that the function defined by $f(x) = x^2$ is uniformly continuous in $[-2, 2]$.

Sol. Uniform continuity : A function f defined on an interval I is said to be uniformly continuous in the interval I, if for each $\epsilon > 0$, there exists some $\delta > 0$ such that

$$|f(x_2) - f(x_1)| < \epsilon, \text{ where } |x_2 - x_1| < \delta \text{ and for all } x_1, x_2 \in I$$

Solution of the problem :

Let $\epsilon > 0$ be any number and $x_1, x_2 \in (-2, 2)$. Then,

$$\begin{aligned} |f(x_2) - f(x_1)| &= |x_2^2 - x_1^2| \\ &= |(x_2 - x_1)(x_2 + x_1)| \\ &= |x_2 - x_1| |x_2 + x_1| \\ &\leq |x_2 - x_1| (|x_2| + |x_1|) \\ &\Rightarrow 4|x_2 - x_1| \text{ as } x_1 < 2, x_2 < 2 \end{aligned}$$

$$|f(x_2) - f(x_1)| < \epsilon, \text{ when } |x_2 - x_1| < \frac{\epsilon}{4}$$

$$|f(x_2) - f(x_1)| < \epsilon, \text{ when } |x_2 - x_1| < \delta$$

where

$$\delta = \frac{\epsilon}{4} \forall x_1, x_2 \in (-2, 2)$$

Hence, the function $f(x)$ is uniformly continuous in $(-2, 2)$.

(b) Prove that $f(x) = \sin x^2$ is not uniformly continuous on $[0, \infty]$.

Sol. $f(x) = \sin x^2$

Let $\epsilon = \frac{1}{2}$ and δ be any positive number. We can choose a positive integer n such that

$$n > \frac{\pi}{\delta^2}$$

Let

$$x_1 = \sqrt{\frac{n\pi}{2}}, x_2 = \sqrt{(n+1)\frac{\pi}{2}} \in [0, \infty], \text{ then}$$

$$\begin{aligned} |f(x_2) - f(x_1)| &= |\sin x_2^2 - \sin x_1^2| \\ &= \left| \sin(n+1)\frac{\pi}{2} - \sin \frac{n\pi}{2} \right| \\ &= \begin{cases} |0 - (\pm 1)| = 1 & \text{if } n \text{ is odd} \\ |(\pm 1) - 0| = 1 & \text{if } n \text{ is even} \end{cases} \end{aligned}$$

$$\therefore |f(x_2) - f(x_1)| = 1 > \epsilon$$

and

$$|x_2 - x_1| = \left| \sqrt{\frac{x_2^2}{2}} - \sqrt{\frac{x_1^2}{2}} \right|$$

$$= \left| \sqrt{\frac{(n+1)\pi}{2}} - \sqrt{\frac{n\pi}{2}} \right|$$

$$< \frac{\pi}{2 \left(2 \sqrt{\frac{n\pi}{2}} \right)}$$

$$< \frac{\pi}{n\pi}$$

$$= \frac{1}{n} < \delta, \text{ using (1)}$$

Hence, $|f(x_2) - f(x_1)| > \epsilon$ when $|x_2 - x_1| < \delta$
 $f(x) = \sin x^2$ is not uniformly continuous on $[0, \infty]$.

(c) Show that the function f defined by

$$f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ -x & \text{if } x \text{ is irrational} \end{cases}$$

is continuous only at $x = 0$.

Sol.

$$f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ -x & \text{if } x \text{ is irrational} \end{cases} \quad \dots(1)$$

Let x be any real number. For each $n \in \mathbb{N}$, $\exists$ a rational number a_n and an irrational number b_n s.t.

$$x - \frac{1}{n} < a_n < x + \frac{1}{n} \text{ and } x - \frac{1}{n} < b_n < x + \frac{1}{n}$$

$$\Rightarrow |a_n - x| < \frac{1}{n} \text{ and } |b_n - x| < \frac{1}{n} \forall n$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = x = \lim_{n \rightarrow \infty} b_n \quad \dots(2)$$

If f is continuous at x , we must have

$$\lim_{n \rightarrow \infty} f(a_n) = f(x) = \lim_{n \rightarrow \infty} f(b_n)$$

$$\text{But, } f(a_n) = a_n \text{ and } f(b_n) = b_n$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = f(x) = \lim_{n \rightarrow \infty} b_n$$

$$\Rightarrow x = f(x) = -x$$

$$\Rightarrow 2x = 0$$

$$\Rightarrow x = 0$$

Thus, 0 is the only possible point of continuity.

Now, we shall show that f is actually continuous at 0.

Let $\epsilon > 0$ be given. Also $f(0) = 0$

For a rational number x , we have:

$$|f(x) - f(0)| = |x - 0| = |x|$$

For an irrational number x , we have

$$|f(x) - f(0)| = |-x - 0| = |-x| = |x|$$

In either case,

$$|f(x) - f(0)| = |x| < \epsilon \text{ whenever } |x| < \epsilon$$

We choose

$$\delta = \epsilon.$$

Then

$$|f(x) - f(0)| < \epsilon, \text{ whenever } |x| < \delta$$

$\Rightarrow f$ is continuous at $x = 0$

Q. 4. (a) State and prove Darboux's Theorem.

Sol. Darboux's Theorem.

If a function f defined in $[a, b]$ is such that

(i) f is derivable in $[a, b]$

(ii) $f'(a)$ and $f'(b)$ are of opposite sign

Then there exists some point $c \in [a, b]$ such that $f'(c) = 0$

Proof:

Let us suppose that $f'(a) > 0$ and $f'(b) < 0$

Since, $f'(a) > 0 \Rightarrow \exists$ some $\delta_1 > 0$ s.t.

$$f(x) > f(a) \forall x \in [a, a + \delta_1] \quad \dots(1)$$

Since, $f'(b) < 0 \Rightarrow \exists$ some $\delta_2 > 0$ s.t.

$$f(x) > f(b) \forall x \in [b - \delta_2, b] \quad \dots(2)$$

Since, f is derivable in $[a, b]$, f is continuous in $[a, b]$ and hence bounded. Theorem and the bounds are attained.

Let $K = \sup f$ and $k = \inf f$. Then, $\exists$ points $c, d \in [a, b]$ s.t.

$$f(c) = K \text{ and } f(d) = k \quad \dots(3)$$

$$\text{Now, } K = \sup f \Rightarrow f(x) \leq K \forall x \in (a, b) \quad \dots(4)$$

We shall show that $c \neq a$ and $c \neq b$

Let if possible $c = a$. Then $f(c) = f(a) = K$. $f'(a)$ using (3)

$$\Rightarrow f(x) > K \forall x \in [a, a + \delta_1] \text{ using (1)}$$

This contradicts (4). Thus $c \neq a$.

Similarly, we can show that $c \neq b$. Thus, $c \in (a, b)$

Finally, we show that $f'(c) = 0$.

Case (1): Let $f'(c) > 0$

$$\Rightarrow \exists \text{ some } \delta_3 > 0 \text{ s.t. } f(x) > f(c) \forall x \in [c, c + \delta_3]$$

$$\Rightarrow f(x) = K \forall x \in [c, c + \delta_3] \text{ using (3)}$$

This contradicts (4) and so $f'(c) \not> 0$.

Case (2): Let $f'(c) < 0$

$$\Rightarrow \exists \text{ some } \delta_4 > 0 \text{ s.t. } f(x) > f(c) \forall x \in [c - \delta_4, c]$$

$$\Rightarrow f(x) > K \forall x \in [c - \delta_4, c], \text{ using (2)}$$

This contradicts (4) and so $f'(c) \not< 0$.

Hence, from case (1) and (2), we have

$$f'(c) = 0.$$

(b) If f and g are two derivable functions at $x = c$, then fg is also derivable at $x = c$ and

$$(fg)'(c) = f(c)g'(c) + g(c)f'(c)$$

Sol. Since, f and g are derivable at $x = c$, therefore:

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

$$\text{and } f'(c) = \frac{f(x) - f(c)}{x - c} \quad \dots(1)$$

Since, f and g are derivable at $x = c$, so f and g are continuous at $x = c$. Thus,

$$\lim_{x \rightarrow c} f(x) = f(c)$$

and
Now,

$$\lim_{x \rightarrow c} g(x) = g(c) \quad \dots(2)$$

$$\begin{aligned} \lim_{x \rightarrow c} \frac{(fg)(x) - (fg)(c)}{x - c} &= \lim_{x \rightarrow c} \frac{f(x)g(x) - f(c)g(c)}{x - c} \\ &= \lim_{x \rightarrow c} \frac{f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c)}{x - c} \\ &= \lim_{x \rightarrow c} \frac{f(x)[g(x) - g(c)] + g(c)[f(x) - f(c)]}{x - c} \\ &= \lim_{x \rightarrow c} \frac{f(x)[g(x) - g(c)]}{x - c} + \lim_{x \rightarrow c} \frac{g(c)[f(x) - f(c)]}{x - c} \\ &= \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c} + g(c) \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \\ &= f(c) \cdot g'(c) + g(c) f'(c) \end{aligned}$$

Which shows that fg is derivable at c and

$$(fg)'(c) = f(c)g'(c) + g(c)f'(c)$$

(c) Define Rolle's Theorem and also give its geometrical interpretation. Proved.

Sol. Rolle's Theorem:

If $f(x)$ be a function such that:

- (i) $f(x)$ is continuous in the closed interval $[a, b]$
- (ii) $f'(x)$ exists for every point in the open interval (a, b)
- (iii) $f(a) = f(b)$

then, there exists at least a point c at which $f'(c) = 0$, where $c \in (a, b)$

Proof:

The function $f(x)$ being continuous in the closed interval $[a, b]$, it is bounded and attains its bounds in the interval $[a, b]$. Let the l.u.b. and g.l.b. of the function $f(x)$ in the interval $[a, b]$ be M and m .

Therefore there exist two number c_1 and c_2 in the closed interval $[a, b]$ s.t. $f(c_1) = M$ and $f(c_2) = m$.

Now, two case arises:

Case I : When $M = m$

Then, $f(x) = M = m$ at each point in $[a, b]$ i.e., $f(x)$ is constant for every $x \in [a, b]$

$$f'(x) = 0 \quad \forall x \in [a, b]$$

Hence, the theorem is true for any value of x in the open interval (a, b) .

Case II : When $M \neq m$

In this case either M or m will be different from $f(a)$, which also equal to $f(b)$.

Let us suppose that $M = f(c_1)$ is different from $f(a)$ and $f(b)$

The number c_1 is different from a and b so lies within the open interval (a, b) . As the function $f(x)$ is differentiable in the open interval (a, b) , so it is also differentiable at $x = c_1$.

Hence,

$$\lim_{x \rightarrow c_1} \frac{f(x) - f(c_1)}{x - c_1} \text{ and } \lim_{x \rightarrow c_1} \frac{f(x) - f(c_1)}{x - c_1}$$

Also, $f(c_1) = M$ being the greatest value of the function $f(x)$ in the open interval (a, b) , so :

$$f(c_1 + h) \leq f(c_1) \text{ and } f(c_1 - h) \leq f(c_1) \quad \dots(1)$$

$$Rf'(c_1) = \lim_{h \rightarrow 0} \frac{f(c_1 + h) - f(c_1)}{h}$$

$$Rf'(c_1) \leq 0 \text{ as } f(c_1 + h) \leq f(c_1) \quad \dots(2)$$

and

$$Lf'(c_1) = \lim_{h \rightarrow 0} \frac{f(c_1 - h) - f(c_1)}{-h}$$

$$Lf'(c_1) \geq 0 \text{ as } f(c_1 - h) \leq f(c_1)$$

From (2) and (3), we have :

$$Rf'(c_1) = Lf'(c_1) \text{ if } Rf'(c_1) = Lf'(c_1) = 0$$

$$f'(c_1) = 0$$

Similarly, when $f(c_2) = m$ and $f(c_2)$ is neither equal to $f(a)$ nor $f(b)$, we can prove that

$$f'(c_2) = 0$$

Thus, we find that,

$$f'(c) = 0, \text{ where } c \in (a, b)$$

Which proves Rolle's theorem.

Geometrical Interpretation of Rolle's Theorem:

If the graph of the function $f(x)$ be drawn between $x = a$ and $x = b$, then it is a continuous curve between $x = a$ and $x = b$ having a unique tangent at all points in the above interval and $f(a) = f(b)$.

Rolle's theorem asserts that there exists at least one point $x = c$ on the curve between $x = a$ and $x = b$ at which the tangent to the curve is parallel to the axis of x .

Q. 6. (a) State and prove Taylor's theorem with Cauchy's form of remainder.

Sol Taylor's Theorem with Cauchy's form of Remainder :
If a function $f(x)$ s.t.

- $f(x), f'(x), f''(x), \dots, f^{(n-1)}(x)$ are continuous in $[a, a+h]$
- $f^{(n)}(x)$ exists in $]a, a+h[$

Then $\exists$ at least one value θ between 0 and 1 s.t.

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots +$$

$$+ \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{h^n}{n!} (1-\theta)^{n-1} f^{(n)}(a+\theta h)$$

Where, $\frac{h^n}{n!} (1-\theta)^{n-1} f^{(n)}(a+\theta h)$ is called Cauchy's remainder after n terms

i.e.

$$R_n = \frac{h^n}{n!} (1-\theta)^{n-1} f^{(n)}(a+\theta h)$$

Proof:

Consider the function defined by :

$$F(x) = f(x) + (a+h-x)f'(x) + \frac{(a+h-x)^2}{2!} f''(x) + \dots + \frac{(a+h-x)^{n-1}}{(n-1)!} f^{(n-1)}(x) + (a+h-x)A \dots (1)$$

where A is a constant which is to be determined s.t.

$$F(a) = F(a+h)$$

$$F(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots +$$

$$+ \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + h \cdot A \dots (2)$$

As we are given that $f(x), f'(x), f''(x), \dots, f^{(n-1)}(x)$ are continuous in $[a, a+h]$ and $f'(x), f''(x), \dots, f^{(n)}(x)$ all exists in $]a, a+h[$.

$\therefore f(x), (a+h-x)f'(x), \frac{(a+h-x)^2}{2!} f''(x), \dots, \frac{(a+h-x)^{n-1}}{(n-1)!} f^{(n-1)}(x)$ are continuous in $[a, a+h]$ and differentiable in $]a, a+h[$.
Thus,

$$f(x) + (a+h-x)f'(x) + \frac{(a+h-x)^2}{2!} f''(x) + \dots + \frac{(a+h-x)^{n-1}}{(n-1)!} f^{(n-1)}(x) + (a+h-x)A$$

is also continuous in $[a, a+h]$ and differentiable in $]a, a+h[$.
 $\Rightarrow F(x)$ is continuous in $[a, a+h]$ and differentiable in $]a, a+h[$ and $F(a) = F(a+h)$.

Thus all the conditions of Rolle's theorem has been satisfied. Therefore at least are value θ between 0 and 1 s.t.

$$F'(a+\theta h) = 0$$

$$F'(x) = f'(x) - f'(x) + (a+h-x)f''(x) - (a+h-x)f''(x) + \dots + \frac{(a+h-x)^{n-2}}{(n-2)!} f^{(n-1)}(x) - A$$

$$F'(x) = \frac{(a+h-x)^{n-1}}{(n-1)!} f^{(n)}(x) - A$$

$$F'(a+\theta h) = \frac{(a+h-\theta h)^{n-1}}{(n-1)!} f^{(n)}(a+\theta h) - A$$

$$F'(a+\theta h) = \frac{(h-\theta h)^{n-1}}{(n-1)!} f^{(n)}(a+\theta h) - A$$

$$\text{Since, } F'(a+\theta h) = 0$$

$$0 = \frac{h^{n-1}(1-\theta)^{n-1}}{(n-1)!} f^{(n)}(a+\theta h) - A$$

$$A = \frac{h^{n-1}(1-\theta)^{n-1}}{(n-1)!} f^{(n)}(a+\theta h)$$

On putty above value of A in Eq. (3):

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{h^n}{n!} (1-\theta)^{n-1} f^{(n)}(a+\theta h) \dots (4)$$

Which proves Taylor's theorem with Cauchy's form of remainder.

(b) Find the Lagrange's and Cauchy's remainder after n times in the expansion of $\sin x$

Sol. Let $f(x) = \sin x$

$$f^n(x) = \sin\left(x + \frac{4\pi}{2}\right)$$

(i) Lagrange's Remainder is calculated as-

$$R_n = \frac{x^n}{n!} f^n(\theta x)$$

$$= \frac{x^n}{n!} \sin\left(\theta x + \frac{4\pi}{2}\right)$$

Ans.

(ii) Cauchy's Remainder is calculated as-

$$R_n = \frac{x^{n+1}}{(n+1)!} (1-\theta)^{n+1} f^{n+1}(\theta x)$$

$$= \frac{x^{n+1}}{(n+1)!} (1-\theta)^{n+1} \sin\left(\theta x + \frac{4\pi}{2}\right)$$

Ans.

(c) Expand $\sin x$ using Maclaurin's series.

Sol. Let

$$y = \sin x$$

$$y_n = \sin\left(x + \frac{4\pi}{2}\right)$$

At $x=0$

$$(y)_0 = \sin 0 = 0$$

$$(y_n)_0 = \sin\left(x + \frac{4\pi}{2}\right)$$

On putting

$$n = 1, 2, 3, 4, \dots \text{ in (4) }$$

$$n = 1 \Rightarrow (y_1)_0 = \sin \frac{\pi}{2}$$

$$n = 2 \Rightarrow (y_2)_0 = \sin \pi = 0$$

$$n = 3 \Rightarrow (y_3)_0 = \sin \frac{3\pi}{2} = -1$$

$$n = 4 \Rightarrow (y_4)_0 = \sin 2\pi = 0$$

Using Maclaurin's series, we have

$$y = (y)_0 + x(y_1)_0 + \frac{x^2}{2!}(y_2)_0 + \dots + \frac{x^n}{n!}(y_n)_0 + \dots$$

$$\sin x = 0 + x(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(-1) + \dots + \frac{x^n}{n!} \sin \frac{n\pi}{2} + \dots$$

$$\Rightarrow \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{x^n}{n!}(-1)^{\frac{n-1}{2}} + \dots$$

Where n is an odd number.

Q. 6. (a) Derive expression for Maclaurian series.

Sol. Maclaurian Series:

Let a function f possess continuous derivative of every order in $[0, a+h]$, then for all $n \in \mathbb{N}$ we have by Taylor's theorem:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^n}{n!}f^{(n)}(a)$$

$$+ \dots + \frac{h^{n+1}}{(n+1)!}f^{(n+1)}(a) + \frac{h^n}{n!}f^{(n)}(1+\theta h)$$

where $0 < \theta < 1$

Let

$$S_n = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^n}{n!}f^{(n)}(a)$$

$$R_n = \frac{h^{n+1}}{(n+1)!}f^{(n+1)}(a+\theta h)$$

Here, R_n is the Taylor's remainder after n terms using (2). Eq. (1) becomes—

$$f(a+h) = S_n + R_n$$

For $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} (R_n) = \lim_{n \rightarrow \infty} \frac{h^{n+1}}{(n+1)!} f^{(n+1)}(a+\theta h) = 0$$

Eq. (3) gives :

$$f(a+h) = \lim_{n \rightarrow \infty} S_n + \lim_{n \rightarrow \infty} R_n$$

$$f(a+h) = \lim_{n \rightarrow \infty} S_n + 0$$

$$f(a+h) = \lim_{n \rightarrow \infty} S_n$$

$$\lim_{n \rightarrow \infty} S_n = f(a+h)$$

Thus, the infinite series $f(a) + hf'(a) + \frac{h^2}{2}f''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + \dots$ converges to $f(a+h)$.

This if

(i) f possesses continuous derivatives of every order in $[a, a+h]$.

(ii) The Taylor's remainder after n terms i.e., $R_n = 0$ as $n \rightarrow \infty$.

Then,

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2}f''(a) + \dots + \frac{h^n}{n!}f^{(n)}(a) + \dots \quad (4)$$

On putting $a = 0$ in (4) and replaug x by h :

$$f(0+x) = f(0) + xf'(0) + \frac{x^2}{2}f''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots$$

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots \quad (5)$$

Which is expression for Maclaurian infinite series.

(b) Expand $e^x \sin x$ Maclarian Theorem.

Sol. Let

$$\begin{aligned} y &= e^x \sin x \\ y_x &= r^n [e^x \sin x] \\ &= r^n e^x \sin(x + n\phi) \end{aligned} \quad (1)$$

where,

$$r = \sqrt{1+1} = \sqrt{2}$$

$$\phi = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

$$y_{a,n} = 2^{n/2} e^x \sin\left(0 + \frac{n\pi}{4}\right)$$

putting

$x = 0$, we get:

$$(y_n)_0 = 2^{n/2} e^0 \sin\left(x + \frac{n\pi}{4}\right)$$

$$(y_n)_0 = 2^{n/2} \sin \frac{n\pi}{4} \quad (2)$$

On putting

$$n = 1, 2, 3, 4, 5, \dots \text{ in Eq. (1)}$$

$$(y_1)_0 = 2^{1/2} \sin \frac{\pi}{4} = 1$$

$$(y_2)_0 = 2^{2/2} \sin \frac{\pi}{2} = 2$$

$$(y_3)_0 = 2^{3/2} \sin \frac{3\pi}{4} = 2$$

$$(y_4)_0 = 2^{4/2} \sin \frac{4\pi}{4} = -4$$

$$(y_5)_0 = 2^{5/2} \sin \frac{5\pi}{4}$$

Using Maclaurien series we get:

$$y = (y_0)_0 + x(y_1)_0 + \frac{x^2}{2!} + \dots$$

$$e^x \sin x = (0) + x(1) + \frac{x^2}{2!}(2) + \frac{x^3}{3!}(3) + \frac{x^4}{4!}(-4) + \dots$$

$$= x + \frac{2x^2}{2!} + \frac{2x^3}{3!} - \frac{4x^4}{4!} + \dots \quad \text{Ans.}$$

(c) Use Taylor's expansion to express $2x^3 + 5x^2 + x - 4$ in ascending power of $(x-1)$

Sol. Let

$$\begin{aligned} f(x) &= 2x^3 + 5x^2 + x - 4 \\ f(x) &= f(1+x-1) \\ &= f(1+x-1) \\ &= f(1+h) \end{aligned} \quad (1)$$

Where

$$\begin{aligned} h &= x-1 \\ f(x) &= 2x^3 + 5x^2 + x - 4 \\ f'(x) &= 6x^2 + 10x + 1 \\ f''(x) &= 2x + 10 \\ f'''(x) &= 12 \\ f^{(4)}(x) &= 0 \\ f^{(5)}(x) &= 0 \end{aligned}$$

On putting $x = 1$, we get

$$\begin{aligned} f(1) &= 2 + 5 + 1 - 4 = 4 \\ f'(1) &= 6 + 10 + 1 = 17 \\ f''(1) &= 12 + 10 = 22 \end{aligned}$$

$$f''(1) = 12$$

$$f'''(1) = 0$$

$$f^{(4)}(1) = 0$$

Using Taylor's series, we get:

$$f(1+h) = f(1) + h f'(1) + \frac{h^2}{2} f''(1) + \frac{h^3}{6} f'''(1) + \dots$$

$$f(1+h) = 4 + h(17) + \frac{h^2}{2}(22) + \frac{h^3}{6}(12) + \frac{h^4}{24}(0) + \dots$$

$$f(1+h) = 4 + 17h + \frac{22}{2}h^2 + \frac{12}{6}h^3 + \dots$$

Replacing h by $x-1$

$$\begin{aligned} f(x) &= 4 + 17(x-1) + \frac{22}{2}(x-1)^2 + \frac{12}{6}(x-1)^3 + \dots \\ &= 4 + 17(x-1) + 11(x-1)^2 + 2(x-1)^3 \end{aligned}$$

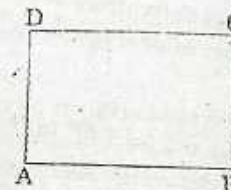
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Name of the Paper : Paper : C-6 : Group Theory-I
 Name of Course : B.Sc. (Hons.) Mathematics II Year (CBCS)
 Semester : III
 Duration : 3 Hours
 Maximum Marks : 75

MODEL TEST PAPER-I

Q. 1. (a) Describe the symmetries of a square.

Sol. Symmetries of a Square



Let us consider all possible motions of the square with vertices A, B, C, D. The final position of the square can be obtained from the original position by the rotation of the square about the axis through the centre, perpendicular to the plane, through an angle of 90° anticlockwise. Let the plane of square be horizontal as above.

The possible motions can be explained as following :

- (i) R_0 = Rotation of 0° about vertical axis in the plane of the square
 - (ii) R_1 = Rotation of one right angle anticlockwise about the vertical axis perpendicular to the plane
 - (iii) R_2 = Rotation of two right angles anticlockwise about the vertical axis perpendicular to the plane of square.
 - (iv) R_3 = Rotation of three right angles anticlockwise about the vertical axis perpendicular to the plane of square.
 - (v) H = Rotation of 180° anticlockwise about the horizontal axis in plane of square.
 - (vi) V = Rotation of 180° anticlockwise about the vertical axis perpendicular to the plane of square.
 - (vii) D = Rotation of 180° anticlockwise about the main diagonal.
 - (viii) D' = Rotation of 180° anticlockwise about the other diagonal.
- Two motions are equivalent, if their net effect is same.

It can be shown that all above eight motions of a square form a group of symmetries and denoted by D_8 . The set of motions of a square hence is given by

$$D_8 = \{R_0, R_1, R_2, R_3, H, V, D, D'\}$$

Here, D_8 is called the dihedral group of order 8.

(b) Prove that a group G is abelian if and only if

$$(a \cdot b)^2 = a^2 \cdot b^2 \quad \forall a, b \in G$$

Sol. Let G be the abelian, then:

$$a \cdot b = b \cdot a \quad \forall a, b \in G$$

Using (1), we have:

$$\begin{aligned} (a \cdot b)^2 &= (a \cdot b) (a \cdot b) \\ &= a \cdot (b \cdot a) \cdot b \\ &= a \cdot (a \cdot b) \cdot b \\ &= (a \cdot a) (b \cdot b) \\ &= a^2 \cdot b^2 \end{aligned}$$

Converse

Let

$$\begin{aligned} (a \cdot b)^2 &= a^2 \cdot b^2 \quad \forall a, b \in G \\ (a \cdot b) \cdot (a \cdot b) &= (a \cdot a) \cdot (b \cdot b) \\ a \cdot b \cdot a \cdot b &= a \cdot a \cdot b \cdot b \end{aligned}$$

On using left cancellation law, we get:

$$b \cdot a \cdot b = a \cdot b \cdot b$$

On using right cancellation law, we get:

$$b \cdot a = a \cdot b$$

Hence, G must be abelian

(c) Prove that a group G is abelian if and only if

$$(a \cdot b)^{-1} = a^{-1} \cdot b^{-1} \text{ for all } a, b \in G$$

Sol. Let a be abelian. We show that

$$\begin{aligned} (a \cdot b)^{-1} &= b^{-1} \cdot a^{-1} \\ (a \cdot b)^{-1} &= a^{-1} \cdot b^{-1} \text{ as } G \text{ is abelian.} \end{aligned}$$

Converse

Let

$$(a \cdot b)^{-1} = a^{-1} \cdot b^{-1}$$

Since, G is a group, we have

$$(b \cdot a)^{-1} = a^{-1} \cdot b^{-1}$$

So, from (1):

$$\begin{aligned} (a \cdot b)^{-1} &= (b \cdot a)^{-1} \\ [(a \cdot b)^{-1}]^{-1} &= [(b \cdot a)^{-1}]^{-1} \\ (a \cdot b) &= (b \cdot a) \end{aligned}$$

Hence, G is abelian.

Q. 2. State and prove Lagrange's Theorem

Sol. Lagrange's Theorem:

The order of a subgroup of a finite group divides the order of the group.

Proof of theorem:

Let

$$O(G) = n$$

$$O(H) = m$$

For $a, b \in G$, we define a relation $\equiv$ as follow:

$$a \equiv b \pmod{H}$$

$$a b^{-1} \in H$$

Then, $\equiv$ is an equivalence relation on G .

Since,

$$(i) \quad a \equiv a \pmod{H}$$

$$(ii) \quad a \equiv b \pmod{H}$$

$$\Rightarrow a b^{-1} \in H$$

$$\Rightarrow (a b^{-1})^{-1} \in H$$

$$\Rightarrow b a^{-1} \in H$$

$$\Rightarrow b \equiv a \pmod{H}$$

$$(iii) \quad a \equiv b \pmod{H} \text{ and } b \equiv c \pmod{H}$$

$$\Rightarrow a b^{-1} \in H \text{ and } b c^{-1} \in H$$

$$\Rightarrow (a b^{-1}) (b c^{-1}) \in H$$

$$\Rightarrow a c^{-1} \in H$$

$$\Rightarrow a \equiv c \pmod{H}$$

Consequently G is expressible as the union of mutually disjoint equivalence classes i.e.,

$$G = \cup [a] \quad \dots (2)$$

$$a \in G$$

We know show that $[a] = Ha$, where

$$[a] = \{x \in G : x \equiv a \pmod{H}\}$$

$$= \{x \in G : x a^{-1} \in H\}$$

Let $x \in [a]$ so that $x a^{-1} \in H$

$$x \in Ha$$

$$[a] \subseteq Ha$$

Conversely, let $y \in Ha$. Then, $y = ha$ for some $h \in H$

$$y a^{-1} = h$$

$$y a^{-1} \in H$$

$$y \in [a]$$

$$Ha \subseteq [a]$$

$$[a] = Ha$$

From (2) and (4), we obtain:

$$G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_r$$

$$a_i \in a \quad (1 \leq i \leq r \leq n) \text{ and } Ha_i \cap Ha_j = \emptyset$$

$$i \neq j$$

Where,

For

Since, G is finite, therefore:

$$O(G) = O(Ha_1) + O(Ha_2) + \dots + O(Ha_r)$$

and

$$H = H e \text{ is a right coset of } H \text{ in } G.$$

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 Since, there exists a one to one correspondence between any two right cosets of H in G .

Therefore, $O(H_i) = O(H_j)$
 $= O(H) = m, 1 \leq i \leq r$
 from (5) and (6):

$$n = m + m + m + \dots + r \text{ times}$$

$$n = mr$$

$$m \text{ divides } n$$

Proved.

Hence, $O(H)$ divides $O(G)$.

(b) Prove that the intersection of two subgroups is also a subgroup.
 Sol. Let H_1 and H_2 be two subgroups of G .

Since, H_1 is a subgroup

Since, H_2 is a subgroup

$$a \in H_1, b \in H_1 \Rightarrow ab^{-1} \in H_1 \quad \dots(1)$$

$$a \in H_2, b \in H_2 \Rightarrow ab^{-1} \in H_2 \quad \dots(2)$$

Now,

and

From (1) and (2), we get

$$ab^{-1} \in H_1, ab^{-1} \in H_2$$

$$ab^{-1} \in H_1 \cap H_2$$

$$b \in H_1, b \in H_2 \Rightarrow b \in H_1 \cap H_2$$

Thus if $a, b \in H_1 \cap H_2$ then $ab^{-1} \in H_1 \cap H_2$ which shows that $H_1 \cap H_2$ is a subgroup of G .

(c) Let H be a subgroup of G and Let $a, b \in G$. Then prove that either

$$aH = bH$$

or

$$aH \cap bH = \phi$$

Sol. Let H be a subgroup of G

Let

$$a, b \in G$$

Case (1) If there is no element common to aH and bH , then $aH \cap bH = \phi$ i.e. will set. Hence, aH and bH are disjoint.

Case (2) If there is an element c common to aH and bH , then we have:

$$c = a \circ h_1, h_1 \in H$$

$$c = b \circ h_2, h_2 \in H$$

$$a \circ h_1 = b \circ h_2$$

$$a \circ h_1 = b \circ h_2$$

$$(a \circ h_1) \circ h_2^{-1} = (b \circ h_2) \circ h_2^{-1}$$

$$a \circ (h_1 \circ h_2^{-1}) = b \circ (h_2 \circ h_2^{-1})$$

$$a \circ (h_1 \circ h_2^{-1}) = b$$

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$$a^{-1} \circ a \circ (h_1 \circ h_2^{-1}) = a^{-1} \circ b$$

$$(a^{-1} \circ a) \circ (h_1 \circ h_2^{-1}) = a^{-1} \circ b$$

$$h_1 \circ h_2^{-1} = a^{-1} \circ b$$

$$\Rightarrow a^{-1} \circ b \in H \text{ as } h_1 \circ h_2^{-1} \in H$$

$$\Rightarrow aH = bH$$

Which proves that if two left cosets which are not disjoint are identical.

Q. 3 (a) If G is group, then $N(a) = \{x \in G : xa = ax\}$ is a subgroup of G . Hence, $N(a)$ is called the normalizes of a in G .

Sol. Let G is a group and

$$N(a) = \{x \in G : xa = ax\}$$

Since,

$$ea = ae, \text{ so } e \in N(a)$$

$\Rightarrow$

$$N(a) \neq \phi$$

Let $x, y \in N(a)$ so that $xa = ax$ and $ya = ay$

Now,

$$ya = ay$$

$\Rightarrow$

$$(ya)^{-1} = (ay)^{-1}$$

$\Rightarrow$

$$a^{-1}y^{-1} = y^{-1}a^{-1}$$

$$a(a^{-1}y^{-1}) = a(y^{-1}a^{-1})$$

$$(a a^{-1})y^{-1} = (a y^{-1})a^{-1}$$

$$(e)y^{-1} = (a y^{-1})a^{-1}$$

$$y^{-1} = (a y^{-1})a^{-1}$$

$$y^{-1}a = a y^{-1}$$

Pre multiply by x :

$$x(y^{-1}a) = x(a y^{-1})$$

$$(x y^{-1})a = (x a) y^{-1}$$

$$(x y^{-1})a = (x a) y^{-1} \text{ using (1)}$$

Using associative law, we get

$$(x y^{-1})a = a(x y^{-1})$$

$$x y^{-1} \in N$$

$\Rightarrow$

$N(a)$ is a subgroup of G .

(b) If $H \subseteq K$ be two subgroup of a finite group G , then show that

$$[G : H] = \frac{O(G)}{O(H)}$$

Sol. Since, $H \subseteq K$ be two subgroup of a finite group G . Here, $H \subseteq K \subseteq G$ mean H is a subgroup of K .

If G is a finite group and H is a subgroup of G , then using Lagrange's theorem we have

$$[G : H] = \frac{O(G)}{O(H)} \quad \dots(1)$$

If G is a finite group and K is a subgroup of G , then using Lagrange's theorem we have :

$$[G : K] = \frac{O(G)}{O(K)} \quad \dots(2)$$

If K is a finite group and H is a subgroup of K , then using Lagrange's theorem we have :

$$[K : H] = \frac{O(K)}{O(H)} \quad \dots(3)$$

From (2) $\times$ (3)

$$[G : K] [K : H] = \frac{O(G)}{O(K)} \times \frac{O(K)}{O(H)}$$

$$[G : K] [K : H] = \frac{O(G)}{O(H)}$$

$$[G : K] [K : H] = [G : H]$$

$$[G : H] = [G : K] [K : H]$$

Proved.

(c) IF H and K are finite subgroups of a group G , then prove that :

$$O(HK) = \frac{O(H)O(K)}{O(H \cap K)}$$

Sol. Let $T = H \cap K$, then T is a subgroup of K .

Since, K is a finite group, then

$$\frac{O(K)}{O(T)} = i_n(T) = m \text{ (let)} \quad \dots(1)$$

Thus, there exists m distinct right cosets of T in K , say $Tk_1, Tk_2, \dots, Tk_m$.
 $k_2, \dots, k_m \in K$.

$$\begin{aligned} K &= Tk_1 \cup Tk_2 \cup \dots \cup Tk_m \\ HK &= H(Tk_1 \cup Tk_2 \cup \dots \cup Tk_m) \\ &= Hk_1 \cup Hk_2 \cup \dots \cup Hk_m \end{aligned} \quad \dots(2)$$

$$T \subseteq H \Rightarrow HT = H$$

Because

Now, we show $Hk_1, Hk_2, \dots, Hk_m$ are all distinct. Let if possible

$$Hk_i = Hk_j \text{ for } 1 \leq i < j \leq m$$

$$k_i k_j^{-1} \in H, \text{ also } k_i k_j^{-1} \in K$$

$$k_i k_j^{-1} \in H \cap K$$

$$k_i k_j^{-1} \in T$$

$$Tk_i = Tk_j$$

Which is a contradiction

Now, from (2) we obtain :

$$\begin{aligned} O(HK) &= O(Hk_1) + O(Hk_2) + \dots + O(Hk_m) \\ &= O(H) + O(H) + \dots + O(H) \text{ m times} \\ &= m O(H) \end{aligned}$$

$$= \frac{O(K)}{O(T)} O(H)$$

$$= \frac{O(K)O(H)}{O(T)}$$

$$= \frac{O(H)O(K)}{O(H \cap K)}$$

$$O(HK) = \frac{O(H)O(K)}{O(H \cap K)}$$

Proved

Q. 4. (a) Prove that of the $\frac{n}{2}$ permutations on n symbols, $\frac{n}{2}$ are even permutations and $\frac{n}{2}$ are odd permutations.

Sol. In order to prove given theorem use are the following facts :

- Product of two odd permutations is an even permutation.
- Product of an even permutation and an odd permutation is an odd permutation.

(iii) A transposition is always an odd permutation.

(iv) A permutation can not be both odd or even.

Let P_n be the set of all permutations on n distinct symbols. Then, P_n contains distinct permutations of degree n . Also P_n is a group w.r.t. permutation multiplication as mentioned.

Out of these $\frac{n}{2}$ permutations, let the even permutations be $E_1, E_2, \dots, E_p$ and the odd permutations be $O_1, O_2, \dots, O_q$.

$$p + q = \frac{n}{2}$$

Let $t \in P_n$ be arbitrary such that t is transposition so that t is an odd permutation.

Let t be operated on each of E_i ($i = 1, 2, \dots, p$) and similarly on each O_i ($i = 1, 2, \dots, q$)

Since (P_n) is a group and therefore

$$t E_i \in P_n$$

and $\alpha_j \in p_n$ for $1 \leq i \leq p$
 $1 < j < q$

The permutation α_i for $i = 1, 2, \dots, p$ are all odd permutations by virtue of (2) and (3).

Also, these permutations are all distinct.

For $\alpha_i = \alpha_j \Rightarrow E_i = E_j$
 $\Rightarrow \alpha_i = \alpha_j \nmid E_i = E_j$

Similarly, the permutations α_j ($j = 1, 2, \dots, q$) are all distinct even permutations.

By virtue of (4), the p even permutations $E_1, E_2, \dots, E_p$ are equal to q even permutations $\alpha_1, \alpha_2, \dots, \alpha_q$.

Similar, q odd permutations $\alpha_1, \alpha_2, \dots, \alpha_q$ are equal to p odd permutations $E_1, E_2, \dots, E_p$.

Consequently $p = q$, also $p + q = \frac{n!}{2}$

$$\Rightarrow p = q = \frac{n!}{2}$$

Which proves theorem.

(b) Let

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{pmatrix}$$

and

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 1 & 2 & 4 & 3 & 5 \end{pmatrix}$$

compute (i) $\alpha\beta$ (ii) $\beta\alpha$ (iii) α^{-1}

Sol. It is given that:

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{pmatrix}$$

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 1 & 2 & 4 & 3 & 5 \end{pmatrix}$$

(i) $\alpha\beta$

$$(\alpha \circ \beta)(1) = \alpha(\beta(1)) = \alpha(6) = 6$$

$$(\alpha \circ \beta)(2) = \alpha(\beta(2)) = \alpha(1) = 2$$

$$(\alpha \circ \beta)(3) = \alpha(\beta(3)) = \alpha(2) = 1$$

$$(\alpha \circ \beta)(4) = \alpha(\beta(4)) = \alpha(4) = 5$$

$$(\alpha \circ \beta)(5) = \alpha(\beta(5)) = \alpha(3) = 3$$

$$(\alpha \circ \beta)(6) = \alpha(\beta(6)) = \alpha(5) = 4$$

$$\alpha\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 1 & 2 & 4 & 3 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 2 & 1 & 5 & 3 & 4 \end{pmatrix}$$

(ii) $\beta\alpha$

$$(\beta \circ \alpha)(1) = \beta(\alpha(1)) = \beta(2) = 1$$

$$(\beta \circ \alpha)(2) = \beta(\alpha(2)) = \beta(1) = 6$$

$$(\beta \circ \alpha)(3) = \beta(\alpha(3)) = \beta(2) = 1$$

$$(\beta \circ \alpha)(4) = \beta(\alpha(4)) = \beta(5) = 3$$

$$(\beta \circ \alpha)(5) = \beta(\alpha(5)) = \beta(4) = 4$$

$$(\beta \circ \alpha)(6) = \beta(\alpha(6)) = \beta(6) = 5$$

$$\beta\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 1 & 2 & 4 & 3 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 6 & 2 & 3 & 4 & 5 \end{pmatrix}$$

(iii) α^{-1}

$$\alpha^{-1} = \begin{pmatrix} 2 & 1 & 3 & 5 & 4 & 6 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{pmatrix}$$

(c) (i) Define even and odd permutations.

(ii) Determine which of the following are even permutation or odd permutation.

$$(i) \alpha = (123)(12)$$

$$(ii) \beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 3 & 6 & 1 & 9 & 7 & 8 \end{pmatrix}$$

Sol. (i) Even and odd permutations:

A permutation is said to be even permutation if it can be expressed as a product an even number of transposition otherwise it is said to be an odd permutation.

(i)

$$\alpha = (123)(12)$$

$$= (12)(13)(12)$$

The number of transpositions = 3 = odd

$\Rightarrow \alpha$ is an odd permutation.

(ii)

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 3 & 6 & 1 & 9 & 7 & 8 \end{pmatrix}$$

Ans.

$$= \begin{pmatrix} 1 & 2 & 5 & 6 & 3 & 4 & 7 & 9 & 8 \\ 2 & 5 & 6 & 1 & 4 & 3 & 9 & 8 & 7 \end{pmatrix}$$

$$= (1 \ 2 \ 5 \ 6) (3 \ 4) (7 \ 9 \ 8)$$

$$= (1 \ 6) (1 \ 5) (1 \ 2) (3 \ 4) (7 \ 8) (7 \ 9)$$

$\Rightarrow$ no. of transpositions = 6

= even

$\Rightarrow \beta$ is an even permutation.

Q. 5. (a) State and prove Cayley's theorem.

Sol. Cayley's Theorem:

Statement :

Every group G is isomorphic to a permutation group.

Proof:

Let G be a group and $A(G)$ the group of all permutations of G i.e. $A(G)$ is the set of all one to one mappings of G onto G . For each $a \in G$, we define a mapping

$$f_a : G \rightarrow G \text{ as } f_a(x) = ax \quad \forall x \in G \quad \dots(1)$$

Then f_a is one to one since,

$$f_a(x) = f_a(y)$$

$\Rightarrow$

$$ax = ay \Rightarrow x = y \quad \forall x, y \in G$$

Now, we choose f_a if onto. For any $g \in G$,

we have,

$$g = e \cdot g = a(a^{-1}g) = ag_1, \text{ where } g_1 = a^{-1}g \in G$$

Using (1)

$$g = f_a(g_1)$$

$\Rightarrow$

f_a is onto

Since, f_a is one to one and onto f_a is a permutation of G .

Thus,

$$f_a \in A(G) \quad \forall a \in G$$

Let

$$G = \{f_a : a \in G\} \quad \dots(2)$$

Now, we proceed to show that $\bar{G}$ is a subgroup of $A(G)$.

Let $f_a \in \bar{G}$. For any $x \in G$, we have

$$(fa fe)x = fa\{fe(x)\}$$

$$= fa(bx)$$

$$= a(bx)$$

$$= (ab)x$$

$$= fae(x)$$

$$fafb = fab \quad \forall a, b \in G \quad \dots(3)$$

From (3), it follows that

$$fa \in \bar{G} \text{ and } fe \in \bar{G} \Rightarrow fafb \in \bar{G}$$

...(4)

Now, we show that $(fa)^{-1} = f_{a^{-1}}$ for all $fa \in \bar{G}$

...(5)

Using (3) we have :

$$fa f_{a^{-1}} = faa^{-1} = fe$$

where

$$fe(x) = ex = x \quad \forall x \in G$$

$\Rightarrow$

$$fe = 1 \text{ is the identity}$$

Similarly

$$f_{a^{-1}}fa = 1. \text{ This proves (3) and so } (fa)^{-1} \in \bar{G} \quad \forall fa \in \bar{G}$$

$\bar{G}$ From (4) and (5), $\bar{G} < A(G)$ and so $\bar{G}$ is a permutation group.

Finally we prove that $G = \bar{G}$

Define a mapping $\psi : G \rightarrow \bar{G}$ as

$$\psi(a) = fa \quad \forall a \in G \quad \dots(6)$$

Clearly, ψ is onto. Now, we show that ψ is a homomorphism.

Let $a, b \in G$ be arbitrary. From (5) and (3) we have,

$$\psi(ab) = fab$$

$$= fafb$$

$$= \psi(a)\psi(b)$$

Thus ψ is a homomorphism. Also ψ is one to one since

$$\psi(a) = \psi(b)$$

$\Rightarrow$

$$fa = fb$$

$\downarrow$

$$fa(x) = fb(x) \quad \forall x \in G$$

$\Rightarrow$

$$ax = bx, \text{ using (1)}$$

$\Rightarrow$

$$a = b \text{ using right cancellation law}$$

Since $\psi : G \rightarrow \bar{G}$ is homomorphism one to one and onto.

Hence,

$$G = \bar{G}$$

Which proves Cayley's theorem.

(b) Show that a group of order 4 is either cyclic or is a direct product of two cyclic subgroups each of order 2.

Sol. Let $O(G) = 4 = p^2$ ($p=2$ is prime), so that G is abelian. If G is cyclic, then there is nothing to prove. Let G is not cyclic. Since $p=2$ $\dim O(G) = 4$, by Cauchy's theorem, there exists an element $a \neq e$ such that $a^2 = 1$.

Let

$$A = \{a, a^2 = e\}$$

$$= \langle a \rangle$$

Since, $A \neq G$ there exist some element $b \in G$ s.t. $b \notin A$

Since $O(b)$ divides $O(G)$, $O(b) = 1$ or 2 or 4

Now,

$$O(b) = 1 \Rightarrow b = e \in A, \text{ a contradiction}$$

$$O(b) = 4 \Rightarrow G = \langle b \rangle \\ \Rightarrow G \text{ is cyclic}$$

which is a contradiction

$$O(b) = 2, \text{ Let } B = \{b, b^2 = e\} = \langle b \rangle$$

Since, G is abelian $A < G$ and $B < G$. Also $A \cap B = \{e\}$

$$AB < G \text{ and } O(AB) = \frac{O(A)O(B)}{O(A \cap B)}$$

$$= \frac{2 \times 2}{1}$$

$$= 4$$

$$= O(G)$$

$$\Rightarrow G = AB, \text{ where } A \cap B = \{e\}$$

and $A < G, B < G$

Hence, G is the internal direct product of A and B where A and B are two cyclic subgroup of G each of order 2.

(c) Show that the quaternion group can not be expressed as the internal direct product of its proper subgroup.

Sol. The quaternion group is:

$$Q = \{\pm 1, \pm i, \pm j, \pm k\}$$

$$i^2 = j^2 = k^2 = -1$$

$$ij = -ji = k$$

$$jk = -kj = i$$

$$ki = -ik = j$$

Q has the following proper normal sub group:

$$N_1 = \{1, -1\}$$

$$N_2 = \{1, -1, i, -i\}$$

$$N_3 = \{1, -1, j, -j\}$$

$$N_4 = \{1, -1, k, -k\}$$

We see that

$$N_i \cap N_j = \{1, -1\} \text{ for } i \neq j$$

Here,

$$N_i \cap \bigcup_{j=1}^4 N_j = \{1, -1\} \neq \{1\} \text{ for each}$$

Hence, Q is not the internal direct product of its normal subgroups.

Q. 6. (a) State and prove first theorem on isomorphism.

Sol. First Theorem on isomorphism:

Statement:

Every homomorphic image of a group G is isomorphic to some quotient group of G .

Proof:

Suppose $f: G \rightarrow G'$ is a group homomorphism with kernel K . K is the normal subgroup of G and

Now, the symbol G/K has a meaning. $K = \{x \in G : f(x) = e'\}$ e' being the identity of G' .

To prove $G/K \cong G'$,

Define a map $g: G/K \rightarrow G'$ s.t. $g(Kx) = f(x) \forall x \in G$ is well defined.

For

$$Ka = Kb : Ka, Kb \in G/K$$

$\Rightarrow$

$$ab^{-1} \in K$$

$\Rightarrow$

$$f(ab^{-1}) = e' \text{ def. of } K$$

$\Rightarrow$

$$f(a)b^{-1} = e' \text{ as } f \text{ is homomorphism}$$

$\Rightarrow$

$$f(a)[f(b)]^{-1} = e'$$

$\Rightarrow$

$$f(a)[f(b)]^{-1}f(b) = e'f(b)$$

$\Rightarrow$

$$f(a)e' = f(b)$$

$\Rightarrow$

$$f(a) = f(b)$$

$\Rightarrow$

$$g(Ka) = g(Kb)$$

Which proves that g is well defined.

g is one-one

$$g(Kb) = g(Ka) : Ka, Kb \in G/K$$

$\Rightarrow$

$$f(a) = f(b)$$

$\Rightarrow$

$$[f(a)][f(b)]^{-1} = e'$$

$\Rightarrow$

$$[f(a)][f(b^{-1})] = e'$$

$\Rightarrow$

$$[f(ab^{-1})] = e'$$

$\Rightarrow$

$$ab^{-1} \in K$$

$\Rightarrow$

$$Ka = Kb$$

$\Rightarrow$

$$g(Ka) = g(Kb) \Rightarrow Ka = Kb$$

$\Rightarrow$

g is one-one

g preserves composition in G/K and G' :

$$g(Ka \cdot Kb) = g(Kab) \text{ as } K \text{ is normal} \Rightarrow (Ka)(Kb) =$$

$$= f(ab)$$

$$= f(a)f(b) \text{ as } f \text{ is homomorphism}$$

$$= g(Ka)g(Kb)$$

$$g(Ka \cdot Kb) = g(Ka)g(Kb)$$

g is onto:

$$\text{any } x \in G' \Rightarrow \exists y \in G \text{ s.t. } f(y) = x$$

For $f: G \rightarrow G'$ is onto

$$\Rightarrow Ky \in G/K, g(Ky) = f(y) = x$$

$\therefore$ Given any $x \in G'$, $\exists Ky \in G/K$ s.t. $g(Ky) = f(y)$
Which proves g is onto.

Hence we can say that g is an isomorphism
Therefore, we have:

$$G/K \cong G'$$

Which proves first theorem on isomorphism

(b) State and prove first theorem on Homomorphism.

Sol. First Theorem on Homomorphism:

Statement:

If f is a Homomorphism of a group G into a group G' with kernel K , then K is a normal subgroup of G .

Proof:

Suppose e and e' are identity elements of the groups G and G' respectively.
Also suppose that $f: G \rightarrow G'$ is a homomorphism with kernel K .

To prove that K is a normal subgroup of G .

We have

$$f(e) = e'$$

$$K = \{x \in G : f(x) = e'\}$$

$$f(e) = e', e \in G, e \in K \Rightarrow K \neq \emptyset$$

Let $a, b \in K$ be an arbitrary, so that $f(a) = f(b) = e'$

$$f(ab^{-1}) = f(a)f(b^{-1})$$

$$= f(a)[f(b)]^{-1}$$

$$= e' [e']^{-1}$$

$$= e' e'$$

$$\Rightarrow f(ab^{-1}) = e'$$

$$\Rightarrow ab^{-1} \in K, a, b \in K$$

Which shows that K is a subgroup of G .

To prove K is normal in G

Let $x \in G$ be arbitrary

$$f(xax^{-1}) = f(x)f(a)f(x^{-1})$$

$$= f(x)e'[f(x)]^{-1}$$

$$= [f(x)e'] [f(x)]^{-1}$$

$$= [f(x)] [f(x)]^{-1}$$

$$= e'$$

$$\Rightarrow f(xax^{-1}) = e'$$

$$\Rightarrow xax^{-1} \in K$$

$$\Rightarrow x \in G, a \in K \Rightarrow xax^{-1} \in K$$

Which proves that K is normal in G .

This concludes the first theorem on homomorphism.

Proved.

(c) Let H and K be subgroup of a group G and N a normal subgroup of G s.t. $HN = KN$. Prove that

$$\frac{K}{(H \cap N)} \cong \frac{K}{(K \cap N)}$$

Sol. Since H is a subgroup of a group G and N is a normal subgroup of G and hence, we get in virtue of second Isomorphism theorem,

$$\frac{HN}{N} \cong \frac{H}{(H \cap N)} \quad (1)$$

Again K is a subgroup of G and N is a normal subgroup of G and therefore we obtain in virtue of second isomorphism theorem:

$$\frac{KN}{N} \cong \frac{K}{(K \cap N)} \quad (2)$$

Since,

$$HN = KN$$

$$\frac{HN}{N} \cong \frac{K}{(K \cap N)} \quad (3)$$

$$\Rightarrow \frac{K}{(K \cap N)} \cong \frac{HN}{N} \quad (4)$$

From (1) and (4):

$$\frac{HN}{N} \cong \frac{K}{(H \cap N)} \cdot \frac{K}{(K \cap N)} = \frac{HN}{N}$$

$$\Rightarrow \frac{K}{(K \cap N)} \cong \frac{H}{(H \cap N)}$$

$$\Rightarrow \frac{H}{(H \cap N)} \cong \frac{K}{(K \cap N)}$$

Proved.

□□□

Proved.

Name of the Paper : Paper : C-6 : Group Theory-I
 Name of Course : B.Sc. (Hons.) Mathematics II Year (CBCS)
 Semester : III
 Duration : 3 Hours
 Maximum Marks : 75

MODEL TEST PAPER-II

Q. 1. (a) (1) Define quaternion group.

(2) Define centre of a group

(3) Define normalize of a group.

(4) Define right coset with example.

Sol. (i) Quaternion Group :

The set $G = \{1, i, j, k, -1, -i, -j, -k\}$, where

$$i^2 = j^2 = k^2 = -1 \text{ and } ij = -ji = k, jk = -kj = i, ki = -ik = j$$

is a non abelian multiplication group of order 8. Thus group is called quaternion group. Here 1 is the identity. The inverse elements of i, j, k are $-i, -j, -k$ respectively.

(2) Centre of a group :

The centre of a group G is the set of those elements of G . Which commute with every element of G . This is denoted by Z and is given by :

$$Z = \{z \in G \mid zx = xz \forall x \in G\}$$

(3) Normalizer of a group : If $a \in G$ then normalizer of a in G is the set of all those elements of a , which commute with a . It is denoted by $N(a)$ and is given by :

$$N(a) = \{x \in G \mid ax = xa\}$$

(4) Right Coset :

Let G be a group and H be a sub group of G then for any $a \in G$, the set $Ha = \{ha \mid h \in H\}$ is called a right coset of H , determined by a .

Example :

Let

$$G = \{-1, 1, -i, i\}$$

$$H = \{-1, 1\}$$

Then

$$H = \{-1, 1\}$$

$$H \times 1 = \{-1, 1\}$$

$$H \times -1 = \{-1, 1\}$$

$$H \times -i = \{-i, i\}$$

Thus, there are only two right coset of H and H_i of H in G .

(b) Let G be a group, prove that

$$(ab)^{-1} = b^{-1}a^{-1} \forall a, b \in G$$

Sol. Let

$$c = a \cdot b$$

$$d = b^{-1}a^{-1}$$

$$c \cdot d = (a \cdot b) \cdot d$$

$$(56)$$

$$\begin{aligned} &= a \cdot (b \cdot d) \\ &= a \cdot [b \cdot (b^{-1} \cdot a^{-1})] \\ &= a \cdot [(b \cdot b^{-1}) \cdot a^{-1}] \\ &= a \cdot [e \cdot a^{-1}] \text{ as } b \cdot b^{-1} = e \\ &= a \cdot a^{-1} \text{ as } e \cdot a^{-1} = a^{-1} \\ &= e \\ d \cdot c &= (b^{-1} \cdot a^{-1}) \cdot c \\ &= b^{-1} \cdot (a^{-1} \cdot c) \\ &= b^{-1} \cdot [a^{-1} \cdot (a \cdot b)] \\ &= b^{-1} \cdot [(a^{-1} \cdot a) \cdot b] \\ &= b^{-1} \cdot [e \cdot b] \text{ as } a^{-1} \cdot a = e \\ &= b^{-1} \cdot b \text{ as } e \cdot b = b \\ &= e \end{aligned}$$

From (1) and (2)

$$c \cdot d = e = d \cdot c$$

$$\Rightarrow c^{-1} = d$$

$$\Rightarrow (a \cdot b)^{-1} = b^{-1}a^{-1}$$

Q. If G is a group such that $(a \cdot b)^n = a^n \cdot b^n$ for three consecutive integers n , and for all $a, b \in G$, show that G is abelian.

Sol. Let G is a group and $a, b \in G$ s.t. $(ab)^n = a^n b^n$ for three consecutive integer n .

Let

$$(a \cdot b)^n = a^n b^n$$

$$(ab)^{n+1} = a^{n+1} b^{n+1}$$

$$(ab)^{n+2} = a^{n+2} b^{n+2}$$

$$(ab)^{n+2} = a^{n+2} b^{n+2}$$

$$(ab)^{n+2} (ab) = (a^{n+1} a) (b^{n+1} b)$$

$$(a^{n+1} b^{n+1}) (ab) = (a^{n+1} a) (b^{n+1} b)$$

$$a^{n+1} (b^{n+1} a) b = a^{n+1} (a b^{n+1}) b$$

On using left cancellation law, we get :

$$(b^{n+1} a) b = (a b^{n+1}) b$$

On using right cancellation law, we get :

$$b^{n+1} a = a b^{n+1}$$

$$a^n (b^{n+1} a) = a^n (a b^{n+1})$$

$$a^n (b^n b a) = a^n (a b^{n+1})$$

$$(a^n b^n) (b a) = (a^n a) b^{n+1}$$

$$(a^n b^n) (b a) = a^{n+1} b^{n+1}$$

$$(ab^n) (b a) = (a b)^{n+1}$$

$$(ab^n) (b a) = (a b)^n (a b)$$

On using left cancellation law, we get:

$$ba = ab$$

Which show that G is an abelian group.

Q. 2. (a) State and prove Fermat's theorem:

Sol. Fermat's Theorem:

If p is a prime number and a is any integer then $a^p \equiv a \pmod{p}$

Proof:

Let G be the set of non-zero residue classes of integers modulo p . If p is a prime number, then with respect to multiplication of residue classes G is a group of order $p-1$. The identity element of the group is $[1]$.

Now suppose a is any integer.

Case (1): p is a divisor of a . In this case $[a] \neq [0]$ and so $[a]$ is not an element of G . But,

$$p \mid a \Rightarrow p \mid ap$$

$$\Rightarrow p \mid (a^p - a)$$

$$\Rightarrow a^p \equiv a \pmod{p}$$

Case (2):

p is not a divisor of a . In this case $[a] \neq [0]$ and so $[a]$ is an element of G . Therefore, we have,

$$[a]^{p(p-1)} = [1]$$

$$\Rightarrow [a]^{p-1} = [1]$$

$$\Rightarrow [a^{p-1}] = [1]$$

$$\Rightarrow a^{p-1} \equiv 1 \pmod{p}$$

$$\Rightarrow a^{p-1} - 1 \text{ is divisible by } p$$

$$\Rightarrow (a^{p-1} - 1) \text{ is divisible by } p$$

$$\Rightarrow a^p - a \text{ is divisible by } p$$

$$\Rightarrow a^p \equiv a \pmod{p}$$

(b) A non empty subset H of a group G is a subgroup of G if and only if $ab^{-1} \in H$ for all $a, b \in H$.

Sol. The condition is necessary:

Let H be a subgroup of G and $a, b \in H$

since, H is itself a group, $b^{-1} \in H$

Hence, $a \in H$ and $b^{-1} \in H \Rightarrow ab^{-1} \in H$

The condition is sufficient:

Let $ab^{-1} \in H \forall a, b \in H$

On putting $b = a$ in Eq. (1), we have:

$$e = aa^{-1} \in H$$

$$\Rightarrow e \in H$$

On putting $a = e$ in Eq. (1), we have

$$ab^{-1} \in H$$

$$b^{-1} \in H \forall b \in H$$

$\Rightarrow$

Proved.

Let $a \in H$ and $b \in H$, so that $a \in H$ and $b^{-1} \in H$.

From (1):

$$a(b^{-1})^{-1} \in H$$

$\Rightarrow$

$$ab \in H$$

Which show that associative law holds in $H \subseteq G$.

Thus $(H, \cdot)$ is a group, which proves that H is a subgroup of G .

(c) Let H and K be two subgroups of a finite group G . Prove that HK is a subgroup of G if and only if $HK = KH$.

Sol. Condition is necessary:

Let H, K be a subgroup of G .

Let $x \in HK$, then

$$x^{-1} \in HK \text{ as } HK < G$$

$\Rightarrow$

$$x^{-1} = hk \forall h \in H, k \in K$$

$$x = (x^{-1})^{-1}$$

$$= (hk)^{-1}$$

$$= k^{-1}h^{-1} \in KH$$

as $k^{-1} \in K$ and $h^{-1} \in H$ i.e. $K < G$ and $H < G$

$\Rightarrow$

$$HK \subseteq KH$$

Let

$$y \in KH$$

Then,

$$y = h_1k_1 \forall k_1 \in K, h_1 \in H$$

$$= (k_1^{-1})^{-1}(h_1^{-1})^{-1}$$

$$= (h_1^{-1}k_1^{-1})^{-1}$$

$$H < G \text{ and } K < G$$

Since,

$$h^{-1} \in H \text{ and } k^{-1} \in K$$

$\Rightarrow$

$$h^{-1}k^{-1} \in HK$$

$\Rightarrow$

$$(h^{-1}k^{-1})^{-1} \in HK \text{ as } HK < G$$

$\Rightarrow$

$$y \in HK$$

$\Rightarrow$

$$KH \subseteq HK$$

From (1) and (2)

$$HK = KH$$

Condition is sufficient:

Let

$$HK = KH$$

Let

$$x, y \in HK \text{ and } x = hk$$

$$y = h'k'$$

Where,

$$h, h' \in H \text{ and } k, k' \in K$$

$$xy = hkh'k'$$

Since,

$$kh' \in KH = HK, \text{ we can write}$$

$$kh' = h_2k_2 \forall h_2 \in H, k_2 \in K$$

From (3):

$$\begin{aligned} xy &= h(h_2 k_2)h^{-1} \\ &= (hh_2)(k_2 h^{-1}) \\ &\in HK \end{aligned}$$

(Since, $hh_2 \in H$ and $k_2 h^{-1} \in K$)

Hence,

$$xy \in HK \quad \forall x, y \in HK$$

$$x^{-1} = (h_1 h)^{-1}$$

$$x^{-1} = h^{-1}h_1^{-1}$$

$$x^{-1} = h^{-1}h_1^{-1} \in KH$$

$$x^{-1} \in KH$$

$$x^{-1} \in HK$$

$$x^{-1} \in HK \quad \forall x \in HK$$

From (4) and (5), we can say that HK is a subgroup of G .

Q. 3. (a) If G is a group then $Z = \{z \in G : xz = zx \quad \forall x \in G\}$ is a subgroup of G . Here, z is called the centre of G .

Sol. Let G is a group and

$$Z = \{z \in G : xz = zx \quad \forall x \in G\}$$

Since,

$$xz = zx \quad \forall x \in G$$

=

$$z \in Z$$

=

$$z \neq \emptyset$$

let $a, b \in Z$ so that $ax = xa$ and $bx = xb \quad \forall x \in G$.

$$bx = xb$$

$$b^{-1}(bx) = b^{-1}(xb)$$

$$x = b^{-1}x$$

$$b^{-1}x = xb^{-1}$$

$$a(b^{-1}x) = a(xb^{-1})$$

$$(ab^{-1})x = (ax)b^{-1}$$

$$(ab^{-1})x = (xa)b^{-1} \text{ using (3)}$$

$$(ab^{-1})x = x(ab^{-1})$$

$$ab^{-1} \in Z$$

Z is a subgroup of G .

(b) If H and K are subgroups of G and given that $O(H) > \sqrt{O(G)}$ and $O(K) > \sqrt{O(G)}$ and $O(K) > \sqrt{O(G)}$, prove that $O(H \cap K) > 1$.

Sol. We show that

$$H \cap K = \{h \in H, h \in K\}$$

If H is a subgroup of G , then,

$$h \in H \implies h^{-1} \in H$$

If K is a subgroup of G , then,

$$h \in K \implies h \in G$$

$$h \in G, k \in G \implies hk \in G$$

$$HK \subset G$$

$$O(HK) \leq O(G)$$

Hence,

Hence,

It is given,

$$O(H) > \sqrt{O(G)}$$

$$O(K) > \sqrt{O(G)}$$

Eq. (1) can be solved as :

$$O(G) \geq O(HK)$$

$$O(G) \geq \frac{O(H)O(K)}{O(H \cap K)}$$

From (2) and (3):

$$O(H)O(K) > \sqrt{O(G)}\sqrt{O(G)}$$

$$O(H)O(K) > \sqrt{O(G)}$$

$$\frac{O(H)O(K)}{O(H \cap K)} > \frac{O(G)}{O(H \cap K)}$$

From (4) and (5):

$$O(G) > \frac{O(G)}{O(H \cap K)}$$

Which is possible if

$$O(H \cap K) > 1$$

(c) If $O(G) = 6$ and $H \neq K$ are subgroups of G , each of order 2, then HK can not be a subgroup of G .

Sol. Since, a group of prime order is cyclic. So we may take

$$H = \{a, a^2 = e\}$$

$$K = \{b, b^2 = e\}$$

and

$$a \neq b \text{ as } H \neq K$$

$\implies$

$$H \cap K = \{e\}$$

Hence,

$$O(HK) = \frac{O(H)O(K)}{O(H \cap K)}$$

$$= \frac{2 \times 2}{1}$$

$$= 4$$

Since, 4 does not divide $O(G) = 6$. So, by Lagrange's theorem, HK can not be a subgroup of G.

Q. 4. (a) Let

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{pmatrix}$$

and

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 5 & 3 & 2 & 6 \end{pmatrix}$$

compute (i) $\alpha\beta$ (ii) $\alpha^{-1}\beta^{-1}$

Sol. It is given that :

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{pmatrix}$$

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 5 & 3 & 2 & 6 \end{pmatrix}$$

(i) $\alpha\beta$

$$(\alpha\beta)(1) = \alpha(\beta(1)) = \alpha(4) = 5$$

$$(\alpha\beta)(2) = \alpha(\beta(2)) = \alpha(1) = 2$$

$$(\alpha\beta)(3) = \alpha(\beta(3)) = \alpha(5) = 4$$

$$(\alpha\beta)(4) = \alpha(\beta(4)) = \alpha(3) = 3$$

$$(\alpha\beta)(5) = \alpha(\beta(5)) = \alpha(2) = 1$$

$$(\alpha\beta)(6) = \alpha(\beta(6)) = \alpha(6) = 6$$

$$\alpha\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 5 & 3 & 2 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 6 & 3 & 1 & 4 \end{pmatrix}$$

Ans.

(ii) $\alpha^{-1}\beta^{-1}$

$$\alpha^{-1} = \begin{pmatrix} 2 & 1 & 3 & 5 & 6 & 4 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 6 & 4 & 5 \end{pmatrix}$$

$$\beta^{-1} = \begin{pmatrix} 4 & 1 & 5 & 3 & 2 & 6 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 5 & 4 & 1 & 3 & 6 \end{pmatrix}$$

$$(\alpha^{-1}\beta^{-1})(1) = \alpha^{-1}(\beta^{-1}(1)) = \alpha^{-1}(2) = 1$$

$$(\alpha^{-1}\beta^{-1})(2) = \alpha^{-1}(\beta^{-1}(2)) = \alpha^{-1}(5) = 4$$

$$(\alpha^{-1}\beta^{-1})(3) = \alpha^{-1}(\beta^{-1}(3)) = \alpha^{-1}(4) = 6$$

$$(\alpha^{-1}\beta^{-1})(4) = \alpha^{-1}(\beta^{-1}(4)) = \alpha^{-1}(1) = 2$$

$$(\alpha^{-1}\beta^{-1})(5) = \alpha^{-1}(\beta^{-1}(5)) = \alpha^{-1}(3) = 1$$

$$(\alpha^{-1}\beta^{-1})(6) = \alpha^{-1}(\beta^{-1}(6)) = \alpha^{-1}(6) = 6$$

$$\alpha^{-1}\beta^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 6 & 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 5 & 4 & 1 & 3 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 4 & 6 & 2 & 3 & 5 \end{pmatrix}$$

Ans.

(b) Write down the following permutations as the product of disjoint cycles.

(i) $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 3 & 4 & 5 & 1 & 6 & 7 & 9 & 8 \end{pmatrix}$

(ii) $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 3 & 1 & 2 \end{pmatrix}$

Sol.

(i) $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 3 & 4 & 5 & 1 & 6 & 7 & 9 & 8 \end{pmatrix}$

$$= \begin{pmatrix} 6 & 7 & 1 & 2 & 3 & 4 & 5 & 8 & 9 \\ 6 & 7 & 2 & 3 & 4 & 5 & 1 & 9 & 8 \end{pmatrix}$$

$$= (6)(7)(12345)(89)$$

$$= (12345)(89)$$

$$(6) = 1 \text{ and } (7) = 1$$

Ans.

(ii) $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 3 & 1 & 2 \end{pmatrix}$

$$= \begin{pmatrix} 1 & 6 & 2 & 5 & 3 & 4 \\ 6 & 2 & 5 & 1 & 4 & 3 \end{pmatrix}$$

$$= (1625)(34)$$

Ans.

(c) Prove that

$$(134)(24) = (1342)$$

Sol. $\alpha = (134) = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 1 & 2 \end{pmatrix}$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix}$$

Let

$$\beta = (24) = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 1 & 3 \end{pmatrix}$$

$$\begin{aligned}
 &= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \\
 (\alpha \circ \beta)(1) &= \alpha(\beta(1)) = \alpha(1) = 3 \\
 (\alpha \circ \beta)(2) &= \alpha(\beta(2)) = \alpha(4) = 1 \\
 (\alpha \circ \beta)(3) &= \alpha(\beta(3)) = \alpha(3) = 4 \\
 (\alpha \circ \beta)(4) &= \alpha(\beta(4)) = \alpha(2) = 2 \\
 (1\ 3\ 4)(2\ 4) &= \alpha\beta \\
 &= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \\
 &= (1\ 3\ 4\ 2)
 \end{aligned}$$

Q. 5. (a) Verify Cayley's theorem for the cyclic group G of order 3. What is the corresponding permutation group isomorphic to G?

Sol. We have $G = \{e, a^2, a^3 = e\}$.

Its permutation group is

$$\bar{G} = \langle f_a, f_{a^2} \rangle$$

Where,

$$\begin{aligned}
 f_a(a) &= ae = a \\
 f_a(a^2) &= a^2e = a^2 \\
 f_a(e) &= ee = e
 \end{aligned}$$

Thus,

Now,

$f_a = I$ is the identity mapping of G.

$$fa(a) = aa = a^2$$

$$fa(a^2) = aa^2 = e$$

$$fa(e) = ae = a$$

and

$$fa^2(a) = a^2a = e$$

$$fa^2(a^2) = a^2a^2 = ab = a$$

$$fa^2(e) = a^2e = a^2$$

It may be observed that

$$\bar{G} = \langle fa \rangle$$

$$fa^2 = faa = fafa = (fa)^2$$

$$fe = fa^2 = (fa)^3$$

$$= \langle a \rangle$$

Since any two cyclic group of the same order are isomorphic i.e.,

$$G = \bar{G}$$

Proved.

(b) If G is internal direct product of its subgroups $H_1, H_2, \dots, H_n$ then prove that

$$G = H_1 \times H_2 \times H_3 \times \dots \times H_n$$

Sol.

Let

$$T = H_1 \times H_2 \times \dots \times H_n$$

Define a mapping $\phi: T \rightarrow G$ as

$$\phi(x) = x_1 x_2 \dots x_n \quad \forall x = (x_1, x_2, \dots, x_n) \in T$$

First of all we show that ϕ is one one

Let

$$x = (x_1, x_2, \dots, x_n) \in T$$

$$y = (y_1, y_2, \dots, y_n) \in T$$

Then,

$$\phi(x) = \phi(y)$$

$\Rightarrow$

$$x_1 x_2 \dots x_n = y_1 y_2 \dots y_n$$

$\Rightarrow$

$$x_i = y_i \quad \forall i = 1, 2, \dots, n$$

$\Rightarrow$

$$(x_1 x_2 \dots x_n) = (y_1 y_2 \dots y_n) \Rightarrow x = y$$

Thus ϕ is one to one.

To show ϕ is onto

Let $g \in G$ be arbitrary. Since, G is the internal direct product of $H_1, H_2, \dots, H_n$. We can uniquely express $g \in G$ as

$$g = h_1 h_2 h_3 \dots h_n \in H_i$$

for

$$1 \leq i \leq n$$

Then,

$$h = (h_1 h_2 h_3 \dots h_n) \in T$$

$\Rightarrow$ This shows that ϕ is onto.

To show that ϕ is homomorphism

Let

$$x = (x_1 x_2 \dots x_n) \in T$$

$$y = (y_1 y_2 \dots y_n) \in T$$

Then

$$xy = (x_1 y_1, x_2 y_2, \dots, x_n y_n)$$

$$\phi(xy) = x_1 y_1 x_2 y_2 \dots x_n y_n$$

By definition of IDP, we have:

$$x_i y_i = y_i x_i \text{ for } i \neq j$$

From (2) and (3):

$$\phi(xy) = x_1 x_2 \dots x_n y_1 y_2 \dots y_n$$

$\Rightarrow$

$$\phi(xy) = \phi(x) \phi(y)$$

$\Rightarrow$

ϕ is a homomorphism

hence,

$$T = G$$

$\Rightarrow$

$$G = T$$

Proved.

(c) Let A and B be cyclic group of order m and n respectively. Prove that $A \times B$ is cyclic if and only if m and n are relatively prime.

Sol. Let

$$A = \langle a \rangle$$

$$a^m = e$$

$$B = \langle b \rangle, b^n = e$$

...

...

Let m and n be relatively prime integers.
We have,

Consider,

$$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$$

$$\begin{aligned}(a, b)^{mn} &= (a^{mn}, b^{mn}) \\ &= (a^m)^n, (b^n)^m \\ &= (e_1^n, e_2^m) \\ &= (e_1, e_2)\end{aligned}$$

$$(a, b)^{mn} = (e_1, e_2), \text{ identity of } A \times B$$

We claim that mn is the smallest positive integer satisfying (3).
Let k be any positive integer s.t.:

$$(a, b)^k = (e_1, e_2)$$

$\Rightarrow$

$$(\alpha^k, b^k) = (e_1, e_2)$$

$\Rightarrow$

$$\alpha^k = e, \text{ and } b^k = e_2$$

Where

$$O(\alpha) = m, O(b) = n$$

$\Rightarrow$

$$m \mid k \text{ and } n \mid k$$

$\Rightarrow$

$$mn \mid k, \text{ since } (m, n) = 1$$

$\Rightarrow$

$$mn < k$$

Which proves our assertion

Hence,

$$\begin{aligned}A \times B &= \langle (a, e), (a, b)^{mn} \rangle \\ &= \langle e_1, e_2 \rangle\end{aligned}$$

and so $A \times B$ is cyclic of order mn

Conversely, let $A \times B$ be a cyclic group of order mn , where,

$$A = \langle a \rangle, a^m = e_1$$

$$B = \langle b \rangle, b^n = e_2$$

Let

$$A \times B = \langle (x, y) \mid x \in A, y \in B \rangle$$

and

$$(x, y)^{mn} = (e_1, e_2)$$

We have to show that $(m, n) = 1$.

Let

$$(m, n) = d \Rightarrow d \mid m \text{ and } d \mid n$$

$\frac{m}{d}$ and $\frac{n}{d}$ are positive integers.

We have

$$(x, y)^{\frac{mn}{d}} = \left((x^m)^{\frac{n}{d}}, (y^n)^{\frac{m}{d}} \right)$$

$$= \left((e_1)^{\frac{n}{d}}, (e_2)^{\frac{m}{d}} \right)$$

$$= (e_1, e_2)$$

Since,

$$O(x, y) = mn \text{ it follows that } mn \text{ divides } \frac{mn}{d}$$

$\Rightarrow$

$$d = 1$$

$\Rightarrow$

$$(m, n) = 1$$

Hence, m and n are relatively prime.

Q. 6. (a) State and prove that Cauchy's theorem for finite abelian group.

Sol. Statement :

Let G be a finite abelian group and let a prime p be a divisor of $O(G)$. Then G contains an element $a \neq e$ s.t. $a^p = e$.

Proof :

Suppose G is a finite abelian group s.t. a prime number p is a divisor of $O(G)$.

To prove that $\exists a \in G$ s.t. $a \neq e$ and $a^p = e$.

We shall prove that result by the result by the method of mathematical induction over $O(G)$ i.e., We suppose that if the theorem is true for every abelian group of order less than $O(G)$, then it is also true for G .

(i) $O(G) = 1$ i.e., if $G = \{e\}$, then there is nothing to prove.

(ii) Suppose G has no proper subgroup and $O(G) > 1$.

Then $O(G)$ is prime.

For every group of composite order possesses proper subgroup.

$O(G)$ is prime $\Rightarrow G$ is cyclic.

$\therefore p \mid O(G)$, p and $O(G)$ are prime

$$\Rightarrow p = O(G)$$

Thus G is cyclic group of order p . Let a be a generator of G so that $O(a) = \text{order of generate} = \text{order of group} = p$ i.e.

$$O(a) = p$$

$\Rightarrow$

$$a^p = e$$

$\Rightarrow$

$$a \neq e$$

(iii) Suppose G has a proper subgroup H and $O(G) > 1$ G is abelian.

$\Rightarrow H$ is normal in G

$\Rightarrow$ the symbol G/H is meaningful

Since G is abelian $\Rightarrow G/H$ is abelian

H is a proper subgroup of G

$\Rightarrow H$ is abelian

$$O(H) < O(G)$$

(a) Suppose $p \mid O(H)$

Hence, our induction hypothesis $\exists$ an element.

$$b \in H \text{ s.t. } b \neq e, b^p = e$$

$$b \in H \Rightarrow b \in G \text{ for } H \subset G$$

Thus, $\exists$ an element $b \in G$ s.t. $b \neq e$ and $b^p = e$

(b) Suppose p is not a divisor of $O(H)$ in this case also, we shall prove that $\exists$ an element $a \in G$ s.t. $a^p = e$

H is a proper subgroup of G

$$\Rightarrow O(H) > 1 \text{ and } O(H) < O(G)$$

$$\Rightarrow \frac{O(G)}{O(H)} < O(G)$$

$$\Rightarrow \frac{O(G)}{O(H)} \nmid O(G)$$

Also p is not divisor of $O(H)$ and $p \mid O(G)$

$$\Rightarrow p \mid O(G/H)$$

$$\text{Finally, } O\left(\frac{G}{H}\right) < O(G) \text{ and } p \mid O(G/H)$$

Hence by our induction hypothesis, $\exists$ an element $He \in \frac{G}{H}$ s.t. $He \neq (He)^p$ and $p \nmid O(HC) = p$.

H being identity element of $\frac{G}{H}$.

$$He \in \frac{G}{H} \Rightarrow e \in G$$

H is normal in G

$$\Rightarrow (He)^p = He^p$$

$$\Rightarrow He^p = H$$

$$\Rightarrow e^p = H$$

$$\Rightarrow (e^p)^{O(H)} = e = \text{identity of } H$$

$$\Rightarrow e^{p \cdot O(H)} = e$$

$$\Rightarrow [e^{O(H)}]^p = e$$

$$\Rightarrow d^p = e, \text{ where } d = e^{O(H)}$$

d will be the required element of G , if we show that $d \neq e$.
If possible let $d = e$

$$d = e \Rightarrow e^{O(H)} = e$$

$$He^{O(H)} = He$$

$$(He)^{O(H)} = H$$

$$O(He) = p, (He)^{O(H)} = H \Rightarrow p \mid O(H)$$

Which is a contradiction

Hence, the assumption $d = e$ is wrong.

Therefore, $d \neq e$.

This completes the proof.

(b) State and prove second Isomorphism theorem.

Sol. Second Isomorphism Theorem:

Statement:

If H and K both are normal subgroups of a group G s.t. $H \subset K$, then

$$\frac{G}{K} \cong \left(\frac{G}{H} \right) / \left(\frac{K}{H} \right)$$

Proof:

Let H and K be normal subgroup of a group G s.t. $H \subset K$, so that the symbols $\frac{G}{K} \cong \left(\frac{G}{H} \right) / \left(\frac{K}{H} \right)$ are meaningful.

Also, $\frac{K}{H}$ is a normal subgroup of $\frac{G}{H}$

Define a map $f: \frac{G}{H} \rightarrow \frac{G}{K}$ by the formula

$$f(Hx) = Kx \quad \forall x \in G$$

Now, we shall show that f is a homomorphism onto.

(i) To prove that f is well defined

$$Hx = Hy : x, y \in G$$

$$\Rightarrow xy^{-1} \in H \in K$$

$$\Rightarrow xy^{-1} \in K$$

$$\Rightarrow Kx = Ky$$

$$\Rightarrow f(Hx) = f(Hy)$$

Which shows that f is well defined.

(ii) To prove f is a homomorphism:

$$x, y \in G \Rightarrow Hx, Hy \in \frac{G}{H}$$

$$f(Hx \cdot Hy) = Kxy$$

$$f(Hx \cdot Hy) = Kx \cdot Ky$$

$$f(Hxy) = b(Hx) b(Hy)$$

f is a homomorphism.

(iii) To prove f is onto:

Given any $Kx \in \frac{G}{K} \Rightarrow x \in G$

$$\Rightarrow Hx \in \frac{G}{H} \text{ s.t. } \phi(Hx) = Kx$$

$\Rightarrow f$ is onto

(iv) To prove that kernel of f is $\frac{K}{H}$

$$\text{So } f = \{ Hx \in \frac{G}{H} \mid f(Hx) = K \}$$

K is the identity of $\frac{G}{K}$

any $Hx \in \text{so } f \Rightarrow f(Hx) = K$

$$\Rightarrow Kx = K$$

$$\Rightarrow x \in K$$

$$\Rightarrow Hx \in \frac{K}{H}$$

$$\text{any } Hx \in \ker f \Rightarrow Hx \in \frac{K}{H}$$

Which proves that $\ker f \subseteq \frac{K}{H}$

$$\text{Again any } Hx \in \frac{K}{H}$$

$$\Rightarrow x \in K$$

$$\Rightarrow Kx = K$$

$$\Rightarrow f(Hx) = K$$

$$\Rightarrow Hx \in \ker f$$

$$\Rightarrow \frac{K}{H} \subseteq \ker f$$

From (1) and (2)

$$\frac{K}{H} = \ker f \quad \dots(2)$$

Finally, we have shown that $f: \frac{G}{H} \xrightarrow{\text{onto}} \frac{G}{K}$ is a homomorphism with $\ker f = \frac{K}{H}$.

$f = \frac{K}{H}$. Also $\frac{K}{H}$ is a normal subgroup of $\frac{G}{H}$.

Hence, by third homomorphism, theorem

We have :

$$\frac{G}{K} \cong \frac{\left(\frac{G}{H} \right)}{\left(\frac{K}{H} \right)}$$

Which proves second Isomorphism theorem.

(b) Stat and prove third theorem on Isomorphism.

Sol. Third theorem on Isomorphism :

Statement :

If N and M are normal subgroup of G , then

$$\frac{NM}{N} \cong \frac{N}{(N \cap M)}$$

Proof :

Let N be a normal subgroup of a group G and H any subgroup of G .

(i) To prove that the symbol $\frac{HN}{N}$ is meaningful.

N is normal in G

$$\Rightarrow NH = HN$$

$\Rightarrow HN$ is a subgroup of G

We have $N \subseteq HN \subseteq G$

Since, N is normal in G , $Nx = xN \forall x \in G$.

In particular this $\Rightarrow Nx = xN \forall x \in HN$ and have the symbol $\frac{HN}{N}$ has a meaning.

(ii) To show that the symbol $\frac{H}{(H \cap N)}$ has a meaning.

N is normal in G and H is a subgroup of G .

$\Rightarrow H \cap N$ is a normal subgroup of H

$\Rightarrow$ The symbol $\frac{H}{(H \cap N)}$ has a meaning

(iii) Define a map $f: H \rightarrow \frac{HN}{N}$ s.t.

$$f(x) = Nx, x \in H$$

To prove that f is a homomorphism onto f is a homomorphism

$x, y \in H \Rightarrow xy \in H$ for H is a subgroup

$$\Rightarrow f(xy) = Nxy$$

$$\Rightarrow f(xy) = (Nx)(Ny)$$

$$\Rightarrow f(xy) = f(x)f(y)$$

$\Rightarrow f$ is a homomorphism.

f is onto:

Let $Nx \in \frac{HN}{N}$ be arbitrary, so that $x \in HN$

$$x \in HN \Rightarrow \exists h \in H, n \in N \text{ s.t. } x = hn$$

$$HN = NH, hn \in HN \Rightarrow \forall nh' \in NH \text{ s.t. } hn = n'h'$$

$$\Rightarrow n, n' \in N; h, h' \in H \text{ s.t. } hn = n'h'$$

$$f(h) = Nh' \text{ by def. of } f$$

$$\Rightarrow (Nn)'h' \text{ for } n' \in N \Rightarrow Nn' = N$$

$$= N(n'h')$$

$$= N(h')$$

$$= Nx$$

Finally, each $Nx \in \frac{HN}{N}$ is f image of some $h' \in H$.

This proves that f is on to.

To show: $\ker f = H \cap N$

$\ker f = \{x \in H : f(x) = N\}$, by definition of kernel any $x \in \ker f$

$$\Rightarrow x \in H, f(x) = N$$

$$\Rightarrow x \in H, Nx = xN$$

$$\Rightarrow x \in H, x \in N$$

$$\Rightarrow x \in H \cap N$$

$$\Rightarrow \ker f \subset H \cap N$$

$$\text{any } y \in H \cap N$$

$$\Rightarrow y \in H, y \in N$$

$$\Rightarrow Ny = N, y \in H$$

$$\Rightarrow f(y) = N, y \in H$$

$$\Rightarrow y \in \ker f$$

$$\Rightarrow H \cap N \subset \ker f$$

From (1) and (2)

$$\ker f = H \cap N$$

Thus, we have shown that $f: H \rightarrow \frac{HN}{N}$ is a homomorphism onto with $\ker f = H \cap N$.

while using third homomorphism theorem, we have final results-

$$\frac{HN}{N} \cong \frac{H}{(H \cap N)}$$

○○○

Name of the Paper : Paper : C-7 : Multivariate Calculus
Name of Course : B.Sc. (Hons.) Mathematics II Year (CBCS)
Semester : III
Duration : 3 Hours
Maximum Marks : 75

MODEL TEST PAPER-I

Q. 1. (a) Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}; (x,y) \neq (0,0)$$

Sol. Let $f(x,y) = \frac{xy^2}{x^2 + y^4}$

Let $(x,y) \rightarrow (0,0)$ along the path $y = \sqrt{x}$, where m is any real number.

When $x \rightarrow 0$, then $y \rightarrow 0$ along the path $y = m\sqrt{x}$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{x(m\sqrt{x})^2}{x^2 + (m\sqrt{x})^4}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{m^2 x^2}{x^2 + m^4 x^2}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{m^2 x^2}{x^2(1 + m^4)}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{m^2}{1 + m^4}$$

$$= \frac{m^2}{1 + m^4}$$

Which is different for different values of m .

$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y)$ does not exist.

$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$ does not exist.

Ans.

(b) Show that the function f defined by

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$.

Sol.
$$f(x, y) = \frac{x^3 - y^3}{x^2 + y^2} \quad ; (x, y) \neq (0, 0)$$

$$= 0 \quad ; (x, y) = (0, 0)$$
 ... (1)

$$|f(x, y) - f(0, 0)| = \left| \frac{x^3 - y^3}{x^2 + y^2} - 0 \right|$$

$$= \left| \frac{x^3 - y^3}{x^2 + y^2} \right|$$
 ... (2)

On putting

We get :

$$x = r \cos \theta \text{ and } y = r \sin \theta$$

$$x^2 + y^2 = r^2$$
 ... (3)

$$|f(x, y) - f(0, 0)| = \frac{|(r \cos \theta)^3 - (r \sin \theta)^3|}{|(r \cos \theta)^2 + (r \sin \theta)^2|}$$

$$= \frac{|r^3 (\cos^3 \theta - \sin^3 \theta)|}{r^2 (\cos^2 \theta + \sin^2 \theta)}$$

$$= r |\cos^3 \theta - \sin^3 \theta|$$

$$\leq r (|\cos^3 \theta| + |\sin^3 \theta|)$$

Because $|\cos^3 \theta| \leq 1$ and $|\sin^3 \theta| \leq 1$

$$\leq r (1 + 1)$$

$$= 2r$$

$$= 2\sqrt{x^2 + y^2}$$

Let $\epsilon > 0$ be any given real number.

$$\therefore |f(x, y) - f(0, 0)| < \epsilon \text{ if } 2\sqrt{x^2 + y^2} < \epsilon$$

$$\Rightarrow |f(x, y) - f(0, 0)| < \epsilon \text{ if } 2\sqrt{x^2 + y^2} < \frac{\epsilon}{2}$$

$$\Rightarrow |f(x, y) - f(0, 0)| < \epsilon, \text{ whenever } \sqrt{(x-0)^2 + (y-0)^2} < \delta$$

Where,

$$\delta = \frac{\epsilon}{2}$$

Hence, by definition of continuity, $f(x, y)$ is continuous at $(0, 0)$. Proved.

(c) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} 1 & ; \text{if } x \text{ is irrational} \\ 0 & ; \text{if } x \text{ is rational} \end{cases}$$

Show that f is continuous everywhere.

Sol.
$$f(x, y) = \begin{cases} 1 & ; \text{if } x \text{ is irrational} \\ 0 & ; \text{if } x \text{ is rational} \end{cases}$$
 ... (1)

Let $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ exists and equal to l .

$\therefore$ For a given $\epsilon > 0$, there exists positive real numbers δ_1, δ_2 such that

$$|f(x, y) - l| < \frac{\epsilon}{2} \text{ whenever } |x - a| < \delta_1$$

$$|y - b| < \delta_2$$

Let x_1 be any rational number and x_2 be any irrational number such that

$$|x_1 - a| < \delta_1 \text{ and } |x_2 - a| < \delta_1$$

$$\therefore |f(x, y) - l| < \frac{\epsilon}{2} \text{ and } |f(x, y) - l| < \frac{\epsilon}{2}$$

$$\therefore |0 - l| < \frac{\epsilon}{2} \text{ and } |1 - l| < \frac{\epsilon}{2}$$

$$|l| < \frac{\epsilon}{2} \text{ and } |1 - l| < \frac{\epsilon}{2}$$

Now,
$$1 = |1 - l + l|$$

$$\leq |1 - l| + |l|$$

$$< \frac{\epsilon}{2} + \frac{\epsilon}{2}$$

$$\Rightarrow 1 < \epsilon$$

Taking $\epsilon = \frac{1}{2}$, we get $1 < \frac{1}{2}$

Which is absurd

Hence, our supposition that $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ is wrong.

$\therefore \lim_{(x, y) \rightarrow (a, b)} f(x, y)$ does not exist.

Hence $f(x, y)$ is discontinuous everywhere

Proved

Q. 2. (a) Find the first order partial derivation of the following :

(i) $\cos^{-1}\left(\frac{y}{x}\right)$

(ii) $\log(x^2 + y^2)$

Sol. Let

$$u = \cos^{-1}\left(\frac{y}{x}\right)$$

The first order partial derivation of the function u are given by $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$. These are calculated as below:

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left[\cos^{-1}\left(\frac{y}{x}\right) \right]$$

$$= \frac{-1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right)$$

$$= \frac{x}{\sqrt{x^2 - y^2}} \left(\frac{-y}{x^2} \right)$$

$$= \frac{y}{x\sqrt{x^2 - y^2}}$$

Ans.

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left[\cos^{-1}\left(\frac{y}{x}\right) \right]$$

$$= \frac{-1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} \cdot \frac{\partial}{\partial y} \left(\frac{y}{x} \right)$$

$$= \frac{-x}{\sqrt{x^2 - y^2}} \left(\frac{1}{x} \right)$$

$$= \frac{-1}{\sqrt{x^2 - y^2}}$$

$$u = \log(x^2 + y^2)$$

(ii) Let

The first order partial derivations of u are given by $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$. These are calculated as:

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} [\log(x^2 + y^2)]$$

$$= \frac{1}{x^2 + y^2} \cdot \frac{\partial}{\partial x} (x^2 + y^2)$$

$$= \frac{1}{x^2 + y^2} (2x + 0)$$

$$= \frac{2x}{x^2 + y^2}$$

Ans.

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} [\log(x^2 + y^2)]$$

$$= \frac{1}{x^2 + y^2} \cdot \frac{\partial}{\partial y} (x^2 + y^2)$$

$$= \frac{1}{x^2 + y^2} (0 + 2y)$$

$$= \frac{2y}{x^2 + y^2}$$

Ans.

(b) The temperature at a point (x, y) on a given metal plate is the xy plane is determined according to the formulas

$$T(x, y) = x^3 + 2xy^2 + y$$

degree. Compute the rate at which the temperature changes with distance if we start at $(2, 1)$ and move

(i) parallel to the vector j

(ii) parallel to the vector i

Sol. The temperature at a point (x, y) on a given metal plate in xy -plane is given by :

$$T(x, y) = x^3 + 2xy^2 + y$$

(i) The rate at which temperature changes with distance and more parallel to the vector j is calculated as partial derivation of T w.r.t. y .

$$\left. \frac{\partial T}{\partial y} \right|_{(2,1)} = \left. \frac{\partial}{\partial y} (x^3 + 2xy^2 + y) \right|_{(2,1)}$$

$$= (0 + 4xy + 1) \Big|_{(2,1)}$$

$$= (4xy + 1) \Big|_{(2,1)}$$

$$= 4 \times 2 \times 1 + 1$$

$$= 8 + 1$$

$$= 9$$

Ans.

(ii) The rate at which temperature T changes with the distance in the direction i is calculated as the partial derivative of T w.r. to x :

$$\frac{\partial T}{\partial x} \Big|_{(2,1)} = \frac{\partial}{\partial x} (x^3 + 2xy^2 + y) \Big|_{(2,1)}$$

$$= (3x^2 + 2y^2 + 0) \Big|_{(2,1)}$$

$$= (3x^2 + 2y^2) \Big|_{(2,1)}$$

$$= 3(2)^2 + 2(1)^2$$

$$= 12 + 2$$

$$= 14$$

Ans.

(c) Find the equation of the tangent plane to the surface

$$z = 10 - x^2 - y^2 \text{ at } P_0(2, 2, 2).$$

Sol.

Let

$$f(x, y) = 10 - x^2 - y^2$$

$$f_x(x, y) = \frac{\partial f}{\partial x}$$

$$= \frac{\partial}{\partial x} (10 - x^2 - y^2)$$

$$= (0 - 2x - 0)$$

$$= -2x$$

$$f_y(x, y) = \frac{\partial f}{\partial y}$$

$$= \frac{\partial}{\partial y} (10 - x^2 - y^2)$$

$$= (0 - 0 - 2y)$$

$$= -2y$$

At $(2, 2)$

$$f'_x(2, 2) = -2(2) = -4$$

$$f'_y(2, 2) = -2(2) = -4$$

The equation of tangent plane at $P_0(2, 2, 2)$ is:

$$z - z_0 = f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0)$$

$$z - 2 = (-4)(x - 2) + (-4)(y - 2)$$

$$z - 2 = -4x + 8 - 4y + 8$$

$$4x + 4y + z + (-2 - 8) = 0$$

$$4x + 4y + z - 18 = 0$$

Ans.

Q. 3. (a) Find the directional derivative of the function $f(x, y) = \frac{e^{-x}}{y}$

at the point $(2, -1)$ in the direction of the vector $\hat{u} = \frac{-\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}}$.

Sol.

$$f(x, y) = \frac{e^{-x}}{y} \quad \dots(1)$$

$$\hat{u} = \frac{-\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}} = u_1 \hat{i} + u_2 \hat{j} \quad \dots(2)$$

$$f(x, y) = \frac{\partial f}{\partial x} \text{ Where } u_1 = \frac{1}{\sqrt{2}}$$

$$u_2 = \frac{1}{\sqrt{2}}$$

$$= \frac{\partial}{\partial x} \left(\frac{e^{-x}}{y} \right)$$

$$= \frac{1}{y} \frac{\partial}{\partial x} (e^{-x})$$

$$= \frac{1}{y} (-e^{-x})$$

$$= \frac{-e^{-x}}{y}$$

$$f'_x(x, y) = \frac{\partial f}{\partial x}$$

$$= \frac{\partial}{\partial y} \left(\frac{e^{-x}}{y} \right)$$

$$\begin{aligned}
 &= e^{-x} \frac{\partial}{\partial y} \left(\frac{1}{y} \right) \\
 &= e^{-x} \left(-\frac{1}{y^2} \right) \\
 &= -\frac{e^{-x}}{y^2}
 \end{aligned}$$

At (2, -1)

$$f_x(2, -1) = \frac{-e^{-x}}{-1} = e^{-2}$$

$$f_y(2, -1) = \frac{-e^{-x}}{(-1)^2} = -e^{-2}$$

The directional derivative of the function $f(x, y)$ at the point $(x_0, y_0) = (2, -1)$ in the direction of $\hat{u}$ be:

$$D_u f(x_0, y_0) = f_x(x_0, y_0) u_1 + f_y(x_0, y_0) u_2$$

$$D_u f(2, -1) = f_x(2, -1) u_1 + f_y(2, -1) u_2$$

$$D_u f(2, -1) = (e^{-x}) \left(\frac{-1}{\sqrt{2}} \right) + (-e^{-2}) \left(\frac{1}{\sqrt{2}} \right)$$

$$D_u f(2, -1) = \frac{-e^{-2}}{\sqrt{2}} - \frac{e^{-2}}{\sqrt{2}}$$

$$= \frac{-e^{-2} - e^{-2}}{\sqrt{2}}$$

$$= \frac{-2e^{-2}}{\sqrt{2}}$$

$$= -\sqrt{2} e^{-2}$$

(b) If $r = |\vec{r}|$, where $\vec{r} = xi + yj + zk$ then prove that

$$(1) \nabla(r) = \frac{\vec{r}}{r}$$

$$(2) \nabla\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$$

$$(3) \nabla(r^4) = 4r^3 \nabla r$$

Sol.

$$\vec{r} = xi + yj + zk$$

$$r = |\vec{r}|$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$r^2 = x^2 + y^2 + z^2$$

d. p. Eq. (2) w. r. t. x:

$$2r \frac{\partial r}{\partial x} = 2x$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

d. f. Eq. (2) w. r. t. y:

$$2r \frac{\partial r}{\partial y} = 2y$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

d. p. Eq. (3) w. r. t. z:

$$2r \frac{\partial r}{\partial z} = 2z$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\nabla(r) = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) r$$

$$= i \left(\frac{\partial r}{\partial x} \right) + j \left(\frac{\partial r}{\partial y} \right) + k \left(\frac{\partial r}{\partial z} \right)$$

$$= i \left(\frac{x}{r} \right) + j \left(\frac{y}{r} \right) + k \left(\frac{z}{r} \right)$$

$$= \frac{1}{r} (ix + jy + kz)$$

$$= \frac{1}{r} \left(\frac{-r}{r^2} \right)$$

$$= \frac{-1}{r^2}$$

(ii)

$$\nabla f(r) = f'(r) \nabla(r)$$

$$\nabla \left(\frac{1}{r} \right) = \left(-\frac{1}{r^2} \right) \nabla(r)$$

$$= \left(-\frac{1}{r^2} \right) \left(\frac{-r}{r} \right)$$

$$= \frac{-r}{r^3}$$

(iii)

$$\nabla f(r) = f'(r) \nabla(r)$$

$$\nabla(r^n) = (nr^{n-1}) \nabla(r)$$

$$= (nr^{n-1}) \left(\frac{-r}{r} \right)$$

$$= nr^{n-2} \frac{-r}{r}$$

Proved

Proved

Proved

(c) Using Lagrange's multiplies method, find the maximum and minimum values of the function,

Sol. Let

$$f(x, y) = 3x + 4y \text{ on the circle } x^2 + y^2 = 1.$$

$$f(x, y) = 3x + 4y \quad \dots(1)$$

$$g(x, y) = x^2 + y^2 - 1 \quad \dots(2)$$

$$f_x(x, y) = \frac{\partial f}{\partial x}$$

$$= \frac{\partial}{\partial x} (3x + 4y)$$

$$= 3 + 0$$

$$= 3$$

$$f_y(x, y) = \frac{\partial f}{\partial y}$$

$$= \frac{\partial}{\partial y} (3x + 4y)$$

$$= 0 + 4$$

$$= 4$$

$$g_x(x, y) = \frac{\partial g}{\partial x}$$

$$= \frac{\partial}{\partial x} (x^2 + y^2 - 1)$$

$$= 2x + 0 - 0$$

$$= 2x$$

$$g_y(x, y) = \frac{\partial g}{\partial y}$$

$$= \frac{\partial}{\partial y} (x^2 + y^2 - 1)$$

$$= 0 + 2y - 0$$

$$= 2y$$

Using Lagrange's multiplies method, we have:

$$f_x(x, y) = \lambda g_x(x, y)$$

$$3 = \lambda (2x)$$

$$x = \frac{3}{2\lambda}$$

and

$$f_y(x, y) = \lambda g_y(x, y)$$

$$4 = \lambda (2y)$$

$$y = \frac{2}{\lambda}$$

The equation $g(x, y) = 0$ gives:

$$x^2 + y^2 = 1$$

$$\left(\frac{3}{2\lambda} \right)^2 + \left(\frac{2}{\lambda} \right)^2 = 1$$

$$\frac{9}{4\lambda^2} + \frac{4}{\lambda^2} = 1$$

$$\frac{9+16}{4\lambda^2} = 1$$

$$\frac{25}{4\lambda^2} = 1$$

$$\lambda^2 = \frac{25}{4}$$

$$\lambda = \pm \frac{5}{2}$$

Thus,

$$x = \frac{3}{2\lambda} = \frac{3}{2} \left(\pm \frac{2}{5} \right) = \pm \frac{3}{5}$$

$$y = \frac{2}{\lambda} = 2 \left(\pm \frac{2}{5} \right) = \pm \frac{4}{5}$$

The function $f(x, y) = 3x + 4y$ has extreme value at (x, y)

$$= \pm \left(\frac{3}{5}, \frac{4}{5} \right)$$

By calculating the value of $3x + 4y$ at the point $\pm \left(\frac{3}{5}, \frac{4}{5} \right)$, we observe that its maximum and minimum values on the circle $x^2 + y^2 = 1$ are ;

$$\text{Max value} = 3 \left(\frac{3}{5} \right) + 4 \left(\frac{4}{5} \right)$$

$$= \frac{9}{5} + \frac{16}{5}$$

$$= \frac{25}{5}$$

$$= 5$$

$$\text{Min value} = 3 \left(-\frac{3}{5} \right) + 4 \left(-\frac{4}{5} \right)$$

$$= -\frac{9}{5} - \frac{16}{5}$$

$$= -\frac{25}{5}$$

$$= -5$$

Q. 4. (a) Examine for maximum and minimum value of the function.

$$f = x^2 + y^2 + 6x + 12$$

$$f = x^2 + y^2 + 6x + 12$$

Sol.

$$\frac{\partial b}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2 + 6x + 12)$$

$$= 2x + 0 + 6 + 0$$

$$= 2x + 6$$

$$\frac{\partial b}{\partial y} = \frac{\partial}{\partial y} (x^2 + y^2 + 6x + 12)$$

$$= 0 = 2y + 0 + 0$$

$$= 2y$$

$$A = \frac{\partial^2 f}{\partial x^2}$$

$$= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$$

$$= \frac{\partial}{\partial x} (2x + 6)$$

$$= 2$$

$$B = \frac{\partial^2 f}{\partial x \partial y}$$

$$= \frac{\partial}{\partial x} \left(\frac{\partial b}{\partial y} \right)$$

$$= \frac{\partial}{\partial x} (2y)$$

$$= 0$$

$$C = \frac{\partial^2 f}{\partial y^2}$$

$$= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)$$

$$= \frac{\partial}{\partial y} (2y)$$

$$= 2$$

On putting $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$, we get :

$$\begin{aligned}\frac{\partial f}{\partial x} &= 0 \\ 2x + 6 &= 0 \\ x &= -3\end{aligned}$$

and

$$\begin{aligned}\frac{\partial f}{\partial y} &= 0 \\ 2y &= 0 \\ y &= 0\end{aligned}$$

Hence $(-3, 0)$ is the extreme point.

At $(-3, 0)$

$$\begin{aligned}A &= 2, B = 0, C = 2 \\ AC - B^2 &= (2)(2) - (0)^2 \\ &= 4 - 0 \\ &= 4\end{aligned}$$

$$\Rightarrow AC - B^2 > 0 \text{ and } A > 0$$

$$\Rightarrow (-3, 0) \text{ is a point of minima.}$$

Min. value of

$$\begin{aligned}f &= x^2 + y^2 + 6x + 12 \\ &= (-3)^2 + (0)^2 + 6(-3) + 12 \\ &= 9 - 18 + 12 \\ &= 21 - 18 \\ &= 3\end{aligned}$$

(b) Examine the differentiability of

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

Sol. The function $f(x, y)$ is given by :

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases} \quad \text{I.(1)}$$

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h}$$

$$\begin{aligned}&= \lim_{h \rightarrow 0} \frac{0 - 0}{h} \\ &= 0\end{aligned}$$

$$\begin{aligned}f_y(0, 0) &= \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} \\ &= \lim_{k \rightarrow 0} \frac{0 - 0}{k} \\ &= 0\end{aligned}$$

Let us assume that f is differentiable at the point $(0, 0)$, then by definition we have:

$$f(h, k) - f(0, 0) = hf_x(0, 0) + kf_y(0, 0) + h\phi(h, k) + k\psi(h, k)$$

Where

$$\phi(h, k) \rightarrow 0 \text{ as } (h, k) \rightarrow (0, 0)$$

$$\psi(h, k) \rightarrow 0 \text{ as } (h, k) \rightarrow (0, 0)$$

$$\frac{hk^2}{h^2 + k^2} - 0 = h(0) + k(0) + h\phi(h, k) + k\psi(h, k)$$

Putting

$$k = h, \text{ we get:}$$

$$\frac{h^3}{h^2 + h^2} = h\phi(h, h) + h\psi(h, h)$$

$$\frac{h}{2} = h[\phi(h, h) + \psi(h, h)]$$

$$\frac{h}{2} = h[\phi(h, h) + \psi(h, h)]$$

$$\frac{1}{2} = [\phi(h, h) + \psi(h, h)]$$

On taking limit as $h \rightarrow 0$, then

$$\phi(h, h) \rightarrow 0$$

and

$$\psi(h, h) \rightarrow 0$$

Hence, from (2)

$$\frac{1}{2} = 0 + 0$$

$$\frac{1}{2} = 0$$

Which is not possible.

Hence, our assumption is wrong

$\therefore f$ is not differentiable at $(0, 0)$.

(c) Define divergence and curl of a vector function.

Sol. (i) Divergence of a vector function.

The divergence of a vector function $\vec{A}(x, y, z) = A_1(x, y, z)\vec{i} + A_2(x, y, z)\vec{j} + A_3(x, y, z)\vec{k}$ where A_1, A_2, A_3 have first order partial derivation is written as $\nabla \cdot \vec{A}$

or $\text{div } \vec{A}$ and is defined as :

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z}$$

The $\text{div } \vec{A} = \nabla \cdot \vec{A}$ is a scalar function

(ii) Curl of a vector function :

The curl of a vector function $\vec{A}(x, y, z) = A_1(x, y, z)\vec{i} + A_2(x, y, z)\vec{j} + A_3(x, y, z)\vec{k}$ where A_1, A_2, A_3 have first order partial derivations is written as $\nabla \times \vec{A}$

or $\text{curl } \vec{A}$ and is defined as :

$$\text{Curl } \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}$$

$$= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \vec{i} + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \vec{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \vec{k}$$

The $\text{curl } \vec{A} = \nabla \times \vec{A}$ is a vector function.

Q. 5. (a) State and prove that Gauss's Divergence Theorem.

Sol. Gauss's Divergence Theorem :

Suppose V is the volume bounded by a closed piecewise smooth surface S .

If $\vec{F}(x, y, z)$ is a vector function of position, which is continuous and has continuous first derivatives in V , then

$$\iiint_V \nabla \cdot \vec{F} dV = \iint_S \vec{F} \cdot \vec{n} ds$$

Where $\vec{n}$ is the outwards drawn unit normal vector to S .

Proof :

Let

$$\vec{F} = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}$$

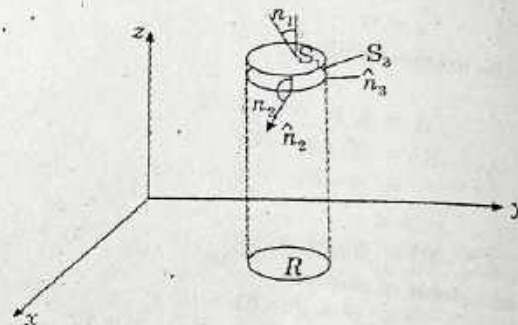
$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

We shall prove the theorem for a special region V which is bounded by a piecewise smooth closed surface S and has the property that any straight line parallel to any one of the co-ordinates axes and intersecting V has only one segment in common with V . If R is the orthogonal projection of S on the x, y plane, then V can be reffered in the form,

$$f(x, y) \leq z \leq g(x, y)$$

where, (x, y) varied in R .

$z = g(x, y)$ represents the upper portion S_1 of S , $z = f(x, y)$ represents the lower portion S_2 of S and there may be a remaining vertical portion $S_3 = S$.



We have,

$$\begin{aligned} \iiint_V \frac{\partial F_3}{\partial z} dV &= \iiint_V \frac{\partial F_3}{\partial z} dx dy dz \\ &= \iint_R \left[\int_{z=f(x,y)}^{z=g(x,y)} \frac{\partial F_3}{\partial z} dz \right] dx dy \\ &= \iint_R [F_3(x, y, z)]_{z=f(x,y)}^{z=g(x,y)} dx dy \end{aligned}$$

$$= \iint_R F_3 \vec{k} \cdot \vec{n} ds = \iiint_V \frac{\partial F_3}{\partial y} dV \quad \dots (3)$$

Similarly, by projecting S on the other co-ordinate planes, we get :

$$\iint_R F_1 \vec{i} \cdot \vec{n} ds = \iiint_V \frac{\partial F_1}{\partial x} dV \quad \dots (4)$$

and

$$\iint_R F_1 i \cdot \hat{n} \, ds = \iiint_V \frac{\partial F_1}{\partial y} dV$$

On adding (3), (4) and (5), we get :

$$\iint_R F_1 i \cdot \hat{n} \, ds + F_2 j \cdot \hat{n} \, ds + F_3 k \cdot \hat{n} \, ds = \iiint_V \left(\frac{\partial F_1}{\partial x} dV + \frac{\partial F_2}{\partial y} dV + \frac{\partial F_3}{\partial z} dV \right)$$

$$\iint_S (F_1 i + F_2 j + F_3 k) \cdot \hat{n} \, ds = \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dV$$

$$\iint_S (\vec{F}) \cdot \hat{n} \, ds = \iiint_V (\nabla \cdot \vec{F}) dV$$

$$\iiint_V (\nabla \cdot \vec{F}) dV = \iint_S (\vec{F} \cdot \hat{n}) \, ds$$

Which is Gauss's divergence theorem.

$$= \iint_R [F_3(x, y, g(x, y)) - F_3(x, y, f(x, y))] dx dy$$

$$= \iint_R F_3[x, y, g(x, y)] dx dy - \iint_R F_3[x, y, f(x, y)] dx dy \quad \dots(2)$$

Now, for the vertical portion S_3 of S the normal $\hat{n}_3$ to S_3 makes a right angle Y with k , therefore

$$\iint_{S_3} F_3 k \cdot \hat{n}_3 dS_3 = 0$$

For the upper portion S_1 of S , the normal $\hat{n}_1$ to S_1 makes an acute angle Y_1 with k . Therefore,

$$k \cdot \hat{n}_1 dS_1 = \cos(Y_1) dS_1 = dx dy$$

$$\iint_{S_1} F_3 k \cdot \hat{n}_1 dS_1 = \iint_R [F_3(x, y, g(x, y))] dx dy$$

For the lower portion S_2 of S , the normal $\hat{n}_2$ to S_2 makes an obtuse angle Y_2 with k . Therefore $k \cdot \hat{n}_2 dS_2 = \cos Y_2 dS_2 = -dx dy$.

Hence,

$$\iint_{S_2} F_3 k \cdot \hat{n}_2 dS_2 = -\iint_R F_3[x, y, f(x, y)] dx dy$$

$$\begin{aligned} \therefore \Rightarrow & \iint_{S_3} F_3 k \cdot \hat{n}_3 dS_3 + \iint_{S_1} F_3 k \cdot \hat{n}_1 dS_1 + \iint_{S_2} F_3 k \cdot \hat{n}_2 dS_2 \\ & = 0 + \iint_R F_3 [x, y, g(x, y)] dx dy - \iint_R F_3 [x, y, f(x, y)] dx dy \end{aligned}$$

(b) State and prove Green's theorem in the plane.

Sol. Green's Theorem:

Let R be a closed bounded region in the x, y plane whose boundary C consists of finitely many smooth curves. Let M and N be continuous functions of x and y having continuous partial derivatives $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ in R .

Then,

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \oint_C (M dx + N dy)$$

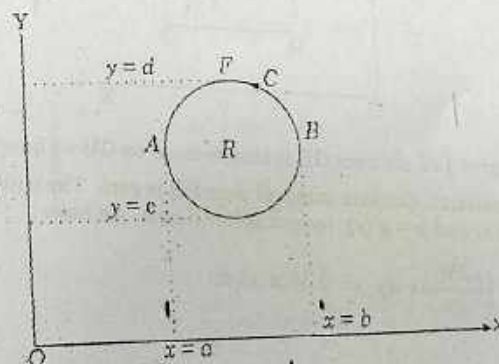
the line integral law is taken along the entire boundary C of R such that R is on the left as one advances in the direction of integration.

Proof:

We shall prove the theorem for a special region R bounded by a closed curve C and having the property that any straight line parallel to any one of the coordinate axes and intersecting R has only one segment in common with R . This means that R can be represented in both of the forms,

$$a \leq x \leq b, f(x) \leq y \leq g(x)$$

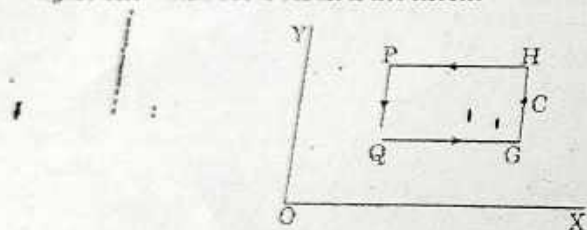
$$\text{and } c \leq y \leq d, p(y) \leq x \leq q(y)$$

In the above figure the equation of the curves AEB and BFA are $y = f(x)$ and $y = g(x)$ respectively. Similarly the equations of the curves FAE and EBF are $x = p(y)$ and $x = q(y)$ respectively.

$$\begin{aligned}
 \iint_R \frac{\partial M}{\partial y} dx dy &= \int_{x=a}^b \left[\int_{y=f(x)}^{y=g(x)} \frac{\partial M}{\partial y} dy \right] dx \\
 &= \int_{x=a}^b [M(x, y)]_{y=f(x)}^{y=g(x)} dx \\
 &= \int_{x=a}^b [M(x, g(x)) - M(x, f(x))] dx \\
 &= - \int_a^b M(x, f(x)) dx + \int_a^b M(x, g(x)) dx \\
 &= - \oint_C M(x, y) dx
 \end{aligned}$$

Now, $y = f(x)$ represents the curve AEB and $y = g(x)$ represents the curve BFA.

If portion of C are segments parallel to y axis such as GH and PG in the figure below then above result is not affected.



The line integral $\int M dx$ over GH is zero because on GH we have $x = \text{constant}$ implies $dx = 0$ similarly, the line integral over PQ is zero. The equations of QG and HP are $y = f(x)$ and $y = g(x)$ respectively. Hence, we have:

$$\iint_R \frac{\partial M}{\partial y} dx dy = \oint_C M(x, y) dx \quad \dots(1)$$

Similarly,

$$\begin{aligned}
 \iint_R \frac{\partial N}{\partial x} dx dy &= \int_{y=c}^d \left[\int_{x=p(y)}^{x=q(y)} \frac{\partial N}{\partial x} dx \right] dy \\
 &= \int_{y=c}^d [N(x, y)]_{x=p(y)}^{x=q(y)} dy
 \end{aligned}$$

$$\begin{aligned}
 &= \int_{y=c}^d [N(q(y), y) - N(p(y), y)] dy \\
 &= \int_c^d N(q(y), y) dy - \int_c^d N(p(y), y) dy \\
 &= \int_c^d N(q(y), y) dy + \int_d^c N(p(y), y) dy \\
 &= \oint_C N(x, y) dy \quad \dots(2)
 \end{aligned}$$

From (1) and (2) on adding we get:

$$- \iint_R \frac{\partial M}{\partial y} dx dy + \iint_R \frac{\partial N}{\partial x} dx dy = \oint_C M(x, y) dx + \oint_C N(x, y) dy$$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \oint_C (M dx + N dy)$$

Which froms Green's theorem in the plane.

(c) Show that the vector field $\vec{F} = yz^2\mathbf{i} + xz^2\mathbf{j} + 2xyz\mathbf{k}$ is conservative and find a scalar potential function f for $\vec{F}$.

$$\text{Sol.} \quad \vec{F} = yz^2\mathbf{i} + xz^2\mathbf{j} + 2xyz\mathbf{k} \quad \dots(1)$$

The component of $\vec{F}$ is:

$$u(x, y, z) = yz^2$$

$$v(x, y, z) = xz^2$$

$$w(x, y, z) = 2xyz$$

$$\frac{\partial u}{\partial y} = z^2 = \frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial z} = 2yz = \frac{\partial w}{\partial x}$$

$$\frac{\partial w}{\partial y} = 2xz = \frac{\partial v}{\partial z}$$

Then, $\vec{F}$ is conservative

To find scalar potential function h such that $\Delta f = \vec{F}$, f must identify,

$$u(x, y, z) = f_x(x, y, z)$$

$$v(x, y, z) = f_y(x, y, z)$$

$$w(x, y, z) = f_z(x, y, z)$$

$$\therefore f(x, y) = \int u(x, y, z) dx$$

$$= \int y z^2 dx$$

$$= x y z^2 + k(y, z)$$

When, k is a constant of integration diff. Eq. (1) w.r.t. y :

$$\frac{\partial f}{\partial y} = x z^2 + \frac{\partial k}{\partial y}$$

$$= v(x, y, z)$$

$$= x z^2$$

$$\frac{\partial f}{\partial y} = 0$$

Therefore, k is a function of z , alone and $f(x, y, z) = x y z^2 + k(z)$
Diff. Eq. (2) w. r. t. z , We get :

$$\frac{\partial f}{\partial z} = 2 x y z + \frac{\partial k}{\partial z} = w(x, y, z) = 2 x y z$$

$$\frac{\partial f}{\partial z} = 0$$

Thus, k is independent of z also.

Hence,

$$f(x, y, z) = x y z^2 + C$$

Ans.

Q. 6. (a) Stoke's theorem, to evaluate $\iint_S (\nabla \times \vec{F}) \cdot \hat{n} ds$, Where $\vec{F} = y\hat{i} + (x-2xyz)\hat{j} - xy\hat{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$ above the xy plane.

Sol.

$$\vec{F} = y\hat{i} + (x-2xyz)\hat{j} - xy\hat{k} \quad \dots(1)$$

The boundary C of the surface S is the circle $x^2 + y^2 = a^2$, $z = 0$. Let $x = a \cos t$, $y = a \sin t$, $z = 0$, $0 \leq t \leq 2\pi$ are parametric equation of C . Then, by stoke's theorem, we have:

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} ds,$$

$$= \int_C \vec{F} \cdot d\vec{r}$$

$$= \int_C [y\hat{i} + (x-2xyz)\hat{j} - xy\hat{k}] [x\hat{i} + y\hat{j} + z\hat{k}]$$

$$= \int_C [y\hat{i} + (x-2xz)\hat{j} - xy\hat{k}] [dx\hat{i} + dy\hat{j} + dz\hat{k}]$$

$$= \int_C [y dx + (x-2xz) dy - xy dz]$$

$$= \int_C [y dx + (x-0) dy - xy(0)]$$

($\because$ on C , $z = 0$, $dz = 0$)

$$= \int_C (y dx + x dy)$$

$$= \int_0^{2\pi} \left(y \frac{dx}{dt} + x \frac{dy}{dt} \right) dt$$

$$= \int_0^{2\pi} [(a \sin t)(-a \sin t) + (a \cos t)(a \cos t)] dt$$

$$= \int_0^{2\pi} a^2 dt [\cos^2 t - \sin^2 t]$$

$$= a^2 \int_0^{2\pi} \cos 2t dt$$

$$= a^2 \left[\frac{\sin 2t}{2} \right]_0^{2\pi}$$

$$= \frac{a^2}{2} [\sin 4\pi \sin 0]$$

$$= \frac{a^2}{2} [0 - 0]$$

$$= 0$$

Ans.

(b) Evaluate $\int_0^e \int_0^{\log y} \int_1^y \log z \, dz \, dx \, dy$

Sol.

$$I = \int_0^e \int_0^{\log y} \int_1^y \log z \, dz \, dx \, dy \quad \dots(1)$$

$$= \int_0^e \int_0^{\log y} \left(\int_1^y \log z \, dz \right) dx \, dy$$

$$= \int_0^e \log z \, dz \int_0^{\log y} 1 \, dx$$

$$= [\log z \cdot z]_1^y - \int_1^y \left(\frac{1}{z} \right) dz$$

$$= e^x \log(e^x) - \log 1 - (z)^{ex} 1$$

$$= e^x(x) - 0 - (e^x - 1)$$

$$= xe^x - e^x + 1$$

Hence,

$$I = \int_0^e \int_0^{\log y} (xe^x - e^x + 1) dx \, dy$$

$$= \int_0^e \int_0^{\log y} [e^x(x-1) + 1] dx \, dy \quad \dots(2)$$

$$\int_0^{\log y} [(x-1)e^x + 1] dx$$

$$= \int_0^{\log y} (x-1)e^x dx + \int_0^{\log y} 1 \cdot dx$$

$$= [(x-1)e^x]_0^{\log y} - \int_0^{\log y} (1)e^x dx + [x]_0^{\log y}$$

$$= (\log y - 1)e^{\log y} + 1 - [e^x]_0^{\log y} + (\log y - 0)$$

$$= (\log y - 1)y + 1 - (e^{\log y} - 1) + \log y$$

$$= y \log y - y + 1 - y + 1 + \log y$$

$$= (y+1) \log y - 2y + 2$$

$$I = \int_0^e [(y+1) \log y - 2y + 2] dy$$

$$= \int_0^e (y+1) \log y \, dy - \int_0^e 2y \, dy + 2 \int_0^e 1 \cdot dy$$

$$= \left[\log y \left(\frac{y^2}{2} + y \right) \right]_0^e - \int_0^e \left(\frac{1}{y} \right) \left(\frac{y^2}{2} + y \right) dy$$

$$- [y^2]_0^e + 2[y]_0^e$$

$$= \left[\left(\frac{e^2}{2} + e \right) - 0 \right] - \int_0^e \left(\frac{y}{2} + 1 \right) dy - (e^2 - 1) + 2(e - 1)$$

$$= \left[\frac{e^2}{2} + e \right] - \left[\frac{y^2}{4} + y \right]_0^e - e^2 + 1 + 2e - 2$$

$$= \left(\frac{e^2}{2} + e \right) - \left(\frac{y^2}{4} + e + \frac{1}{4} - 1 \right) - e^2 + 2e - 1$$

$$= \left(\frac{e^2}{2} - \frac{e^2}{4} \right) + (e - e) + \frac{1}{4} + 1 - e^2 + 2e - 1$$

$$= \frac{e^2}{4} + 0 + \frac{1}{4} - e^2 + 2e$$

$$= \frac{-3e^2}{4} + 2e + \frac{1}{4}$$

$$= \frac{1}{4} (-3e^2 + 8e + 1)$$

$$= \frac{1}{4} (1 + 8e - 3e^2)$$

Ans.

(c) Show that a constant vector field has divergence and curl zero.

Sol. Let

$$\vec{F} = ai + bj + ck$$

= constant field

Here, a, b, c are constants

(i)

$$\text{div } \vec{F} = (\nabla \cdot \vec{F})$$

$$= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (ai + bj + ck)$$

$$= \frac{\partial}{\partial x}(a) + \frac{\partial}{\partial y}(b) + \frac{\partial}{\partial z}(c)$$

$$= 0 + 0 + 0$$

$$= 0$$

(ii) Curl $\vec{F}$

$$= (\nabla \times \vec{F})$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a & b & c \end{vmatrix}$$

$$= i \left[\frac{\partial}{\partial y}(c) - \frac{\partial}{\partial z}(b) \right] - j \left[\frac{\partial}{\partial x}(c) - \frac{\partial}{\partial y}(a) \right] + k \left[\frac{\partial}{\partial x}(b) - \frac{\partial}{\partial y}(a) \right]$$

$$= i [0 - 0] - j [0 - 0] + k [0 - 0]$$

$$= 0 + 0 + 0$$

$$= 0$$

○○○

Name of the Paper : Paper : C-7 : Multivariate Calculus
 Name of Course : B.Sc. (Hons.) Mathematics II Year (CBCS)
 Semester : III
 Duration : 3 Hours
 Maximum Marks : 75

MODEL TEST PAPER-III

Q. 1. (a) Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} ; (x,y) \neq (0,0)$$

Sol. Let

$$f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$$

Let $(x,y) \rightarrow (0,0)$ along the path $y = mx$, where m is any real number.When $x \rightarrow 0$, then $y \rightarrow 0$ along the path $y = mx$.

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 - (mx)^2}{x^2 + (mx)^2}$$

$$= \lim_{x \rightarrow 0} \frac{x^2(1 - m^2)}{x^2(1 + m^2)}$$

$$= \lim_{x \rightarrow 0} \frac{1 - m^2}{1 + m^2}$$

$$= \frac{1 - m^2}{1 + m^2}$$

Which is different for different values of m .

$$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) \text{ does not exist.}$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} \text{ does not exist.}$$

Ans. :

(b) Prove that the function f defined by

$$f(x,y) = \begin{cases} y \sin\left(\frac{1}{x}\right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

is continuous at the origin.

(99)

Sol.
$$f(x,y) = \begin{cases} y \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases} \quad \dots(1)$$

Let $\epsilon > 0$ be any given real number
Now,

$$\begin{aligned} |f(x,y) - f(0,0)| &= \left| y \sin\left(\frac{1}{x}\right) - 0 \right| \\ &= \left| y \sin\left(\frac{1}{x}\right) \right| \\ &= |y| \left| \sin\left(\frac{1}{x}\right) \right| \\ &\leq |y| \text{ because } \left| \sin\frac{1}{x} \right| \leq 1 \end{aligned}$$

$$\therefore |f(x,y) - f(0,0)| \leq |y - 0|$$

Let $|x - 0| < \epsilon$ and $|y - 0| < \epsilon$, then

$$|f(x,y) - f(0,0)| < \epsilon \text{ whenever } |x - 0| < \delta$$

and $|y - 0| < \delta$,

Where $\delta = \epsilon$

$\therefore f(x,y)$ is continuous at $(0,0)$

Proved.

(c) Show that $f(x,y) = \sqrt{|x,y|}$ is continuous at the origin.

Sol.
$$f(x,y) = \sqrt{|x,y|}$$

...(1)

Let $\epsilon > 0$ be any given real number,

Then

$$\begin{aligned} |f(x,y) - f(0,0)| &= |\sqrt{|xy|} - 0| \\ &= \sqrt{|xy|} \\ &= \sqrt{|x|} \sqrt{|y|} \end{aligned}$$

$$|f(x,y) - f(0,0)| < \epsilon \text{ if } \sqrt{|x|} \sqrt{|y|} < \epsilon$$

$$|f(x,y) - f(0,0)| < \epsilon \text{ if } \sqrt{|x|} < \sqrt{\epsilon} \text{ and } \sqrt{|y|} < \sqrt{\epsilon}$$

$$|f(x,y) - f(0,0)| < \epsilon \text{ if } |x| < \epsilon \text{ and } |y| < \epsilon$$

$$|f(x,y) - f(0,0)| < \epsilon \text{ whenever } |x - 0| < \epsilon \text{ and } |y - 0| < \epsilon$$

Taking

$$\epsilon = \delta.$$

We have :

$$|f(x,y) - f(0,0)| < \epsilon, \text{ whenever } |x - 0| < \delta \text{ and } |y - 0| < \delta$$

Hence, by definition of continuity $f(x,y)$ is continuous at the origin.

Proved.

Q. 2. (a) Find all the four second order partial derivations of the following function,

$$u = e^{2x+3y}$$

Sol.

$$u = e^{2x+3y}$$

The four second order partial derivation are given by

...(1)

$$\frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial y^2}, \frac{\partial^2 u}{\partial x \partial y} \text{ and } \frac{\partial^2 u}{\partial y \partial x}$$

$$(i) \quad \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (e^{2x+3y})$$

$$= e^{2x+3y} \frac{\partial}{\partial x} (2x+3y)$$

$$= e^{2x+3y} (2+0)$$

$$= 2e^{2x+3y}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right)$$

$$= \frac{\partial}{\partial x} (2e^{2x+3y})$$

$$= 2e^{3y} \frac{\partial}{\partial x} (e^{2x})$$

$$= 2e^{3y} \left(e^{2x} \frac{\partial}{\partial x} (2x) \right)$$

$$= 2e^{3y} (e^{2x} (2))$$

$$= 4e^{2x+3y}$$

Ans.

(ii)

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (e^{2x+3y})$$

$$= \frac{\partial}{\partial y} (e^{2x} e^{3y})$$

$$= e^{2x} \frac{\partial}{\partial y} (e^{3y})$$

$$= e^{2x} \left[e^{3y} \frac{\partial}{\partial y} (3y) \right]$$

$$= e^{2x} [e^{3y} (3)]$$

$$= 3e^{2x} e^{3y}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right)$$

$$= \frac{\partial}{\partial y} (3e^{2x} e^{3y})$$

$$= 3e^{2x} \frac{\partial}{\partial y} (e^{3y})$$

$$= 3e^{2x} \left[e^{3y} \frac{\partial}{\partial y} (3y) \right]$$

$$= 3e^{2x} [e^{3y} (3)]$$

$$= 3e^{2x+3y}$$

(iii)

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

$$= \frac{\partial}{\partial x} [3e^{2x} e^{3y}]$$

$$= 3e^{3y} \frac{\partial}{\partial x} (e^{2x})$$

$$= 3e^{3y} [e^{2x} (2)]$$

$$= 6e^{2x+3y}$$

(iv)

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right)$$

$$= \frac{\partial}{\partial y} [2e^{2x} e^{3y}]$$

$$= 2e^{2x} \frac{\partial}{\partial y} (e^{3y})$$

$$= 2e^{2x} \left[e^{3y} \frac{\partial}{\partial y} (3y) \right]$$

$$= 6e^{2x+3y}$$

Ans.

(b) Biologists have studied the oxygen consumption of certain furry mammals. They have found that if the mammal's body temperature is T degree celsius, fur temperature is t degree celsius and the mammal does not sweat, then its relative oxygen consumption can be modelled by $C(m, t, T) = \sigma (T - t) m^{-0.67}$

(kg/h), where m is the mammal's mass in Kg and σ is a physical constant. Compute the rate at which the oxygen consumption changes with respect to

(a) the mass m

(b) the body temperature T

(c) the fur temperature t .

Sol. The relative oxygen consumption C is given by :

$$C(m, t, T) = \sigma (T - t) m^{-0.67} \quad \dots (1)$$

(a) The rate of change of oxygen consumption w. r. t. mass m is calculated

Ans.

$$C_m(m, t, T) = \frac{\partial C}{\partial m}$$

$$= \frac{\partial}{\partial m} [\sigma (T - t) m^{-0.67}]$$

$$= \sigma (T - t) \frac{\partial}{\partial m} (m^{-0.67})$$

$$= \sigma (T - t) [(-0.67) m^{-0.67-1}]$$

$$= -0.67 \sigma (T - t) e^{-1.67}$$

Ans.

(b) The rate of change of oxygen consumption w.r.t. body temperature T is calculated as:

Ans.

$$C_T(m, t, T) = \frac{\partial C}{\partial T}$$

$$= \frac{\partial}{\partial T} [\sigma (T - t) m^{-0.67}]$$

$$= \sigma m^{-0.67} \frac{\partial}{\partial T} (T - t)$$

$$= \sigma m^{-0.67} (1-0)$$

$$= \sigma m^{-0.67}$$

(c) The rate of change of oxygen consumption w.r.t. body temperature t is calculated as

$$C_t(m, t, T) = \frac{\partial C}{\partial t}$$

$$= \frac{\partial}{\partial t} [\sigma (T-t) m^{-0.67}]$$

$$= \sigma m^{-0.67} \frac{\partial}{\partial t} [(T-t)]$$

$$= \sigma m^{-0.67} [1-0]$$

$$= \sigma m^{-0.67}$$

(c) Find the total diff of the following function.

(i) $f(x, y, z) = 3x^3 - 2y^4 + 5z$

(ii) $f(x, y, z) = \sin x + \sin y + \cos z$

Sol. $f(x, y, z) = 3x^3 - 2y^4 + 5z$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (3x^3 - 2y^4 + 5z)$$

$$= 9x^2 - 0 + 0$$

$$= 9x^2$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (3x^3 - 2y^4 + 5z)$$

$$= (0 - 8y^3 + 0)$$

$$= -8y^3$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z} (3x^3 - 2y^4 + 5z)$$

$$= (0 - 0 + 5)$$

$$= 5$$

The total differentiation of f is calculated as :

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$= (9x^2) dx + (-8y^3) dy + (5) dz$$

$$= 9x^2 dx - 8y^3 dy + 5 dz$$

$$f(x, y, z) = \sin x + \sin y + \cos z$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (\sin x + \sin y + \cos z)$$

$$= (\cos x + 0 + 0)$$

$$= \cos x$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (\sin x + \sin y + \cos z)$$

$$= (0 + \cos y + 0)$$

$$= \cos y$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z} (\sin x + \sin y + \cos z)$$

$$= (0 + 0 - \sin z)$$

$$= -\sin z$$

The total differentiation of f is calculated as :

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$= (\cos x) dx + (\cos y) dy + (-\sin z) dz$$

$$= \cos x dx + \cos y dy - \sin z dz$$

Q. 3.(a) Find the directional derivative of $f(x, y) = \sqrt{xy} e^y$ at the point $(1, 1)$ in the direction of the negative y -axis

Sol.

$$f(x, y) = \sqrt{xy} e^y$$

$$f_x(x, y) = \frac{\partial}{\partial x} (\sqrt{xy} e^y)$$

$$= \sqrt{y} e^y \frac{\partial}{\partial x} (\sqrt{x})$$

$$= \sqrt{y} e^y \left(\frac{1}{2\sqrt{x}} \right)$$

$$= \frac{\sqrt{y}}{2\sqrt{x}} e^y$$

$$f_y(x, y) = \frac{\partial}{\partial y} (\sqrt{xy} e^y)$$

$$= \sqrt{x} \frac{\partial}{\partial y} (\sqrt{y} e^y)$$

$$= \sqrt{x} \left(\sqrt{y} \cdot e^y + \frac{1}{2\sqrt{y}} e^y \right)$$

$$= \sqrt{xy} e^y + \frac{\sqrt{x}}{2\sqrt{y}} e^y$$

At (1, 1)

$$f_x(1, 1) = \frac{\sqrt{1}}{2\sqrt{1}} e^1 = \frac{e}{2}$$

$$f_y(1, 1) = \sqrt{1 \times 1} e^1 + \frac{\sqrt{1}}{2\sqrt{1}} e^1$$

$$= e + \frac{e}{2}$$

$$= \frac{3e}{2}$$

Since, the negative y-axis where an angle $\frac{3\pi}{2}$ with the positive direction of the x-axis.

The directional derivative of f at $(x_0, y_0) = (1, 1)$ in the direction of ϕ angle from positive x-axis is :

$$D_n f(x_0, y_0) = f_x(x_0, y_0) \cos \phi + f_y(x_0, y_0) \sin \phi$$

$$D_n f(1, 1) = f_x(1, 1) \cos + \frac{3\pi}{2} + f_y(1, 1) \sin \frac{3\pi}{2}$$

$$= \left(\frac{e}{2}\right)(0) + \left(\frac{3e}{2}\right)(-1)$$

$$= 0 - \frac{3e}{2}$$

$$= -\frac{3}{2}e$$

(b) If $r = |\vec{r}|$, where $\vec{r} = xi + yj + zk$ prove that

$$(1) \nabla f(r) = f'(r) \nabla(r)$$

Sol.

$$\vec{r} = xi + yj + zk$$

...(1)

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\nabla f(r) = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) f(r)$$

$$= i \frac{\partial}{\partial x} [f(r)] + j \frac{\partial}{\partial y} [f(r)] + k \frac{\partial}{\partial z} [f(r)]$$

$$= \left[f'(r) \frac{\partial r}{\partial x} \right] + j \left[f'(r) \frac{\partial r}{\partial y} \right] + k \left[f'(r) \frac{\partial r}{\partial z} \right]$$

$$= f'(r) \left(i \frac{\partial r}{\partial x} + j \frac{\partial r}{\partial y} + k \frac{\partial r}{\partial z} \right)$$

$$= f'(r) \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) r$$

$$= f'(r) \nabla(r)$$

Proved.

(c) Maximize the function $f(x, y, z) = x^2 + 2y - z^2$ subject to the constants $2x - y = 0$ and $y + z = 0$

Sol. Let

$$g_1(x, y) = 2x - y \quad \dots(1)$$

$$g_2(x, y) = y + z \quad \dots(2)$$

$$f(x, y) = x^2 + 2y - z^2 \quad \dots(3)$$

$$f_x = \frac{\partial f}{\partial x}$$

$$= \frac{\partial}{\partial x} (x^2 + 2y - z^2)$$

$$= 2x$$

$$f_y = \frac{\partial f}{\partial y}$$

$$= \frac{\partial}{\partial y} (x^2 + 2y - z^2)$$

$$= 2$$

$$f_z = \frac{\partial f}{\partial z}$$

$$= \frac{\partial}{\partial z} (x^2 + 2y - z^2)$$

$$= -2z$$

$$g_{1x} = \frac{\partial g_1}{\partial x}$$

$$= \frac{\partial}{\partial x}(2x - y)$$

$$= 2$$

$$g_{1y} = \frac{\partial g_1}{\partial y}$$

$$= \frac{\partial}{\partial y}(2x - y)$$

$$= -1$$

$$g_{1z} = \frac{\partial}{\partial z}(2x - y)$$

$$= 0$$

$$g_{2x} = \frac{\partial g_2}{\partial x}$$

$$= \frac{\partial}{\partial x}(y + z)$$

$$= 0$$

$$g_{2y} = \frac{\partial g_2}{\partial y}$$

$$= \frac{\partial}{\partial y}(y + z)$$

$$= 1$$

$$g_{2z} = \frac{\partial g_2}{\partial z}$$

$$= \frac{\partial}{\partial z}(y + z) = 1$$

Using Lagrange's method of multipliers we have :

$$f_x = \lambda g_{1x} + \mu g_{2x}$$

$$2x = \lambda(2) + \mu(0)$$

$$x = \lambda$$

$$f_y = \lambda g_{1y} + \mu g_{2y}$$

$$2 = \lambda(-1) + \mu(1)$$

and

... (4)

$$2 = -\lambda + \mu$$

$$-\lambda + \mu = 2$$

$$f_z = \lambda g_{1z} + \mu g_{2z}$$

$$-2z = \lambda(0) + \mu(1)$$

$$z = -\frac{\mu}{2}$$

and

From (1)

$$2x - y = 0$$

$$y = 2x$$

$$y = 2(\lambda) = 2\lambda$$

and

$$y + z = 0$$

$$2\lambda - \frac{\mu}{2} = 0$$

$$4\lambda - \mu = 0$$

From (5) + (6) :

$$3\lambda = 2$$

$$\lambda = \frac{2}{3}$$

From (5)

$$-\lambda + \mu = 2$$

$$-\frac{2}{3} + \mu = 2$$

$\Rightarrow$

$$\mu = 2 + \frac{2}{3} = \frac{8}{3}$$

For

$$\lambda = \frac{2}{3}, \mu = \frac{8}{3}$$

We have :

$$x = \lambda = \frac{2}{3}$$

$$y = 2\lambda = 2\left(\frac{2}{3}\right) = \frac{4}{3}$$

$$z = -\frac{\mu}{2} = -\frac{1}{2}\left(\frac{8}{3}\right) = -\frac{4}{3}$$

The corresponding point on $f(x, y, z)$ is $\left(\frac{2}{3}, \frac{4}{3}, -\frac{4}{3}\right)$

and

$$\begin{aligned}
 & f\left(\frac{2}{3}, \frac{4}{3}, -\frac{4}{3}\right) \\
 &= x^2 + 2y - z^2 \\
 &= \left(\frac{2}{3}\right)^2 + 2\left(\frac{4}{3}\right) - \left(-\frac{4}{3}\right)^2 \\
 &= \frac{4}{9} + \frac{8}{3} - \frac{16}{9} \\
 &= \frac{4 + 24 - 16}{9} \\
 &= \frac{28 - 16}{9} \\
 &= \frac{12}{9} \\
 &= \frac{4}{3}
 \end{aligned}$$

Ans.

Q. 4. (a) Examine for maximum and minimum value of the function.

Sol.

$$f = x^2y + xy^2 + 12xy$$

$$f = x^2y + xy^2 + 12xy \quad \dots(1)$$

$$\begin{aligned}
 \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(x^2y + xy^2 + 12xy) \\
 &= 2xy + y^2 + 12y
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}(x^2y + xy^2 + 12xy) \\
 &= x^2 + 2xy + 12x
 \end{aligned}$$

$$A = \frac{\partial^2 f}{\partial x^2}$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$

$$= \frac{\partial}{\partial x}(2xy + y^2 + 12y)$$

$$= 2y + 0 + 0$$

$$= 2y$$

$$B = \frac{\partial^2 f}{\partial x \partial y}$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

$$= \frac{\partial}{\partial x}(x^2 + 2xy + 12x)$$

$$= 2x + 2y + 12$$

$$C = \frac{\partial^2 f}{\partial y^2}$$

$$= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

$$= \frac{\partial}{\partial y}(x^2 + 2xy + 12x)$$

$$= 0 + 2x + 0$$

$$= 2x$$

On putting $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$, we get:

$$\frac{\partial f}{\partial x} = 0$$

$$2xy + y^2 + 12x = 0$$

$$y(2x + y + 12) = 0 \quad \dots(2)$$

and

$$\frac{\partial f}{\partial y} = 0$$

$$x^2 + 2xy + 12x = 0$$

$$x(x + 2y + 12) = 0 \quad \dots(3)$$

On solving (2) and (3) we get:

$$x = 0, y = 0 \text{ and } x = 4, y = 4$$

Hence, stationary points are (0,0) R (4,4)

At (0,0)

$$A = 2y = 0$$

$$B = 2x + 2y + 12 = 0 + 0 + 12 = 12$$

$$C = 2x = 0$$

$$AC - B^2 = (0)(0) - (12)^2$$

$$= -144$$

$$= 0$$

$$= AC - B^2 < 0$$

$(0,0)$ is a saddle point

At $(4,4)$

$$A = 2y = 2(4) = 8 > 0$$

$$B = 2x + 2y - 12 = 2 \cdot 4 + 2(4) - 12 = 4$$

$$C = 2x = 2(4) = 8$$

$$AC - B^2 = (8)(8) - (4)^2$$

$$= 64 - 16$$

$$= 48$$

$$AC - B^2 > 0 \text{ and } A > 0$$

$(4,4)$ is a point of minima.

The min value of f be:

$$= f(4,4)$$

$$= x^2y + xy^2 - 12xy$$

$$= (4)^2(4) + 4(4)^2 - 12(4)(4)$$

$$= (4)^2(4 + 4 - 12)$$

$$= 16(-4)$$

$$= -64$$

(b) Show that the function,

$$f(x,y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{y}{x}\right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

is differentiable at $(0,0)$

Sol.

$$f(x,y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{y}{x}\right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{0 - 0}{h}$$

$$= 0$$

$$f_y(0,0) = \lim_{k \rightarrow 0} \frac{f(0,k) - f(0,0)}{k}$$

Ans.

...(1)

...(2)

$$= \lim_{h \rightarrow 0} \frac{0 - 0}{0}$$

$$= 0$$

If f is differentiable at $(0,0)$, then:

$$f(h,k) - f(0,0) = h f_x(0,0) + k f_y(0,0) + h \phi(h,k) + k \psi(h,k)$$

Where,

$$\phi = (h,k) \rightarrow 0 \text{ as } (h,k) \rightarrow (0,0)$$

$$\psi = (h,k) \rightarrow 0 \text{ as } (h,k) \rightarrow (0,0)$$

$$(h^2 + k^2) \sin\left(\frac{k}{h}\right) - 0 = h(0) + k(0) + h \phi(h,k) + k \psi(h,k)$$

$$h^2 \sin\left(\frac{k}{h}\right) + k^2 \sin\left(\frac{k}{h}\right) = h \phi(h,k) + k \psi(h,k)$$

Comparing both sides, we have

$$h \phi(h,k) = h^2 \sin\left(\frac{k}{h}\right)$$

$$\phi(h,k) = h \sin\left(\frac{k}{h}\right)$$

and

$$k \phi(h,k) = k^2 \sin\left(\frac{k}{h}\right)$$

$$\phi(h,k) = k \sin\left(\frac{k}{h}\right)$$

If $(h,k) \rightarrow (0,0)$, then

$$\phi(h,k) \rightarrow 0$$

and

$$\psi(h,k) \rightarrow 0$$

Hence, f is differentiable at $(0,0)$.

(c) Show that

(i) $\text{div } \vec{r} = 3$

(ii) $\text{curl } \vec{r} = 0$

Where $r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$

Sol.

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

(i) $\text{div } \vec{r}$

$$= \nabla \cdot \vec{r}$$

...(1)

$$= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (ix + jy + kz)$$

$$= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$$

$$= 1 + 1 + 1$$

$$= 3$$

(2)

$$\vec{r} = xi + yj + zk$$

$$\frac{\partial \vec{r}}{\partial x} = \frac{\partial}{\partial x}(xi + yj + zk) = i + 0 + 0 = i$$

$$\frac{\partial \vec{r}}{\partial y} = \frac{\partial}{\partial y}(xi + yj + zk) = 0 + j + 0 = j$$

$$\frac{\partial \vec{r}}{\partial z} = \frac{\partial}{\partial z}(xi + yj + zk) = 0 + 0 + k = k$$

Curl $\vec{r}$

$$= \nabla \times \vec{r}$$

$$= \left(i + \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times \vec{r}$$

$$= i \times \frac{\partial \vec{r}}{\partial x} + j \times \frac{\partial \vec{r}}{\partial y} + k \times \frac{\partial \vec{r}}{\partial z}$$

$$= i \times i + j \times j + k \times k$$

$$= 0 + 0 + 0$$

$$= 0$$

Proved.

Q. 5. (a) State and prove Stoke's theorem:

Sol. Stoke's Theorem:

Let S be a piece wise smooth open surface bounded by a piece wise smooth simple closed curve C . Let $\vec{F}(x, y, z)$ be a continuous vector function, which has continuous first partial derivatives in a region of space, which contain S in its interior, then:

$$\oint_C \vec{F} \cdot d\vec{r} = \iiint_S (\nabla \times \vec{F}) \cdot \vec{n} \, dS = \iiint_S (\text{Curl } \vec{F}) \cdot d\vec{S}$$

Where, C is traversed in the positive direction. The direction of C is called positive if an observer walking on the boundary of S in this direction with his head pointing in the direction of outward drawn normal $\vec{n}$ to S , has the surface on the left.

Proof: Let $\vec{F} = F_1i + F_2j + F_3k$. Let outward drawn normal vector $\vec{n}$ to S makes angles α, β, γ with x, y, z axis respectively. Then

$$\vec{n} = \cos \alpha i + \cos \beta j + \cos \gamma k.$$

Let S be a surface, which is such that its projections on the xy, yz and xz planes are regions bounded by simple closed curves. Surface S can be represented simultaneously in the forms,

$$z = f(x, y)$$

$$y = g(x, z)$$

$$x = h(z, y)$$

Which f, g, h are continuous functions and have continuous first partial derivatives.

Consider the integral $\iiint_S (\nabla \times \vec{F}) \cdot \vec{n} \, dS$

$$\nabla \times (\vec{F}_1i) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & 0 & 0 \end{vmatrix}$$

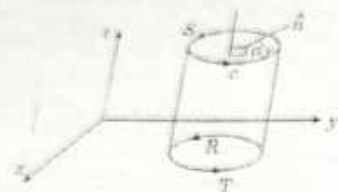
$$= \frac{\partial F_1}{\partial z} j - \frac{\partial F_1}{\partial y} k$$

$$[\nabla \times (\vec{F}_1i)] \cdot \vec{n} = \left(\frac{\partial F_1}{\partial z} j - \frac{\partial F_1}{\partial y} k \right) \cdot \vec{n}$$

$$= \frac{\partial F_1}{\partial z} (j \cdot \vec{n}) - \frac{\partial F_1}{\partial y} (k \cdot \vec{n})$$

$$= \frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \mu$$

$$\iiint_S [\nabla \times (\vec{F}_1i)] \cdot \vec{n} \, dS = \iiint_S \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \mu \right) dS$$



We have prove that

$$\iiint_V \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dz = \oint_C F_1 dx$$

Let R be the orthogonal projection of S on the xy-plane and let T be its boundary which is oriented as shown in fig. Using the representation $z = f(x, y)$ of S, we may write the line integral over C as a line integral over T. Thus,

$$\begin{aligned} \oint_C F_1(x, y, z) dx &= \oint_T F_1(x, y, f(x, y)) dx \\ &= \oint_T [F_1(x, y, f(x, y)) dx + 0 dy] \end{aligned}$$

$$= - \iint_R \frac{\partial F_1}{\partial y} dx dy$$

...(1)

Using Green's theorem in plane for the region R

$$\text{But, } \frac{\partial F_1}{\partial y} [x, y, f(x, y)] = \frac{\partial F_1(x, y, z)}{\partial y} + \frac{\partial F_1(x, y, z)}{\partial z} \frac{\partial f}{\partial y}$$

$$\oint_C F_1(x, y, z) dx = - \iint_R \left(\frac{\partial F_1}{\partial y} + \frac{\partial F_1}{\partial z} \frac{\partial f}{\partial y} \right) dx dy$$

...(2)

Now, the equation $z = f(x, y)$ of the surface S can be written as:
 $\phi(x, y, z) = z - f(x, y) = 0$

We have,

$$\text{grad } \phi = -\frac{\partial f}{\partial x} i - \frac{\partial f}{\partial y} j + k$$

$$|\text{grad } \phi| = u$$

Let
 Since, grad ϕ is normal to S, therefore, we get:

$$u = \pm \frac{\text{grad } \phi}{a}$$

But, the component of both $\hat{n}$ and grad ϕ in positive direction of z-axis are positive.

$$\text{Therefore, } \hat{n} = \pm \frac{\text{grad } \phi}{a}$$

$$\cos \alpha i + \cos \beta j + \cos \gamma k = -\frac{1}{a} \frac{\partial f}{\partial x} i - \frac{1}{a} \frac{\partial f}{\partial y} j + \frac{1}{a} k$$

$$\cos \alpha = -\frac{1}{a} \frac{\partial f}{\partial x}$$

$$\cos \beta = -\frac{1}{a} \frac{\partial f}{\partial y}$$

$$\cos \gamma = \frac{1}{a}$$

Now,

$$ds = \frac{dx dy}{\cos \gamma} = a dx dy$$

$$\iiint_S \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dz$$

$$= \iint_R \left[\frac{\partial F_1}{\partial z} \left(-\frac{1}{a} \frac{\partial f}{\partial y} \right) - \frac{\partial F_1}{\partial y} \frac{1}{a} \right] a dx dy$$

$$= - \iint_R \left[\frac{\partial F_1}{\partial y} + \frac{\partial F_1}{\partial z} \frac{\partial f}{\partial y} \right] dx dy$$

...(2)

From (1) + (2)

$$\oint_C F_1 dx = \iint_S \left[\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right] dy$$

$$= \iint_S [\nabla \times F_1] \cdot \hat{n} dy$$

...(3)

Similarly, by projections on the other co-ordinates planes, we get:

$$\oint_C F_2 dy = \iint_S [\nabla \times F_2] \cdot \hat{n} ds$$

...(4)

and

$$\oint_C F_3 dz = \iint_S [\nabla \times F_3] \cdot \hat{n} ds$$

...(5)

On adding (3), (4) and (5) :

$$\oint_C (F_1 dx + F_2 dy + F_3 dz) = \iint_S [\nabla \times (F_1 i + F_2 j + F_3 k)] \cdot \hat{n} \, ds$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, ds$$

Which proves Stoke's theorem

(c) Using Fundamental theorem of line integrals, find the work done by the vector Field?

$$\vec{F} = yzi + xzj + xyk$$

along any smooth curve C joining the points A (-1, 3, 9) to B (1, 6, -4).

Sol.

$$\vec{F} = yzi + xzj + xyk$$

$$= \Delta(x, y, z)$$

$$= \Delta f$$

By fundamental theorem, the work done by the conservative field $\vec{F}$ along C is calculated as :

$$u = \int_A^B \vec{F} \cdot d\vec{R}$$

$$= \int_A^B \nabla f \cdot d\vec{R}$$

$$= f(B) - f(A)$$

$$= [x, y, z]_{(1, 6, -4)} [x, y, z]_{(-1, 3, 9)}$$

$$= [1 \times 6 \times -4] - [-1 \times 3 \times 9]$$

$$= -24 + 27$$

$$= 3$$

Ans.

(c) Show that $f(x, y) = e^x \sin y$ is harmonic.

Sol.

$$f(x, y) = e^x \sin y$$

$$f(x, y) = \frac{\partial f}{\partial x}$$

$$= \frac{\partial}{\partial x} (e^x \sin y)$$

$$= \sin y \frac{\partial}{\partial x} (e^x)$$

$$= \sin y e^x$$

$$f_y(x, y) = \frac{\partial f}{\partial y}$$

$$= \frac{\partial}{\partial y} (e^x \sin y)$$

$$= e^x \frac{\partial}{\partial y} (\sin y)$$

$$= e^x \cos y$$

$$f_{xx}(x, y) = \frac{\partial^2 f}{\partial x^2}$$

$$= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$$

$$= \frac{\partial}{\partial x} (\sin y e^x)$$

$$= \sin y e^x$$

$$f_{yy}(x, y) = \frac{\partial^2 f}{\partial y^2}$$

$$= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)$$

$$= \frac{\partial}{\partial y} (e^x \cos y)$$

$$= e^x (-\sin y)$$

$$= -e^x (\sin y)$$

The Laplacian of f is given by :

$$\nabla^2 f(x, y) = f_{xx}(x, y) + f_{yy}(x, y)$$

$$= (e^x \sin y) + (-e^x \sin y)$$

$$= 0$$

$\Rightarrow f$ is harmonic

Proved.

Q. 6. (a) Verify Green's theorem in the plane for $\oint_C (xy + y^2) dx + x^2 dy$.

where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.

Sol.



Using Green's theorem in plane we have :

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \oint_C (M dx + N dy)$$

Here,

$$M = xy + y^2$$

$$N = x^2$$

The curves $y = x$ and $y = x^2$ intersect at $(0, 0)$ and $(1, 1)$. The positive direction in traversing C is as shown in fig.

$$= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \iint_R \left[\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(xy + y^2) \right] dx dy$$

$$= \iint_R [(2x) - (x + 2y)] dx dy$$

$$= \iint_R (x - 2y) dx dy$$

$$= \int_{x=0}^1 \int_{y=x^2}^x (x - 2y) dy dx$$

$$= \int_0^1 \left[xy - y^2 \right]_{y=x^2}^x dx$$

$$= \int_0^1 \left[(x^2 - x^2) - (x^3 - x^4) \right] dx$$

$$= \int_0^1 (x^4 - x^3) dx$$

$$= \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1$$

$$= \left(\frac{1}{5} - \frac{1}{4} \right) - (0, 0)$$

$$= -\frac{1}{20}$$

Let us now evaluating the line integral along C . Along $y = x^2$, $dy = 2x dx$. Therefore, along $y = x^2$, the line integral equals,

$$= \int_0^1 \left[(x)(x^2) + (x^2)^2 \right] dx + x^2 (2x) dx$$

$$= \int_0^1 (x^3 + 3x^3) dx$$

$$= \left[\frac{x^4}{4} + \frac{9x^4}{4} \right]_0^1$$

$$= \left(\frac{1}{4} + \frac{9}{4} \right) - (0, 0)$$

$$= \frac{10}{4}$$

Along $y = x$, $dy = dx$. Therefore, along $y = x$, the line integral equals

$$= \int_0^1 \left[(x)(x) + x^2 \right] dx + x^2 dx$$

$$= \int_0^1 3x^2 dx$$

$$= \left[x^3 \right]_0^1$$

$$= [1 - 0]$$

$$= 1$$

Therefore, the required line integral $= \frac{10}{4} - 1 = \frac{1}{2}$

Hence, Green's theorem verified.

(b) If $\vec{F} = axi + byj + czk$; a, b, c are constants show that

$$\iint_S \vec{F} \cdot \hat{n} dS = \frac{4}{3}(a + b + c), \text{ where, } S \text{ is the surface of a unit sphere.}$$

Sol

From Gauss's divergence theorem, we have

$$\iint_S \vec{F} \cdot \hat{n} \, dS = \iiint_V (\nabla \cdot \vec{F}) \, dV$$

Where, V is the volume enclosed by

$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} \, dS &= \iiint_V (\nabla \cdot \vec{F}) \, dV \\ &= \iiint_V \nabla \cdot (axi + byj + czk) \, dV \\ &= \iiint_V \left[\frac{\partial}{\partial x}(ax) + \frac{\partial}{\partial y}(by) + \frac{\partial}{\partial z}(z) \right] \, dV \\ &= \iiint_V (a + b + c) \, dV \\ &= (a + b + c) \iiint_V dV \\ &= (a + b + c) \frac{4\pi}{3} \\ &= \frac{4\pi}{3} (a + b + c) \end{aligned}$$

Because, volume enclosed by a sphere of unit radius = $\frac{4\pi}{3} (1)^3 = \frac{4\pi}{3}$.

(c) Evaluate.

$$\int_0^3 \int_1^2 xy(1+x+y) \, dy \, dx$$

Sol.

$$\begin{aligned} &\int_0^3 \int_1^2 xy(1+x+y) \, dy \, dx \\ &= \int_0^3 \left[\int_1^2 xy(1+x+y) \, dy \right] \, dx \\ &= \int_0^3 \left[\int_1^2 (xy + x^2y + xy^2) \, dy \right] \, dx \end{aligned}$$

$$\begin{aligned} &= \int_0^3 \left[\int_1^2 (xy + x^2y + xy^2) \, dy \right] \, dx \\ &= \int_0^3 \left[x \left[\frac{y^2}{2} + x^2y + \frac{y^3}{3} \right]_1^2 \right] \, dx \\ &= \int_0^3 \left[x \left(\frac{4}{2} + x^2 \left(\frac{4}{2} - \frac{1}{2} \right) + \frac{8}{3} - \frac{1}{3} \right) \right] \, dx \\ &= \int_0^3 \left[x \left(\frac{3}{2} + x^2 \left(\frac{3}{2} \right) + x \left(\frac{7}{3} \right) \right) \right] \, dx \\ &= \int_0^3 \left[x \left(\frac{3}{2} + \frac{3}{2}x^2 + \frac{7}{3}x \right) \right] \, dx \\ &= \int_0^3 \left[\frac{3}{2}x + \frac{3}{2}x^3 + \frac{7}{3}x^2 \right] \, dx \\ &= \left[\frac{3}{2} \left(\frac{x^2}{2} \right) + \frac{3}{2} \left(\frac{x^4}{4} \right) + \frac{7}{3} \left(\frac{x^3}{3} \right) \right]_0^3 \\ &= \left[\frac{27}{4} + \frac{27}{2} + \frac{21}{2} \right] - [0 - 0] \\ &= \frac{69}{4} + \frac{27}{2} = \frac{69 + 54}{2} \\ &= \frac{123}{2} \end{aligned}$$

Ans.

○○○

Paper Code : 32351301
 Name of the Paper : C5 : Theory of Real Functions
 Name of Course : B.Sc. (Hons.) Mathematics II Year
 Semester : III (CBCS)
 Duration : 3 Hours
 Maximum Marks : 75

Nov/Dec. 2016

Q. 1. (a) Use the $\epsilon - \delta$ definition of limit of a function to find lim

$$\lim_{x \rightarrow 1} \frac{x^2 - x + 1}{x + 1} \quad (5)$$

Ans. We know that $\lim_{x \rightarrow a} f(x) = l$, if given $\epsilon > 0$ there exists a positive real number δ (depending upon ϵ) such that

$$|f(x) - l| < \epsilon \text{ for } 0 < |x - a| < \delta \quad \dots(1)$$

$$\text{Let } f(x) = \frac{x^2 - x + 1}{x + 1} \quad \dots(2)$$

$$l = \frac{1}{2}$$

Here, we choose a real number δ such that equation (1) hold true.

$$|f(x) - l| < \epsilon$$

$$\Rightarrow \left| \left(\frac{x^2 - x + 1}{x + 1} \right) - \frac{1}{2} \right| < \epsilon$$

$$\Rightarrow \left| \frac{2(x^2 - x + 1) - (x + 1)}{2(x + 1)} \right| < \epsilon$$

$$\Rightarrow \left| \frac{2x^2 + 2x + 2 - x - 1}{2(x + 1)} \right| < \epsilon$$

$$\Rightarrow \left| \frac{2x^2 - 3x + 1}{2(x + 1)} \right| < \epsilon$$

$$\Rightarrow \frac{2x^2 - 2x - x + 1}{2(x + 1)} < \epsilon$$

$$\Rightarrow \left| \frac{2x(x - 1) - (x - 1)}{2(x + 1)} \right| < \epsilon$$

$$\Rightarrow \left| \frac{(2x-1)(x-1)}{2(x+1)} \right| < \epsilon$$

$$\Rightarrow |(2x-1)(x-1)| < \epsilon$$

$$\Rightarrow |2x-1| |x-1| < \epsilon$$

$$\Rightarrow |x-1| < \epsilon$$

Let we choose $\delta = \epsilon$ such that

$$\left| \frac{x^2 - x + 1}{x+1} - \frac{1}{2} \right| < \epsilon, \text{ where } |x-1| < \delta$$

$$\Rightarrow \left| f(x) - \frac{1}{2} \right| < \epsilon, \text{ where } 0 < |x-1| < \delta$$

$$\Rightarrow \lim_{x \rightarrow 1} f(x) = \frac{1}{2}$$

$$\Rightarrow \lim_{x \rightarrow 1} \left(\frac{x^2 - x + 1}{x+1} \right) = \frac{1}{2}$$

(b) State Sequential Criterion for Limits. Using Sequential

Criterion for limits, prove that $\lim_{x \rightarrow 0} \sin \frac{1}{x^2}$ does not exist.

Ans. Sequential Criterion for Limits:

Let $f: A \rightarrow \mathbb{R}$ and let c be a Cluster point of A , then the following are equivalent

$$(i) \lim_{x \rightarrow c} f = L$$

(ii) For every sequence (x_n) in A that converges to c such that $x_n \neq c$ for all $n \in \mathbb{N}$, the sequence $(f(x_n))$ converges to L .

(c) State Squeeze Theorem. Use the theorem to show that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

Ans. Squeeze Theorem

Let $A \subseteq \mathbb{R}$, let $f, g, h: A \rightarrow \mathbb{R}$ and let $c \in \mathbb{R}$ be a cluster point of A . If $f(x) \leq g(x)$

$\leq h(x)$ for all $x \in A$, $x \neq c$ and if $\lim_{x \rightarrow c} h = L$, then

$$\lim_{x \rightarrow c} g = L$$

$$\text{To prove } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Let $f(x) = \sin x$

Case (1): When $x > 0$

On applying Taylor's theorem to the function $f(x) = \sin x$ in $[0, x]$, with remainder after three terms, we have:

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2} f''(0) + \frac{x^3}{6} f'''(\theta x), 0 < \theta < 1$$

$$\sin x = (\sin 0) + (x)(\cos 0) + \frac{(0)^2}{2} (-\sin 0) + \frac{x^3}{6} (-\cos \theta x)$$

$$\sin x = x - \frac{x^3}{6} \cos \theta x \quad \dots(2)$$

We have $\cos(\theta x) \leq 1$, whatever θx may be. Therefore, for $x > 0$

$$-\frac{x^3}{6} \cos(\theta x) \geq -\frac{x^3}{6}$$

$$\Rightarrow x - \frac{x^3}{6} \cos(\theta x) \geq x - \frac{x^3}{6}$$

$$\Rightarrow \sin x \geq x - \frac{x^3}{6}$$

$$\Rightarrow x - \frac{x^3}{6} \leq \sin x \quad \dots(3)$$

Applying again Taylor's theorem to the function $f(x) = \sin x$ in $[0, x]$ with remainder after five terms, we have:

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} \cos(\theta_1 x), 0 < \theta_1 < 1 \quad \dots(4)$$

We know that $\cos(\theta_1 x) \leq 1$, whatever $\theta_1 x$ may be. For $x > 0$, we get:

$$x - \frac{x^3}{6} + \frac{x^5}{120} \cos(\theta_1 x) \leq x - \frac{x^3}{6} + \frac{x^5}{120} \quad \dots(5)$$

Using (4), Eqn. (5) becomes:

$$\sin x \leq x - \frac{x^3}{6} + \frac{x^5}{120} \quad \dots(6)$$

From (3) and (6), we get:

$$x - \frac{x^3}{6} \leq \sin x \leq x - \frac{x^3}{6} + \frac{x^5}{120}, x > 0 \quad \dots(7)$$

Case (2): When $x < 0$ In this case, $x < 0 \Rightarrow -x > 0$

Putting

 $y = -x$ in Eqn. (7) we get:

$$y - \frac{y^3}{6} \leq 0 \text{ and } y \leq y - \frac{y^2}{6} + \frac{y^3}{120} \quad \dots(8)$$

On putting

 $y = -x > 0$ in Eqn. (8):

$$-\left(x - \frac{x^3}{6}\right) \geq \sin x \geq x - \frac{x^2}{6} + \frac{x^3}{120}; x < 0 \quad \dots(9)$$

From (7) and (9), we have:

$$x - \frac{x^3}{6} \leq \sin x \leq x; x > 0 \quad \dots(10.1)$$

and

$$x \leq \sin x \leq x - \frac{x^3}{6}; x < 0 \quad \dots(10.2)$$

Which following:

$$x - \frac{x^3}{6} \leq \sin x \leq x$$

$$1 - \frac{x^2}{6} \leq \frac{\sin x}{x} \leq 1; x \neq 0 \quad \dots(11)$$

But, $\lim_{x \rightarrow 0} \left(1 - \frac{x^2}{6}\right) = 1 - \frac{0}{6} = 1$

Therefore, using the squeeze theorem,

We have:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Proved.

(d) Let f be defined on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ and let c be a cluster point of A . If $\lim_{x \rightarrow c} f(x) > 0$, then show that there exists a neighbourhood $V_\delta(c)$ of c such that $f(x) > 0$ for all $x \in A \cap V_\delta(c)$, $x \neq c$. (5)

Ans. Let $L = \lim_{x \rightarrow c} f(x)$ and suppose that $L > 0$

We know from theorem that if let $A \subseteq \mathbb{R}$ and c be a cluster point of A . Then for a function $f: A \rightarrow \mathbb{R}$, a real number L is said to be a limit of f at c if given any $\epsilon > 0$ there exists a $\delta > 0$, such that if $x \in A$ and $0 < |x - c| < \delta$,

Then

$$|f(x) - L| < \epsilon$$

We take $\epsilon = \frac{L}{2}$ in above theorem and obtain a number $\delta > 0$ such that if $0 < |x - c| < \delta$ and $x \in A$, then $|f(x) - L| < \frac{L}{2}$.

Therefore it follows that if $x \in A \cap V_\delta(c)$

$$f(x) > \frac{L}{2}$$

 $\Rightarrow$

$$f(x) > \frac{L}{2} > 0$$

 $\Rightarrow$

$$f(x) > 0$$

Q. 2. (a) Let f be defined on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ and let c be a cluster point of $A \cap (c, \infty)$ and $A \cap (-\infty, c)$. Then show that $\lim_{x \rightarrow c} f(x) = L$ exists if and only

if $\lim_{x \rightarrow c^+} f(x) = L = \lim_{x \rightarrow c^-} f(x)$

Ans. Condition is necessary:

$$\text{Let } \lim_{x \rightarrow c} f(x) = L$$

Then for any $\epsilon > 0$, there exists some $\delta > 0$ such that

$$|f(x) - L| < \epsilon, \text{ when } 0 < |x - c| < \delta$$

$$\Rightarrow |f(x) - L| < \epsilon, \text{ when } c - \delta < x < c + \delta, x \neq c$$

It follows that

$$|f(x) - L| < \epsilon, \text{ when } c - \delta < x < c$$

and

$$|f(x) - L| < \epsilon, \text{ when } c < x < c + \delta$$

From (1) and (2), we see that:

$\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$ both exist and are equal to L

Condition is sufficient:

$$\text{Let } \lim_{x \rightarrow c^+} f(x) = l = \lim_{x \rightarrow c^-} f(x)$$

Then, for any $\epsilon > 0$, there exist some $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$|f(x) - l| < \epsilon, \text{ when } c < x < c + \delta_1$$

and

$$|f(x) - l| < \epsilon, \text{ when } c - \delta_2 < x < c$$

Let

$$\delta = \min(\delta_1, \delta_2)$$

Then,

$$\delta \leq \delta_1 \text{ and } \delta \leq \delta_2$$

 $\Rightarrow$

$$c + \delta \leq c + \delta_1 \text{ and } c - \delta \geq c - \delta_2 \text{ i.e., } c - \delta_2 \leq c - \delta \leq c + \delta \leq c + \delta_1$$

From (3) and (5), we have

and
 $\Rightarrow |f(x) - l| < \epsilon$, when $c - \delta < x < c + \delta$
 $\Rightarrow |f(x) - l| < \epsilon$, when $c - \delta < x < c$
 $\Rightarrow |f(x) - l| < \epsilon$, when $c - \delta < x < c + \delta$, $x \neq c$
 $\Rightarrow \lim_{x \rightarrow c} f(x) = l$

(b) Prove that

(i) $\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$

(ii) $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$

Ans. (i) $\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$

We have,

Def: The expression $\lim_{x \rightarrow \infty} f(x) = l$ means given any $\epsilon > 0$, there exist some $M > 0$ such that,

$$|f(x) - l| < \epsilon, \text{ when } x > M$$

Let $\epsilon > 0$ be given. Choose $M = \frac{1}{\epsilon^2}$

Then:

$$x^2 > M$$

$$\Rightarrow \left| \frac{1}{x^2} \right| < M = \epsilon$$

$$\Rightarrow \left| \frac{1}{x^2} - 0 \right| < \epsilon$$

Hence using the above definition,

We have:

$$\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$$

(ii) $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} \right) = \infty$

We have:

Def: The expression $\lim_{x \rightarrow a} f(x) = +\infty$ means given any $M > 0$, there exist some $\delta > 0$ such that,

Here, $f(x) > M$, when $|x - a| < \delta$
 $-\delta < x - a < \delta$

$$\Rightarrow \frac{1}{x^2} > \delta$$

$$\Rightarrow \frac{1}{x^2} > M$$

Hence using above definition, we get:

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$$

(c) Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$g(x) = \begin{cases} x, & \text{for } x \text{ rational} \\ 0, & \text{for } x \text{ irrational} \end{cases}$$

Find all the points at which g is continuous.

Ans. The function $g(x)$ is defined as:

$$g(x) = \begin{cases} x & : x \text{ is rational} \\ 0 & : x \text{ is irrational} \end{cases}$$

Case (1):

Let $a \neq 0$ be any rational number, so that

$$g(a) = a$$

Then, any nbd $\left(a - \frac{1}{n}, a + \frac{1}{n} \right)$ of a contains an irrational number a_n for each $n \in \mathbb{N}$.

$\in \mathbb{N}$ i.e.

$$a_n \in \left(a - \frac{1}{n}, a + \frac{1}{n} \right)$$

$$\Rightarrow |a_n - a| < \frac{1}{n}$$

$$\Rightarrow |a_n - a| \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow a_n \rightarrow a$$

$$\text{Now } g(a_n) = 0 \forall n \text{ (as } a_n \text{ is irrational)}$$

$$\Rightarrow \lim_{n \rightarrow \infty} g(a_n) = 0 \neq g(a) \text{ (as } g(a) = a \neq 0)$$

Thus g is discontinuous at every non zero rational point.

Case (2):

Let $b \neq 0$ be any rational number, so that

$$g(b) = 0$$

Then, there exists a rational number $b_n, \forall n \in \mathbb{N}$ such that

$$|b_n - b| < \frac{1}{n}$$

$$\Rightarrow |b_n - b| \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow b_n \rightarrow b$$

Now $g(b_n) = b_n, \forall n$ (as b_n is rational)

$$\Rightarrow \lim_{n \rightarrow \infty} g(b_n) = \lim_{n \rightarrow \infty} b_n = b = 0$$

$\Rightarrow g(b_n) \rightarrow 0$ does not converge to $g(b)$ as $g(b) = 0$

Thus g is discontinuous at every non zero irrational point.

We have $g(0) = 0$

$$\text{and } |g(x) - g(0)| = |g(x) - 0| \\ = |g(x)|$$

$$= \begin{cases} (x) & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

Let $\epsilon > 0$, then $|g(x) - g(0)| < \epsilon$ for $|x - 0| < \delta$ ($\delta = \epsilon$)

Hence g is continuous at $x = 0$.

(d) Determine the points of continuity of the function $f(x) = [x], x \in \mathbb{R}$, where $[x]$ denotes the greatest integer $n \in \mathbb{Z}$ such that $n \leq x$. (5)

Ans. The given function $f(x)$ is:

$$f(x) = [x], x \in \mathbb{R} \quad \dots(1)$$

By definition, we have

$$[x] = 0 \text{ for } 0 \leq x < 1$$

$$[x] = 1 \text{ for } 1 \leq x < 2$$

$$[x] = 2 \text{ for } 2 \leq x < 3$$

$$[x] = -1 \text{ for } -1 \leq x < 0$$

$$[x] = -2 \text{ for } -2 \leq x < -1 \text{ and so on}$$

At $x = 0$,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} [x] = -1$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} [x] = 0$$

$$\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$$

$\Rightarrow f(x)$ is discontinuous at $x = 0$

At $x = 1$:

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} [x] = 0$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} [x] = 1$$

$$\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$$

$\Rightarrow f(x)$ is discontinuous at $x = 1$

Similarly, f is discontinuous at 2, 3, 4,

At $x = -1$:

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} [x] = -2$$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} [x] = -1$$

$$\lim_{x \rightarrow -1^-} f(x) \neq \lim_{x \rightarrow -1^+} f(x)$$

$\Rightarrow f(x)$ is discontinuous at $x = -1$.

Similarly $f(x)$ is discontinuous at $-2, -3, -4, \dots$

Let $\alpha \in \mathbb{R} - \mathbb{Z}$ be any real number but not an integer. Then, there exists an integer n such that $n \leq \alpha < n + 1$.

Then $[x] = n$ for $n \leq x < n + 1$.

$$\text{Now } \lim_{x \rightarrow \alpha^-} f(x) = \lim_{x \rightarrow \alpha^-} [x] = f(x) = n$$

$\Rightarrow f$ is continuous at α .

Q. 3. (a) Let $A, B \subseteq \mathbb{R}$. Let $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$. If f is continuous at $c \in A$ and g is continuous at $b = f(c) \in B$ then show that the composite function $g \circ f: A \rightarrow \mathbb{R}$ is continuous at c . (5)

Ans. Let W be an ϵ -neighbourhood of $g(b)$.

Since g is continuous at b , there is a δ -neighbourhood V of $b = f(c)$ such that if $y \in B \cap V$, then $g(y) \in W$. Since f is continuous at c , there is a Y -neighbourhood U of c such that if $x \in A \cap U$, then $f(x) \in V$. Since, $f(A) \subseteq B$, it follows that if $x \in A \cap U$, then $f(x) \in B \cap V$, so that $g \circ f(x) = g(f(x)) \in W$. But since W is an arbitrary ϵ -neighbourhood of $g(b)$, this implies that $g \circ f$ is continuous at c .

(b) Let f, g be continuous from $\mathbb{R}$ to $\mathbb{R}$ and suppose that $f(r) = g(r)$ for all rational numbers r . Show that $f(x) = g(x)$ for all $x \in \mathbb{R}$. (5)

Ans. Consult text book

(c) Let f be a continuous real valued function defined on $[a, b]$. By

assuming that f is a bounded function, show that f attains its maximum value on $[a, b]$.

Ans. Consult text book

(d) Suppose that f is continuous on $[0, 2]$ and that $f(0) = f(2)$. Prove that there exist x, y in $[0, 2]$ such that $|y - x| = 1$ and $f(x) = f(y)$. (5)

Ans. Consult text book

Q. 4. (a) Let I be a closed and bounded interval and let $f: I \rightarrow \mathbb{R}$ be continuous on I . Then prove that f is uniformly continuous on I . (5)

Ans. If f is not uniformly continuous on I , then there exists $\epsilon_0 > 0$ and two sequences $\langle x_n \rangle$ and $\langle u_n \rangle$ in I such that $|x_n - u_n| < \frac{1}{n}$ and $|f(x_n) - f(u_n)| \geq \epsilon_0$ for all $n \in \mathbb{N}$. Since I is bounded, the sequence $\langle x_n \rangle$ is bounded. By Bolzano-Weierstrass theorem, there is a subsequence $\langle x_{n_k} \rangle$ of $\langle x_n \rangle$ that converges to an element z . Since I is closed the limit z belongs to I . It is clear that the corresponding subsequences $\langle u_{n_k} \rangle$ also converges to z .

$$\text{Since, } |u_{n_k} - z| = |u_{n_k} - x_{n_k}| + |x_{n_k} - z|$$

Now, if f is continuous at the point z then both of the sequences $\langle f(x_{n_k}) \rangle$ and $\langle f(u_{n_k}) \rangle$ must converge to $f(z)$. But this is not possible since,

$$|f(x_{n_k}) - f(u_{n_k})| \geq \epsilon_0$$

For all $n \in \mathbb{N}$. Thus the hypothesis that f is not uniformly continuous on the closed bounded interval I implies that f is not continuous at some point $z \in I$. Consequently if f is continuous at every point of I , then f is uniformly continuous on I .

(b) Show that the function $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$. (5)

Ans. The function $f(x)$ is defined as:

$$f(x) = x^2 \quad \dots(1)$$

Let $\epsilon = \frac{1}{2}$ and δ be any positive number. We can choose a positive integer n such that:

$$n > \frac{1}{4\delta^2} \text{ or } \frac{1}{4n} < \delta^2 \quad \dots(1)$$

Let

$$x_1 = \sqrt{n}$$

$$x_2 = \sqrt{n+1} \in [0, \infty).$$

Then

$$|f(x_2) - f(x_1)| = |x_2^2 - x_1^2|$$

$$= |(\sqrt{n+1})^2 - (\sqrt{n})^2|$$

$$= |(n+1) - n|$$

$$= 1 > \epsilon$$

and

$$|x_2 - x_1| = |\sqrt{n+1} - \sqrt{n}|$$

$$= \frac{|\sqrt{n+1} - \sqrt{n}|(\sqrt{n+1} + \sqrt{n})}{(\sqrt{n+1} + \sqrt{n})}$$

$$= \frac{|(\sqrt{n+1})^2 - (\sqrt{n})^2|}{(\sqrt{n+1} + \sqrt{n})}$$

$$= \frac{|(n+1) - n|}{(\sqrt{n+1} + \sqrt{n})}$$

$$= \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$< \frac{1}{2\sqrt{n}}$$

$$< \delta \quad \text{using (1)}$$

Because,

$$\sqrt{n+1} + \sqrt{n} > \sqrt{n} + \sqrt{n}$$

$\Rightarrow$

$$\sqrt{n+1} + \sqrt{n} > 2\sqrt{n}$$

Thus, $|f(x_2) - f(x_1)| > \epsilon$, when $|x_2 - x_1| < \delta$.

Hence, f is not uniformly continuous on $\mathbb{R}$ i.e. $[0, \infty)$. (5)

(c) State and prove Caratheodory's Theorem

Ans. Caratheodory's Theorem

Statement:

Let f be defined on an interval I containing the point c . Then, f is differentiable at c if and only if there exists a function ϕ on I that is continuous at c and satisfies

$$f(x) - f(c) = \phi(x)(x - c) \text{ for } x \in I$$

In this case

$$\phi(c) = f'(c)$$

Proof:

Condition is necessary:

If $f'(c)$ exists, we have define ϕ by

$$\phi(x) = \frac{f(x) - f(c)}{x - c}, \quad \begin{matrix} \text{for } x \neq c, x \in \mathbb{R} \\ \text{for } x = c \end{matrix}$$

$$= f'(c)$$

Also,

$$f(x) - f(c) = \phi(x)(x - c) \quad \dots(2)$$

The continuity of ϕ follows from the fact that $\lim_{x \rightarrow c} \phi(x) = f'(c)$. If $x = c$, then both sides of (2) equal 0, while if $x \neq c$, then

Multiplication of $f(x)$ by $x - c$ gives (2) for all other $x \in I$

Condition is Sufficient:

Now, assume that a function ϕ that is continuous at c and satisfying (2) exists. If we divide (2) by $x - c = 0$, then the continuity of ϕ implies that

$$\phi(c) = \lim_{x \rightarrow c} \phi(x)$$

$$= \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists. Therefore f is differentiable at c and $f'(c) = \phi(c)$

(d) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^2, & \text{for } x \text{ rational} \\ 0, & \text{for } x \text{ irrational} \end{cases}$$

Show that, f is differentiable at $x = 0$ and find $f'(0)$.

Ans. The given function $f(x)$ is defined as:

$$\left. \begin{aligned} f(x) &= x^2 : x \text{ is rational} \\ &= 0^2 : x \text{ is irrational} \end{aligned} \right\} \quad \dots(1)$$

We have,

$$f(0) = 0$$

$$\frac{f(x) - f(0)}{x - 0} = \frac{f(x) - 0}{x - 0}$$

$$= \frac{f(x)}{x}$$

$$= \left. \begin{aligned} \frac{x^2}{x} : x \text{ is rational} \\ \frac{0}{x} : x \text{ is irrational} \end{aligned} \right\}$$

$$\left. \begin{aligned} &= x : x \text{ is rational} \\ &= 0 : x \text{ is irrational} \end{aligned} \right\}$$

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = 0$$

$$f'(0) = 0$$

Q.5. (a) Let f be continuous on $[a, b]$ and differentiable on (a, b) . Prove that if $f'(x) < 0 \forall x \in (a, b)$, then f is strictly decreasing on $[a, b]$. Is the converse true? Justify.

Ans. Since,

- (1) $f(x)$ is continuous in $[a, b]$
- (2) $f(x)$ is differentiable in (a, b)
- (3) $f'(x) < 0$ in $a < x < b$

Let x_1, x_2 be any two points belonging to the closed interval $[a, b]$ such that $x_2 > x_1$.

Applying Lagrange's mean value theorem to $f(x)$ in the interval $[x_1, x_2]$

We have:

$$f(x_2) - f(x_1) = (x_2 - x_1) f'(c)$$

Where,

$$x_1 < c < x_2$$

Now, $x_2 - x_1 > 0$. Since by hypothesis $f'(x)$ is negative for every x in (a, b) therefore $f'(c) < 0$. Hence from (1), we have:

$$f(x_2) - f(x_1) < 0$$

$$\Rightarrow f(x_2) < f(x_1)$$

Thus, $f(x)$ is a decreasing function of x in $[a, b]$

(b) Let $g: [-1, 1] \rightarrow \mathbb{R}$ be defined by

$$g(x) = \begin{cases} 2, & 0 < x \leq 1 \\ 0, & x = 0 \\ 1, & -1 \leq x < 0 \end{cases}$$

Show that g is not the derivative on $[-1, 1]$ of any function. (5)

Ans. Do yourself

(c) Find the Taylor series for $\sin x$ and indicate why it converges to $\sin x \forall x \in \mathbb{R}$.

Ans. Let

$$f(x) = \sin x \forall x \in \mathbb{R}$$

$$f^n(x) = D^n[f(x)]$$

$$= \sin \left(x + \frac{n\pi}{2} \right) \forall n \in \mathbb{N}$$

Thus f has derivational of all order.

$$|f_n(\phi x)| = \left| \sin\left(\phi x + \frac{n\pi}{2}\right) \right| \leq \forall n \text{ and}$$

$$\forall x \in \mathbb{R}$$

$$|R_n| = \left| \frac{x^4}{4} f''''(\phi x) \right|$$

$$\leq \frac{|x^4|}{4} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Thus $R_n \rightarrow 0$ as $n \rightarrow \infty$ and so by Taylor series:

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\sin x = 0 + x - 1 + \frac{x^2}{2!} (0) + \frac{x^3}{3!} (-1) + \frac{x^4}{4!} (0) + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

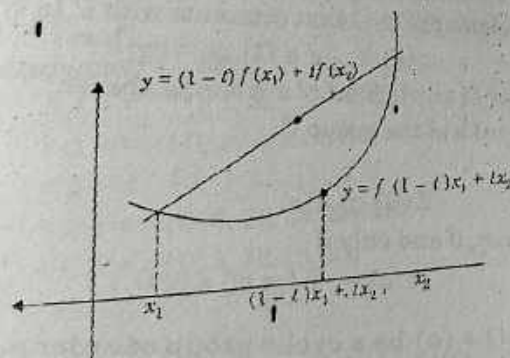
(d) Define a convex function on $[a, b]$. Show that the function $f(x) = |x|$, $x \in [-1, 1]$ is convex but not differentiable on $[-1, 1]$. (5)

Ans. Convex Function:

Let $I \subseteq \mathbb{R}$ be an interval. A function $f: I \rightarrow \mathbb{R}$ is said to be convex on I if for any t satisfying $0 \leq t \leq 1$ and any points x_1 and x_2 in I , we have:

$$f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2)$$

If $x_1 < x_2$, then as t range from 0 to 1, the point $(1-t)x_1 + tx_2$ transerses the interval from x_1 to x_2 . Thus if f is convex on I and if $x_1, x_2 \in I$, then the chord joining any two points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ on the graph of f lies above the graph of f .



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Name of the Paper : C6 : Group Theory I

Name of Course : B.Sc. (Hons.) Mathematics II Year

Semester : III (CBCS)

Duration : 3 Hour

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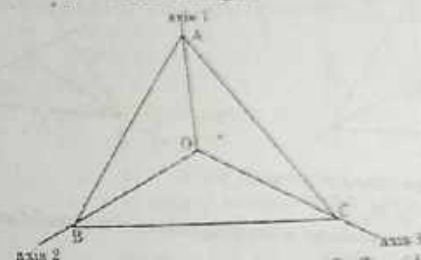
5 Dec. 2016

Attempt any two parts from each

All questions are compulsory.

Q. 1. (a) Describe the elements of the dihedral group D_3 and that they form a group under composition of mappings. (6)

Ans. Symmetries of equilateral triangle:



Let ABC be an equilateral triangle with centre O. Consider the following motions:

R_0 = Rotation about the (axis-1) centre through O

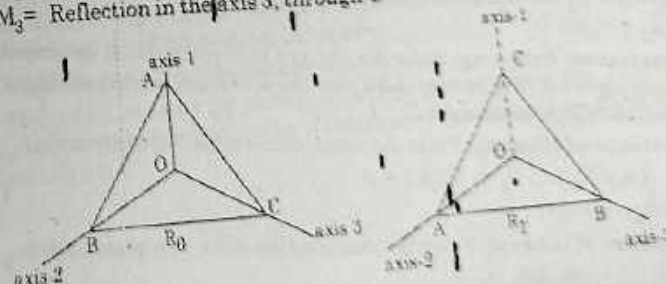
R_1 = Rotation through $\frac{2\pi}{3}$ anticlockwise about the line through O perpendicular to the plane of the triangle

R_2 = Rotation through $\frac{4\pi}{3}$ anticlockwise about the line through O perpendicular to the plane of the triangle.

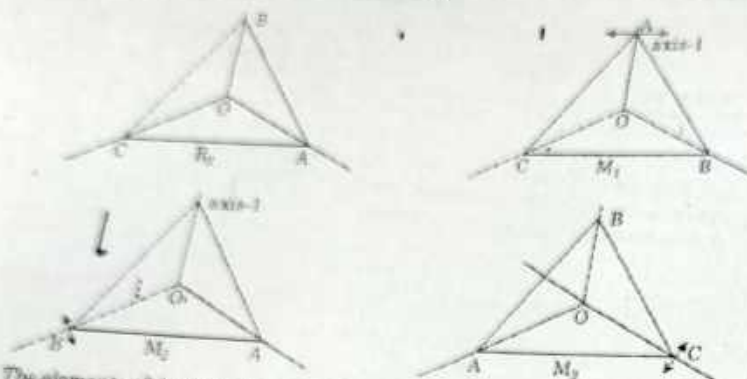
M_1 = Reflection in the axis 1 through A

M_2 = Reflection in the axis 2 through B

M_3 = Reflection in the axis 3 through C



(139)



The elements of the dihedral group D_3 are:
 $D_3 = \{R_0, R_1, R_2, M_1, M_2, M_3\}$

The binary operation considered is the composition of motions of an equilateral triangle. The composition table is:

Δ	R_0	R_1	R_2	M_1	M_2	M_3
R_0	R_0	R_1	R_2	M_1	M_2	M_3
R_1	R_1	R_2	R_0	M_2	M_1	M_3
R_2	R_2	R_0	R_1	M_3	M_1	M_2
M_1	M_1	M_3	M_2	R_0	R_1	R_2
M_2	M_2	M_1	M_3	R_2	R_0	R_1
M_3	M_3	M_2	M_1	R_1	R_2	R_0

From the composition table we can verify:

(i) **Closure Property:** Since all entries in the composition table are elements of the set D_3 , therefore closure property is verified i.e.,
 $x \Delta y \in D_3 \forall x, y \in D_3$

(ii) **Associative Property:** Since we observe that six motions described above are mapping of $\{A, B, C\}$ onto itself and the operation is composition of mappings. Therefore D_3 is associative.

(iii) **Existence of Identity:** From the composition table, we observe that
 $\forall x \in D_3, x \Delta R_0 = R_0 \Delta x = x$
 Therefore R_0 is identity.

(iv) **Existence of inverse:** From the composition table, we observe that,
 $R_0 \Delta R_0 = R_0$
 $R_1 \Delta R_2 = R_2 \Delta R_1$

$$M_1 \Delta M_1 = R_0$$

$$M_2 \Delta M_2 = R_0$$

$$M_3 \Delta M_3 = R_0$$

i.e., R_0, M_1, M_2, M_3 are self inverses, while R_1 and R_2 are inverses to each others.

Therefore, on the basis of above properties, we conclude that D_3 is a group with respect to motion of an equilateral triangle.

(b) Prove that if H and K are subgroups of G , then so is $H \cap K$. Is $H \cup K$ a subgroup of G ? Justify your answer.

Ans. (i) Let H and K are any two subgroups of G .

We have to show that $H \cap K$ is a subgroup of G , obviously $H \cap K$ is non-empty, since $e \in H \cap K$. Let $a, b \in H \cap K$, so that $a, b \in H$ and $a, b \in K$.

$$\text{Since } H < G, ab^{-1} \in H$$

$$\text{Similarly } ab^{-1} \in K$$

From (1) and (2):

$$ab^{-1} \in H \cap K \forall a, b \in H \cap K$$

Hence, by theorem we can say that $H \cap K$ is a subgroup of G .

(ii) The union of two subgroups of a group G need not to be a subgroup of G .
 Let $G = (I, +)$, then

$$H = \{\dots, -4, -2, 0, 2, 4, \dots\}$$

and

$$K = \{\dots, -3, 0, 3, 6, \dots\}$$

are subgroups of G , but

$$H \cup K = \{\dots, -6, -4, -3, -2, 0, 2, 3, 4, 5, \dots\}$$

is not a group of G , because

$$2, 3 \in H \cup K$$

But

$$2 + 3 = 5 \notin H \cup K.$$

(c) Define Centralizer $C(G)$ of a group G . Is $C(G)$ Abelian? Justify your answer.

Ans. Centralizer of a group:

Let a be a fixed element of a group G . The centralizer of a in G is denoted by $c(a)$ is the set of all elements in G that commute with a . In symbols:

$$c(a) = \{g \in G \mid ga = ag\}$$

Is the centralizer of an element of a group be abelian?
 For every element a of the group G ,
 We have:

$$Z(G) \subseteq c(a)$$

Also, G is abelian, if and only if

$$c(a) = G \text{ for all } a \text{ in } G.$$

Q. 2. (a) Let $G = \langle a \rangle$ be a cyclic group of order n . Show that $G = \langle a^k \rangle$ if and only if $\gcd(k, n) = 1$.

Ans. Let

Then:

We show that a^k is a generator of G if and only if $\gcd(k, n) = 1$, i.e. k and n are relatively prime.

Suppose,

Since,

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

Converse:

Let k and n be relatively prime integer ($0 < k < n$). Then there exist integers r and s such that $kr + ns = 1$.

$$a = a^1$$

$$a = a^{kr+ns}$$

$$a = a^{kr} \cdot a^{ns}$$

$$a = (a^k)^r (a^n)^s$$

$$a = (a^k)^r (e)^s \text{ as } a^n = e$$

$$a = (a^k)^r$$

Since for any $x \in G = \langle a \rangle$, we have:

$$x = a^k (1 \leq k \leq n)$$

$\Rightarrow$

$$x = (a^k)^r$$

Hence,

$$G = \langle a^k \rangle$$

It follows that a^k is a generator of

$$G = \langle a \rangle : a^n = e \text{ iff } \gcd(k, n) = 1.$$

(b) List the subgroups of Z_{30} and their generators.

Ans. Subgroups of Z_{30} :

The list of subgroups of Z_{30} are:

$$\langle 1 \rangle = \{0, 1, 2, \dots, 29\}$$

$$\langle 2 \rangle = \{0, 2, 4, \dots, 28\}$$

$$\langle 3 \rangle = \{0, 3, 6, \dots, 27\}$$

$$\langle 5 \rangle = \{0, 5, 10, 15, 20, 25\}$$

$$\langle 6 \rangle = \{0, 6, 12, 18, 24\}$$

$$\langle 10 \rangle = \{0, 10, 20\}$$

$$\langle 15 \rangle = \{0, 15\}$$

$$\langle 30 \rangle = \{0\}$$

order 30

order 15

order 10

order 6

order 5

order 3

order 2

order 1

The generators of Z_{30} are

1, 2, 3, 5, 6, 10, 15, 30.

(c) (i) Let G be an Abelian group with identity e . Prove that

$H = \{x \in G \mid x^2 = e\}$ is a subgroup of G . (3)

(ii) Let a and b be elements of a group. If $|a| = 10$ and $|b| = 21$.

show that $\langle a \rangle \cap \langle b \rangle = \{e\}$. (3)

Ans. (i) G be an abelian group with identity e .

$$H = \{x \in G \mid x^2 = e\} \quad \dots (1)$$

Here,

$$e^2 = e, \text{ so that } H \text{ is non-empty}$$

Now, we assume that a and b belong to H . This means:

$$a^2 = e$$

$$b^2 = e \quad \dots (2)$$

Since G is abelian, we have:

$$(ab^{-1})^2 = (ab^{-1})(ab^{-1})$$

$$= a^2 (b^{-1})^2$$

$$= a^2 (b^2)^{-1}$$

$$= (e)(e)^{-1}, \text{ using (1) and (2)}$$

$$= ee^{-1}$$

$$= e$$

$$\Rightarrow (ab^{-1})^2 = e$$

$$\Rightarrow ab^{-1} \in H$$

Hence, we can say that H is a subgroup of G .

$$(ii) \quad |a| = 10$$

$$|b| = 21$$

Which gives:

$$\langle a \rangle = \langle 1, 2, 5 \rangle$$

$$\langle b \rangle = \langle 1, 3, 7 \rangle$$

Since 10 and 21 are relative prime. So,

We have:

$$\langle a \rangle \cap \langle b \rangle = \{e\}$$

Q. 3. (a) Prove that every permutation on a finite set can be written as a cycle or product of disjoint cycles. (6)

Ans. Let f be a permutation on a set $S = \{a_1, a_2, \dots, a_n\}$

First of all, we look for such elements $a_m \in S$ such that $f(a_m) = a_m$. In their way, we can write down all cycles of length one. Then, we start with an element $a_i \in S$ satisfying.

$$f(a_i) = a_i + 1$$

$$f(a_i + 1) = a_i + 2,$$

$$\vdots$$

$$f(a_i) = a_i$$

In this way, we get a cycle $(a_i, a_i + 1, \dots, a_j)$.
 Suppose $a_k \in S$ be another element which have not been included in the cycles obtained above. Proceeding as above, we shall get another cycles of the form $(a_k, a_{k+1}, \dots, a_l)$.
 In this manner, all the elements of S will be exhausted after a finite number of steps and we shall get,

$$f = (a_i, a_i + 1, \dots, a_j) (a_k, a_{k+1}, \dots, a_l) \dots$$

which is a product of disjoint cycles.
 Hence, the theorem is proved.

(b) Prove that the set of even permutations in S_n forms a subgroup of S_n .
 Ans. Let A_n be the set of all even permutations in S_n and

$$O(S_n) = \{e\}$$

Now, we have to prove that A_n is a subgroup of S_n .
 Since the identity permutation I is an even permutation i.e. $I \in A_n$, it means that A_n is non-empty.

Let $f, g \in A_n$, so that f and g are even permutations.
 Since the product of two even permutations is even. Then $f \circ g \in A_n$.
 Consequently A_n is a subgroup of S_n as S_n is finite.
 Hence we proved that the set of even permutations (A_n) in S_n forms a subgroup of S_n .

(c) Let $\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 1 & 3 & 5 & 4 & 7 & 6 & 8 \end{bmatrix}$ and $\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 3 & 8 & 7 & 6 & 5 & 4 & 4 \end{bmatrix}$
 Write α and β as

- (i) products of disjoint cycles.
 (ii) products of 2-cycles.

Ans. Let

$$\alpha = (1\ 2)(3\ 4)(5\ 6)(7\ 8) \quad (6)$$

$$\beta = (1\ 3)(2\ 8)(4\ 7)(5\ 6)$$

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 1 & 3 & 5 & 4 & 7 & 6 & 8 \end{bmatrix}$$

We can express α as a product of disjoint cycles as follows.
 $\alpha = (1\ 2)(4\ 5)(6\ 7)$

$$\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 3 & 8 & 7 & 6 & 5 & 4 & 4 \end{bmatrix}$$

Ans.

We can express β as product of disjoint cycles as follows:

$$= (2\ 3\ 8\ 4\ 7)(5\ 6)$$

(ii)

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 1 & 3 & 5 & 4 & 7 & 6 & 8 \end{bmatrix}$$

$$= (1\ 2)(4\ 5)(6\ 7)$$

$$= \text{product of 2-cycles.}$$

$$\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 3 & 8 & 7 & 6 & 5 & 2 & 4 \end{bmatrix}$$

$$= (2\ 3\ 4\ 7)(5\ 6)$$

$$= (2\ 7)(2\ 4)(2\ 3)(5\ 6)$$

$$= \text{Products of 2-cycles}$$

Q. 4. (a) Prove that any finite cyclic group of order n is isomorphic to Z_n and any infinite cyclic group is isomorphic to Z .
 Ans. Let

$$G = \langle a \rangle \text{ be a cyclic group of order } n \text{ such that}$$

$$G = \langle a \rangle = \{e, a^2, \dots, a^{n-1}, a^n = e\}$$

Also,

$$Z_n = \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}$$

The mapping $f: Z_n \rightarrow G$ defined by

$$f(\bar{i}) = a^i \text{ for } i = 0, 1, 2, \dots, n-1$$

is obviously one to one and onto.

Also f is a homomorphism, since,

$$f(\bar{i} + \bar{j}) = f(\overline{i+j})$$

$$= a^{i+j}$$

$$= f(\bar{i})f(\bar{j})$$

Hence,

$\Rightarrow$

$$Z_n = G$$

$$G = Z_n$$

Which proves that any finite cyclic group of order n is isomorphic to Z_n .
 (ii) Let

$$G = \langle a \rangle \text{ be any infinite group s.t.}$$

$$G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$$

Define a mapping $f: \mathbb{Z} \rightarrow G$ as

$$f(n) = a^n \forall n \in \mathbb{Z}$$

Then, f is homomorphism, since for $m, n \in \mathbb{Z}$.

$$f(m+n) = a^{m+n}$$

$$= a^m a^n$$

$$= f(m)f(n), \text{ using (1)}$$

Now, we show that f is onto. Let $g \in G$ be arbitrary.

Then,

We have

$$g = \alpha^i \text{ for some } i \in \mathbb{Z}, \text{ using (1)}$$

$$f(i) = g$$

 $\Rightarrow f$ is ontoWe show that f is one to one. Let $m, n \in \mathbb{Z}$.

Then,

$$f(m) = f(n)$$

$$\alpha^m = \alpha^n$$

$$\alpha^{m-n} = e$$

$$m - n = 0, \text{ Since } G \text{ is infinite}$$

$$m = n$$

 f is one to one

Hence,

$$Z = G \text{ or } G = Z$$

Which proves that any infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$ (b) Let $\bar{G}$ denote the left regular representation of the group G (as defined in Cayley's Theorem). Calculate $\overline{U(12)}$.Ans. $U(12)$ is calculated as—

$$U(12) = \{1, 5, 7, 11\}$$

On writing the permutations of $U(12)$ in array form, we have

$$T_1 = \begin{pmatrix} 1 & 5 & 7 & 11 \\ 1 & 5 & 7 & 11 \end{pmatrix}$$

$$T_5 = \begin{pmatrix} 1 & 5 & 7 & 11 \\ 5 & 1 & 11 & 7 \end{pmatrix}$$

$$T_7 = \begin{pmatrix} 1 & 5 & 7 & 11 \\ 7 & 11 & 1 & 5 \end{pmatrix}$$

$$T_{11} = \begin{pmatrix} 1 & 5 & 7 & 11 \\ 11 & 7 & 5 & 1 \end{pmatrix}$$

The Cayley's table for $U(12)$ and its left regular representation $\overline{U(12)}$ are as below:

$U(12)$	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	1	1	5
11	11	7	5	1

and

$\overline{U(12)}$	T_1	T_5	T_7	T_{11}
T_1	T_1	T_5	T_7	T_{11}
T_5	T_5	T_1	T_{11}	T_7
T_7	T_7	T_{11}	T_1	T_5
T_{11}	T_{11}	T_7	T_5	T_1

(c) Give statements only, of Lagrange's Theorem and its converse. Is the converse true? Justify your answer. (5'2)

Ans. Lagrange's Theorem. If G is a finite group and H is a subgroup of G , then order of H is a divisor of order of G .Converse of Lagrange's Theorem. If m divide order of the finite group G , then G need not have a subgroup of order m .Example: Let S_4 be the set of all permutations defined on the set $\{1, 2, 3, 4\}$ and A_4 be the set of all even permutations in S_4 . Then, A_4 is a subgroup of S_4 .

$$\text{where } |A_4| = \frac{1}{2}|S_4| = \frac{24}{2} = 12$$

Obviously G divides $|A_4|$, but A_4 has no subgroup of order 6.Q. 5. (a) Let H and K be subgroups of a finite group G . Prove that

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} \quad (5'2)$$

Ans. Let H and K be subgroup of a finite group G .Let $T = H \cap K$. Then, T is a subgroup of K . $\therefore T \subseteq K$. Since, K is a finite group.

$$i_k(T) = \frac{|K|}{|T|} = m \text{ (Let)} \quad (1)$$

Thus, there exist m distinct right cosets of T in K say, $T_{k_1}, T_{k_2}, \dots, T_{k_m}$ where $k_1, k_2, \dots, k_m \in K$. So

$$\begin{aligned} K &= T_{k_1} \cup T_{k_2} \cup \dots \cup T_{k_m} \\ HK &= H(T_{k_1} \cup T_{k_2} \cup \dots \cup T_{k_m}) \\ &= HT_{k_1} \cup HT_{k_2} \cup \dots \cup HT_{k_m} \\ &= Hk_1 \cup Hk_2 \cup \dots \cup Hk_m \end{aligned} \quad (2)$$

$$(\because T \subseteq H \Rightarrow HT = H)$$

Now, we show that $Hk_1, Hk_2, \dots, Hk_m$ are all distinct.Let if possible, $Hk_i = Hk_j, \forall 1 \leq i < j \leq m$

$$\Rightarrow k_i k_j^{-1} \in H \text{ Also, } k_i k_j^{-1} \in K (\because K < G)$$

$$\Rightarrow k_i k_j^{-1} \in H \cap K = T$$

$$\Rightarrow f(Hx) = f(My)$$

$$\Rightarrow f \text{ is well defined.}$$

(ii) To prove f is homomorphism

$$\Rightarrow x, y \in G \Rightarrow xy \in G$$

$$\Rightarrow f(Hxy) = Kxy$$

$$\Rightarrow f(Hxy) = Kx, Ky$$

$$\Rightarrow f(Hxy) = f(Hx)f(Hy)$$

$$\Rightarrow f(Hxy) = f(Hx)f(Hy)$$

$$\Rightarrow f \text{ is a homomorphism.}$$

(iii) To prove f is onto

$$\text{Given any } Kx \in \frac{G}{K}$$

$$\Rightarrow x \in G$$

$$\Rightarrow Hx \in \frac{G}{H} \text{ s.t. } f(Hx) = Kx$$

(iv) To prove that kernel of f is $\frac{K}{H}$:

$$\text{Ker } f = \{Hx \in \frac{G}{H} : f(Hx) = K\}$$

K is the identity of $\frac{G}{K}$

$$\text{any } Hx \in \ker f \Rightarrow f(Hx) = K$$

$$\Rightarrow Kx = K, \text{ by def of } f$$

$$\Rightarrow x \in K$$

$$\Rightarrow Hx \in \frac{K}{H}$$

$$\text{any } Hx \in \ker f \Rightarrow Hx \in \frac{K}{H}$$

$$\text{Thus } \ker f \subseteq \frac{K}{H}$$

$$\Rightarrow x \in K$$

$$\Rightarrow Kx = K$$

$$\Rightarrow f(Hx) = K$$

$$\Rightarrow Hx \in \ker f$$

$$\Rightarrow \frac{K}{H} \subseteq \ker f$$

From (1) and (2):

$$\frac{K}{H} = \ker f$$

Finally, we have shown that $f: \frac{G}{H} \xrightarrow{\text{onto}} \frac{G}{K}$ is a homomorphism with $\ker f = \frac{K}{H}$.

Also $\frac{K}{H}$ is a normal subgroup of $\frac{G}{H}$.

Using third homomorphism theorem.

We have:

$$\frac{G}{K} = \frac{\left(\frac{G}{H}\right)}{\left(\frac{K}{H}\right)}$$

Which proves the second isomorphism theorem.

(b) Let G be a subgroup of some dihedral group. For each x in G , define

$$\varphi(x) = \begin{cases} +1 & \text{if } x \text{ is a rotation} \\ -1 & \text{if } x \text{ is a reflection} \end{cases}$$

Prove that φ is a homomorphism from G to the multiplicative group $\{+1, -1\}$. What is the kernel of φ ? (6)

Ans. Consult text book.

(c) Determine all homomorphisms from Z_{12} to Z_{30} . (6)

Ans. We have the property P_1 and P_2 as

P_1 : Let ϕ be a homomorphism from a group G to a group $\bar{G}$ and let g be an element of G , then $\phi(g^n) = (\phi(g))^n$

P_2 : Let ϕ be a homomorphism from a group G to a group $\bar{G}$ let g be an element of G . Then, if $|g| = n$, then $|\phi(g)|$ divides n .

By the property P_1 such a homomorphism is completely specified by the image of 1. If 1 maps to a , then x maps to $x \cdot a$. Lagrange's theorem and property P_2 require $|a|$ divide both 12 and 30. So $|a| = 1, 2, 3, 6$.

Thus $a = 0, 15, 10, 20, 5$, or 25. This gives a list of candidates for homomorphism. ○○○

Paper Code : 32351303
 Name of the Paper : C7: Multivariate Calculus
 Name of Course : B.Sc. (Hons.) Mathematics II Year
 Semester : III (CBCS)
 Duration : 3 Hours
 Maximum Marks : 75

9 Dec. 2016

All sections are compulsory. Attempt any five questions from each section.
 All questions carry equal marks.

SECTION-I

Q. 1. Show that f is continuous at $(0, 0)$ where

$$f(x, y) = \begin{cases} y \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Ans. The function $f(x, y)$ is defined as:

$$f(x, y) = y \sin \left(\frac{1}{x} \right); x \neq 0$$

$$= 0; x = 0$$

Let $\epsilon > 0$ be any given real number,

$$\text{Now, } |f(x, y) - f(0, 0)| = \left| y \sin \left(\frac{1}{x} \right) - 0 \right|$$

$$= \left| y \sin \left(\frac{1}{x} \right) \right|$$

$$= |y| \left| \sin \left(\frac{1}{x} \right) \right|$$

$$\leq |y| \text{ because } \left| \sin \frac{1}{x} \right| \leq 1$$

$$\Rightarrow |f(x, y) - f(0, 0)| \leq |y|$$

$$\Rightarrow |f(x, y) - f(0, 0)| \leq |y - 0|$$

Let $|x - 0| < \epsilon$ and $|y - 0| < \epsilon$.

Then,

$$|f(x, y) - f(0, 0)| < \epsilon, \text{ whenever } |x - 0| < \delta$$

and

$$|y - 0| < \delta$$

Where,

$$\delta = \epsilon$$

$\therefore f(x, y)$ is continuous at $(0, 0)$

Q. 2. A closed box is found to have length 2 ft, width 4 ft and height 3 ft, where the measurement of each dimension is made with a maximum

(152)

possible error of ± 0.02 ft. The top of the box is made from material that costs \$2/ft², and the material for the sides and bottom costs only \$1.50/ft². What is the maximum error involved in the computation of the cost of the box?

Ans. The length of box (x) = 2 ft
 width of box (y) = 4 ft
 Height of box (z) = 3 ft

The surface area of the box with length x , breadth (y) and height (z) is calculated as:

$$S = xy + xy + 2xz + 2yz$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 top bottom (four side)

Since the top costs \$2/ft² and bottom costs \$1.50/ft². Therefore the total cost be:

$$C(x, y, z) = 2(xy) + 1.5(xy) + 1.5(2xz + 2yz)$$

$$= 2xy + 1.5(xy + 2xz + 2yz)$$

The partial derivatives of C are:

$$C_x = \frac{\partial C}{\partial x}$$

$$= 2y + 1.5(y + 2z + 0)$$

$$= 2y + 1.5y + 3z$$

$$C_y = \frac{\partial C}{\partial y}$$

$$= 2x + 1.5(x + 0 + 2z)$$

$$= 2x + 1.5x + 3z$$

$$C_z = \frac{\partial C}{\partial z}$$

$$= 3x + 3y$$

At $(2, 4, 3)$:

$$C_x(2, 4, 3) = 3.5(4) + 3(3) = 14 + 9 = 23$$

$$C_y(2, 4, 3) = 2.5(2) + 3(3) = 7 + 9 = 16$$

$$C_z(2, 4, 3) = 3(2) + 3(4) = 6 + 12 = 18$$

The error in dimension is ± 0.02 ft.

Then $\Delta x = \Delta y = \Delta z = 0.02$.
 The change in the total cost is approximated by:

$$\begin{aligned}\Delta C &= C_x(2, 4, 3)\Delta x + C_y(2, 4, 3)\Delta y + C_z(2, 4, 3)\Delta z \\ &= (23)(0.02) + (16)(0.02) + (18)(0.02) \\ &= 0.46 + 0.32 + 0.36 \\ &= \$1.14\end{aligned}$$

Q. 3. If $z = f(u - v, v - u)$, show that

$$\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} = 0$$

Ans.

Let

$$z = f(u - v, v - u)$$

$$u - v = X$$

$$v - u = Y$$

$$z = f(X, Y)$$

$\Rightarrow z$ is a composite function of X and Y .

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial u} + \frac{\partial z}{\partial Y} \cdot \frac{\partial Y}{\partial u}$$

$$= \frac{\partial z}{\partial X}(1-0) + \frac{\partial z}{\partial Y}(0-1)$$

$$= \frac{\partial z}{\partial X} - \frac{\partial z}{\partial Y}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial v} + \frac{\partial z}{\partial Y} \cdot \frac{\partial Y}{\partial v}$$

$$= \frac{\partial z}{\partial X}(0-1) + \frac{\partial z}{\partial Y}(1-0)$$

$$= -\frac{\partial z}{\partial X} + \frac{\partial z}{\partial Y}$$

$\therefore$ From (3) and (4) we have:

$$\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} = \left(\frac{\partial z}{\partial X} - \frac{\partial z}{\partial Y} \right) + \left(-\frac{\partial z}{\partial X} + \frac{\partial z}{\partial Y} \right)$$

$$= \left(\frac{\partial z}{\partial X} - \frac{\partial z}{\partial X} \right) + \left(-\frac{\partial z}{\partial Y} + \frac{\partial z}{\partial Y} \right)$$

$$= (0) + (0)$$

$$= 0$$

Q. 4. Suppose at point $P_0(-1, 2)$, a certain function $f(x, y)$ has directional derivative 8 in the direction of $v_1 = 3i - 4j$ and 1 in the direction

of $v_2 = 12i + 5j$. What is the directional derivative of f at P_0 in the direction of $v = 3i - 5j$?

Ans. Let $f(x, y)$ be a function that is differentiable at $P_0(x_0, y_0)$. Then, the directional derivative of the unit vector $u = u_1i + u_2j$ is given by $D_u f(x_0, y_0) = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2$.

For condition (1)

$P_0(-1, 2)$ has a directional derivative 8 in the direction of $v_1 = 3i - 4j$. It means:

$$8 = f_x(3) + f_y(-4)$$

$$3f_x - 4f_y - 8 = 0 \quad \dots(1)$$

For Condition (2)

$P_0(-1, 2)$ has a directional derivative 1 in the direction of $v_2 = 12i + 5j$. It means:

$$1 = f_x(12) + f_y(5)$$

$$12f_x + 5f_y - 1 = 0 \quad \dots(2)$$

From (2) and (3): Using cross multiplication

We have:

$$\begin{array}{ccccc} f_x & f_y & 1 & f_x & f_y \\ 3 & -4 & -8 & 3 & -4 \\ 12 & 5 & -1 & 12 & 5 \end{array}$$

$$\frac{f_x}{-4+40} = \frac{f_y}{-96+3} = \frac{1}{-15+48}$$

$$\frac{f_x}{44} = \frac{f_y}{-93} = \frac{1}{63}$$

$$f_x = \frac{44}{63}$$

$$f_y = \frac{-93}{63} = \frac{-31}{21}$$

Hence, the directional derivative of f at $P_0(-1, 2)$ in the direction of $v = 3i - 5j$ will be:

$$D_u f(x_0, y_0) = f_x(3) + f_y(-5)$$

$$= \left(\frac{44}{63} \right)(3) + \left(\frac{-31}{21} \right)(-5)$$

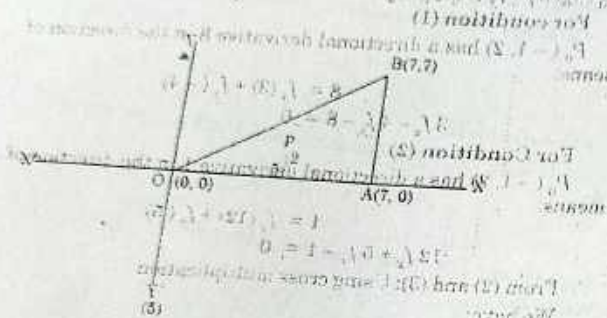
$$= \frac{44}{21} + \frac{155}{21}$$

$$= \frac{199}{21}$$

Ans.

Q. 5. Find the absolute extrema of $f(x, y) = xy - 2x - 5y$ on the closed bounded set S where S is the triangular region with vertices $(0, 0)$, $(7, 0)$ and $(7, 7)$.

Ans. $f(x, y) = xy - 2x - 5y$
Here, S is closed bounded set i.e. triangular region with vertices $(0, 0)$, $(7, 0)$ and $(7, 7)$.



From (1)

$$f_x = \frac{\partial f}{\partial x} = (1)y - 2(1) = y - 2$$

$$f_y = \frac{\partial f}{\partial y} = x(1) - 2(0) - 5(1) = x - 5$$

When
We have:

$$f_x = 0 \Rightarrow y = 2$$

When
We have:

$$f_y = 0 \Rightarrow x = 5$$

Let $P = (5, 2)$. Since $P(5, 2)$ lies with the closed triangular region. It means at theoretical point P , there is neither maxima nor minima.

On line AB

$$f(x, y) = x(0) - 2x - 5(0) = -2x$$

On line AB

$$f(x, y) = 7y - 14 + 5y = 2y - 14$$

On line AB

$$f_{\max} = 0 \text{ and } f_{\min} = -14$$

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$$f(x, y) = (x)(y) - 2x - 5y$$

$$= x^2 - 2x - 2x$$

$$= x^2 - 2x - 5y$$

$$\Rightarrow f_{\max} = 0 \text{ and } f_{\min} = -14$$

Hence, We have:

absolute maxima in $S = 0$

absolute minima in $S = -14$

Q. 6. Find the point of intersection of the plane $x + 2y + z = 10$ and the paraboloid $z = x^2 + y^2$ that is closest to the origin. You may assume that such a point exists.

Ans. The equation of the plane is:

$$x + 2y + z = 10$$

The equation of the paraboloid is:

$$z = x^2 + y^2$$

The distance from a point (x, y, z) to the origin is $S = \sqrt{x^2 + y^2 + z^2}$, but instead of minimizing this quality it is easier to minimize its square. Hence, We will minimize,

Subject to the joint constraints

$$g(x, y, z) = x + 2y + z = 10$$

and

$$h(x, y, z) = x^2 + y^2 - z = 0$$

From (3):

$$f_x = \frac{\partial b}{\partial x} = 2x$$

$$f_y = \frac{\partial b}{\partial y} = 2y$$

$$f_z = \frac{\partial b}{\partial z} = 2z$$

From (4):

$$g_x = \frac{\partial g}{\partial x} = 1$$

$$g_y = \frac{\partial g}{\partial y} = 2$$

$$g_z = \frac{\partial g}{\partial z} = 1$$

From (5):

$$h_x = \frac{\partial u}{\partial x} = 2x$$

$$h_y = \frac{\partial u}{\partial y} = 2y$$

$$h_z = \frac{\partial u}{\partial z} = -1$$

and

add to

multiplied

To apply the method of Lagrange's multipliers, we use the formulas:
 $\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0) + \mu \nabla h(x_0, y_0, z_0)$
 Which leads to the following systems of

(1) Equations:

(2)...

and, $\frac{z}{x} +$

$$2x = \lambda(1) + \mu(2x)$$

$$2y = \lambda(2) + \mu(2y)$$

$$2z = \lambda(1) + \mu(-1)$$

$$x + 2y + z = 10$$

$$z = x^2 + y^2$$

(3) This is not a linear system, so solving it require ingenuity.

Multiply the first equation by (2) and subtract the second equation, we get.

$$(1) \dots 4x - 2y = (4x - 2y)\mu$$

$$(1) \dots (4x - 2y) - (4x - 2y)\mu = 0$$

$$(4x - 2y)(1 - \mu) = 0$$

$$4x - 2y = 0$$

or

$$1 - \mu = 0$$

Case (1): When $4x - 2y = 0$

If $4x - 2y = 0$, then $y = 2x$.

Substitute this in to the two constant equations:

$$x + 2y + z = 10$$

$$x + 2(2x) + z = 10$$

$$3x + z = 10$$

$$z = 10 - 3x$$

$$x^2 + y^2 - z = 0$$

$$x^2 + (2x)^2 - z = 0$$

$$5x^2 - z = 0$$

$$z = 5x^2$$

$$\Rightarrow z = 10 - 3x = 5x^2$$

$$5x^2 = 5x - 10 = 0$$

$$x^2 + x - 2 = 0$$

$$x^2 + 2x - x - 2 = 0$$

$$x(x+2) - 1(x-2) = 0$$

$$(x+2)(x-1) = 0$$

$$x = 1, -2$$

Which gives

$$x = 1$$

$$y = 2x = 2(1) = 2$$

$$z = 5x^2 = 5(1)^2 = 5$$

$$x = -2$$

$$y = 2x = 2(-2) = -4$$

$$z = 5x^2 = 5(-2)^2 = 20$$

Thus, the points (1, 2, 5) and (-2, -4, 20) are candidates for all minima distance.

Case (2): When $1 - \mu = 0$

If $1 - \mu = 0$, then $\mu = 1$. We look at the system of equations involving x, y, z, λ and μ :

$$2x = \lambda(1) + \mu(2x)$$

$$2y = \lambda(2) + \mu(2y)$$

$$2z = \lambda(1) + \mu(-1)$$

$$x + 2y + z = 10$$

$$z = x^2 + y^2$$

The top equation becomes so that

$$2x = \lambda + 2\mu x$$

$$2x = \lambda + 2(1)x$$

$$2x = \lambda + 2x$$

$$\lambda = 0$$

And also,

$$2z = \lambda - 1$$

$$2z = 0 - 1$$

$$z = -\frac{1}{2}$$

Next, turn to the constraint equations:

$$x + 2y + z = 10$$

$$x + 2y - \frac{1}{2} = 10$$

$$x + 2y = \frac{1}{2} + 10$$

$$x + 2y = \frac{21}{2}$$

$$x^2 + y^2 - z = 0$$

$$x^2 + y^2 + \frac{1}{2} = 0$$

$$x^2 + y^2 = -\frac{1}{2}$$

There is no relation because $x^2 + y^2$ can not equal to a negative number
 We check the candidates for minimal distance:

$$f(x, y, z) = x^2 + y^2 + z^2, \text{ so that}$$

$$f(1, 2, 5) = (1)^2 + (2)^2 + (5)^2 = 1 + 4 + 25 = 30$$

$$f(-2, -4, 20) = (-2)^2 + (-4)^2 + (20)^2 = 4 + 16 + 400 = 420$$

Because $\sqrt{x^2 + y^2 + z^2}$ represents the square to the distance the minimal distance is $\sqrt{36}$ and the point on the intersection of the two surface nearest to the origin is (1, 2, 5).

$$z = 1 - x - y \quad z = 1 + x + y$$

SECTION-III

Q. 7. Compute $\iint_R \frac{1}{x^2 + y^2} dA$ where R is the rectangle $1 \leq x \leq 3, 1 \leq y \leq 2$.

Ans. Do yourself.

Q. 8. Evaluate the integral $\int_0^{\pi/4} \int_0^{\sqrt{2} \cos \theta} \sec^2 x \, dx \, dy$

Ans. Do yourself

Q. 9. Evaluate $\iint_D \frac{1}{x} dA$, where D is the region that lies inside the circle $r = 3 \cos \theta$ and outside the cardioid $r = 1 + \cos \theta$.

Ans.

$$r = 3 \cos \theta \quad r = 1 + \cos \theta$$

$$r = 1 + \cos \theta$$

From (1) and (2):

$$3 \cos \theta = 1 + \cos \theta$$

$$2 \cos \theta = 1 \quad \cos \theta = \frac{1}{2}$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = \pm \frac{\pi}{3}$$

Hence the point of intersection of curve (1) and (2) are $\theta = \frac{\pi}{3}$ and $\theta = -\frac{\pi}{3}$.

D is the region such that for each fix angle θ between $-\frac{\pi}{3}$ and $\frac{\pi}{3}$, r varies from $1 + \cos \theta$ (the cardioid) to $3 \cos \theta$ (the circle) this gives us the limits of integration and in polar form the integrand becomes:

$$\frac{1}{x} = \frac{1}{r \cos \theta} = \frac{1}{r} \sec \theta$$

Thus, $\iint_D \frac{1}{x} dA = \int_{-\pi/3}^{\pi/3} \int_{1+\cos \theta}^{3 \cos \theta} \frac{1}{r} \sec \theta \, r \, dr \, d\theta$

$$= \int_{-\pi/3}^{\pi/3} \left[\ln r \right]_{1+\cos \theta}^{3 \cos \theta} \sec \theta \, d\theta$$

Q. 10. Find the volume V of the solid, using spherical coordinates, bounded by the sphere $(x-1)^2 + (y-2)^2 + (z-3)^2 = 1$.

Ans. Consult text book

Q. 11. Find the volume of the solid D where D is the intersection of the solid sphere $x^2 + y^2 + z^2 \leq 9$ and the solid cylinder $x^2 + y^2 \leq 1$.

Ans. Consult text book

Q. 12. Evaluate $\iint_D e^{-(4x^2+5y^2)} dA$, using change of variable, where D is the elliptical disk $\frac{x^2}{5} + \frac{y^2}{4} \leq 1$.

Ans. Consult text book

SECTION-III

Q. 13. Let $F(x, y) = e^x i + xe^y j$. Verify that the vector field F is conservative on the entire xy -plane. Use a potential function of F to find the work done by the field F on a particle that moves in anticlockwise direction from $(1, 0)$ to $(-1, 0)$ along the semi-circular path

$$y = \sqrt{1-x^2}; -1 \leq x \leq 1$$

Ans. (i)

$$F(x, y) = e^x i + xe^y j$$

Let

$$u(x, y) = e^x$$

$$v(x, y) = xe^y$$

...(1)

Eqn (1) becomes:

$$F(x, y) = u(x, y) i + v(x, y) j$$

...(2)

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(e^x) = e^x$$

...(3)

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x}(xe^y) = (1)e^y = e^y$$

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = e^y$$

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

$\Rightarrow \vec{F}$ is conservative

(ii) In the x, y plane we have:

$$\vec{r} = xi + yj$$

$$d\vec{r} = dx i + dy j$$

$$\begin{aligned} \vec{F} \cdot d\vec{r} &= (e^x i + xe^y j) \cdot (dx i + dy j) \\ &= (e^x)(dx) + (xe^y)(dy) \end{aligned}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (e^x dx + xe^y dy)$$

Ans.

Along the curve C ,

$$y = \sqrt{1-x^2}; -1 \leq x \leq 1, \text{ so}$$

$$dy = \frac{-2x}{2\sqrt{1-x^2}} dx$$

$$dx = \frac{-x dx}{\sqrt{1-x^2}}$$

The required both done (w) will be

$$W = \int_C \vec{F} \cdot d\vec{r}$$

$$= \int_{x=1}^{-1} [e^x dx + xe^y dy]$$

$$= \int_{-1}^1 \left[e^x dx + xe^y \left(\frac{-x dx}{\sqrt{1-x^2}} \right) \right]$$

$$= \int_{-1}^1 e^x \left[\frac{1-x^2}{\sqrt{1-x^2}} \right] dx$$

$$= \int_{-1}^1 e^x \sqrt{1-x^2} \left(\frac{1-x^2}{\sqrt{1-x^2}} \right) dx$$

Q. 14. State Green's theorem for simply connected regions. Verify Green's theorem for the field $F(x, y) = (x-y)i + xj$ and the region R bounded by the unit circle $C: r(t) = \cos t i + \sin t j; 0 \leq t \leq 2\pi$.

Aps. Green's Theorem for simply connected regions. Let D be a simply connected region that is bounded by the positively oriented piecewise smooth Jordan curve C . Then if the vector field $\vec{F}(x, y) = M(x, y)i + N(x, y)j$ is continuously differentiable on D , we have:

$$\oint_C (Mdx + Ndy) = \iint_D \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

Q. 15. Let E be the solid bounded by the xy -plane and the paraboloid $z = 4 - x^2 - y^2$. Let S be the boundary of E (that is, piece of the paraboloid and a disk in the xy -plane), oriented with the outward-pointing normal $\vec{n}$.

$$\text{If } F(x, y, z) = (xz \sin(yz) + x^3)i + \cos(yz)j + (3zy^2 - e^{x^2+y^2})k,$$

Find $\iiint_S \vec{F} \cdot \vec{n} \, ds$ over the surface S using divergence theorem.

Ans. Do yourself.

Q.16. Evaluate $\iint_S (x+y+z) \, dS$ over the surface S defined parametrically by $R(u, v) = (2u+v)\mathbf{i} + (u-2v)\mathbf{j} + (u+3v)\mathbf{k}$ for $0 \leq u \leq 1$; $0 \leq v \leq 2$.

Ans. $R(u, v) = (2u+v)\mathbf{i} + (u-2v)\mathbf{j} + (u+3v)\mathbf{k}$... (1)

$\iint_S (x+y+z) \, dS = \iint_D (R \cdot (R_u \times R_v)) \, du \, dv$... (2)

First we need to find the component parts for the surface integral on RHS. Using $R = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, we see that

$$x = 2u + v$$

$$y = u - 2v$$

$$z = u + 3v$$

$$f(x, y, z) = x + y + z$$

$$f(R) = f(2u+v, u-2v, u+3v)$$

$$= (2u+v) + (u-2v) + (u+3v)$$

$$= 4u + 2v$$

$$R_u = \frac{\partial R}{\partial u} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$$

$$R_v = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$$

$$R_u \times R_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix}$$

$$= (6+2)\mathbf{i} + (6-1)\mathbf{j} + (-4-1)\mathbf{k}$$

$$= 8\mathbf{i} + 5\mathbf{j} - 5\mathbf{k}$$

Q.17. State and prove the Stokes' theorem.

Q.17. State and prove the Stokes' theorem.

Ans. Stokes' theorem

Statement:

Let S be a piecewise smooth open surface bounded by a piecewise smooth simple closed curve C . Let $\vec{F}(x, y, z)$ be a continuous vector function which has continuous first partial derivatives in a region of space which contains in its interior then,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, dS = \iint_S (\text{curl } \vec{F}) \cdot d\vec{S}$$

where C is traversed in the positive direction. The direction of C is called positive if an observer walking on the boundary of S in this direction with his head pointing in the direction of outward normal $\vec{n}$ to S has the surface on left.

Proof:

Let

$$\vec{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$$

Let outward drawn normal vector $\vec{n}$ to S makes an angles α, β, γ with x, y and z axis respectively.

Then,

$$\vec{n} = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}$$

Let S be a surface which is such that its projection on the xy, yz and xz planes are regions bounded by simple closed curve. Surface S can be represented simultaneously in the forms:

Let H be the orthogonal projection of S on the xy plane. Let C be the boundary of H as shown in figure. Let $\vec{r}$ be the position vector of a point (x, y, z) in S . Then we may write the line integral as

$$x = h(z, y), \quad y = g(z, x), \quad z = f(x, y)$$

Where f, g, h are continuous functions and have continuous first partial derivative

$$\text{Consider the integral } \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, dS$$

$$\nabla \times (\vec{F}_1) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & 0 & 0 \end{vmatrix}$$

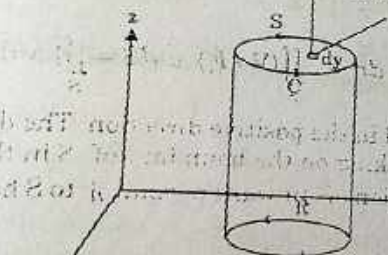
$$= \frac{\partial F_1}{\partial z} j - \frac{\partial F_1}{\partial y} k$$

$$[\nabla \times \vec{F}_1] \cdot \hat{n} = \left(\frac{\partial F_1}{\partial z} j - \frac{\partial F_1}{\partial y} k \right) \cdot \hat{n}$$

$$= \frac{\partial F_1}{\partial z} (j \cdot \hat{n}) - \frac{\partial F_1}{\partial y} (k \cdot \hat{n})$$

$$= \frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma$$

$$\iiint_S [\nabla \times (\vec{F}_1)] \cdot \hat{n} dS = \iint_S \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dS$$



We have prove that

$$\iiint_R \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dS = \oint_T F_1 dy$$

Let R be the orthogonal projection of S on the xy-plane and let T be its boundary while is oriented as shown in figure. using the representation $z = f(x, y)$ of S, we may write the line integral once as a line integral on T. Thus

$$\oint_C F_1(x, y, z) dx = \oint_T F_1(x, y, f(x, y)) dx$$

$$= \oint_T [F_1(x, y, f(x, y)) dx + 0 dy]$$

$$F = \iint_R \frac{\partial F_1}{\partial y} dx dy \quad (1)$$

Using Green's theorem in plane for the region R.

$$\text{But, } \frac{\partial F_1}{\partial y} [x, y, f(x, y)] = \frac{\partial F_1}{\partial y} (x, y, z) + \frac{\partial F_1}{\partial z} (x, y, z) \frac{\partial f}{\partial y}$$

$$\oint_C F_1(x, y, z) dx = - \iint_R \left(\frac{\partial F_1}{\partial y} + \frac{\partial F_1}{\partial z} \frac{\partial f}{\partial y} \right) dx dy \quad (2)$$

Now the equation $z = f(x, y)$ of the surface S can be written as:

$$\phi(x, y, z) = z - f(x, y) = 0$$

We have,

$$\text{grad } \phi = - \frac{\partial f}{\partial x} i - \frac{\partial f}{\partial y} j + k$$

Let

$$|\text{grad } \phi| = \alpha$$

Since grad ϕ is normal to S. Therefore, we get.

$$\hat{n} = \pm \frac{\text{grad } \phi}{\alpha}$$

But the component of both $\hat{n}$ and grad ϕ in positive direction of z-axis are positive

Therefore,

$$\hat{n} = \pm \frac{\text{grad } \phi}{\alpha}$$

$$\cos \alpha i + \cos \beta j + \cos \gamma k = - \frac{1}{\alpha} \frac{\partial f}{\partial x} i - \frac{1}{\alpha} \frac{\partial f}{\partial y} j + \frac{1}{\alpha} k$$

$$\cos \alpha = - \frac{1}{\alpha} \frac{\partial f}{\partial x}$$

$$\cos \beta = - \frac{1}{\alpha} \frac{\partial f}{\partial y}$$

$$\cos \gamma = \frac{1}{\alpha}$$

Now,

$$dS = \frac{dx dy}{\cos \gamma} = \alpha dx dy$$

$$\iint_S \left(\frac{\partial F_1}{\partial z} \cos \beta - \frac{\partial F_1}{\partial y} \cos \gamma \right) dx dy$$

$$\frac{\partial}{\partial x} \left[\frac{\partial F_1}{\partial x} \left(-\frac{1}{\alpha} \frac{\partial b}{\partial y} \right) - \frac{\partial F_2}{\partial y} \frac{1}{\alpha} \right] \alpha \, dx \, dy$$

$$= \iint_R \left[\frac{\partial F_1}{\partial x} \frac{\partial F_2}{\partial y} - \frac{\partial F_2}{\partial x} \frac{\partial F_1}{\partial y} \right] \alpha \, dx \, dy$$

From (1) + (2)

$$\oint_C F_1 \, dx + F_2 \, dy = \iint_S \left[\frac{\partial F_1}{\partial x} \cos \beta - \frac{\partial F_2}{\partial y} \cos \gamma \right] \, dS$$

$$= \iint_S (\nabla \times F_1) \cdot \hat{n} \, dS$$

Similarly, by projection on the other co-ordinates planes, we get:

$$\oint_C F_2 \, dy = \iint_S (\nabla \times F_2) \cdot \hat{n} \, dS \quad \dots (4)$$

and

$$\oint_C F_3 \, dz = \iint_S (\nabla \times F_3) \cdot \hat{n} \, dS \quad \dots (5)$$

On adding (3), (4) and (5):

$$\oint_C (F_1 \, dx + F_2 \, dy + F_3 \, dz) = \iint_S [\nabla \times (F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k})] \cdot \hat{n} \, dS$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$$

Which proves stoker's theorem.

Q. 15. State fundamental theorem of line integrals. If a semi-circular wire has the equation $y = \sqrt{25 - x^2}$ its mass density function is $\psi(x, y) = 15 - y$. Find the mass of the wire.

Ans. Fundamental theorem of line integrals:

Let C be a piece wise smooth curve that is parameterized by the vector function $R(t)$ for $a \leq t \leq b$ and let $\vec{F}$ be a vector field that is continuous on C . If f is a scalar function such that $\vec{F} = \nabla f$.

Then

$$\oint_C \vec{F} \cdot d\vec{R} = f(Q) - f(P)$$

Where,

$$Q = R(b)$$

$$P = R(a)$$

are end point of C .

ooo

Sr. No. of Q. Paper	: 32351301	NOV/DEC, 2017
Paper Code	: THEORY OF REAL FUNCTIONS	
Name of the Paper	: B.Sc. (Hons.) Mathematics II Year	
Name of Course	: III	
Semester	: 3 Hours	
Duration	Maximum Marks : 75	

All questions are compulsory. Attempt any three parts from each question.

Q. 1. (u) use the $\epsilon - \delta$ definition of the limit to find $\lim_{x \rightarrow 2} f(x)$ where

$$f(x) = \frac{1}{1-x}$$

Ans. We show that $\lim_{x \rightarrow a} f(x) = l$, if given $\epsilon > 0$, there exists a positive real number δ (depending upon ϵ) such that

$$|f(x) - l| < \epsilon \text{ for } 0 < |x - a| < \delta$$

Here,

$$f(x) = \frac{1}{1-x}$$

$$l = -1$$

Now, we choose a real numbers δ such that the equation (1) hold true, i.e.

$$|f(x) - l| < \epsilon$$

$$\left| \frac{1}{1-x} - (-1) \right| < \epsilon$$

$$\left| \frac{1}{1-x} + 1 \right| < \epsilon$$

$$\left| \frac{2-x}{1-x} \right| < \epsilon$$

$$\left| \frac{2-x}{1-x} \right| < \epsilon$$

$$\left| \frac{x-2}{x-1} \right| < \epsilon$$

Since, if x is near 2, then $x - 2$ is near zero. If x is near 2, and away from 1,

then $\frac{1}{|x-1|}$ is not very large.

If $\delta \leq \frac{1}{2}$ and $0 < |x - 2| < \delta$, then

$$2 - \delta < x < 2 + \delta, x = 2$$

$$2 - \frac{3}{2} < x < 2 + \frac{1}{2}$$

$$\frac{3}{2} < x < \frac{5}{2}$$

$$\frac{3}{2} - 1 < x - 1 < \frac{5}{2} - 1$$

$$\frac{1}{2} < x - 1 < \frac{3}{2}$$

$$|x - 1| > \frac{1}{2}$$

$$\frac{1}{|x - 1|} < 2$$

Using (4) Equation (3) gives:

$$|x - 2| < \frac{\epsilon}{2}$$

Let we choose $\delta = \frac{\epsilon}{2}$ such that

$$\frac{|x - 2|}{|x - 1|} < \epsilon, \text{ where } |x - 2| < \delta$$

$$|f(x) - (-1)| < \epsilon, \text{ where } 0 < |x - 2| < \delta$$

$$\Rightarrow \lim_{x \rightarrow 2} f(x) = -1$$

(b) State and prove sequential Criterion for limits

Ans. Sequential criterion for Limits:

Statement: Let $f: A \rightarrow R$ and C be a cluster point of A , then the following are equivalent:

$$(i) \lim_{x \rightarrow c} f = L$$

(ii) For every sequence (x_n) in A that converges to c such that $x_n \neq c$ for all $n \in N$, the sequence $(f(x_n))$ converges to L .

Proof:

(i) $\Rightarrow$ (ii)

Assume f has limit L at c and suppose (x_n) is a sequence in A with $\lim (x_n) = c$ and $x_n \neq c$ for all n . We must prove that the sequence $(f(x_n))$ converges to L . Let $\epsilon > 0$ be given, then by definition of limit, there exists $\delta > 0$ such that if $x \in A$

satisfies $0 < |x - c| < \delta$, then $f(x)$ satisfies $|f(x) - L| < \epsilon$. We now apply definition of convergent sequence for the given δ to obtain a natural number $K(\delta)$ such that if $n > K(\delta)$, then $|x_n - c| < \delta$. But for each such x_n we have $|f(x_n) - L| < \epsilon$.

Thus, if $n > K(\delta)$, then $|f(x_n) - L| < \epsilon$.

Therefore, the sequence $(f(x_n))$ converges to L .

(ii) $\Rightarrow$ (i):

If (i) is not true, then there exists an ϵ_0 neighborhood $V_{\epsilon_0}(L)$ such that no matter what δ -neighborhood of c , we pick, there thus will be at least one number x_n in $A \cap V_\delta(c)$ with $x_n \neq c$ such that $f(x_n) \notin V_{\epsilon_0}(L)$. Hence for every $n \in N$, the

$\left(\frac{1}{n}\right)$ -neighborhood of c contains a number x_n such that

$$0 < |x_n - c| < \frac{1}{n} \text{ and } x_n \in A \text{ but such that}$$

$$|f(x_n) - L| \geq \epsilon_0 \text{ for all } n \in N$$

We conclude that the sequence (x_n) in $A \setminus \{c\}$ converges to c , but the sequence $(f(x_n))$ does not converge to L . Therefore, we have shown that if (i) is not true, then (ii) is not true. We conclude that (ii) implies (i).

(c) State Squeeze Theorem. For $n \in N, n \geq 3$ derive the inequality.

$-x^2 \leq x^n \leq x^2$ for $-1 < x < 1$. Hence prove that $\lim_{x \rightarrow 0} x^n = 0$ for $n \geq 3$.

assuming that $\lim_{x \rightarrow 0} x^2 = 0$.

(5)

Ans. Squeeze Theorem:

Let $A \subseteq R$, let $f, g, h: A \rightarrow R$ and let $c \in R$ be a cluster point of A . If $f(x) \leq g(x) \leq h(x)$ for all $x \in A, x \neq c$ and if $\lim_{x \rightarrow c} f = L = \lim_{x \rightarrow c} h$, then $\lim_{x \rightarrow c} g = L$.

To prove $\lim_{x \rightarrow 0} x^n = 0$:

We prove the inequality $-x^2 \leq x^n \leq x^2$ for $0 < |x| < 1, n \geq 3$ with the help of mathematical induction.

Let $P(n): -x^2 \leq x^n \leq x^2$ for $0 < |x| < 1, n \geq 3$ and $n \in N$

For $n = 3; -1 < x < 1, x^2 \geq 0 \forall x \in R$

$$\Rightarrow -1 x^2 \leq x, x^2 \leq 1 - x^2$$

$$\Rightarrow -x^2 \leq x^3 \leq x^2$$

$\Rightarrow P(n)$ is true for $n = 3$.

Assume that $P(n)$ is true for $n = m$.

$$\text{i.e., } -x^2 \leq x^m \leq x^2$$

As $0 \leq x \leq 1$, we have

$$-x^2 \leq x \leq x^2, x \leq x^3 \leq x$$

$$= -x^2 \leq x^{m+1} \leq x^2$$

As $0 \leq x \leq 1$, we have

$$0 \leq x^2 \leq x^2$$

$$\Rightarrow x^2 \leq x^2$$

$$\Rightarrow -x^2 \leq -x^2$$

Combining these three, we get:

$$-x^2 \leq x^{m+1} \leq x^2$$

Hence, we can say that $P(n)$ is true for $n = m + 1$, where $0 < x < 1$.

Similarly, we can do for $-1 < x < 1$.

By method of mathematical induction,

We have:

$$-x^2 \leq x^n \leq x^2 \quad \forall n \geq 3$$

where

$$-1 < x < 1$$

As $\lim_{x \rightarrow 0} x^2 = 0$. Then, by squeeze theorem,

We have:

$$\lim_{x \rightarrow 0} x^{m+1} = \lim_{x \rightarrow 0} x^2$$

$$= 0$$

Ans.

(d) Let f, g be defined on $A \subseteq \mathbb{R}$ to $\mathbb{R}$, and let c be a cluster point A .

Suppose that f is bounded on a neighbourhood of c and that $\lim_{x \rightarrow c} g = 0$.

Prove that $\lim_{x \rightarrow c} fg = 0$.

(5)

Ans. $f, g: A \rightarrow \mathbb{R}$ where $A \subseteq \mathbb{R}$

Since c is a cluster point of A

$$\Rightarrow [N(c, \delta) \cap A] - \{c\} \neq \emptyset$$

Also, f is bounded on a n.b.d of c

$$\Rightarrow f(x) \leq k \text{ for } x \in N(c, \delta)$$

Also

$$\lim_{x \rightarrow c} g = 0$$

$$\lim_{x \rightarrow c} fg = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x)$$

$$\leq (K) \lim_{x \rightarrow c} g(x)$$

$$= K(0)$$

$$= 0$$

$$\Rightarrow \lim_{x \rightarrow c} fg = 0 \text{ for } x \in N(c, \delta)$$

Ans.

Q. 2. (a) Let $c \in \mathbb{R}$ and let f be defined for $x \in (c, \infty)$ and $f(x) > 0$ for $x \in (c, \infty)$. Show that $\lim_{x \rightarrow c} f = \infty$ if and only if $\lim_{x \rightarrow c} 1/f = 0$

Ans. (i) $c \in \mathbb{R}$

(ii) f be defined for $x \in (c, \infty)$

(iii) $f(x) > 0 \quad \forall x \in (c, \infty)$

$$\text{Let } \lim_{x \rightarrow c} f = \infty$$

$$\Rightarrow \forall \epsilon > 0 \exists \delta > 0 \text{ s.t. } c < x < c + \delta$$

$$f(x) > \frac{1}{\epsilon}$$

$\Rightarrow$

$$0 < \frac{1}{f(x)} < \epsilon$$

$\Rightarrow$

$$\lim_{x \rightarrow c} \frac{1}{f(x)} < \epsilon$$

$\Rightarrow$

$$\lim_{x \rightarrow c} \frac{1}{f} < \epsilon$$

$\Rightarrow$

(b) Prove that

$$(i) \lim_{x \rightarrow 0} \frac{1}{\sqrt{x}} = +\infty; x \neq 0$$

$$(ii) \lim_{x \rightarrow 0} -e^x = 0, x \neq 0$$

$$\text{Ans. (i) } \lim_{x \rightarrow 0} \frac{1}{\sqrt{x}} : \text{ Let } f(x) = \frac{1}{\sqrt{x}}$$

$$\text{r.h.l.} = \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{0+h}}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{|h|}}$$

$$= \lim_{x \rightarrow 0} \frac{1}{0}$$

$$= \infty$$

$$\text{l.h.l.} = \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{x \rightarrow 0^-} \frac{1}{\sqrt{|x|}}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{|0-h|}}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{|-h|}}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{h}} = \frac{1}{0} = \infty$$

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = \infty$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) = \infty$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{1}{\sqrt{|x|}} = \infty$$

$$(ii) \lim_{x \rightarrow 0} e^x = 0:$$

$$\text{Let } f(x) = \frac{1}{e^x}$$

$$= \lim_{x \rightarrow 0} e^{1/x}$$

$$= \lim_{h \rightarrow 0} e^{\frac{1}{0-h}}$$

$$= \lim_{h \rightarrow 0} e^{-\frac{1}{h}}$$

$$= e^{-\infty}$$

$$= \frac{1}{e^{\infty}}$$

$$= \frac{1}{\infty}$$

$$= 0$$

(c) Let $A = \mathbb{R}$ and Let f be Dirichlet's function defined by.

$$g(x) = 1; \text{ for } x \text{ rational}$$

$$= -1; \text{ for } x \text{ irrational}$$

Show that f is discontinuous at any point of $\mathbb{R}$.

Ans. The $g(x)$ is defined as:

$$g(x) = 1; \text{ for } x \text{ rational}$$

$$= -1; \text{ for } x \text{ irrational}$$

Case (1):

Let a be any rational number, so that

$$g(a) = 1$$

Then, any $n \in \mathbb{N}$ $\left[a - \frac{1}{n}, a + \frac{1}{n} \right]$ of a contains an irrational number a_n for each $n \in \mathbb{N}$

$$\text{i.e., } a_n \in \left[a - \frac{1}{n}, a + \frac{1}{n} \right] \forall n \in \mathbb{N}$$

$$\Rightarrow a - \frac{1}{n} < a_n < a + \frac{1}{n} \forall n \in \mathbb{N}$$

$$\Rightarrow |a_n - a| < \frac{1}{n} \forall n \in \mathbb{N}$$

$$\Rightarrow |a_n - a| \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow a_n - a \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow \langle a_n \rangle \text{ converges to } a$$

Now, $g(a_n) = -1$ for all n ($\because a_n$ is irrational and $g(a) = 1$).

$$\lim_{n \rightarrow \infty} g(a_n) = -1 \neq g(a)$$

$\Rightarrow \langle g(a_n) \rangle$ does not converge to $g(a)$ where $a_n \rightarrow a$.
Thus, g is discontinuous at all rational points a .

Case (2):

Let b any irrational point, so that

$$g(b) = -1$$

As argued above we can choose a rational number b_n such that

$$\begin{aligned} |b_n - b| &< \frac{1}{n} \quad \forall n \in \mathbb{N} \\ \Rightarrow b_n &\rightarrow b \text{ as } n \rightarrow \infty \\ \Rightarrow \langle b_n \rangle &\text{converges to } b \\ \text{Now, } g(b_n) &= 1 \text{ for all } n \text{ (} b_n \text{ is rational) and } g(b) = -1 \\ \lim_{n \rightarrow \infty} g(b_n) &= 1 \neq g(b) \end{aligned}$$

$\Rightarrow \langle g(b_n) \rangle$ does not converge to $g(b)$ where $b_n \rightarrow b$.
Thus, the function g is discontinuous at all rational points b .
From (1) and (2), we can say that the function g is discontinuous at all points of $\mathbb{R}$.

(d) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous at c and let $f(c) > 0$. Show that there exists a neighbourhood $V_\delta(c)$ of c such that if $x \in V_\delta(c)$ then $f(x) > 0$. (5)

Ans. Consult text book

Q. 3. (a) Determine the points of continuity of the function $f(x) = x - [x]$, $x \in \mathbb{R}$, where $[x]$ denotes the greatest integer $n \in \mathbb{Z}$ such that $n \leq x$. (5)

Ans. The given function $f(x)$ is:

$$f(x) = x - [x] \quad \dots (1)$$

By definition of $[x]$, we have:

$$\begin{aligned} [x] &= 0 & \text{for } 0 \leq x < 1 \\ [x] &= 1 & \text{for } 1 \leq x < 2 \\ [x] &= 2 & \text{for } 2 \leq x < 3 \\ [x] &= -1 & \text{for } -1 \leq x < 0 \\ [x] &= -2 & \text{for } -2 \leq x < -1 \end{aligned} \quad \text{etc.}$$

Hence the function $f(x)$ is defined as:

$$\begin{aligned} f(x) &= x - [x] = x - 0 = x & ; 0 \leq x < 1 \\ f(x) &= x - [x] = x - 1 & ; 1 \leq x < 2 \\ f(x) &= x - [x] = x - 2 & ; 2 \leq x < 3 \\ f(x) &= x - [x] = x - (-1) = x + 1 & ; -1 \leq x < 0 \\ f(x) &= x - [x] = x - (-2) = x + 2 & ; -2 \leq x < -1 \end{aligned} \quad \text{etc.}$$

At $x = 0$

$$\begin{aligned} \lim_{x \rightarrow 0^+} g(x) &= \lim_{x \rightarrow 0^+} (x) \\ &= \lim_{h \rightarrow 0} (0 + h) \\ &= \lim_{h \rightarrow 0} h \\ &= 0 \end{aligned}$$

$$\begin{aligned} \lim_{h \rightarrow 0^+} f(x) &= \lim_{h \rightarrow 0} (x + 1) \\ &= \lim_{h \rightarrow 0} (0 + h) + 1 \\ &= \lim_{h \rightarrow 0} -h + 1 \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

$$\lim_{h \rightarrow 0^+} f(x) \neq \lim_{h \rightarrow 0^+} f(x)$$

$\Rightarrow f(x)$ is discontinuous at $x = 0$.

At $x = 1$:

$$\begin{aligned} \lim_{h \rightarrow 1^+} f(x) &= \lim_{h \rightarrow 1^+} (x - 1) \\ &= \lim_{h \rightarrow 0} (1 + h) - 1 \\ &= \lim_{h \rightarrow 0} h \\ &= 0 \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (x) \\ &= \lim_{h \rightarrow 0} (1 - h) \\ &= 1 - 0 \\ &= 1 \end{aligned}$$

$$\Rightarrow \lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x)$$

$\Rightarrow f(x)$ is discontinuous at $x = 1$

At $x = -1$:

$$\begin{aligned} \lim_{x \rightarrow -1^+} f(x) &= \lim_{x \rightarrow -1^+} (x + 1) \\ &= \lim_{h \rightarrow 0} (-1 + h) + 1 \\ &= \lim_{h \rightarrow 0} h \\ &= 0 \end{aligned}$$

$$\begin{aligned} \lim_{h \rightarrow -1^-} f(x) &= \lim_{x \rightarrow -1^-} (x + 2) \\ &= \lim_{h \rightarrow 0} (1 - h) + 2 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} (-1 - h) + 2 \\
 &= 1 - 0 \\
 &= 1
 \end{aligned}$$

$$\Rightarrow \lim_{x \rightarrow -1^+} f(x) \neq \lim_{x \rightarrow -1^-} f(x)$$

$\Rightarrow f(x)$ is discontinuous at $x = -1$

Similarly, we can prove that $f(x)$ is discontinuous at $x = \pm 2, \pm 3, \dots$

It may be noted that $f'(x) = 1$ for all $x \in \mathbb{R}$ except at the points $0, \pm 1, \pm 2, \pm 3, \dots$

3. (b) Let $A, B \subseteq \mathbb{R}$, let $f: A \rightarrow \mathbb{R}$ be continuous on A , and let $g: B \rightarrow \mathbb{R}$ be continuous on B . If $f(A) \subseteq B$, show that the composite function $g \circ f: A \rightarrow \mathbb{R}$ is continuous on A .

Ans. let W be an ϵ neighbourhood of $g(b)$.

Since g is continuous at b , there is a δ -neighbourhood V of $b = f(c)$ such that if $y \in V$, then $g(y) \in W$. Since f is continuous at c there is a Y neighbourhood $U \cap U$, then $f(x) \in B \cap V$, so that $g \circ f(x) = g(f(x)) \in W$. But, since W is an arbitrary neighbourhood of $g(b)$, this implies that $g \circ f$ is continuous at c .

(c) Let f be a continuous real valued function defined on $[a, b]$. Show that f is a bounded function.

Ans. let if possible, f be not bounded above in $[a, b]$. Then, for each positive integer n , we can find $x_n \in [a, b]$ such that:

$$f(x_n) > n \quad \forall n \in \mathbb{N} \quad \dots(1)$$

Since, $x_n \in [a, b]$ for each $n \in \mathbb{N}$, $a \leq x_n \leq b \quad \forall n \in \mathbb{N}$

Thus, $\langle x_n \rangle$ is a bounded sequence and so it must have a limit point say p as per Bolzano Weierstrass Theorem.

obviously p is a limit point of $[a, b]$.

Now $[a, b]$ being a closed interval is a closed set and so $p \in [a, b]$

Since p is a limit point of the sequence, $\langle x_n \rangle$ therefore, there exists a subsequence $\langle x_{n_i} \rangle$ of $\langle x_n \rangle$ such that $x_{n_i} \rightarrow p$... (2)

From (1), $f(x_{n_i}) > n_i$ for all i

$\Rightarrow \langle f(x_{n_i}) \rangle$ diverges to ∞

$\Rightarrow f(x_{n_i})$ does not converge to $f(p)$... (3)

From (2) and (3) it follows that f is not continuous at the point $p \in [a, b]$.

This is contrary to the hypothesis that f is continuous at every point of $[a, b]$.

Hence f is bounded above on $[a, b]$.

Now, we show that f is bounded below on $[a, b]$.

Since, f is continuous in $[a, b]$ - f is also continuous in $[a, b]$ and so $-f$ is

bounded above (as proved above). Consequently there exists some $k \in \mathbb{R}$ such that

$$\begin{aligned}
 &\Rightarrow -f(x) \leq k \quad \forall x \in [a, b] \\
 &\Rightarrow f(x) \geq -k \\
 &\Rightarrow -k \leq f(x) \quad \forall x \in [a, b]
 \end{aligned}$$

$\Rightarrow f$ is bounded below in $[a, b]$

Hence, f is bounded in $[a, b]$.

(d) Prove that a polynomial of odd degree has at least one real root.

Ans. Let $f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n$ ($a_0 > 0$) be a given polynomial of odd degree. (5)

Eq. (1) can be solved as:

$$f(x) = x^n \left[a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \dots + \frac{a_n}{x^n} \right]$$

As $x \rightarrow \infty$ or $-\infty$, the bracket $\rightarrow a_0$

$f(x)$ and $a_0 x^n$ are of like signs.

$$f(0) = a_n$$

If a_n is +ve and $x \rightarrow -\infty$, then

$\Rightarrow f(x) \rightarrow$ a negative number (n being odd)

$\Rightarrow f(0) f(-\infty) < 0$

$\Rightarrow f$ has a real root in $(-\infty, 0)$.

If a_n is -ve and $x \rightarrow +\infty$, then

$\Rightarrow f(0) f(+\infty) < 0$

$\Rightarrow f$ has a real root in $(0, \infty)$

Hence, in any case, f has at least one real root.

It proves that a polynomial of odd degree has at least one real root.

Q. 4. (a) Define uniform continuity of a function on a set $A \subseteq \mathbb{R}$. Show that every uniformly continuous function on A is continuous on A . Is the converse true? Justify your answer. (5)

Ans. Uniform Continuity of a Function: A function f defined on an interval I is said to be uniformly continuous in the interval I if for each $\epsilon > 0$, there exists some $\delta > 0$ such that

$$|f(x_2) - f(x_1)| < \epsilon$$

when $|x_2 - x_1| < \delta$ and for all $x_1, x_2 \in I$.

Solution of given Problem:

(i) let a function f be uniformly continuous on an interval I . Then, for any $\epsilon > 0$, there exists some $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon$$

when $|x - y| < \delta \quad \forall x, y \in I$

Let c be any point of I . Taking $y = x$ in Eq. (1),
We have

$$|f(x) - f(c)| < \epsilon \quad |x - c| < \delta$$

Thus f is continuous at the point c .
Since $c \in I$ is arbitrary, it follows that f is continuous at each point of I .
Hence, f is continuous on the interval I .

(ii) The converse of the theorem is not true. A function which is continuous on an interval may not be uniformly continuous on that interval.

Since $f(x) = \frac{1}{x}$ is continuous on $[0, 1]$.

Let $\epsilon = \frac{1}{2}$ and δ be any positive number. We can choose a positive integer $n, n > \frac{1}{\delta}$... (1)

$$n > \frac{1}{\delta}$$

Let
Then

$$x_1 = \frac{1}{n}, x_2 = \frac{1}{n+1} \in [0, 1].$$

$$\begin{aligned} |f(x_2) - f(x_1)| &= \left| \frac{1}{x_2} - \frac{1}{x_1} \right| \\ &= |(n+1) - (n)| \\ &= 1 > \epsilon \end{aligned}$$

and

$$\begin{aligned} |x_2 - x_1| &= \left| \frac{1}{n+1} - \frac{1}{n} \right| \\ &= \frac{1}{n(n+1)} < \frac{1}{n} < \delta, \text{ by (1)} \end{aligned}$$

$$|f(x_2) - f(x_1)| > \epsilon \text{ and, when } |x_2 - x_1| < \delta$$

$$f(x) = \frac{1}{x} \text{ is not uniformly continuous on } [0, 1].$$

(b) Show that the function $\sqrt{x}$ is uniformly continuous on $[0, \infty)$.
(5)

Ans. Let

$$f(x) = \sqrt{x}$$

Where, $x \in [0, \infty)$

Let x_1 and $x_2 \in [0, \infty)$ and $x_1 > x_2$

$$f(x_1) - f(x_2) = \sqrt{x_1} - \sqrt{x_2}$$

$$= \frac{(\sqrt{x_1} - \sqrt{x_2})(\sqrt{x_1} + \sqrt{x_2})}{(\sqrt{x_1} + \sqrt{x_2})}$$

$$(\sqrt{x_1} + \sqrt{x_2})$$

$$\leq \frac{x_1 - x_2}{\sqrt{x_1}}$$

$$\leq \frac{x_1 - x_2}{\sqrt{x_1 - x_2}}$$

$$= \sqrt{x_1 - x_2}$$

$$|f(x_1) - f(x_2)| \leq \sqrt{x_1 - x_2} < \epsilon$$

$$\Rightarrow |f(x_1) - f(x_2)| \leq \epsilon \text{ if } |x_1 - x_2| < \delta = \epsilon^2$$

$$\Rightarrow |f(x_1) - f(x_2)| < \epsilon \text{ if } |x_1 - x_2| < \delta$$

$\Rightarrow f(x)$ is uniformly continuous on $[0, \infty)$.

(c) Let I, J be intervals in $\mathbb{R}$, let $g: I \rightarrow \mathbb{R}$ and $f: J \rightarrow \mathbb{R}$ be functions such that $f(J) \subseteq I$ and let $c \in J$. If f is differentiable at c and if g is differentiable at $f(c)$, show that the composite function $g \circ f$ is differentiable at c and $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$.

Ans.

$$\begin{aligned} g: I &\rightarrow \mathbb{R} \\ f: J &\rightarrow \mathbb{R} \\ g(J) &\subseteq I \\ c &\in J \end{aligned}$$

Since, $f'(c)$ exists

Caratheodory's Theorem implies that there exists a function ϕ on J such that ϕ is continuous at c and $f(x) - f(c) = \phi(x)(x - c)$ for $x \in J$ and where $\phi(c) = f'(c)$.

Also, since $g'(f(c))$ exists, there is a function Ψ defined on I such that Ψ is continuous at $d := f(c)$ and $g(y) - g(d) = \Psi(y)(y - d)$ for $y \in I$, where $\Psi(d) = g'(f(c))$. Substitution of $y = f(x)$ and $d = f(c)$, then produces, $g(f(x)) - g(f(c))$

$$= \Psi(f(x))(f(x) - f(c))$$

$$= [(\Psi \circ f)(x)] \cdot \phi(x) \cdot (x - c)$$

for all $x \in J$ such that $f(x) \in I$. Since, the function $(\Psi \circ f) \cdot \phi$ is continuous at c and its value at c is $g'(f(c)) \cdot f'(c)$, Caratheodory's theorem gives

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$$

(d) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Show that f is differentiable at $x = 0$ and find $f'(0)$.

Ans. The given function $f(x)$ is defined as :

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

At $x = 0$

$$f(0) = 0$$

$$Rf'(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0^+} \frac{x^2 \sin\left(\frac{1}{x}\right) - 0}{x - 0}$$

$$= \lim_{x \rightarrow 0^+} x \sin\left(\frac{1}{x}\right)$$

$$= \lim_{h \rightarrow 0} (0 + h) \sin\left(\frac{1}{0 + h}\right)$$

$$= \lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right)$$

$$= 0 \times \sin(\infty)$$

$$= \times (\text{a finite value lying between } -1 \text{ to } 1)$$

$$= 0$$

$$Lf'(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0^-} \frac{x^2 \sin\left(\frac{1}{x}\right) - 0}{x - 0}$$

$$= \lim_{x \rightarrow 0^-} x \sin\left(\frac{1}{x}\right)$$

$$= \lim_{h \rightarrow 0} (0 - h) \sin\left(\frac{1}{0 - h}\right)$$

$$= \lim_{h \rightarrow 0} (-h) \sin\left(-\frac{1}{h}\right)$$

$$= \lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right)$$

$$= 0 \times \sin(\infty)$$

$$= 0$$

$$Rf'(0) = Lf'(0) = 0$$

$\Rightarrow f(x)$ is differentiable at $x = 0$ and $f'(0) = 0$

Q. 5. (a) Let f be continuous on $[a, b]$ and differentiable on (a, b) .
Ans. (1) f is increasing on $[a, b]$ if and only if $f'(x) \geq 0 \forall x \in (a, b)$. (5)

(2) f is differentiable on $[a, b]$

Let x_1 and x_2 be two any points of $[a, b]$ s.t. $x_1 < x_2$.

Let $f'(x) > 0 \forall x \in [a, b]$

Then by Lagrange's mean value theorem, we have, some $c \in (a, b)$ s.t.

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$\Rightarrow$

Since, $f(x_2) - f(x_1) = (x_2 - x_1) f'(c)$

$\Rightarrow f'(c) > 0$ and $x_2 - x_1 > 0$

$\Rightarrow f(x_2) - f(x_1) > 0$

$\Rightarrow f(x_2) > f(x_1) \forall x_2 > x_1 : x_1, x_2 \in (a, b)$

Hence, f is strictly increasing in (a, b) .

(b) State Darboux's Theorem. Suppose that if $f : [0, 2] \rightarrow \mathbb{R}$ is continuous on $[0, 2]$ and differentiable on $(0, 2)$, and that $f(0) = 0, f(1) = 1, f(2) = 1$.

(i) Show that there exists $c_1 \in (0, 1)$ such that $f'(c_1) = 1$

(ii) Show that there exists $c_2 \in (1, 2)$ such that $f'(c_2) = 0$

(iii) Show that there exists $c \in (0, 2)$ such that $f'(c) = 1/3$ (5)

Ans. Darboux's Theorem. If a function f defined in $[a, b]$ is such that

(i) f is derivable in $[a, b]$

(ii) $f'(a)$ and $f'(b)$ are of opposite signs then there exists some point $c \in (a, b)$ such that $f'(c) = 0$

(i) Let $f(x) = ax^2 + bx + c$

$$f(0) = 0$$

$$f(1) = 1$$

$$f(2) = 1$$

From (2):

$$f(0) = 0$$

$$a(0)^2 + b(0) + c = 0$$

$$c = 0$$

From (2):

$$f(1) = 1$$

$$a(1)^2 + b(1) + 0 = 1$$

$$a + b = 1$$

From (2):

$$f(2) = 1$$

$$a(2)^2 + b(2) + 0 = 1$$

$$4a + 2b = 1$$

$$2a + b = \frac{1}{2}$$

From (3.3) - (3.2):

$$(2a+b) - (a+b) = \frac{1}{2} - 1$$

$$a = -\frac{1}{2}$$

From (3.2):

$$a+b=1$$

$$\frac{1}{2} + b = 1$$

$$b = \frac{3}{2}$$

$$a = -\frac{1}{2}, b = \frac{3}{2}, c = 0$$

$$f(x) = x^2 + bx + c$$

$$= -\frac{1}{2}x^2 + \frac{3}{2}x + 0$$

$$= -\frac{1}{2}x^2 + \frac{3}{2}x$$

$$f'(x) = \frac{3}{2} - x$$

if

$$f'(x) = 1$$

$$\frac{3}{2} - x = 1$$

$$x = \frac{3}{2} - 1 = \frac{1}{2}$$

$$c_1 = \frac{1}{2} \in (0, 1) \forall f'(c_1) = 1$$

(ii)

$$f'(x) = 0$$

$$\frac{3}{2} - x = 0$$

$$x = \frac{3}{2}$$

$$c_2 = \frac{3}{2} \in (1, 2) \forall f'(c_2) = 0$$

(iii)

$$f'(x) = \frac{1}{3}$$

$$\frac{3}{2} - x = \frac{1}{3}$$

$$x = \frac{3}{2} - \frac{1}{3}$$

$$x = \frac{9-2}{6}$$

$$x = \frac{7}{6}$$

$$c_2 = \frac{7}{6} \in (0, 2) \forall f'(c) = \frac{1}{3}$$

(c) Find the Taylor series for $\cos x$ and indicate why it converges to $\cos x \forall x \in \mathbb{R}$. (5)

Ans. Let

$$f(x) = \cos x \forall x \in \mathbb{R}$$

$$f^n(x) = D^n[f(x)]$$

$$= D^n(\cos x)$$

$$= \cos\left(x + \frac{n\pi}{2}\right) \forall x \in \mathbb{R}$$

There, f has derivational of all order Lagrange's Remainder after n terms,

$$R_n(x) = \frac{x^n}{[n]} f^n(\phi x) \forall 0 < \phi < 1$$

$$R_n(x) = \frac{x^n}{[n]} \cos\left(\phi x + \frac{n\pi}{2}\right) \forall x \in \mathbb{R}$$

$$\Rightarrow |R_n(x)| \leq \left| \frac{x^n}{[n]} \right|$$

which gives

$$\lim_{x \rightarrow \infty} \frac{x^n}{[n]} = 0$$

and

$$\lim_{x \rightarrow \infty} R_n(x) = 0$$

Taking $R_n \rightarrow 0$ or $n \rightarrow \infty$. Then By Taylor's theorem, we have

$$f(x) = f(0) + xf'(0) + \frac{x^2}{[2]} f''(0) + \dots$$

$$\cos x = 1 + x(0) + \frac{x^2}{[2(-1)]} + \frac{x^3}{[3]}(0) + \frac{x^4}{[4]}(1) + \dots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$$

(d) Define a convex function on $[a, b]$. Check the convexity of the following functions on given intervals:

(i) $f(x) = x - \sin x, x \in [0, \pi]$

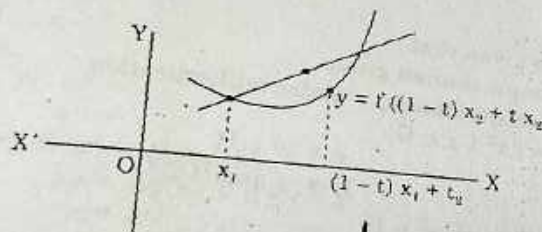
(ii) $g(x) = x^3 + 2x, x \in [-1, 1]$.

Ans. Convex Function:

Let $I \subseteq \mathbb{R}$ be an interval. A function $f: I \rightarrow \mathbb{R}$ is said to be convex on I if for any t satisfying $0 \leq t \leq 1$ and any points x_1 and x_2 in I , we have

$$f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2)$$

If $x_1 < x_2$, then as t range from 0 to 1, the point $(x_1, f(x_1)) \rightarrow (x_2, f(x_2))$ transveres the interval from x_1 to x_2 . Thus, if f is convex on I and if $x_1, x_2 \in I$, then the chord joining any two points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ on the graph of f lies above the graph of f .



(i) $f(x) = x - \sin x \forall x \in [0, \pi]$

$$f'(x) = 1 - \cos x$$

$$f''(x) = \sin x$$

$$f''(x) = \sin x \geq 0 \text{ on } [0, \pi]$$

$\Rightarrow f(x)$ is a convex function on $[0, \pi]$

(ii) $f(x) = x^3 + 2x \forall x \in [-1, 1]$

$$f'(x) = 3x^2 + 2$$

$$f''(x) = 6x$$

$$f''(x) = 6x < 0 \text{ as } x \in [-1, 0]$$

$$f''(x) = 6x > 0 \text{ as } x \in (0, 1)$$

and

$\Rightarrow f(x)$ is concave in $(-1, 0)$

and $f(x)$ is convex in $(0, 1)$.

Ans.

St. No. of Q. Paper : 6624
 Paper Code : 32351302
 Name of the Paper : GROUP THEORY-I
 Name of Course : B.Sc. (Hons.) Mathematics II Year
 Semester : III
 Duration : 3 Hours

NOV/DEC. 2017

Maximum Marks : 75

Attempt any two parts from each question.
 All questions are compulsory.

Q. 1. (a) For a fixed point (a, b) in $\mathbb{R}^2$, define $T_{(a,b)}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $(x, y) \rightarrow (x+a, y+b)$.

Show that $T(\mathbb{R}^2) = \{T_{(a,b)} \mid a, b \in \mathbb{R}\}$ is a group under function composition. (6)

Ans. It is given that:

(1) (a, b) is a fixed point in $\mathbb{R}^2$

(2) $T(a, b): \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$T_{(a,b)}(x, y) \rightarrow (x+a, y+b)$$

Let

$$G = T(\mathbb{R}^2) = \{T_{(a,b)} \mid a, b \in \mathbb{R}\}$$

where

$$G \neq \phi \text{ as } T_{(0,0)} \in G$$

(I) If

$$T_{(a,b)}, T_{(c,d)} \in G, \text{ then}$$

$$(T_{(a,b)} \circ T_{(c,d)})(x, y) = T_{(a,b)}(T_{(c,d)}(x, y))$$

$$= T_{(a,b)}(x+c, y+d)$$

$$= T_{(a+c, b+d)}(x, y)$$

$$= T_{(a+c, b+d)}(x, y)$$

$$= T_{(a+c, b+d)}(x, y)$$

$$T_{(a,b)} \circ T_{(c,d)} = T_{(a+c, b+d)}$$

$\Rightarrow$ closure law holds in G

(II) If

$$T_{(a,b)} \in G$$

$$T_{(c,d)} \in G$$

$$T_{(a,b)} \in G$$

Then we have,

$$(T_{(a,b)} \circ T_{(c,d)} \circ T_{(a,b)}) = T_{(a,b)} \circ (T_{(c,d)} \circ T_{(a,b)})$$

$\Rightarrow$ Associative law holds in G .

(III) If $T_{(a,b)} \in G, T_{(0,0)} \in G$, we have

$$(T_{(a,b)} \circ T_{(0,0)})(x, y) = T_{(a,b)}(T_{(0,0)}(x, y))$$

$$= T_{(a,b)}(x+0, y+0)$$

$$= T_{(a,b)}(x, y)$$

$$T_{(0,0)} \circ T_{(a,b)}(x, y) = T_{(0,0)}(T_{(a,b)}(x, y))$$

$$= T_{(0,0)}(x+a, y+b)$$

$$= T_{(x+a, y+b)}(x, y)$$

$$= T_{(x+a, y+b)}(x, y)$$

$$\Rightarrow (T_{(a,b)}, T_{(0,0)}) = T_{(0,0)} = (T_{(0,0)}, T_{(0,0)})$$

$\Rightarrow$ Identity $T_{(0,0)}$ exists in G .

$$(IV) \quad [T_{(a,b)}]^{-1} = T_{(-a,-b)}$$

$\Rightarrow$ Inverse exists in G .

From (I), (II), (III), (IV), we can say that G is a group under function composition.

(b) (i) Find the inverse of $\begin{pmatrix} 2 & 6 \\ 3 & 5 \end{pmatrix}$ in $GL(2, Z_{11})$. (4)

(ii) Let G be an Abelian group under multiplication with identity e . Show that

$H = \{x^2 \mid x \in G\}$ is a subgroup of G . (2)

Ans. (i) Let $A = \begin{pmatrix} 2 & 6 \\ 3 & 5 \end{pmatrix}$

The determinant of A be:

$$\begin{aligned} |A| &= \begin{vmatrix} 2 & 6 \\ 3 & 5 \end{vmatrix} \\ &= 2 \times 5 - 6 \times 3 \\ &= 10 - 18 \\ &= -8 \\ &= 3 \pmod{11} \\ &\neq 0 \end{aligned}$$

$\Rightarrow A$ is non singular.

The cofactor matrix of A be:

$$\begin{aligned} \text{cof}(A) &= \begin{pmatrix} (5) & -(3) \\ -(6) & (2) \end{pmatrix} \\ &= \begin{pmatrix} 5 & -3 \\ -6 & 2 \end{pmatrix} \end{aligned}$$

The adjoint of A be:

$$\begin{aligned} \text{adj}(A) &= (\text{cof } A) \\ &= \begin{pmatrix} 5 & -3 \\ -6 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 5 & -6 \\ -3 & 2 \end{pmatrix} \end{aligned}$$

The inverse of A with be:

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

$$= -\frac{1}{8} \begin{pmatrix} 5 & -6 \\ -3 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{5}{8} & \frac{6}{8} \\ \frac{3}{8} & -\frac{2}{8} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{5}{8} & \frac{3}{4} \\ \frac{3}{8} & -\frac{1}{4} \end{pmatrix}$$

(ii) It is given that

(1) G be an abelian group

(2) G has an identity e under multiplication

(3) $H = \{x^2 \mid x \in G\}$

$$e = e^2 \in H$$

$$H \neq \emptyset \text{ and } H \subseteq G$$

If $x, y \in G \Rightarrow x^2, y^2 \in H$

$$x^2, (y^2)^{-1} = x^2 y^{-2}$$

$$= (xy^{-1})^2 \text{ as } G \text{ is abelian}$$

$\Rightarrow$

$$x, y \in G \Rightarrow x, y^{-1} \in G \Rightarrow xy^{-1} \in G$$

$\Rightarrow$

$$(xy^{-1})^2 \in H$$

$\Rightarrow$

$$x^2, (y^2)^{-1} \in H$$

$\Rightarrow$

$$x^2, y^2 \in H \text{ as } x, y \in G$$

$\Rightarrow H$ is a subgroup of G .

(c) (i) Let G be a group. Show that $Z(G) = Z(G) = \bigcap_{a \in G} C(a)$

where $Z(G)$ is the Center of G and $C(a)$ is the Centralizer of a . (4)

(ii) Let G be the group of nonzero real numbers under multiplication. Show that

$$H = \{x \in G \mid x = 1 \text{ or } x \text{ is irrational}\}$$

and

$$K = \{x \in G \mid x > 1\} \text{ are not subgroups of } G. (2)$$

Ans. (ii) It is given that

(1) $Z(G)$ is center of G

(2) $C(a)$ is centralizes of a

and

$$Z(G) = \{x \in G \mid xy = yx \forall y \in G\}$$

Let $x \in Z(G)$

$$C(a) = \{x \in G \mid xa = ax\}$$

$\Rightarrow$

$$xy = yx \quad \forall y \in G$$

$\Rightarrow$

$$xa = ax \quad \forall a \in G$$

$$|a^k| = r \Rightarrow r \left(\frac{n}{k} \right)$$

From (1) and (2):

$$r = \frac{n}{d}$$

$$|a^k| = \frac{n}{\gcd(n, k)}$$

$\Rightarrow$ The subgroup $\langle a^k \rangle$ has $r = \frac{n}{d}$ elements

Subgroup $\langle a^{\gcd(n, k)} \rangle$ has $\frac{n}{d}$ elements as $|0| = n$.

Hence, by uniqueness of a cyclic subgroup of order r in a cyclic group of order n .

We have:

$$\langle a^k \rangle = \langle a^{\gcd(n, k)} \rangle$$

Q. 3. (a) Define the Alternating Group A_n . Show that it forms a subgroup of the Permutation Group S_n and $|A_n| = \frac{n!}{2}$. (8)

Ans. **Alternative Group:** The group of even permutations of n symbols is denoted by A_n and is called the alternate group of degree n .

For $n > 1$, A_n has order $\frac{n!}{2}$.

Solution of given problem: Let P_n be the symmetric group of degree n i.e. the elements of P_n are permutations of degree n . If A_n is the set of all even permutations of degree n , then $A_n \leq P_n$ and A_n is closed with respect to multiplication of permutations. We remember that the product of two even permutations is also an even permutation. Therefore, A_n is a subgroup of P_n . Hence A_n is itself a group with respect to multiplication of permutations. Hence, A_n is called alternating group of degree n or order $\frac{n!}{2}$.

(b) Prove that every group is isomorphic to a group of permutations. (6)

Ans. Let G be a group and $A(G)$ the group of all permutations of G i.e. $A(G)$ is the set of all one to one mappings of G onto G . For each $a \in G$, we define mapping

$$f_a : G \rightarrow G \text{ as } f_a(x) = ax \quad \forall x \in G$$

Then, f_a is one-to-one, since,

$$f_a(x) = f_a(y)$$

$$ax = ay$$

...(1)

$$x = y \quad \forall x, y \in G$$

$\Rightarrow$ (Using left cancellation law in G)

Now, we show that f_a is onto. For any $g \in G$, we have:

$$g = e \cdot g$$

$$= a(a^{-1}g)$$

$$= ag, \text{ where } g' = a^{-1}g \in G$$

Using (1), we have:

$$g = f_a(g') \text{ } \therefore f_a \text{ is out}$$

Since, f_a is one-to-one and onto, f_a is a permutation of G . Thus,

$$f_a \in A(G) \quad \forall a \in G$$

$$\bar{G} = \{f_a : a \in G\}$$

Let

Now, we proceed to show that $\bar{G}$ is a subgroup of $A(G)$

Let

$$f_a, f_b \in \bar{G}$$

For any $x \in G$, we have:

$$(f_a f_b)(x) = f_a\{f_b(x)\}$$

$$= f_a(bx),$$

$$= a(bx),$$

$$= (ab)x,$$

$$= f_{ab}(x)$$

$$f_a f_b = f_{ab} \quad \forall a, b \in G$$

From (3) it follows that

$$f_a \in \bar{G} \text{ and } f_b \in \bar{G}$$

$\Rightarrow$ Now, we show that

$$f_a f_b \in \bar{G}$$

$$(f_a)^{-1} = f_{a^{-1}} \quad \forall f_a \in \bar{G}$$

Using (3), we have:

$$f_a f_a^{-1} = f_a a^{-1}$$

$$= f_e$$

where,

$$f_e(x) = ex = x \quad \forall x \in G$$

$\Rightarrow$

$$f_e = I \text{ is identity}$$

Similarly,

$$f_{a^{-1}} f_a = I$$

This proves (5) and so:

$$(f_a)^{-1} \in \bar{G} \quad \forall f_a \in \bar{G}$$

From (4) and (5), $\bar{G} < A(G)$ and so $\bar{G}$ is a permutation group.

Finally, we prove that $G = \bar{G}$

Define a mapping, $\Psi : G \rightarrow \bar{G}$ as

$$\Psi(a) = f_a \quad \forall a \in G$$

Clearly, Ψ is onto. Now, we show that Ψ is a homomorphism. Let $a, b \in G$ be arbitrary.

Finally, from (6) and (3), we obtain:

$$\begin{aligned}\Psi(a, b) &= f_a b \\ &= f_a f_b \\ &= \Psi(a) \Psi(b)\end{aligned}$$

Thus, Ψ is a homomorphism. Also Ψ is one-to-one.

$$\begin{aligned}\text{Since, } \Psi(a) &= \Psi(b) \\ f_a &= f_b \\ f_a(x) &= f_b(x) \quad \forall x \in G \\ ax &= bx, \text{ using (1)} \\ a &= b, \text{ using right cancellation law in } G.\end{aligned}$$

Since, $\Psi: G \rightarrow \bar{G}$ is a homomorphism, one to one and onto, we have

$$G \cong \bar{G}$$

Hence, we can say that every group G is isomorphic to a permutation group.

(c) Prove that $U(10)$ is not isomorphic to $U(12)$.
Ans. $U(n)$ represents the group of positive integers $< n$ and coprime with n c.r.t. multiplication mod n . (6)

$$U(10) = \{1, 3, 7, 9\} \text{ w.r.t. } \odot_{10}$$

The Cayley table for $U(10)$ be:

$\odot_{10}$	1	3	7	9
1	1	3	7	9
3	3	9	1	7
7	7	1	9	3
9	9	7	3	1

$$U(12) = \{1, 5, 7, 11\} \text{ w.r.t. } \odot_{12}$$

The Cayley table for $U(12)$ be:

$\odot_{12}$	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	11	1	5
11	11	7	5	1

Now, $U(10)$ and $U(12)$ can not be isomorphic because under an isomorphism, element of order m in one group is mapped an element of the same order in a second group. Here, in $U(10)$ 3 is of order 4, where as $U(12)$ has no element of order 4. (6½)

Q. 4. (a) State and prove Orbit Stabilizer Theorem. (6½)

Ans. Orbit Stabilizer Theorem:

Statement:

Let G be a finite group of permutations of a set S . Then, for any i from S ,

Proof:

By Lagrange's theorem, $|G| / |\text{stab}_G(i)|$ is the number of distinct left cosets of $\text{stab}_G(i)$ in G . Thus, it suffices to establish a one-to-one Correspondence between the left cosets of $\text{stab}_G(i)$ and the elements in the orbit of i . To do this we define a correspondence T by $T(\alpha \text{stab}_G(i)) = \alpha(i)$. To show that T is a well defined function, we must show that

$$\begin{aligned}\Rightarrow \alpha \text{stab}_G(i) &= \beta \text{stab}_G(i) \\ \alpha(i) &= \beta(i) \\ \text{But, } \alpha \text{stab}_G(i) &= \beta \text{stab}_G(i) \\ \alpha^{-1}\beta &\in \text{stab}_G(i) \\ \Rightarrow (\alpha^{-1}\beta)(i) &= i \\ \Rightarrow \beta(i) &= \alpha(i)\end{aligned}$$

Reversing the argument from the last step to the first step shows that T is also one-to-one.

We conclude the proof by showing that T is onto $\text{orb}_G(i)$.

Let $j \in \text{orb}_G(i)$, then $\alpha(i) = j$ for some $\alpha \in G$ and clearly $T(\alpha \text{stab}_G(i)) = \alpha(i) = j$, so that T is onto.

(b) (i) Prove that $aH = H$ if and only if $a \in H$.

(ii) Prove that $aH = bH$ or $aH \cap bH = \emptyset$

$$\begin{aligned}\text{Ans: (i) Let } aH &= H, \text{ then} \\ aH &= \{ah \mid h \in H\}\end{aligned}$$

and $e \in H$, so

$$\begin{aligned}\Rightarrow a &= ae \in aH \\ a &\in H\end{aligned}$$

Converse:

$$\begin{aligned}\text{Let } a &\in H \\ \text{again } h &\in H \quad \forall h \in H, ah \in H \\ \Rightarrow aH &\subseteq H \\ h &= eh \\ &= (a a^{-1})h \quad \text{as } a^{-1} \in H \\ &= a(a^{-1}h) \quad h \in H, \Rightarrow a^{-1}h \in H \\ \Rightarrow H &\subseteq aH \\ \Rightarrow aH &= H\end{aligned}$$

$$\begin{aligned}\Rightarrow h &= eh \\ &= (a a^{-1})h \quad \text{as } a^{-1} \in H \\ &= a(a^{-1}h) \quad h \in H, \Rightarrow a^{-1}h \in H\end{aligned}$$

$$\begin{aligned}\Rightarrow H &\subseteq aH \\ \Rightarrow aH &= H\end{aligned}$$

(ii) Let $aH \cap bH \neq \emptyset$

$$\begin{aligned}\Rightarrow \exists x &\in aH \cap bH \\ \Rightarrow x &= ah_1 = bh_2 \quad \forall h_1, h_2 \in H \\ \Rightarrow b^{-1}a &= h_2 h_1^{-1} \in H \\ \Rightarrow b^{-1}aH &= H \\ \Rightarrow aH &= bH\end{aligned}$$

as H is a sub group using part (i)

(c) (i) Prove that order of $U(n)$ is even when $n > 2$

(ii) Prove that a group of prime order is cyclic.

Ans. (i) For $n > 2$,

For $n = 1$	$U(1) = \{1\}$
For $n = 2$	$U(2) = \{1, 3\}$
For $n = 3$	$U(3) = \{1, 2, 3, 4\}$
For $n = 4$	$U(4) = \{1, 5\}$
For $n = 5$	$U(5) = \{1, 2, 3, 4, 5, 6\}$
For $n = 6$	$U(6) = \{1, 3, 5, 7\}$

and so on.

Hence, we can say that $U(n)$ is always even, when $n > 2$

(ii) Let G be a group of order p , where p is a prime.

Here, $p \geq 2$

$$\Rightarrow |G| = p \geq 2$$

Let e be an element of G and $H = \langle e \rangle$ be the cyclic subgroup generated by e of G .

By Lagrange's theorem

$$|H| \text{ divides } |G| = p$$

$$\Rightarrow |H| = 1 \text{ or } p$$

$$\text{But, } H \neq 1 \text{ as } e \neq e \in H$$

$$\Rightarrow |H| = p = |G|$$

$$\Rightarrow |H| = |G|$$

$$\Rightarrow G = H$$

$$\Rightarrow G = H = \langle e \rangle$$

$\Rightarrow G$ is cyclic

Q. 5. (a) Let H and K be subgroups of a finite group G and let

$$HK = \{hk | h \in H, k \in K\}$$

$$KH = \{kh | k \in K, h \in H\}$$

and

Prove that HK is a group if and only if $HK = KH$.

Ans. It is given that

(1) H and K be subgroup of G

(2) $HK = \{hk | h \in H, k \in K\}$

(3) $KH = \{kh | k \in K, h \in H\}$

Let HK be a subgroup of G . We shall prove that $HK = KH$.

Let $x \in HK$ be arbitrary, then

$$\Rightarrow x^{-1} \in HK \quad (\because HK < G)$$

$$\Rightarrow x^{-1} = hk \text{ for some } h \in H, k \in K$$

$$\Rightarrow x = (x^{-1})^{-1}$$

$$\Rightarrow x = (hk)^{-1}$$

$$\Rightarrow x = k^{-1}h^{-1} \in KH \text{ (as } k^{-1} \in K, h^{-1} \in H)$$

$$K < G \text{ and } H < G$$

$$HK \subseteq KH$$

Suppose that $y \in KH$ is arbitrary, then

$$y = k_1 h_1 \text{ for } k_1 \in K \text{ and } h_1 \in H$$

$$= (k_1^{-1})^{-1} (h_1^{-1})^{-1}$$

$$= (h_1^{-1} k_1^{-1})^{-1}$$

Since $H < G$ and $K < G$, therefore,

...(2)

$$\Rightarrow h_1^{-1} k_1^{-1} \in H \text{ and } k_1^{-1} \in K$$

$$\Rightarrow h_1^{-1} k_1^{-1} \in HK$$

$$\Rightarrow (h_1^{-1} k_1^{-1})^{-1} \in HK$$

$$\Rightarrow y \in HK, \text{ using (2)}$$

$$\Rightarrow KH \subseteq HK$$

From (1) and (3):

$$HK = KH$$

Condition is sufficient:

$$HK = KH$$

Suppose that

We shall prove that HK is a subgroup of G .

Let $x, y \in HK$. Thus,

$$x = hk,$$

$$y = h'k' \quad \forall h, h' \in H, k, k' \in K$$

$$xy = hkh'k'$$

We have,

$$kh' \in KH = HK, \text{ we can write}$$

Since,

$$kh' = h_2 k_2 \text{ for } h_2 \in H \text{ and } k_2 \in K$$

Putting in (5),

$$xy = h(h_2 k_2)k'$$

$$= (hh_2)(k_2 k') \in HK$$

as $hh_2 \in H$ and $k_2 k' \in K, H < G, K < G$

Thus,

$$xy \in HK \quad \forall x, y \in HK$$

Also,

$$x^{-1} = (hk)^{-1} = k^{-1}h^{-1} \in KH = HK$$

$$x^{-1} \in HK \quad \forall x \in HK$$

From (6) and (7), it follows that HK is a subgroup of G .

(b) Let ϕ be a homomorphism from a group G to a group $\bar{G}$ and let g be an element of G . Prove that

(i) If $\phi(g) = g'$, then $(\phi^{-1}(g')) = \{x \in G | \phi(x) = g'\} = g \text{Ker } \phi$

(ii) If $|\text{Ker } \phi| = n$, then ϕ is an n -to-1 mapping from G onto $\phi(G)$.

Ans. It is given that:

(1) ϕ is a homomorphism from G to $\bar{G}$

(2) $g \in G$,

(i) $\phi(g) = g'$

Then, by definition,

$$\phi^{-1}(g') = \{x \in G | \phi(x) = g'\}$$

To show: $\phi^{-1}(g') = g \text{Ker } \phi$

Let

$$y \in \phi^{-1}(g')$$

$\Rightarrow$

$$\phi(y) = g' = \phi(g)$$

$\Rightarrow$

$$g^{-1}y \in \text{Ker } \phi$$

$\Rightarrow$

$$y \in g \text{Ker } \phi$$

$\Rightarrow$

$$\phi^{-1}(g') \subseteq g \text{Ker } \phi$$

Conversely, let $x \in g \text{Ker } \phi$

$$\begin{aligned}
 & \Rightarrow x = g x, \forall x \in \ker \phi \\
 & \Rightarrow \phi(x_1) = e \\
 & \Rightarrow \phi(x) = \phi(g x_1) \\
 & \Rightarrow \phi(x) = \phi(g) \phi(x_1) \\
 & \Rightarrow \phi(x) = g' \cdot e \\
 & \Rightarrow \phi(x) = g'
 \end{aligned}$$

$$\begin{aligned}
 & \Rightarrow x \in \phi^{-1}(g') \\
 & \Rightarrow g \ker \phi \subseteq \phi^{-1}(g')
 \end{aligned}$$

From (1) and (2) we have:

$$\phi^{-1}(g') = g \ker \phi \quad \dots(2)$$

$$\begin{aligned}
 \text{(ii) Given: } & |\ker \phi| = 1 \\
 & \ker \phi = \{e\} \quad \dots(3)
 \end{aligned}$$

Now, $\phi: G \rightarrow \bar{G}$ is a group homomorphism.

$\Rightarrow \phi: G \rightarrow \phi(G) = \{\phi(g) \mid g \in G\}$ is onto

$$\phi(g_1) = \phi(g_2)$$

$$\phi(g_1 g_2^{-1}) = e$$

$$g_1 g_2^{-1} \in \ker \phi = \{e\}$$

$$g_1 g_2^{-1} = e$$

$$g_1 = g_2$$

$\Rightarrow \phi$ is one-one

Hence, $\phi: G \rightarrow \phi(G)$ is one-to-one and onto homomorphism.

(c) (i) Prove that A_n is normal in S_n . (3/4)

(ii) If G is a non-Abelian group of order p^3 (p is prime) and $Z(G) \neq \{e\}$, prove that $|Z(G)| = p$. (3)

Ans. (i) Since, A_n is the subset of S_n containing of all even permutations

$$d(S_n) = \lfloor n \rfloor$$

A_n is non-empty, since the identity permutation I is an even permutation $I \in A_n$.

Let $f, g \in A_n$ so that f and g are even permutations. Since, the product of two even permutations is even, $fog \in A_n$. Consequently, A_n is a subgroup of S_n .

Let $W = \{1, -1\}$ be the group under multiplication

Define: $S_n \rightarrow W$ as follows

$$\Psi(f) = 1; f \text{ is an even permutation} \quad \dots(1)$$

$$= -1; f \text{ is an odd permutation} \quad \dots(2)$$

We shall prove that Ψ is a homomorphism.

Let $f, g \in S_n$ be arbitrary

Case: 1. Let f, g be both even permutation

Then, $\Psi(f) = \Psi(g) = 1$, Using (3)

Since fog is even so by (1)

$$\Psi(fog) = 1$$

$$= 1 \times 1$$

$$= \Psi(f) \Psi(g)$$

Case: 2. Let f, g be both even permutation

Then, $\Psi(f) = \Psi(g) = -1$, Using (2)
Since product of two odd permutations is even, by (1),

$$\begin{aligned}
 \Psi(fog) &= 1 \\
 &= (-1) \times (-1) \\
 &= \Psi(f) \Psi(g)
 \end{aligned}$$

Case: 3. Let f be even and g be odd

Then, $\Psi(f) = 1$ and $\Psi(g) = -1$

Since fog is odd,

$$\begin{aligned}
 \Psi(fog) &= -1, \text{ by (2)} \\
 &= 1 \times -1 \\
 &= \Psi(f) \Psi(g)
 \end{aligned}$$

Similarly $\Psi(fog) = \Psi(f) \Psi(g)$, where f is odd and g is even
In any case, Ψ is a homomorphism.

Obviously, Ψ is onto, since S_n must contain both even and odd permutation ($n \geq 2$).

By fundamental theorem of homomorphism,

$$\frac{S_n}{\ker \Psi} \cong W \quad \dots(3)$$

where, $\ker \Psi = \{f \in S_n \mid \Psi(f) = 1\}$

Now, $f \in \ker \Psi$

$$\Leftrightarrow \Psi(f) = 1$$

$$\Leftrightarrow f \text{ is an even permutation}$$

$$\Leftrightarrow f \in A_n$$

$$\therefore \ker \Psi = A_n$$

Hence, $\ker \Psi = A_n$ is a normal subgroup.

(ii) It is given that

(1) G is a non abelian group of order p^3

(2) p is prime

(3) $Z(G) \neq \{e\}$

Using Lagrange's Theorem, we have:

$$Z(G) \text{ divides } |G| = p^3$$

$$\Rightarrow |Z(G)| \text{ can be } 1, p, p^2, p^3$$

$$\text{As } Z(G) \neq \{e\}$$

$$= Z \mid Z(G) \neq 1$$

Also, $Z(G) \neq G$ as G is non abelian

$Z(G)$ is abelian

$$|Z(G)| = p^2$$

Hence, $|Z(G)| = p$ or p^2

Now for an $a \in G - Z(G)$, consider the

Subgroup $N(a) = \{x \in G \mid xa = ax\}$

$Z(G) \subseteq N(a)$ and $a \in Z(G)$
 $Z(G)$ is a proper subgroup of $N(a)$
 If $Z(G) = N(a)$, then $N(a) = G$
 $a \in Z(G)$

Hence, $Z(G) = G$ (6/12)

Q. 6. (a) State and prove The First Isomorphism Theorem.

Ans. First Isomorphism Theorem:

Statement: Every homomorphism image of a group G is isomorphic to same quotient group of G .

Proof:

Suppose $f: G \rightarrow G'$ is a group homomorphism with kernel K , so that K is the normal subgroup of G and

$K = \{x \in G : f(x) = e'\}$, e' being the identity of G' .

Now, the symbol a/K has a meaning

To prove $a/K \cong G'$

Define a map $g: G/K \rightarrow G'$ s.t. $g(k, x) = f(x) \forall x \in G$ is well defined.

For $Ka = Kb : Ka, Kb \in a/K$
 $ab^{-1} \in K$ [ss $Ha = Hb \Rightarrow ab^{-1} \in H$]

$f(ab^{-1}) \in e'$, def of K

$f(a)f(b)^{-1} = e'$ as f is homomorphism

$f(a)[f(b)]^{-1} = e'$

$f(a)[f(b)]^{-1} = e' f(b)$

$f(a)e' = f(b)$

$f(a) = f(b)$

$g(Ka) = g(Kb)$

Which proves that g is well defined.

g is one-one

$g(Kb) = g(Ka) : Ka, Kb \in G/K$

$f(a) = f(b)$

$[f(a)][f(b)]^{-1} = e'$

$[f(a)][f(b)^{-1}] = e'$

$[f(ab^{-1})] = e'$

$ab^{-1} \in K$

$Ka = Kb$

$g(Ka) = g(Kb)$

$Ka = Kb$

$\Rightarrow g$ is one-one

g preserves composition in G/K and G' :

$g(Ka \cdot Kb) = g(Kab)$ as K is normal

$$\begin{aligned} (Ka)(Kb) &= Kab \\ &= f(ab) \\ &= f(a)f(b) \text{ as } f \text{ is homomorphism} \\ &= g(Ka)g(Kb) \\ g(Ka \cdot Kb) &= g(Ka)g(Kb) \end{aligned}$$

$\Rightarrow g$ is onto:

any $x \in G' \Rightarrow \exists y \in G$ s.t. $f(y) = x$

For $f: G \rightarrow G'$ is onto

$$Ky \in a/K, g(Ky) = f(y) = x$$

$\Rightarrow$ Given any $x \in G', \exists Ky \in G/K$ s.t. $g(Ky) = f(y) = x$

Which proves g is onto.

Hence, we can say that g is an isomorphism

Therefore, we have:

$$G/K \cong G'$$

Which proves first theorem of isomorphism.

(b) Let G be a group and let $Z(G)$ be the center of G . Prove that if $G/Z(G)$ is cyclic, then G is Abelian.

Ans. It is given that

- (1) G be a group
- (2) $Z(G)$ be the center of G
- (3) $G/Z(G)$ be cyclic

$$\Rightarrow \exists x \in G \text{ s.t. } G/Z = \langle zx \rangle$$

Now for any $a, b \in G$

$$za, zb \in a/Z = \langle zx \rangle$$

$\Rightarrow \exists$ integers m and n s.t.

$$za = (zx)^m$$

$$= zx^m$$

$$= z x^m$$

$$zb = (zx)^n$$

$$= z x^n$$

$$\Rightarrow a = z_1 x^m, b = z_2 x^n \text{ for some } z_1, z_2 \in Z$$

Now,

$$ab = z_1 x^m z_2 x^n$$

$$= z_1 z_2 x^{m+n}$$

$$= z_2 z_1 x^{m+n}$$

$$= z_2 z_1 x^{n+m}$$

$$= z_2 x^n z_1 x^m$$

$$= ba \quad \forall ab \in G$$

$\Rightarrow G$ is an abelian group.

(c) Let $4Z = \{0, \pm 4, \pm 8, \dots\}$. Find $Z/4Z$.

Ans.

$$4Z = \{0, 1, 4, \pm 8, \dots\}$$

When we divide $4Z$ by 4, the possible non-negative integer are 0, 1, 2, 3.

$$\frac{Z}{4Z} = \{4Z, 4Z+1, 4Z+2, 4Z+3\}$$

Sr. No. of Q. Paper : 6625
 Paper Code : 32351303
 Name of the Paper : MULTIVARIATE CALCULUS
 Name of Course : B.Sc. (Hons.) Mathematics II Year
 Semester : III
 Duration : 3 Hours

NOV/DEC 2017

Attempt any Five questions from each Section. Maximum Marks : 75
 All questions carry equal marks.

SECTION-I

Q. 1. (a) Let f be the function defined by:

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^2} & \text{for } (x, y) \neq (0, 0) \\ 0 & \text{for } (x, y) = (0, 0) \end{cases}$$

Is f continuous at $(0, 0)$? Explain.

Ans. The function $f(x, y)$ is defined as:

$$f(x, y) = \frac{xy^2}{x^2 + y^2} \quad ; (x, y) \neq (0, 0)$$

$$= 0$$

$$; (x, y) = (0, 0)$$

$$\text{at } (0, 0)$$

$$f(0, 0) = 0$$

Let $(x, y) \rightarrow (0, 0)$ along the path $y = mx$, where m is any real numbers, where $x \rightarrow 0$, then $y \rightarrow 0$ along the path $y = mx$.

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{(x, y) \rightarrow (0, 0)} \frac{x(mx)^3}{x^2 + (mx)^6}$$

$$= \lim_{(x, y) \rightarrow (0, 0)} \frac{m^3 x^4}{x^2 + m^6 x^6}$$

$$= \lim_{(x, y) \rightarrow (0, 0)} \frac{m^3 x^4}{x^2 (1 + m^6 x^4)}$$

$$= \lim_{(x, y) \rightarrow (0, 0)} \frac{m^3 x^2}{1 + m^6 x^4}$$

$$= \frac{m^3 (0)^2}{1 + m^6 (0)^4}$$

$$= 0$$

$$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0) = 0$$

$$\Rightarrow f(x, y) \text{ is continuous at } (0, 0)$$

(202)

Q. 2. Find the equation for each horizontal tangent plane to the surface:

Ans. Let

$$z = 5 - x^2 - y^2 + 4y$$

$$f_x(x, y) = \frac{\partial f}{\partial x} = 0 - 2x - 0 + 0 = -2x$$

$$f_y(x, y) = \frac{\partial f}{\partial y} = 0 - 0 - 2y + 4 = -2y + 4$$

Let required point P be (x_0, y_0, z_0)

$$a = f_x(x_0, y_0)$$

$$b = f_y(x_0, y_0) = -2y_0 + 4$$

Equation of tangent to the surface $f(x, y)$ at $P(x_0, y_0, z_0)$ be:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$z - z_0 = a(x - x_0) + b(y - y_0)$$

$$z - z_0 = (-2x_0)(x - x_0) + (-2y_0 + 4)(y - y_0)$$

$$z - z_0 = -2x_0x + 2x_0^2 + (-2y_0 + 4)y + 2y_0^2 - 4y_0$$

$$2xx_0 + (2y_0 - 4)y + z = 2x_0^2 + 2y_0^2 - 4y_0 + z_0$$

Q. 3. Let f and g be twice differentiable functions of one variable and let $u(x, t) = f(x + ct) + g(x - ct)$ for a constant c . Show that:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Ans.

$$u(x, t) = f(x + ct) + g(x - ct)$$

Diff. Eq. partially w.r.t. t :

$$\frac{\partial u}{\partial t} = f'(x + ct) \cdot c + g'(x - ct) \cdot (-1) = cf'(x + ct) - cg'(x - ct)$$

Diff. Eq. (2) partially w.r.t. t :

$$\frac{\partial^2 u}{\partial t^2} = c^2 f''(x + ct) + c^2 g''(x - ct)$$

Diff. Eq. (1) partially w.r.t. x :

$$\frac{\partial u}{\partial x} = f'(x + ct) + g'(x - ct)$$

Diff. Eq. partially w.r.t. x :

$$\frac{\partial^2 u}{\partial x^2} = f''(x + ct) + g''(x - ct)$$

Using Eq. (5), Eq. (3) can be solved as:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \{ f''(x + ct) + g''(x - ct) \}$$

Ans.

$$\frac{\partial^2 u}{\partial x^2} = e^{1/x} \frac{\partial^2 u}{\partial x^2}$$

Q. 4. Let f have continuous partial derivatives and suppose the maximal directional derivative of f at $P_0(1, 2)$ has magnitude 50 and is attained in the direction from P_0 towards $Q(3, -4)$. Use this information to find $\Delta f(1, 2)$.

Ans. We know that the directional derivative of f at the point $P_0(x_0, y_0)$ in the direction of unit vector $\hat{u}$ is given by

$$D_{\hat{u}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \hat{u} \quad \dots(1)$$

Which gives:

$$\nabla f(x, y) = f_x(x, y)\hat{i} + f_y(x, y)\hat{j} \quad \dots(2)$$

at $P_0(1, 2)$

$$\nabla f(1, 2) = f_x(1, 2)\hat{i} + f_y(1, 2)\hat{j} \quad \dots(3)$$

The maximum value of directional derivative is:
at $P_0(1, 2)$

$$\|\nabla f(1, 2)\| = 50$$

$$\sqrt{f_x^2(1, 2) + f_y^2(1, 2)} = 50$$

$$f_x^2(1, 2) + f_y^2(1, 2) = 2500 \quad \dots(4)$$

The maximum value of directional derivative occurs in the direction of $\nabla f(1, 2)$. The unit vector in the direction of this vector,

$$\hat{u} = \frac{\nabla f(1, 2)}{\|\nabla f(1, 2)\|} = \frac{(3-1)\hat{i} + (-4-2)\hat{j}}{\|(3-1)\hat{i} + (-4-2)\hat{j}\|}$$

$$\frac{f_x(1, 2)\hat{i} + f_y(1, 2)\hat{j}}{\sqrt{f_x^2(1, 2) + f_y^2(1, 2)}} = \frac{2\hat{i} - 6\hat{j}}{\sqrt{4 + 36}}$$

$$\frac{f_x(1, 2)\hat{i} + f_y(1, 2)\hat{j}}{50} = \frac{2\hat{i} - 6\hat{j}}{\sqrt{40}}$$

$$f_x(1, 2)\hat{i} + f_y(1, 2)\hat{j} = \frac{50}{\sqrt{40}}(2\hat{i} - 6\hat{j})$$

$$f_x(1, 2)\hat{i} + f_y(1, 2)\hat{j} = \frac{50}{2\sqrt{10}}(2\hat{i} - 6\hat{j})$$

$$f_x(1, 2)\hat{i} + f_y(1, 2)\hat{j} = \frac{25}{\sqrt{10}}(2\hat{i} - 6\hat{j})$$

On comparing coefficients of $\hat{i}$ and $\hat{j}$ on both sides we get:

$$f_x(1, 2) = \frac{50}{\sqrt{10}}\hat{i}$$

$$f_y(1, 2) = -\frac{150}{\sqrt{10}}\hat{j}$$

$$\nabla f(1, 2) = f_x(1, 2)\hat{i} + f_y(1, 2)\hat{j} = \frac{25}{\sqrt{10}}(2\hat{i} - 6\hat{j})$$

Hence,

Q. 5. Find the absolute extrema of $f(x, y) = x^2 + xy + y^2$ on the closed bounded set S where S is the disk $x^2 + y^2 \leq 1$.
Ans. Consult text book.

Q. 6. Find the point on the plane $2x + y + z = 1$ that is nearest to the origin.

Ans. Let $P(x, y, z)$ be the point on the plane
 $2x + y + z = 1$

that is nearest to $(0, 0)$

it means that the problem is to find out the minimum value of the function

$$\begin{aligned} |\overline{OP}| &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \\ &= \sqrt{(x - 0)^2 + (y - 0)^2 + (z - 0)^2} \\ &= \sqrt{x^2 + y^2 + z^2} \end{aligned}$$

Subject to the constraint that
 $2x + y + z = 1$

Since $|\overline{OP}|$ has minimum value, whenever the function,

$$\begin{aligned} f(x, y, z) &= x^2 + y^2 + z^2 \\ g(x, y, z) &= 2x + y + z - 1 \end{aligned} \quad \dots(2)$$

Let

Then,

$$\begin{aligned} f_x &= 2x \\ f_y &= 2y \\ f_z &= 2z \\ g_x &= 2 \\ g_y &= 1 \\ g_z &= 1 \end{aligned}$$

By Lagrange's multipliers, we have:

$$f_x(x, y) = \lambda g_x(x, y)$$

$$2x = \lambda(2)$$

$$x = \lambda$$

$$f_y(x, y) = \lambda g_y(x, y)$$

$$2y = \lambda(1)$$

$$y = \frac{\lambda}{2}$$

$$f_z(x, y) = \lambda g_z(x, y)$$

$$2z = \lambda(1)$$

$$z = \frac{\lambda}{2}$$

On putting above values of (x, y, z) in $g(x, y, z) = 0$

We get :

$$\begin{aligned} g(x, y, z) &= 0 \\ 2x + y + z - 1 &= 0 \\ 2\left(\frac{\lambda}{2}\right) + \frac{\lambda}{2} - 1 &= 0 \\ 2\lambda + \lambda - 1 &= 0 \\ 3\lambda - 1 &= 0 \end{aligned}$$

$$\lambda = \frac{1}{3}$$

$$x = \frac{1}{3}, y = \frac{1}{6}, z = \frac{1}{6}$$

Hence,

Thus, the point $\left(\frac{1}{3}, \frac{1}{6}, \frac{1}{6}\right)$ is closed to the origin.

SECTION-II

Q. 7. Find the area of the region D by setting double integral, where D is bounded by the parabola $y = x^2 - 2$ and the line $y = x$.

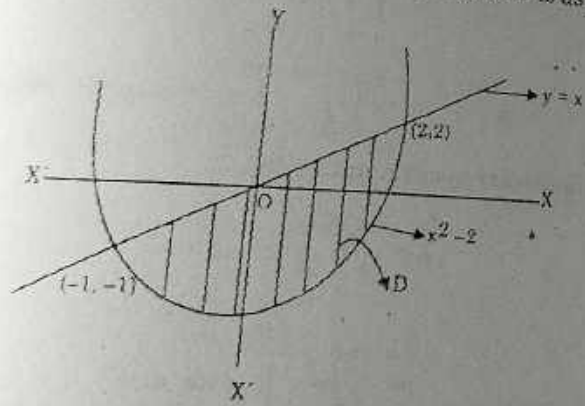
$$\begin{aligned} \text{Ans.} \quad y &= x^2 - 2 \\ y &= x \end{aligned}$$

Eq. (1) represents a parabola and Eq. (2) represents a straight line. The parabola and the line intersect, where :

$$\begin{aligned} x^2 - 2 &= x \\ x^2 - x - 2 &= 0 \\ x^2 - 2x + x - 2 &= 0 \\ x(x - 2) + 1(x - 2) &= 0 \\ (x - 2)(x + 1) &= 0 \\ x &= 2, -1 \end{aligned}$$

Thus, at $x = -1, y = x = -1$
at $x = 2, y = x = 2$

Hence, they intersect at $(-1, -1)$ and $(2, 2)$. The graph of D is as shown below :



The region D can be described as the set of all points (x, y) such that for each fixed x in $[-1, 2]$, y varies from $x^2 - 2$ to x . The area A of D is given by :

$$\begin{aligned} A &= \int_{-1}^2 \int_{x^2-2}^x dy dx \\ &= \int_{-1}^2 [y]_{x^2-2}^x dx \\ &= \int_{-1}^2 [(x) - (x^2 - 2)] dx \\ &= \int_{-1}^2 (x - x^2 + 2) dx \\ &= \left[\frac{x^2}{2} - \frac{x^3}{3} + 2x \right]_{-1}^2 \\ &= \frac{9}{2} \end{aligned}$$

Ans.

Q. 8. Write an equivalent integral with the order of integration reversed and then compute the integral :

$$\int_0^1 \int_0^{1-x} xy dy dx$$

Ans. Consult text book

Q. 9. Calculate the Jacobian of transformation from rectangular to polar coordinates and hence evaluate the integral :

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{1}{\sqrt{9-x^2-y^2}} dx dy$$

Ans. Jacobian of transformation from rectangular to polar coordinates :

Let f be a continuous function on a region D in the xy -plane and let T be a one-to-one transformation that maps the region G in the uv -plane into D under the change of variable $x = g(u, v)$, $y = h(u, v)$ for the functions g and h , continuously differentiable function f .

Then,

$$\iint_D f(x, y) dy dx = \iint_G f(g(u, v), h(u, v)) |J(u, v)| du dv \quad \dots (1)$$

where, $J(u, v)$ is called Jacobian and is given by :

$$\begin{aligned} \frac{\partial(x, y)}{\partial(u, v)} &= \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \\ &= \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \\ &= \frac{\partial(x, y)}{\partial(u, v)} \end{aligned} \quad (2)$$

We can change the variables from rectangular to polar by making substitution $x = r \cos \theta$, $y = r \sin \theta$. Then, the Jacobian of the change of variable

$$\begin{aligned} \frac{\partial(x, y)}{\partial(r, \theta)} &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} \\ &= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} \\ &= r \cos^2 \theta + r \sin^2 \theta \\ &= r \end{aligned} \quad (3)$$

$$\text{Then, } \iint_D f(x, y) dx dy = \iint_C f(r \cos \theta, r \sin \theta) r dr d\theta \quad (4)$$

Q. 10. Find the volume V of the solid bounded above by the cylinder $y^2 + z = 4$ and below $x^2 + 3y^2 = z$.

Ans. Do your self

Q. 11. Evaluate the integral below, where D is the region bounded above by the sphere $x^2 + y^2 + z^2 = 2$ and below by the paraboloid $z = x^2 + y^2$:

$$\iiint_D z dx dy dz$$

Ans. Let D is the region bounded above by the sphere $x^2 + y^2 + z^2 = 2$ and below by the paraboloid $z = x^2 + y^2$.

$$\text{Thus: } x^2 + y^2 + z^2 = 2$$

$$z = x^2 + y^2 \quad (1)$$

Using (2), Eq. (1) can be solved as:

$$z + z^2 = 2$$

$$z^2 + z - 2 = 0$$

$$z^2 + 2z - z - 2 = 0$$

$$z(z+2) - 1(z+2) = 0$$

$$z = 1, -2$$

Since,

$$z = -2 \text{ rejected as } z \geq 0$$

$$z = 1$$

It follows that the projected region in the $x-y$ plane is the disc $x^2 + y^2 \leq 1$. Here, The region of D integration is D , where:

$$D = \{(x, y, z) \text{ s.t.}$$

$$x^2 + y^2 \leq z \leq \sqrt{2 - x^2 - y^2}$$

$$-\sqrt{1 - x^2} \leq y \leq \sqrt{1 - x^2} \text{ and } -1 \leq x \leq 1\}$$

$$I = \iiint_D z dz dy dx$$

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^{\sqrt{2-x^2-y^2}} z dz dy dx$$

It may seen that transforming to spherical co-ordinates may result in a less complicated integration, since the upper boundary of D has the simple form, $P = \sqrt{2}$, but the lower boundary $z = x^2 + y^2$ has the form,

$$z = x^2 + y^2 = r^2$$

$$P \cos \phi = (P \sin \phi)^2$$

$$P = \text{get its cosec } \phi$$

Now,

$$0 \leq \phi \leq \frac{\pi}{4}$$

But

$$0 \leq P \leq \sqrt{2}$$

$$0 \leq P \leq \cot \phi \operatorname{cosec} \phi \text{ for}$$

$$\frac{\pi}{4} \leq \phi \leq \frac{\pi}{2}$$

$$I = \int_0^{\pi/4} \int_0^{\cot \phi \operatorname{cosec} \phi} \int_0^{\sqrt{2}} (P \cos \phi)^2 P^2 \sin \phi dP d\phi d\theta$$

$$+ \int_{\pi/4}^{\pi/2} \int_0^{\cot \phi \operatorname{cosec} \phi} \int_0^{\sqrt{2}} (P \cos \phi)^2 P^2 \sin \phi dP d\phi d\theta$$

If we are cylindercial co-ordinator, then:

$$I = \int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^{\sqrt{2-r^2}} zr dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^{\sqrt{2}} r \left[\frac{z^2}{2} \right]_{r^2}^{\sqrt{2-r^2}} dr d\theta$$

$$\begin{aligned}
 &= \int_0^{2\pi} \int_0^1 r \left[\frac{z^2}{2} \right]_z^{\sqrt{2-r^2}} dr d\theta \\
 &= \int_0^{2\pi} \int_0^1 \frac{r}{2} [(2-r^2) - r^2] dr d\theta \\
 &= \frac{1}{2} \int_0^{2\pi} \left[r - \frac{r^3}{3} - \frac{r^3}{3} \right]_0^1 d\theta \\
 &= \frac{1}{2} \int_0^{2\pi} \left[r - \frac{2r^3}{3} \right]_0^1 d\theta \\
 &= \frac{1}{2} \int_0^{2\pi} \left[1 - \frac{2}{3} \right] d\theta \\
 &= \frac{1}{2} \int_0^{2\pi} \frac{1}{3} d\theta \\
 &= \frac{1}{6} \int_0^{2\pi} d\theta \\
 &= \frac{1}{6} [2\pi]_0^{2\pi} \\
 &= \frac{1}{6} \cdot 2\pi \\
 &= \frac{\pi}{3}
 \end{aligned}$$

Q. 12. Let D be the region in the xy plane that is bounded by the coordinate axes and the line $x+y=1$. Use the suitable change of variable to compute the integral:

$$\iint_D \left(\frac{x-y}{x+y} \right)^6 dy dx.$$

Ans.

$$x+y=1$$

$$I = \iint_D \left(\frac{x-y}{x+y} \right)^6 dy dx \quad \dots(1)$$

Let

$$u = x-y \quad \dots(2)$$

$$v = x+y \quad \dots(3)$$

from (3) + (4):

$$u+v = 2x \quad \dots(4)$$

$$x = \frac{u+v}{2}$$

from (3) - (4):

$$u-v = 2y$$

$$y = \frac{u-v}{2}$$

The Jacobian is:

$$J = \frac{\partial(x,y)}{\partial(u,v)}$$

$$= \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \frac{\partial}{\partial u} \left(\frac{u+v}{2} \right) \frac{\partial}{\partial v} \left(\frac{u+v}{2} \right) - \frac{\partial}{\partial u} \left(\frac{u-v}{2} \right) \frac{\partial}{\partial v} \left(\frac{u-v}{2} \right)$$

$$= \frac{1}{2} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

$$= \frac{1}{2} \times \frac{1}{2} - \left(-\frac{1}{2} \right) \times \frac{1}{2}$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

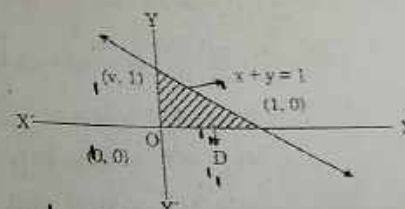


Fig. The region of integration D

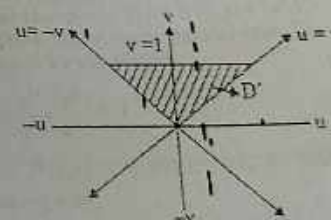


Fig. The transformed region D'

To find the image D' of D in the uv -plane note that the boundary lines $x=0$ and $y=0$ for D map into the lines $u=-v$ and $u=v$ respectively, while $x+y=1$ map into $v=1$. Therefore, the transformed region of integration D' is the triangular region shown in above figure with vertices $(0,0)$, $(1,1)$ and $(-1,1)$.

$$I = \iint_{D'} \left(\frac{x-y}{x+y} \right)^6 dy dx$$

$$= \int_0^1 \left[\frac{1}{2} u^2 \right]_0^1 du$$

$$= \frac{1}{2} \int_0^1 u^2 du$$

$$= \frac{1}{2} \left[\frac{u^3}{3} \right]_0^1$$

$$= \frac{1}{2} \left[\frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{3} \right]$$

Ans.

SECTION-III

Q. 13. State Green's Theorem for simply connected regions. Use Green's theorem to find the work done by the force field $F(x, y) = (e^x - y^2) i - (\cos y + x^2) j$ along the circle $x^2 + y^2 = 1$ in anticlockwise direction.

Ans. Green's Theorem for simply Connected Regions. Let D be a simply connected region that is bounded by the positively oriented piecewise smooth Jordan curves. Then if the vector field $\vec{F}(x, y) = M(x, y) i + N(x, y) j$ is continuously differentiable on D , we have:

$$\oint_C (M dx + N dy) = \iint_D \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

Q. 14. Give the geometrical interpretation of the surface integral $\iint_S xz \, ds$ over the surface S which is the part of the plane $x + y + z = 1$ that lies in the first octant.

Ans. Surface Integral. Any surface integral which is to be evaluated over a surface is called a surface integral. Suppose S is a surface of finite area. Suppose $f(x, y, z)$ is a single valued function of position defined over S . Sub divide the area S into elements of areas $\delta S_1, \delta S_2, \dots, \delta S_n$. In each part δS_i , we choose an arbitrary point P_i , whose co-ordinates are (x_i, y_i, z_i) . We define $f(P_i) = f(x_i, y_i, z_i)$. From the sum

$$\sum_{i=1}^n f(P_i) \delta S_i$$

Now, we take the sum limit of this sum as $n \rightarrow \infty$ in sub a way that the largest of the area δS_i approaches zero. This limit if it exists is called the surface integral of $f(x, y, z)$ over S and is denoted by,

$$\iint_S f(x, y, z) \, ds$$

It can be shown that if the surface S is piecewise smooth and the function $f(x, y, z)$ is continuous over S , then the above limit exists i.e., is independent of the choice of sub division and points P_i .

Q. 15. Verify Stokes' theorem for the vector field $F(x, y, z) = 2zi + 3xj + 5yk$ taking surface σ to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward orientation and C to be positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of σ in the xy -plane.

Ans. Consult text book

Q. 16. State and prove Divergence theorem.

Ans. Gauss's Divergence Theorem. Suppose V is the volume bounded by a closed piecewise smooth surface S . If $\vec{F}(x, y, z)$ is a vector function of position, which is continuous and his continuous first derivatives is v , then

$$\iiint_V \nabla \cdot \vec{F} \, dv = \iint_S \vec{F} \cdot \vec{n} \, ds$$

where $\vec{n}$ is the outwards drawn unit normal vector to S .

Proof:

Let

$$\vec{F} = F_1 i + F_2 j + F_3 k$$

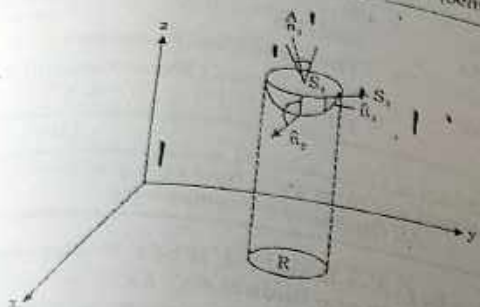
$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

We shall prove the theorem for a special region V which is bounded by a piecewise smooth closed surface S and has the property that any straight line parallel to any one of the co-ordinates axes and intersecting V has only one segment in common with V . If R is the orthogonal projection of S on the xy plane then V can be represented in the form,

$$f(x, y) \leq z \leq g(x, y)$$

where, (x, y) various in R

$z = g(x, y)$ represents the upper position S_1 of S . $z = f(x, y)$ represents the lower position S_2 of S and there may be a remaining vertical position $S_3 = S$.



We have,

$$\begin{aligned} \iiint_V \frac{\partial F_3}{\partial z} dv &= \iiint_V \frac{\partial F_3}{\partial z} dx dy dz \\ &= \iint_R \left[\int_{z=f(x,y)}^{g(x,y)} \frac{\partial F_3}{\partial z} dz \right] dx dy \\ &= \iint_R [F_3(x,y,z)]_{z=f(x,y)}^{z=g(x,y)} dx dy \\ &= \iint_R [F_3(x,y,g(x,y)) - F_3(x,y,f(x,y))] dx dy \\ &= \iint_R F_3(x,y,g(x,y)) dx dy - \iint_R F_3(x,y,f(x,y)) dx dy \quad \dots (2) \end{aligned}$$

Now, for the vertical position S_3 of S , the normal $\hat{u}_3$ to S_3 makes a right angle with k , therefore:

$$\iint_{S_3} F_3 k \cdot \hat{u}_3 dS_3 = 0$$

For the upper position S_1 of S the normal $\hat{u}_1$ to S_1 makes an acute angle y_1 with k . Therefore,

$$k \cdot \hat{u}_1 dS_1 = \cos Y_1 dS_1 = dx dy$$

$$\iint_{S_1} F_3 k \cdot \hat{u}_1 dS_1 = \iint_R [F_3(x,y,g(x,y))] dx dy$$

For the lower position S_2 of S the normal $\hat{u}_2$ to S_2 makes an obtuse angle y_2 with k . Therefore,

$$k \cdot \hat{u}_2 dS_2 = \cos y_2 dS_2 = -dx dy$$

Hence,

$$\iint_{S_2} F_3 k \cdot \hat{u}_2 dS_2 = -\iint_R F_3(x,y,g(x,y)) dx dy$$

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$$\begin{aligned} \iint_{S_1} F_3 k \cdot \hat{u}_1 dS_1 + \iint_{S_2} F_3 k \cdot \hat{u}_2 dS_2 + \iint_{S_3} F_3 k \cdot \hat{u}_3 dS_3 &= 0 \\ &= \iint_R F_3(x,y,g(x,y)) dx dy - \iint_R F_3(x,y,f(x,y)) dx dy \end{aligned}$$

$$\iint_S F_3 k \cdot \hat{u} ds = \iiint_V \frac{\partial F_3}{\partial z} dV \quad \dots (3)$$

Similarly by projecting S on the other co-ordinate planes, we get

$$\iint_S F_1 j \cdot \hat{n} ds = \iiint_V \frac{\partial F_1}{\partial x} dV \quad \dots (4)$$

$$\iint_S F_2 i \cdot \hat{n} ds = \iiint_V \frac{\partial F_2}{\partial y} dV \quad \dots (5)$$

On adding (3), (4) and (5), we get:

$$\iint_S (F_1 i + F_2 j + F_3 k) \cdot \hat{n} ds = \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dV$$

$$\iint_S (F_1 i + F_2 j + F_3 k) \cdot \hat{n} ds = \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dV$$

$$\iint_S (\vec{F} \cdot \hat{n}) ds = \iiint_V (\nabla \cdot \vec{F}) dV$$

$$\iint_S (\vec{F} \cdot \hat{n}) ds = \iiint_V (\nabla \cdot \vec{F}) dV$$

$$\iint_S (\nabla \cdot \vec{F}) dV = \iint_S (\vec{F} \cdot \hat{n}) ds$$

Which is Gauss's divergence theorem.

Q. 17. Verify that the vector field $\vec{F}(x,y) = (e^x \sin y - y) i + (e^x \cos y - x - 2) j$ is conservative using cross partial test. Use a line integral to find the area enclosed by the ellipse.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{Ans. } \vec{F}(x,y) = (e^x \sin y - y) i + (e^x \cos y - x - 2) j \quad \dots (1)$$

On comparing above with

$$\vec{F}(x,y) = u(x,y) i + v(x,y) j \quad \dots (2)$$

We get:

$$u(x,y) = e^x \sin y - y \quad \dots (3)$$

$$v(x,y) = e^x \cos y - x - 2 \quad \dots (4)$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y} (e^x \sin y - y) \\ &= e^x \cos y - 1 \end{aligned} \quad \dots (5)$$

$$\begin{aligned}\frac{\partial v}{\partial x} &= \frac{\partial}{\partial x}(e^x \cos y - x - 2) \\ &= e^x \cos y - 1 - 0 \\ &= e^x \cos y - 1\end{aligned}\quad \dots(6)$$

From (5) and (6), we have:

$$\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \quad \dots(7)$$

By cross partial test, we can say that $\vec{F}$ is conservative.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Equation of ellipse be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots(1)$$

The elliptical path E is given parametrically by $x = a \cos \theta$, $y = b \sin \theta$; $0 \leq \theta \leq 2\pi$.

We have:

$$\begin{aligned}dx &= -a \sin \theta d\theta \\ dy &= b \cos \theta d\theta\end{aligned}$$

If A is the area of the ellipse, then:

$$\begin{aligned}A &= \frac{1}{2} \oint_C (-y dx + x dy) \\ &= \frac{1}{2} \int_0^{2\pi} [-(b \sin \theta)(-a \sin \theta d\theta) + (a \cos \theta)(b \cos \theta d\theta)] \\ &= \frac{1}{2} \int_0^{2\pi} ab (\sin^2 \theta + \cos^2 \theta) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} ab d\theta \\ &= \frac{1}{2} ab (2\pi - 0) \\ &= \pi ab\end{aligned}$$

Q. 18. Let E be the solid unit cube with opposing corners at the origin and $(1, 1, 1)$ with faces parallel to co-ordinate planes. Let S be the boundary surface of E oriented with the outward pointing normal. If $F(x, y, z) = 2xyi + 3yzj + xz \sin^{-1} t$, find the integral $\iint_S F \cdot n ds$ over surface S using divergence theorem.

Ans. Consult text book.

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Sr. No. of Q. Paper : 90
Unique Paper Code : 32351301
Name of the Paper : Theory of Real Functions
Name of the course : B. Sc. (Hons.) Mathematics II Year
Semester : III

Maximum Marks : 75

Duration : 3 Hours

Attempt any three parts from each question.
All Questions are compulsory.

Q. 1. (a) Let $A \subseteq \mathbb{R}$, $f: A \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$ be a cluster point of A . Then prove that f can have only one limit at c .

Ans. We know that let $A \subseteq \mathbb{R}$ and let c be a cluster point of A . For a function $f: A \rightarrow \mathbb{R}$, a real number L is said to be a limit of f at c , if given any $\epsilon > 0$, there exist a $\delta > 0$ such that if $x \in A$ and $0 < |x - c| < \delta$, then $|f(x) - L| < \epsilon$.

Suppose that the numbers L and L' satisfy above definition.

For any $\epsilon > 0$, there exists $\delta (\epsilon/2) > 0$ such that if $x \in A$ and $0 < |x - c| < \delta$

($\epsilon/2$), then $|f(x) - L| < \frac{\epsilon}{2}$.

Also, there exist $\delta' (\epsilon/2)$ such as that if $x \in A$ and $0 < |x - c| < \delta' (\epsilon/2)$, then

$|f(x) - L'| < \frac{\epsilon}{2}$.

Now, let $\delta = \inf \{ \delta (\epsilon/2), \delta' (\epsilon/2) \}$.

Then, if $x \in A$ and $0 < |x - c| < \delta$, the triangle Inequality implies that

$$|L - L'| \leq |L - f(x)| + |f(x) - L'| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Since, $\epsilon > 0$ is arbitrary, we conclude that

$$L - L' = 0$$

$$\Rightarrow L = L'$$

which proved that f can have one limit at c .

(b) Use the $\epsilon - \delta$ definition of the limit to prove that $\lim_{x \rightarrow c} x^3 = c^3$ for

any $c \in \mathbb{R}$.

Ans. Let, $\epsilon > 0$ we given.

To find : $\delta > 0$, s.t. if, $0 < |x - c| < \delta$, then $|x^3 - c^3| < \epsilon$

Consider, $|x^3 - c^3| = |(x - c)(x^2 + cx + c^2)|$

$$= |x - c|, |x^2 + cx + c^2| \quad \dots(i)$$

if, $|x - c| < 1$... (ii)

$$|x| = |(x - c) + c|$$

(217)

now,

$$\begin{aligned} |x^2 + cx + c^2| &\leq |x|^2 + |cx| + |c^2| \\ &= |x^2| + |c| |x| + |c^2| \\ &\leq (1 + |c| + 1)^2 + |c| (1 + |c|) + |c^2| \\ &\leq 1 + 3|c| + 3|c| = k \text{ (let)} \end{aligned}$$

using (i) and (ii),

$$\begin{aligned} |x^3 - c^3| &= |x-c| |x^2 + cx + c^2| \\ &< K |x-c| < \epsilon, \text{ when } |x-c| < \frac{\epsilon}{K} \end{aligned}$$

$$\text{Now, } |x^3 - c^3| < \epsilon \text{ if } K|x-c| < \epsilon, |x-c| < \frac{\epsilon}{K}$$

$$\therefore \delta = \min. \left\{ 1, \frac{\epsilon}{K} \right\}$$

$$\text{since, } 0 < |x-c| < \delta$$

$$\Rightarrow |x^3 - c^3| < \epsilon$$

$$\therefore \lim_{x \rightarrow c} x^3 = c^3$$

(c) State divergence criterion for limit of a function. Show that $\lim_{x \rightarrow 0} (x + \operatorname{sgn}(x))$ does not exist.

Ans. Divergence Criteria for limit of a function:

Let, $A \subseteq \mathbb{R}$, let $f: A \rightarrow B$ and let $c \in \mathbb{R}$ be a cluster point of A .

(a) If $L \in \mathbb{R}$, then f does not have limit L at c , if and only if there exists a sequence (x_n) in A with $x_n \neq c$ for all $n \in \mathbb{N}$ such that the sequence (x_n) converges to c , but the sequence $(f(x_n))$ does not converge to L .

(b) The function f does not have a limit at c if and only if, there exists a sequence (x_n) in A with $x_n \neq c$ for all $n \in \mathbb{N}$ such that the sequence (x_n) converges to c , but the sequence $(f(x_n))$ does not converge in $\mathbb{R}$.

Let the Σ function sgn be defined by

$$\operatorname{sgn}(x) = \begin{cases} +1 & \text{for } x > 0 \\ +0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases}$$

$$\text{Hence, } \operatorname{sgn}(x) = \frac{x}{|x|} \text{ for } x \neq 0.$$

We shall show that sgn does not have a limit at $x = 0$. We shall do this by showing that there is a sequence (x_n) such that $\lim_{n \rightarrow \infty} x_n = 0$ but such that $\operatorname{sgn}(x_n)$ does not converge.

$$\text{Let, } (x_n) = \frac{(-1)^n}{n} \text{ for } n \in \mathbb{N}, \text{ so that } \lim_{n \rightarrow \infty} x_n = 0$$

However, since, $\operatorname{sgn}(x_n) = (-1)^n$ for $n \in \mathbb{N}$

It follows that $(\operatorname{sgn}(x_n))$ does not converge. Therefore, $\lim_{x \rightarrow 0} \operatorname{sgn}(x)$ does not exist.

(d) Prove that:

$$(i) \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$

$$(ii) \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$$

$$\text{Ans (i) } \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$

Let, $M > 0$ be any large number

$$\text{Let, } \delta = \frac{1}{M}$$

$$\text{Then, } 0 < x < \delta \Rightarrow \frac{1}{x} > \frac{1}{\delta} = \frac{1}{\frac{1}{M}} = M$$

$$\text{We know that the expression } \lim_{x \rightarrow 0} f(x) = \infty$$

means given any $M > 0$ (however large) there exist some $\delta > 0$ such that $f(x) > M$ when $|x - a| < \delta$.

$$\text{Hence, by above definition, we have } \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty.$$

$$(ii) \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$$

$$\text{We have, def: The expression } \lim_{x \rightarrow \infty} f(x) = l$$

means given any $\epsilon > 0$, there exist some $M > 0$ such that

$$|f(x) - l| < \epsilon, \text{ when } x > M$$

$$\text{Let, } \epsilon > 0 \text{ the given. Choose } M = \frac{1}{\epsilon^2}.$$

then, $x^2 > M$

$$\Rightarrow \left| \frac{1}{x^2} \right| < M = \varepsilon$$

$$\Rightarrow \left| \frac{1}{x^2} - 0 \right| < \varepsilon$$

Hence, using the above definition, we have $\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$.

Q. 2. (a) Let $A \subseteq \mathbb{R}$, $f, g, h: A \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$ be a cluster point of A . If $f(x) \leq g(x) \leq h(x)$ for all $x \in A, x \neq c$ and if $\lim_{x \rightarrow c} f(x) = L = \lim_{x \rightarrow c} h(x)$, then prove that $\lim_{x \rightarrow c} g(x) = L$.

Ans. Squeeze Theorem: Let $A \subseteq \mathbb{R}$, $f, g, h: A \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$ be a cluster point of A . If $f(x) \leq g(x) \leq h(x)$ for all $x \in A, x \neq c$ and if $\lim_{x \rightarrow c} f(x) = L = \lim_{x \rightarrow c} h(x)$,

then $\lim_{x \rightarrow c} g(x) = L$.

Proof: $\lim_{x \rightarrow c} f(x) = L = \lim_{x \rightarrow c} h(x)$

since, $\lim_{x \rightarrow c} f(x) = L$ for $\varepsilon > 0 \exists \delta_1 > 0$ s.t.

$|f(x) - L| < \varepsilon$, whenever $0 < |x - c| < \delta_1$

Since, $\lim_{x \rightarrow c} h(x) = L$ for $\varepsilon > 0 \exists \delta_2 > 0$ s.t.

$|h(x) - L| < \varepsilon$ whenever $0 < |x - c| < \delta_2$

Choose, $\delta = \min\{\delta_1, \delta_2\}$

From (i), (ii) and (iii), we get:

$L - \varepsilon < f(x) < L + \varepsilon$, whenever $0 < |x - c| < \delta$

and, $L - \varepsilon < h(x) < L + \varepsilon$, whenever $0 < |x - c| < \delta$

and, $f(x) \leq g(x) \leq h(x)$

From (iv), (v) and (vi), we get:

$\Rightarrow L - \varepsilon < f(x) \leq g(x) \leq h(x) < L + \varepsilon$ whenever $0 < |x - c| < \delta$

$\Rightarrow L - \varepsilon < g(x) < L + \varepsilon$ Whenever $0 < |x - c| < \delta$

$\Rightarrow \lim_{x \rightarrow c} g(x) = L$ Proved.

(b) State and prove sequential criterion for continuity of a real valued function.

Ans. Sequential criterion for continuity
A function $f: A \rightarrow \mathbb{R}$ is continuous at the point $c \in A$ if and only if for every sequence (x_n) in A that converges to c , the sequence $(f(x_n))$ converges to $f(c)$.

Proof: Consult text book.

(c) Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 2x & : \text{if } x \text{ is rational} \\ x+3 & : \text{if } x \text{ is irrational} \end{cases}$$

Find all the points at which f is continuous.

Ans. Consult text book.

(d) Let $x \rightarrow [x]$ denote the greatest integer function. Determine the points of continuity of the function $f(x) = x - [x]$, $x \in \mathbb{R}$.

Ans. The given function $f(x)$ is defined as:

$$f(x) = x - [x]$$

by definition of $[x]$, we have

$$[x] = 0$$

$$[x] = 1$$

$$[x] = 2$$

$$[x] = -1$$

$$[x] = -2$$

$$\text{for } 0 \leq x < 1$$

$$\text{for } 1 \leq x < 2$$

$$\text{for } 2 \leq x < 3$$

$$\text{for } -1 \leq x < 0$$

$$\text{for } -2 \leq x < -1 \text{ etc.}$$

Hence, the function $f(x)$ is defined as

$$f(x) = x - [x] = x - 0 = x \quad ; 0 \leq x < 1$$

$$f(x) = x - [x] = x - 1 \quad ; 1 \leq x < 2$$

$$f(x) = x - [x] = x - 2 \quad ; 2 \leq x < 3$$

$$f(x) = x - [x] = x - (-1) = x + 1 \quad ; -1 \leq x < 0$$

$$f(x) = x - [x] = x - (-2) = x + 2 \quad ; -2 \leq x < -1$$

At $x = 0$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x)$$

$$= \lim_{h \rightarrow 0^+} (0+h) = \lim_{h \rightarrow 0^+} (h) = 0$$

$$\lim_{h \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x+1)$$

$$= \lim_{x \rightarrow 0} (x+1)$$

$$= \lim_{h \rightarrow 0} (-h+1)$$

$$= 0+1$$

$$= 1$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$$

$\Rightarrow f(x)$ is discontinuous at $x = 1$.

At $x = 1$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x-1)$$

$$= \lim_{h \rightarrow 0} (1+h)-1$$

$$= \lim_{h \rightarrow 0} h = 0$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x)$$

$$= \lim_{h \rightarrow 0} (1-h)$$

$$= 1-0=1$$

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$$

$\Rightarrow f(x)$ is a discontinuity at $x = 1$.

At $x = -1$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} (x+1)$$

$$= \lim_{h \rightarrow 0} (-1+h)+1$$

$$= \lim_{h \rightarrow 0} h = 0$$

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} (x+2)$$

$$= \lim_{h \rightarrow 0} (-1-h)+2$$

$$= \lim_{h \rightarrow 0} (1-h) = 1-0$$

$$= 1-0=1$$

$$\Rightarrow \lim_{x \rightarrow -1^+} f(x) \neq \lim_{x \rightarrow -1^-} f(x)$$

$\Rightarrow f(x)$ is discontinuous at $x = -1$.

similarly, we can prove that $f(x)$ is discontinuous at $x = \pm 2, \pm 3, \dots$

It may be noted that $f(x) = 1$ for all $x \in \mathbb{R}$ except at the point $0, \pm 1, \pm 2, \pm 3, \dots$

3. (a) Let f be a continuous real valued function defined on $[a, b]$. By assuming that f is a bounded function show that f attains its bound on $[a, b]$.

Ans. Since, f is continuous in $[a, b]$, therefore, it is bounded in $[a, b]$.

Let, $M = \sup f$ and $m = \inf f$

$$m \leq f(x) \leq M \quad \forall x \in [a, b]$$

We shall show that there exist points, $\alpha, \beta \in [a, b]$ such that

$$f(\alpha) = M \text{ and } f(\beta) = m$$

we shall prove it by contradiction.

Let, $f(x) \neq M \quad \forall x \in [a, b]$

$$\Rightarrow M - f(x) \neq 0 \quad \forall x \in [a, b]$$

Since f is given to be continuous in $[a, b]$ and M , being a constant function, is also continuous in $[a, b]$, therefore, $M - f(x)$ is continuous in $[a, b]$.

Since, $M - f(x) \neq 0 \quad \forall x \in [a, b]$, therefore:

$$\frac{1}{M - f(x)} \text{ is a continuous in } [a, b]$$

$$\Rightarrow \frac{1}{M - f(x)} \text{ is bounded in } [a, b]$$

$$\text{Let, } k = \sup \frac{1}{M - f(x)}$$

$$k = \inf \frac{1}{M - f(x)}$$

$$\Rightarrow \frac{1}{M - f(x)} \leq k \quad \forall x \in [a, b]$$

$$\Rightarrow M - f(x) \geq \frac{1}{k} \quad \forall x \in [a, b]$$

$$\Rightarrow M - \frac{1}{k} \geq f(x) \quad \forall x \in [a, b]$$

$$f(x) \leq M - \frac{1}{k} \quad \forall x \in [a, b]$$

$M - \frac{1}{k} (< M)$ is an upper bound off which contradicts the fact that M is the l.u.b. of f .

Hence, there exists some $\alpha \in [a, b]$ such that $f(\alpha) = M$.

Similarly, there exist some $\beta \in [a, b]$ such that $f(\beta) = m$.

Hence, we proved that f is continuous on a closed bounded interval $[a, b]$, then it attains its bound in $[a, b]$.

(b) State Bolzano's intermediate value theorem and show that the function $f(x) = xe^x - 2$ has a root ϵ in the interval $[0, 1]$.

Ans. Bolzano's Intermediate Theorem:

If a function $f(x)$ is continuous in $[a, b]$ and $f(a) \neq f(b)$, then $f(x)$ assumes every value between $f(a)$ and $f(b)$.

Do yourself.

(c) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $\mathbb{R}$ and suppose that $f(r) = 0$ for every rational number r . Show that $f(x) = 0$ for all $x \in \mathbb{R}$.

Ans. Consult text book.

(d) Define uniform continuity of a function. Prove that if a function is continuous on a closed and bounded interval I , then it is uniformly continuous on I .

Ans. Uniform Continuity of a function: A function f defined on an interval I is said to be uniformly continuous in the interval I , if for each $\epsilon > 0$, there exists some $\delta > 0$ such that

$$|f(x_1) - f(x_2)| < \epsilon$$

When $|x_1 - x_2| < \delta$ and for all $x_1, x_2 \in I$.

Solution of the given problem:

If f is not uniformly continuous on I , then there exist $\epsilon > 0$ and two sequence

$\langle x_n \rangle$ and $\langle u_n \rangle$ in I such that $|x_n - u_n| < \frac{1}{n}$ and $|f(x_n) - f(u_n)| > \epsilon$, to for all $n \in \mathbb{N}$.

Since, I is bounded, the sequence $\langle x_n \rangle$ is bounded. By Bolzano Weierstrass Theorem, there is a subsequence $\langle x_{n_k} \rangle$ of $\langle x_n \rangle$ that converges to an element z . Since, I is closed, the limit z belongs to I . It is clear that the corresponding subsequence $\langle u_{n_k} \rangle$ also converges to z , since,

$$|u_{n_k} - z| \leq |u_{n_k} - x_{n_k}| + |x_{n_k} - z|$$

Now, if f is continuous at the point z , then both of the sequence $\langle f(x_{n_k}) \rangle$ and $\langle f(u_{n_k}) \rangle$ must converge to $f(z)$. But, this is not possible, since,

$$|f(x_{n_k}) - f(u_{n_k})| \geq \epsilon$$

for all $k \in \mathbb{N}$. Thus the hypothesis, that f is not uniformly continuous on the closed bounded interval I implies that f is not continuous at some point $z \in I$. Consequently, if f is continuous at every point of I , then f is uniformly continuous on I .

4. (a) Show that the function $f(x) = 1/x^2$ is uniformly continuous on $B =]0, \infty[$ but it is not uniformly continuous on $B =]0, \infty[$.

$$(i) f(x) = \frac{1}{x^2}$$

For $x, y \geq a > 0$, we have

$$|f(x) - f(y)| = \left| \frac{1}{x^2} - \frac{1}{y^2} \right|$$

$$= \left| \left(\frac{1}{x} - \frac{1}{y} \right) \left(\frac{1}{x} + \frac{1}{y} \right) \right|$$

$$= \left| \frac{1}{x} - \frac{1}{y} \right| \left| \frac{1}{x} + \frac{1}{y} \right|$$

$$= \left| \frac{y-x}{xy} \right| \left| \frac{1}{x} + \frac{1}{y} \right| = \frac{|y-x|}{|xy|} \left| \frac{1}{x} + \frac{1}{y} \right|$$

$$\leq \frac{2}{a} \frac{|x-y|}{xy} \quad \text{as } x, y \geq a > 0$$

$$\leq \frac{2}{a^3} |x-y|$$

because $\therefore x \geq a, y \geq a$

$$\Rightarrow xy \geq a^2$$

$$\Rightarrow \frac{1}{xy} \leq \frac{1}{a^2}$$

Let, $\epsilon > 0$ be given. Let $\delta = \epsilon a^3/2$.

Then, $|f(x) - f(y)| < \epsilon$, unliar $|x-y| < \delta \forall x, y \geq a$

Hence, f is uniformly continuous on $[a, \infty[$.

$$(ii) f(x) = \frac{1}{x^2}$$

Let, $\epsilon = \frac{1}{2}$ and δ be any positive number. We can always choose a positive integer in such that:

$$n > \frac{1}{2\delta}$$

$$\frac{1}{2n} < \delta$$

Let,

$$x_1 = \frac{1}{\sqrt{n}}$$

$$x_2 = \frac{1}{\sqrt{n+1}} \in (0, \alpha). \text{ Then,}$$

$$\begin{aligned} |f(x_1) - f(x_2)| &= \left| \frac{1}{x_1^2} - \frac{1}{x_2^2} \right| \\ &= |n - (n+1)| = 1 > \epsilon \end{aligned}$$

$$\begin{aligned} \text{and } |x_1 - x_2| &= \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \\ &= \frac{(\sqrt{n+1} - \sqrt{n})}{\sqrt{n}\sqrt{n+1}} \\ &= \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n}\sqrt{n+1}(\sqrt{n+1} + \sqrt{n})} \\ &= \frac{(\sqrt{n+1})^2 - (\sqrt{n})^2}{\sqrt{n}\sqrt{n+1}(\sqrt{n+1} + \sqrt{n})} \\ &= \frac{(n+1) - n}{\sqrt{n}\sqrt{n+1}(\sqrt{n+1} + \sqrt{n})} \end{aligned}$$

$$= \frac{1}{\sqrt{n}\sqrt{n+1}(\sqrt{n+1} + \sqrt{n})}$$

$$< \frac{1}{\sqrt{n}\sqrt{2n}} \text{ because } \sqrt{n}\sqrt{n+1} > \sqrt{n}$$

$$\sqrt{n}\sqrt{n+1} > \sqrt{n} = \frac{1}{2\epsilon}$$

$$= < \delta, \text{ using (i)}$$

$$= |f(x_1) - f(x_2)| > \epsilon, \text{ when } |x_1 - x_2| < \delta$$

Hence, f is not uniformly continuous on $[0, \infty)$.

(b) Determine where the following function $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable, $f(x) = |x-1| + |x+1|$.

$$\text{Ans. } f(x) = |x-1| + |x+1|$$

$$\begin{array}{cc} I & II \\ x < -1 & x > 1 \end{array}$$

For $x < -1$

$$\begin{aligned} f(x) &= |x+1| + |x-1| \\ &= -(x+1) - (x-1) \\ &= -x-1-x+1 \\ &= -2x \end{aligned}$$

For $-1 < x < 1$

$$\begin{aligned} f(x) &= |x+1| + |x-1| \\ &= (x+1) - (x-1) \\ &= x+1-x+1 \\ &= 2 \end{aligned}$$

For $x > 1$

$$\begin{aligned} f(x) &= |x+1| + |x-1| \\ &= (x+1) + (x-1) \\ &= 2x \end{aligned}$$

At $x = -1$

$$\begin{aligned} f(x) &= |x+1| + |x-1| \\ &= |-1+1| + |-1-1| \\ &= |0| + |-2| \\ &= 0+2 \\ &= 2 \end{aligned}$$

At $x = 1$

$$f(x) = (x+1) + (x-1)$$

$$= (1+1) + (1-1)$$

$$= (2) + (0)$$

$$= 2+0$$

$$= 2$$

Hence, we have:

$$f(x) = -2x$$

$$= 2$$

$$= 2x$$

$$\text{At } x = -1$$

$$Rf'(-1) = \lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)}$$

$$= \lim_{x \rightarrow -1} \frac{2-2}{x+2}$$

$$= \lim_{x \rightarrow -1} \frac{0}{x+1}$$

$$= 0$$

$$Lf' = \lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)}$$

$$= \lim_{x \rightarrow -1} \frac{(-2x) - (-2)}{x+1}$$

$$= \lim_{x \rightarrow -1} \frac{-2(x+1)}{(x+1)}$$

$$= -2$$

$$\Rightarrow Rf'(-1) \neq Lf'(-1)$$

$\Rightarrow f(x)$ is not derivable at $x = -1$.

$$\text{At } x = 1$$

$$Rf'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{2x-2}{x-1}$$

$$; x \leq -1 \dots (i)$$

$$; -1 \leq x \leq 1 \dots (ii)$$

$$; x \geq 1 \dots (iii)$$

$$= \lim_{x \rightarrow 1} \frac{2(x-2)}{(x-1)}$$

$$= 2$$

$$Lf'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{2-2}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{0}{x-1}$$

$$= 0$$

$$\Rightarrow Rf'(1) \neq Lf'(1)$$

$\Rightarrow f(x)$ is not derivable at $x = 1$.

From (i):

$$f'(x) = -2; x < -1$$

Form (ii)

$$f'(x) = 0; -1 < x < 1$$

From (iii)

$$f'(x) = 2; x > 1$$

Hence, $f(x)$ is derivative or all x except $x = -1$ and $x = 1$.

(c) Let f be defined on an interval I containing the point c . Then prove that f is differentiable at c if and only if there exist a function ϕ on I that is continuous at c and satisfies $f(x) - f(c) = \phi(c)(x-c)$ for all $x \in I$. In this case, we have $\phi(c) = f'(c)$. Using the above result find the function ϕ for $f(x) = x^3, x \in \mathbb{R}$.

Ans. Caratheodory's Theorem :

Let f defined on an interval I containing the point c , then f is differentiable at c if and only if there exists a function ϕ on I that is continuous at c and satisfies

$$f(x) - f(c) = \phi(x)(x-c) \text{ for } x \in I$$

In this case, we have, $\phi(c) = f'(c)$.

Proof: Condition is necessary

Let f be defined on an interval I .

Let f is differentiable at 1 .

$$f(x) - f(c) = \phi(x)(x-c)$$

If $f'(c)$ exist, we can define ϕ by,

$$\phi(x) = \frac{f(x) - f(c)}{x - c} \text{ for } x \neq c, x \in I$$

$$= f'(c) \text{ for } x = c$$

The continuity of ϕ follows from the fact that $\lim_{x \rightarrow c} \phi(x) = f'(c)$.

If $x = c$, then both sides of (1) equals 0, while if $x \neq c$, then multiplication of (1) by $(x - c)$ gives (1) for all other $x \in I$.
Condition is sufficient: Now assume that function ϕ is continuous at c and satisfying (1) exists.
If we divide (1) by $(x - c) \neq 0$, then the continuity of ϕ implies that

$$\begin{aligned} \phi(1) &= \lim_{x \rightarrow c} \phi(x) \\ &= \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \end{aligned}$$

exists. Therefore f is differentiable at c and $f'(c) = \phi(x)$.

Solution of given problem:

$$f(x) = x^3 \text{ for } x \in \mathbb{R}$$

For $c \in \mathbb{R}$ we see from the factorization:

$$x^3 - c^3 = (x^2 + cx + c^2)(x - c) \quad \dots(i)$$

$$\text{since, } f(x) - f(c) = \phi(x)(x - c)$$

$$x^3 - c^3 = \phi(x)(x - c) \quad \dots(ii)$$

$$(x^2 + cx + c^2)(x - c) = \phi(x)(x - c)$$

$$\phi(x) = x^2 + cx + c^2$$

Hence, $\phi(x)$ satisfies the condition of Caratheodory's Theorem.

Therefore, we conclude that f is differentiable at $c \in \mathbb{R}$ and

$$\begin{aligned} f'(x) &= \phi(x) \\ &= (c^2) + (cc) + 1 \\ &= c^2 + c^2 + c^2 = 3c^2 \end{aligned} \quad \dots(iii)$$

(d) State and prove Mean Value Theorem.

Ans. Lagrange's Mean Value Theorem:

Let, f be a function defined on $[a, b]$ such that

(i) f is continuous in $[a, b]$

(ii) f is derivable in $]a, b[$

then, there exist some $c \in]a, b[$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof:

Let $f(x)$ be a function defined on $[a, b]$, $f(x)$ is continuous on $[a, b]$ and $f(x)$ is derivable in $]a, b[$.

Let us define a function $\phi(x)$ on $[a, b]$.

$$\text{s.t. } \phi(x) = f(x) - Ax$$

Where, A is a constant to be chosen s.t.

$$\phi(a) = \phi(b) = 0$$

$$f(a) - Aa = f(b) - Ab$$

$$-Ab + Aa = f(a) - f(b)$$

$$-A(b-a) = f(b) - f(a)$$

$$A = \frac{f(b) - f(a)}{b - a}$$

Now, $\phi(x)$ satisfies the following conditions:

(i) $\phi(x)$ is continuous in $[a, b]$, since $f(x)$ and Ax are both continuous in $[a, b]$, i.e. $f(x) + Ax$ is continuous in $[a, b]$.

(ii) $\phi(x)$ is derivable in $]a, b[$, since $f(x)$ and Ax are both derivable in $]a, b[$, i.e. $f(x) + Ax$ is derivable in $]a, b[$.

(iii) $\phi(a) = \phi(b)$

Since, all the conditions of Rolle's Theorem are satisfied, there exists some $c \in]a, b[$ s.t.

$$\phi'(c) = 0$$

$$\text{From (i), } \phi(x) = f(x) - Ax$$

equation (iii) can be solved as

$$\phi'(x) = 0$$

$$f'(x) - A = 0$$

$$f'(c) = A$$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Which proves Lagrange's Mean Value Theorem.

Q. 5. (a) State Darboux's theorem. Suppose that $f: [0, 2] \rightarrow \mathbb{R}$ is continuous on $[0, 2]$ and differentiable on $]0, 2[$ and that $f(0) = 0$, $f'(1) = 1$, $f(2) = 1$. (i) Show that there exists $c_1 \in (0, 1)$ such that $f'(c_1) = 1$. (ii) Show that there exists $c_2 \in (1, 2)$ such that $f'(c_2) = 0$. (iii) Show that there exists $c \in (0, 2)$ such that $f'(c) = 1/2$.

Ans. Darboux's Theorem:

If a function f defined in $[a, b]$ is such that

- (i) f is derivable in (a, b)
 (ii) $f'(a)$ and $f'(b)$ are opposite signs then there exists some point $c \in (a, b)$ such that $f'(c) = 0$.

Solution of given problem

$$\begin{aligned} f(0) &= 0 \\ f(1) &= 1 \\ f(2) &= 1 \end{aligned}$$

Let $f(x) = ax^2 + bx + c$

At $x = 0$ Equation (ii) given:

$$f(0) = a(0)^2 + b(0) + c$$

$$0 = 0 + 0 + c$$

$$c = 0$$

At $x = 1$ Equation (ii) given:

$$f(1) = a(1)^2 + b(1) + c$$

$$1 = a + b + c$$

$$1 = a + b + 0$$

$$a + b = 1$$

At $x = 2$ Equation (ii) given:

$$f(2) = a(2)^2 + b(2) + c$$

$$f(2) = 4a + 2b + c$$

$$1 = 4a + 2b + 0$$

$$1 = 2(2a + b)$$

$$2a + b = \frac{1}{2}$$

From (iii.2) and (iii.3)

$$(2a + b) - (a + b) = \frac{1}{2} - 1$$

$$a = -\frac{1}{2}$$

from (iii.2)

$$a + b = 1$$

$$-\frac{1}{2} + b = 1$$

$$b = \frac{3}{2}$$

Hence, $a = -\frac{1}{2}, b = \frac{3}{2}, c = 0$

$$f(x) = ax^2 + bx + c$$

$$= -\frac{1}{2}x^2 + \frac{3}{2}x + 0$$

$$= -\frac{1}{2}x^2 + \frac{3}{2}x$$

$$f'(x) = -\frac{1}{2}(2x) + \frac{3}{2}(1)$$

$$= -x + \frac{3}{2} = \frac{3}{2} - x$$

(i) If, $f'(x) = 1$

$$\frac{3}{2} - x = 1$$

$$x = \frac{3}{2} - 1$$

$$x = \frac{1}{2}$$

$$\Rightarrow c_1 = \frac{1}{2} \in (0, 1) \forall f'(c_1) = 1 \text{ Answer.}$$

(ii) If, $f'(x) = 0$

$$\frac{3}{2} - x = 0$$

$$x = \frac{3}{2}$$

$$\Rightarrow c_2 = \frac{3}{2} \in (1, 2) \forall f'(c_2) = 0 \text{ Answer.}$$

(iii) If, $f'(x) = \frac{1}{10}$

$$\frac{3}{2} - x = \frac{1}{10}$$

$$x = \frac{3}{2} - \frac{1}{10}$$

$$x = \frac{15-1}{10}$$

$$x = \frac{14}{10}$$

$$x = \frac{7}{5}$$

$$c_3 = \frac{7}{5} \in (0, 2) \quad \forall f'(c_3) = \frac{1}{10}$$

(b) Let $f: \rightarrow \mathbb{R}$ be differentiable on the interval I . Then prove that f is increasing on I if and only if $f'(x) \geq 0$ for all $x \in I$.

Ans. (i) f is continuous on $[a, b]$

(ii) f is differentiable on (a, b)

Let, x_1 and x_2 be two any point of $[a, b]$ i.e. $x_1 < x_2$.

Let, $f'(x) > 0 \quad \forall x \in [a, b]$

Then, by Lagrange's Mean Value Theorem, we have, some $c \in (x_1, x_2)$ s.t.

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\Rightarrow f(x_2) - f(x_1) = (x_2 - x_1) f'(c)$$

since, $f'(c) > 0$ and $x_2 - x_1 > 0$

$$\Rightarrow f(x_2) - f(x_1) > 0$$

$$\Rightarrow f(x_2) > f(x_1) \quad \forall x_2 > x_1 : x_1, x_2 \in (a, b)$$

Hence, f is strictly increasing in (a, b) .

(c) State Taylor's theorem. Use it to prove that $1 - x^2/2 \leq \cos x$ for all $x \in \mathbb{R}$.

Ans. (Taylor's Theorem with Lagrange form of Remainder):

If a function $f(x)$ is such that

(i) $f(x), f'(x), f''(x), \dots, f^{(n-1)}(x)$ are continuous in $[a, a+h]$

(ii) $f^{(n)}(x)$ exists in $]a, a+h[$

then $\exists$ at least one value θ between 0 and 1 s.t.

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{h^n}{n!} f^{(n)}(a+\theta h)$$

Where, $\frac{h^n}{n!} f^{(n)}(a+\theta h)$ is called Lagrange's form of Remainder after n terms i.e.

$$R_n = \frac{h^n}{n!} f^{(n)}(a+\theta h)$$

Solution of the given problem:

$$\text{To prove: } 1 - \frac{x^2}{2} \leq \cos x : x \in \mathbb{R}$$

Case (1) Let $x = 0$

Then, there is nothing to prove because, the expression $1 - \frac{x^2}{2}$ and $\cos x$ have the value 1.

Case (2) Let $x > 1$

Applying Taylor's Theorem,

$$f(x) = f(0) + f'(0)x + \frac{x^2}{2!} f''(\theta x) : 0 < \theta < 1$$

putting $f(x) = \cos x$, we get

$$\cos x = \cos 0 + x(-\sin 0) + \frac{x^2}{2}(-\cos \theta x)$$

$$\cos x = 1 - 0 - \frac{x^2}{2} \cos(\theta x)$$

$$\cos x = 1 - \frac{x^2}{2} \cos(\theta x) : 0 < \theta < 1 \quad \dots (1)$$

We know that

$$\cos(\theta x) \leq 1 \quad \forall \cos(\theta x)$$

$$\Rightarrow -\frac{x^2}{2} \cos(\theta x) \geq -\frac{x^2}{2}$$

$$\Rightarrow 1 - \frac{x^2}{2} \cos(\theta x) \geq 1 - \frac{x^2}{2}$$

$$\cos x \geq 1 - \frac{x^2}{2}$$

$$1 - \frac{x^2}{2} \leq \cos x$$

Case (3): Let $x < 0$

Then, $-x > 0$ or $y > 0$, where $y = 0$, where $y = -x$

By case (2)

$$1 - \frac{y^2}{2} \leq \cos y \quad (y > 0)$$

putting $y = -x$ in above equation

$$1 - \frac{(-x)^2}{2} \leq \cos(-x)$$

$$1 - \frac{x^2}{2} \leq \cos x \quad \forall x \in \mathbb{R}$$

From (i) (ii) and (iii) we conclude.

$$1 - \frac{x^2}{2} \leq \cos x \text{ Proved.}$$

[We know that if $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = l$ and $|l| < 1$, then $\lim_{n \rightarrow \infty} a_n = 0$]

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{a_n}{n!} = 0$$

$$\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{x^n}{n!} e^{\theta x}$$

$$= \left(\lim_{n \rightarrow \infty} \frac{x^n}{n!} \right) e^{\theta x}$$

$$= 0 \times e^{\theta x}$$

$$= 0$$

Thus, the condition of Taylor series are satisfied.

For all $x \in \mathbb{R}$,
 $e^x = f(x)$

$$f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

$$= 1 + x(1) + \frac{x^2}{2!}(1) + \dots + \frac{x^n}{n!}(1) + \dots$$

$$= 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots$$

$$= \sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ Answer.}$$

(d) Find the Taylor series for e^x and state why it converges to e^x for all $x \in \mathbb{R}$.

Ans. Let $f(x) = e^x$

$$f^n(x) = e^x$$

$$f^n(0) = e^0 = 1$$

$$f^n(0) = e^0 = 1$$

Hence, f and all its derivations exist and are continuous for every real value of x . Lagrange's form of remainder i.e.

$$R_n = \frac{x^n}{n!} f^{(n)}(\theta x); 0 < \theta < 1 = \frac{x^n}{n!} e^{\theta x}$$

$$\text{Let } a_n = \frac{x^n}{n!} \quad \forall n \in \mathbb{N}$$

$$a_{n+1} = \frac{x^{n+1}}{(n+1)!}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{x^{n+1}}{(n+1)!} \times \frac{n!}{x^n}$$

$$= \lim_{n \rightarrow \infty} \frac{x^{n+1}}{(n+1)!} \times \frac{n!}{x^n}$$

$$= \lim_{n \rightarrow \infty} \frac{x}{n+1} = 0 < 1$$

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Maximum Marks : 75

Attempt any three parts from each question.
 All Questions are compulsory.

Q.1. (a) Define a group. Give an example of :

- (i) an abelian group consisting of eight elements,
- (ii) a non-abelian group consisting of six elements,
- (iii) an infinite abelian group, and
- (iv) an infinite non-abelian group.

Sol. Group : A non empty set G with binary composition $*$ is called a group, if the following conditions are satisfied :

G1 : Closure Law

$$a * b \in G \quad \forall a, b \in G$$

G2 : Associative law

$$a * (b * c) = (a * b) * c \quad \forall a, b, c \in G$$

G3 : Existence of identity

There exists an element $e \in G$ s.t.

$$a * e = e * a = a \quad \forall a \in G$$

Here, e is called identity in G .

G4 : Existence of inverse

There exist an element $a' \in G$ s.t.

$$a * a' = a' * a = e \quad \forall a \in G$$

Here, a' is called inverse of a and is written as $a = a^{-1}$

An abelian group consisting of eight elements

$$G = \{ \dots \}$$

(ii) A non abelian group consisting of six elements

An infinite abelian group

(iii) The set $(1, +)$ is an infinite abelian group

An infinite abelian group

(iv) The set S of all 2×2 non singular matrices w.r.t. multiplication of two matrices

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(b) Show that the set $\{5, 15, 25, 35\}$ is a group under multiplication modulo 40. What is the identity element of this group? Find the inverse of each element.

Sol. Let $G = \{5, 15, 25, 35\}$

The composition table of G w.r.t. $\odot_{40}$ is as follows : =

$\odot_{40}$	5	15	25	35
5	25	35	5	15
15	35	25	15	5
25	5	15	25	35
35	15	5	35	25

Hence, $G, \odot_{40}$ is a group.

- (i) Here, $5 \odot_{40} 25 = 5$
 $15 \odot_{40} 25 = 15$
 $25 \odot_{40} 25 = 25$
 $35 \odot_{40} 25 = 35$

Hence, identity element of G is 25. Answer

- (ii) $5 \odot_{40} 5 = 25$
 $15 \odot_{40} 15 = 25$
 $25 \odot_{40} 25 = 25$
 $35 \odot_{40} 25 = 25$

Inverse element of 5 = 15

Inverse element of 15 = 15

Inverse element of 25 = 25

Inverse element of 35 = 35

(c) Prove that the intersection of an arbitrary family of subgroup of a group G is again a subgroup of G . What can you say about the union of two subgroups? Justify your answer.

Sol. Let $\{H_\lambda : \lambda \in \Delta\}$ be an arbitrary family of subgroup of a group G . We shall show that $\bigcap_{\lambda \in \Delta} H_\lambda$ is subgroup of G . Obviously, $\bigcap_{\lambda \in \Delta} H_\lambda$ is non empty, since $e \in H_\lambda$ for each $\lambda \in \Delta$.

Let a, b be any two element of $\bigcap_{\lambda \in \Delta} H_\lambda$. Then, $a, b \in H_\lambda$ for each $\lambda \in \Delta$. Since, H_λ is a sub-group of G ; i.e.

$$ab^{-1} \in H_\lambda \text{ for each } \lambda \in \Delta$$

$$\text{Implies } ab^{-1} \in \bigcap_{\lambda \in \Delta} H_\lambda$$

Hence, $\bigcap_{\lambda \in \Lambda} H_\lambda$ is a subgroup of G .

(iii) The union of two subgroups of a group G need not to be a subgroup of G .
Let $G = (\mathbb{Z}, +)$, then

$$H = \{\dots, -4, -2, 0, 2, 4, \dots\}$$

and $K = \{\dots, -6, -3, 0, 3, 6, \dots\}$ are subgroups of G .

But, $U K = \{\dots, -6, -4, -3, -2, 0, 2, 3, 4, 6, \dots\}$

is not a subgroup of G , since $2, 3 \in H \cup K$ but $2+3=5 \notin H \cup K$.

Q.2. (a) (i) Prove that in $(\mathbb{Z}, +)$, the group of integers under addition, every non-zero element is of infinite order.

Sol. $(\mathbb{Z}, +)$ is the group of integers under addition.

(i) 0 is the identity element of $\mathbb{Z}$

$$1 \cdot 0 = 0$$

$$\Rightarrow 0(0) = 1$$

(ii) $na = 0 \forall n \in \mathbb{N}$ if $a = 0$ and $a \in \mathbb{Z}$

$$0(0) = x \forall a \in \mathbb{Z} \text{ s.t. } a \neq 0.$$

Which conclude that in $(\mathbb{Z}, +)$ the group of integers under addition, every non-zero element from is of infinite order. proved.

(ii) Let G be a group and $a \in G$. If $|a| = n$ and k is a positive divisor of n , then prove that $|a^{n/k}| = k$.

Sol. Consult text book.

(b) Prove that the order of a cyclic group is equal to the order of its generator.

Sol. Let a be a generator of a group $G = \langle a \rangle$ and

Let a be of finite order n so that

$$O(a) = n$$

$$a^n = e$$

$$a^r \neq e \text{ for } 0 < r < n.$$

To prove, that $O(G) = O(a)$ i.e. to prove $O(a) = n$

This will prove in two steps

Step (I)

To prove that G contains n elements.

The elements of the cyclic group G are given below :

$$a, a^2, a^3, a^4, \dots, a^n = e$$

If possible, let G contains an element a^m these elements, where $m > n$.

By division algorithm,

$$m = nq + r \quad \forall 0 < r < n \text{ and } q, r \in \mathbb{N}$$

$$a^m = a^{nq+r}$$

$$= a^{nq} a^r$$

$$= (a^n)^q a^r$$

$$= (e)^q a^r$$

$$= e a^r$$

$$= a^r$$

implies $a^m = a^r \quad \forall 0 < r < n$

a^r is already contained in the set of n elements and so a^m is also contained. Consequently G contains n elements.

Step (II)

To prove that any two elements of G are not equal.

For this, we have to show that $ar = as$ where $r \neq s$, where $0 < r < n$, $0 < s < n$.

To be particular at this point, let $r < s < n$

Then, $s - r > 0$

$$a^r = a^s$$

$$\Rightarrow a^r e a^r = a^s$$

$$\Rightarrow a^{2r} = e$$

$$\Rightarrow 0 < 2r \leq s \cdot r \text{ and } s - r < n$$

$$\Rightarrow 0(0) < n$$

A contradiction. For $0(a) = n$

Hence, $ar \neq as$, when $r \neq s$

Thus, we have shown that G contains n distinct elements and hence, $O(G) = n$. This concludes that problem.

(c) Define a cyclic group. If $G = \langle a \rangle$ is a finite cyclic group of order n , then prove that the order of any subgroup of G is a divisor of n , and for each positive divisor k of n , G has exactly one subgroup of order k , namely, $\langle a^{n/k} \rangle$.

Sol. Cyclic Group :

A group G is called a cyclic group, if there exists an element $a \in G$ such that each element of G is expressible as

$$x = a^k = a \cdot a \cdot \dots \cdot a \text{ (n times)}$$

Where, n some integer.

The element $a \in G$ is called generator of G and G is written as $G = \langle a \rangle$ or $G = \{a\}$.

For example, a group $(G, +)$ is cyclic, if there exists some element a of G s.t. each element x of G is expressible of

$$x = na = a + a + \dots + a \text{ (n times)}$$

for some integer n .

(ii) Let $G = \langle a \rangle$; $|a| = n$

$$\Rightarrow G = \{e, a^r : r \in \mathbb{Z}\}$$

G is a finite cyclic group of order n . and $H \leq G$.
We know that subgroup of a cyclic group is cyclic.

$$\text{Let } H = \langle a^m \rangle$$

$$(a^m)^n = (a^n)^m$$

$$(a^m)^n = (e)^m$$

$$(a^m)^n = e$$

$$\Rightarrow |a^m|/n$$

$$\Rightarrow |H|/|G| \text{ are } |H| = |\langle a^m \rangle|$$

Which proves that the order of any subgroup of G is a divisor of n .
(iii) Let k be a positive divisor of n .
We will prove that G has exactly one subgroup of order k —namely $\langle a^{n/k} \rangle$.

$$(a^{n/k})^k = a^n = e; \quad n \text{ is least positive integer.}$$

$$\text{Also, } (a^{n/k})^t \neq e \text{ for } t < k$$

$$\Rightarrow |a^{n/k}| = k$$

$$\Rightarrow \langle a^{n/k} \rangle = H$$

Now, we will prove that

$\langle a^{n/k} \rangle$ only one subgroup of order k .

Let H be any subgroup of G of order k s.t. $|H| = k$.

$$\Rightarrow H = \langle a^m \rangle, \quad m \text{ is least positive integer.}$$

Using Divisor algorithm for $n \div m$, $\exists q, r \in \mathbb{Z}$ s.t.

$$n = mq + r; \quad 0 \leq r < m$$

$$\therefore a^n = e$$

$$\Rightarrow a^{mq+r} = e$$

$$\Rightarrow a^{mq+r} = e$$

$$\Rightarrow a^r = a^{-mq}$$

$$\Rightarrow a^r = (a^m)^{-q} \in H$$

$$\Rightarrow a^r \in H; \quad 0 \leq r < m$$

which is a contradiction.

$$\Rightarrow r = 0$$

$$\Rightarrow n = mq$$

$$\Rightarrow m = \frac{n}{q}$$

$$\Rightarrow k = |m| = |\langle a^m \rangle| = \frac{n}{q}$$

$$m = \frac{n}{k}$$

$$H = \langle a^m \rangle$$

$$H = \langle a^{n/k} \rangle \text{ Proved.}$$

Q.3. (a) Prove that if the identity permutation $e = \beta_1 \dots \beta_r$, where the β_i are 2-cycles then r is even.

Sol. We will prove this lemma by induction of r .

Case (I)

If

$$r = 1$$

$$\beta_1 = (a, b) = e$$

Case (II)

If

$$r = 2, \beta_1 = \beta_2 = (a, b)$$

$$\beta_1 \beta_2 = (a, b)(a, b)$$

this forming 2-cycles. Hence, given lemma is true for $r = 2$.

Case (III)

We assume that the lemma is true for all the expression of 2-cycles, which are in k —

numbers for $2 < k < r$.

$$(i) \beta_1 = \beta_2 = (a, b)$$

$$(ii) \beta_1 = (a, b) \beta_2 = (c, d)$$

$$(iii) \beta_1 = (a, b) \beta_2 = (a, c)$$

$$(iv) \beta_1 = (a, b) \beta_2 = (b, c)$$

Case (i)

$$\beta_1 = \beta_2 = (a, b)$$

$$\beta_1 \beta_2 = e$$

$$e = \beta_1 \beta_2 \dots \beta_r$$

$$= \beta_3 \dots \beta_r$$

No of 2 cycles are $r-2$

According to induction

$$r-2 = 2k \text{ is even for some } k \in \mathbb{Z}$$

$$\Rightarrow r = 2k + 2$$

$$\Rightarrow r = 2(k+1)$$

$\Rightarrow r$ is even

Case (ii)

$$\beta_1 = (a, b), \beta_2 = (c, d)$$

$$e = \beta_1 \beta_2 \dots \beta_r$$

$$(a, b)(c, d) = (a, b)(c, d) \beta_3 \beta_4 \dots \beta_r$$

$$= \beta_3 \beta_4 \beta_1 \beta_2 \beta_5 \dots \beta_r$$

$$= \beta_1 \beta_2 \beta_3 \beta_4 \dots \beta_n$$

$$= \beta_1 \beta_2 \dots \beta_n$$

No. of 2-cycles on LHS is $2 + \text{no. of cycles in } \epsilon$

No. of 2-cycle on RHS $= r - 2$

$$\Rightarrow 2 + \text{no. of 2 cycles in } \epsilon = r - 2$$

$$\Rightarrow \text{No. of 2 cycles in } \epsilon = r - 4$$

according to our assumption,

$r - 4$ is even.

$\Rightarrow$

$$r - 4 = 2k \text{ for some } k \in \mathbb{Z}$$

$$r = 2k + 4$$

$$r = 2(k + 2)$$

$\Rightarrow r$ is even.

Case (iii)

$$\beta_1 = (a b) \beta_2 = (a c)$$

$$\beta_1 \beta_2 = (a b)(a c)$$

$$= (b c)(a b)$$

$$(a b)(a c) \epsilon = \beta_1 \beta_2 \beta_3 \beta_4 \dots \beta_n$$

$$= (a b)(a c)(a b)(a c) \beta_3 \dots \beta_n$$

$$= (b c)(a b)(a b)(a c) \beta_3 \dots \beta_n$$

$$= (b c)(a c) \beta_3 \dots \beta_n$$

No. of 2-cycles on LHS $= 2 + \text{No. of 2-cycles in } \epsilon$

No. of 2-cycles on RHS $= r$

$$\Rightarrow \text{No. of 2-cycles in } \epsilon = r - 2$$

According to our assumption, $r - 2$ is even.

$$r - 2 = 2k \quad \forall k \in \mathbb{Z}$$

$$\text{implies } r = 2k + 2$$

$$\text{implies } r \text{ is even.}$$

Case (iv)

$$\text{Let } \beta_1 = (a b) \beta_2 = (b c)$$

$$(a b)(a c) \epsilon = \beta_1 \beta_2 \beta_3 \beta_4 \dots \beta_n$$

$$= (a b)(b c)(a b)(b c) \beta_3 \dots \beta_n$$

$$= (a b)(b c)(b c)(a c) \beta_3 \dots \beta_n$$

$$= (a b)(a c) \beta_3 \dots \beta_n$$

No. of 2-cycles on LHS $= 2 + \text{No. of 2-cycles in } \epsilon$

No. of 2-cycles on RHS $= r$

$$\text{implies } 2 + \text{no. 2-cycles in } \epsilon = r$$

$$\text{No. of 2 cycles in } \epsilon = r - 2$$

According to our assumption, $r - 2$ is even

$$\therefore r - 2 = \text{even}$$

$$\Rightarrow r - 2 = k \quad \forall k \in \mathbb{Z}$$

$$\Rightarrow r = 2(k + 1)$$

$\Rightarrow r$ is even.

Hence Proved.

(b) Show that for $n \geq 3$, $Z(S_n) = \{1\}$.

Sol.(b) We will prove by example.

let $x \in Z(S_n)$

$$x \cdot a = a \cdot x \quad \forall a \in S_n$$

$$\text{Let } S_3 = \{(1), (12), (13), (23), (123), (321)\}$$

$$\text{Let } (12) \in S_3$$

$$x(12) = (12)x$$

so if $x = (13)$, then

$$(13)(12) \neq (12)(13)$$

if $x = (23)$, then,

$$(23)(12) \neq (12)(23)$$

if $x = (123)$ then

$$(123)(12) \neq (12)(123)$$

if $x = (321)$, then

$$(321)(12) \neq (12)(321)$$

But, if $x = (1)$, then only

$$(1)(12) = (12)(1)$$

$$\text{so, } Z(S_n) = \{1\}$$

(ii) $S_n = \{1, \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7, \alpha_8, \alpha_9, \alpha_{10}, \alpha_{11}, \alpha_{12}, \alpha_{13}, \alpha_{14}, \alpha_{15}, \alpha_{16}, \alpha_{17}, \alpha_{18}, \alpha_{19}, \alpha_{20}, \alpha_{21}, \alpha_{22}, \alpha_{23}\}$ so for let $\alpha = (123)$ then

$$(123)x = x(123) \quad \forall x \in S_n$$

But there is no other element in S_n other than 1, which will comment will (123) . So,

$$Z(S_n) = \{1\}$$

In general, we have for $n \geq 3$,

$$Z(S_n) = \{1\}$$

(c) Prove that :

(i) a group of prime order has no proper, non-trivial subgroup. State its converse. Is it true?

(ii) a group of prime order is cyclic and any non-identity element can be taken as its generator.

Sol. (i) It is given that G has no proper subgroup. Let $a \in G$ be such that

$a \neq e$. Then, $H = \langle a \rangle$ is a subgroup of G and $H \neq \{e\}$. Since, G has no proper subgroup.

$H = G$ i.e. $G = \langle a \rangle$

If $O(a)$ is infinite, then,

$$K = \langle a^2 \rangle = \{a^2, a^4, a^6, \dots\}$$

is a proper subgroup of $G = \langle a \rangle$, which is contrary to the given hypothesis. Hence, $O(a)$ must be finite and so $G = \langle a \rangle$ is a finite group. If $O(G)$ is composite, then G must have a non-trivial subgroup, which is again contrary to the given hypothesis. Hence, G is a finite of prime order.

(ii) Let $O(a) = p$, p having prime.

Let, H be any subgroup of G .

By Lagrange's Theorem, $O(H)$ divides $O(G) = p$

Constantly,

$$O(H) = 1 \text{ or } O(H) = p$$

implies $H = \langle e \rangle$ or $H = a$... (i)

It follows that a group of prime order has no non-trivial (proper) subgroup. Here $\{e\}$ and G are called trivial subgroup of G .

Let $H = \langle a \rangle$, where $a \neq e \in G$

Then, H is a subgroup of G and $H = \langle e \rangle$, since $a \neq e \in H$. Hence, by (i).

$$G = H = \langle a \rangle$$

Which proves that G is cyclic.

Q. 4. (a) Let G be a finite group of permutations of a set S . Then prove that for any i from S .

$$|G| = |orb_G(i)| |stab_G(i)|.$$

Sol. Let G be a finite group of permutations of set S , for $i \in S$.

$$stab_G(i) \leq G$$

Then by Lagrange's Theorem :

$\frac{|G|}{|stab_G(i)|}$ shows that the number of distinct left cosets of $stab_G(i)$. Thus, it

is sufficient to prove the result that there is one-to-one correspondence between left cosets of $stab_G(i)$ elements in the $orb_G(i)$.

$$\frac{|G|}{|stab_G(i)|} = |\{\alpha stab_G(i) \mid \alpha \in G\}|$$

$$\& Orb_G(i) = \{\phi^\alpha(i) \in S \mid \phi^\alpha \in G\}$$

Define, $f: Orb_G(i) \rightarrow S$ s.t.

$$f(\alpha stab_G(i)) = \alpha(i)$$

Well defined

$$\text{Let } \alpha stab_G(i) = \beta stab_G(i)$$

$$\Rightarrow \beta^{-1} \alpha stab_G(i) = stab_G(i)$$

$$\Rightarrow \beta^{-1} \alpha(i) = i$$

$$\Rightarrow \alpha(i) = \beta(i)$$

$$\Rightarrow f(\alpha stab_G(i)) = f(\beta stab_G(i))$$

One-to-One

$$\text{Let, } f(\alpha stab_G(i)) = f(\beta stab_G(i))$$

$$\Rightarrow \alpha(i) = \beta(i)$$

$$\Rightarrow \beta^{-1} \alpha(i) = i$$

$$\Rightarrow \beta^{-1} \alpha \in stab_G(i)$$

$$\Rightarrow \beta^{-1} \alpha stab_G(i) = stab_G(i)$$

$$\Rightarrow \alpha stab_G(i) = \beta stab_G(i)$$

f is well defined and one-to-one

onto :

for $\alpha(i) \in Orb_G(i) \exists \alpha \in G$ s.t.

$$|\{stab_G(i)\}| = \frac{|G|}{|stab_G(i)|} \text{ and}$$

thus,

$$f(\alpha stab_G(i)) = \alpha(i)$$

$\Rightarrow f$ is onto

Hence,

$$|\{\alpha stab_G(i) \mid \alpha \in G\}| = |Orb_G(i)|$$

$$\Rightarrow |stab_G(i)| = |Orb_G(i)|$$

$$\Rightarrow \frac{|G|}{|stab_G(i)|} = |Orb_G(i)|$$

$$\Rightarrow |G| = |stab_G(i)| |Orb_G(i)|$$

Proved

(b) (i) Prove that the center $Z(G)$ of a group G is a subgroup of G and is normal in G .

Sol. (i) Let G is a group and

$$Z = \{z \in G : xz = zx \forall x \in G\}$$

Since,

$$ex = xe \forall x \in G$$

$$\Rightarrow e \in Z$$

$$\Rightarrow z = d$$

... (ii)

Let $a, b \in Z$ so that $ax = xa$ and $bx = xb \forall x \in G$

$$ba = ab$$

$$b^{-1}(ba) = b^{-1}(ab)$$

$$x = b^{-1}ax$$

$$b^{-1}ax = x b^{-1}a$$

$$a(b^{-1}x) = a(x b^{-1})$$

$$(ab^{-1})x = (ax)b^{-1}$$

$$(ab^{-1})x = (x a)b^{-1}$$

$$(ab^{-1})x = x(ab^{-1})$$

$$ab^{-1} \in Z$$

$\Rightarrow Z$ is a subgroup of G Proved.

We show that $Z < G$.

Let, $g \in G$ and $A \in Z$

Since, $a \in Z, ag = ga \forall g \in G$

implies $gAg^{-1} = a \in Z$

$g = g^{-1} \in Z \forall g \in G$ and $a \in Z$

implies $Z < G$.

implies Z is a normal in G .

Proved.

(ii) If H is a subgroup of G such that H is contained in the center, $Z(G)$, then prove that H is a normal subgroup of G . Is the converse true? Justify your answer.

Sol. Consult Text Book.

(c) Let N be a normal subgroup of a group G and let H be a subgroup of G . If N is a subgroup of H , prove that H/N is a normal subgroup of G/N if and only if H is a normal subgroup of G .

Sol. Consult Text Book.

Q. 5. (a) Let C be the complex number and :

$$M = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} : a, b \in \mathbb{R} \right\}.$$

Prove that C and M are isomorphic under addition and $C^* = C/\{0\}$ and $M^* = M/\{0\}$ are isomorphic under multiplication.

Sol.

$$M = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

(I) To prove C and M are isomorphic under addition
 $f: C \rightarrow M$ s.t

$$f(a+ib) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

(i) Homomorphism or preserving operation :

$$f(a_1+ib_1) = \begin{bmatrix} a_1 & -b_1 \\ b_1 & a_1 \end{bmatrix}$$

$$f(a_2+ib_2) = \begin{bmatrix} a_2 & -b_2 \\ b_2 & a_2 \end{bmatrix}$$

$$f(a_1+ib_1) + f(a_2+ib_2) = f((a_1+a_2) + i(b_1+b_2))$$

$$= \begin{bmatrix} a_1+a_2 & -(b_1+b_2) \\ b_1+b_2 & a_1+a_2 \end{bmatrix}$$

$$= \begin{bmatrix} a_1 & -b_1 \\ b_1 & a_1 \end{bmatrix} + \begin{bmatrix} a_2 & -b_2 \\ b_2 & a_2 \end{bmatrix}$$

$$= f(a_1+ib_1) + f(a_2+ib_2)$$

$\Rightarrow f$ preserving the operation.

(ii) One-to-one

Let $f(a_1+ib_1) = f(a_2+ib_2)$

$$\Rightarrow \begin{bmatrix} a_1 & -b_1 \\ b_1 & a_1 \end{bmatrix} = \begin{bmatrix} a_2 & -b_2 \\ b_2 & a_2 \end{bmatrix}$$

$$\Rightarrow a_1 = a_2 \text{ and } b_1 = b_2$$

$$\Rightarrow a_1+ib_1 = a_2+ib_2$$

$\Rightarrow f$ is one-to-one.

(iii) Onto

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \in M$$

$$\Rightarrow a, b \in \mathbb{R}$$

$$\Rightarrow a+ib \in \mathbb{C} \text{ s.t.}$$

$$f(a+ib) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$\Rightarrow f$ is onto

From (i), (ii) and (iii), we Conclude that C and M are isomorphic under addition

(II) To prove C^* and M^* are isomorphic under multiplication.

$$\text{Let } f: C^* \rightarrow M^* \text{ s.t. } f(a + ib) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

(i) Homomorphism or preserving operation

$$\text{Let } a_1 + ib_1 \in C$$

$$a_2 + ib_2 \in C$$

$$f(a_1 + ib_1) = \begin{bmatrix} a_1 & -b_1 \\ b_1 & a_1 \end{bmatrix}$$

$$f(a_2 + ib_2) = \begin{bmatrix} a_2 & -b_2 \\ b_2 & a_2 \end{bmatrix}$$

$$\begin{aligned} f[(a_1 + ib_1)(a_2 + ib_2)] &= f[(a_1 a_2 - b_1 b_2) + i(a_1 b_2 + b_1 a_2)] \\ &= \begin{bmatrix} a_1 a_2 - b_1 b_2 & -(a_1 b_2 + b_1 a_2) \\ a_1 b_2 + a_2 b_1 & a_1 a_2 - b_1 b_1 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} a_1 & -b_1 \\ b_1 & a_1 \end{bmatrix} \begin{bmatrix} a_2 & -b_2 \\ b_2 & a_2 \end{bmatrix}$$

$$= f(a_1 + ib_1) \cdot f(a_2 + ib_2)$$

$\Rightarrow f$ preserving operation

(ii) **One-to-one**: Same as part (ii) alone

f is one to one

(iii) **Onto**: Same as part (iii) alone

f is onto

Hence, from (i), (ii), (iii), we conclude that C^* and M^* are isomorphic under multiplication.

(b) Prove that an infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$. Hence show that every subgroup of an infinite cyclic group is isomorphic to the group itself.

Sol. Let

$$G = \langle a \rangle \text{ be any infinite group s.t.}$$

$$G = \langle a \rangle = \{ a^n : n \in \mathbb{Z} \}$$

Define a mapping $f: \mathbb{Z} \rightarrow G$ as

$$f(n) = a^n \quad \forall n \in \mathbb{Z} \quad \dots(i)$$

Then f is homomorphism, since $m, n \in \mathbb{Z}$

$$f(m+n) = a^{m+n}$$

$$= a^m a^n$$

$$= f(m)f(n), \quad \text{using (i)}$$

Now, we show that f is onto. Let $g \in G$ be arbitrary. Then

$$g = a^i \text{ for some } i \in \mathbb{Z}, \text{ using (i)}$$

$$f(i) = g$$

We have,

$\Rightarrow f$ is onto

We show that f is one-to-one. Let $m, n \in \mathbb{Z}$. Then

$$f(m) = f(n)$$

$\Rightarrow$

$$a^m = a^n$$

$\Rightarrow$

$$a^{m-n} = a^0$$

$\Rightarrow$

$$m - n = 0, \text{ since } C \text{ is infinite}$$

$\Rightarrow$

$$m = n$$

$\Rightarrow f$ is one-to-one.

Hence, $\mathbb{Z} \cong G$ or $G \cong \mathbb{Z}$

Which prove that any infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$

(c) Let G be a group of permutations. For each $\sigma \in G$, define.

$$\text{sgn}(\sigma) = \begin{cases} 1, & \text{if } \sigma \text{ is an even permutation,} \\ -1, & \text{if } \sigma \text{ is an odd permutation} \end{cases}$$

Prove that sgn is a homomorphism from G to $\{1, -1\}$. What is the kernel?

Sol. Let $G \rightarrow \{1, -1\}$ defined by

$$\text{sgn}(\sigma) = 1; \sigma \text{ is an even permutation}$$

$$= -1; \sigma \text{ is odd permutation}$$

Will defined

Let,

$$\alpha = \beta \quad \forall \alpha, \beta \in S_n$$

α & β are even or α & β are odd

If α & β are even;

$$\text{sgn}(\sigma) = 1 = \text{sgn}(\beta)$$

If α & β are odd;

$$\text{sgn}(\sigma) = -1 = \text{sgn}(\beta)$$

Hence, sgn is well defined

Homomorphism:

(1) When α & β are even:

$\Rightarrow$

$$\alpha\beta = \text{even}$$

$$\text{sgn}(\alpha\beta) = 1$$

$$= 1 \cdot 1 = \text{sgn}(\alpha) \text{sgn}(\beta)$$

(2) When α is even β is odd

$$\begin{aligned}\alpha\beta &= (\text{even})(\text{odd}) \\ &= (\text{odd}) \\ \text{sgn}(\alpha\beta) &= -1 \\ &= (1)(-1) \\ &= (\text{sgn } \alpha)(\text{sgn } \beta)\end{aligned}$$

(3) When α is odd β is even

$$\begin{aligned}\alpha\beta &= (\text{odd})(\text{even}) \\ &= (\text{even}) \\ \text{sgn}(\alpha\beta) &= 1 \\ &= (-1)(1) \\ &= (\text{sgn } \alpha)(\text{sgn } \beta)\end{aligned}$$

(4) When α and β both are odd

$$\begin{aligned}\alpha\beta &= (\text{odd})(\text{odd}) \\ &= (\text{odd}) \\ \text{sgn}(\alpha\beta) &= -1 \\ &= (-1)(-1) \\ &= (\text{sgn } \alpha)(\text{sgn } \beta)\end{aligned}$$

From (1), (2), (3), (4), we get

$$\text{sgn}(\alpha\beta) = (\text{sgn } \alpha)(\text{sgn } \beta)$$

 $\Rightarrow \text{sgn}(\sigma)$ is homomorphism

Then

$$\begin{aligned}(\text{sgn}) &= \{\sigma \in S_n \mid \text{sgn}(\sigma) = 1\} \\ &= \{\sigma \mid \sigma \text{ is an even permutation in } G\}\end{aligned}$$

Q. 6. (a) Let ϕ be a homomorphism from a group G to a group $\bar{G}$. Let g be an element of G . Then:(i) $\phi(g^n) = \phi(g)^n$ for all $n \in \mathbb{Z}$ (ii) ϕ is one-one if and only if $\ker(\phi) = \{e\}$, where e is the identity of G .Sol. $g \in G$

$$\begin{aligned}g^n &= g \cdot g \cdot g \cdots g \text{ (n times)} \\ \phi(g^n) &= \phi(g \cdot g \cdot g \cdots g) \\ &= \phi(g) \cdot \phi(g) \cdot \phi(g) \cdots \phi(g) \text{ (n times)}\end{aligned}$$

Since, ϕ preserve operation as this homomorphism.

$$\phi(g^n) = (\phi(g))^n$$

(b) State and prove the First Isomorphism Theorem.

Sol. First Theorem on Isomorphism:

Statement: Every homomorphic image of a group G is isomorphic to same quotient group of G .

• Proof.

Suppose $f: G \rightarrow H$ is a group homomorphism with kernel K , so that K is the normal subgroup of G and $K = \{x \in G : f(x) = e'\}$ e' having the identity of G' .Now, the symbol G/K has a meaningTo prove $G/K \cong G'$ Define a map $g: G/K \rightarrow G'$ s.t. $g(Kx) = f(x) \forall x \in G$ g is well definedFor $Ka = Kb : Ka, Kb \in a/K$ $\Rightarrow ab^{-1} \in K$ (as $Ka = Kb \Leftrightarrow ab^{-1} \in K$) $\Rightarrow f(ab^{-1}) \in e'$ def of K as f is homomorphism $\Rightarrow f(a)f(b^{-1}) = e'$ $\Rightarrow f(a)[f(b)]^{-1} = e'$ $\Rightarrow f(a)[f(b)]^{-1}f(b) = e' f(b)$ $\Rightarrow f(a)e' = f(b)$ $\Rightarrow f(a) = f(b)$ $\Rightarrow g(Ka) = g(Kb)$ which prove g is well defined g is one-one

$$g(Ke) = g(Ka)Ka, K \in G/K$$

 $\Rightarrow f(a) = f(e)$ $\Rightarrow [f(a)]^{-1}f(e) = e'$ $\Rightarrow f(a)f(b^{-1}) = e'$ $\Rightarrow f(a)b^{-1} = e'$ $\Rightarrow ab^{-1} \in K$ $\Rightarrow Ka = Kb$ $\Rightarrow g(Ka) = g(Kb)$ as $Ka = Kb$ $\Rightarrow g$ is one to one g preserves composition in G/K and G'

$$g(Ka \cdot Kb) = g(Kab)$$

as K is normal

$$= f(ab)$$

$$(Ka)(Kb) = (Kab)$$

$$= f(a)f(b)$$

as f is homomorphism

$$= g(Ka)g(Kb)$$

$$\Rightarrow g(Ka \cdot Kb) = g(Ka)g(Kb)$$

g is onto :

any $x \in G \Rightarrow \exists y \in G$ s.t. $f(y) = x$

For $f: G \rightarrow G'$ is onto

$\Rightarrow Ky \in a/K, g(Ky) = f(y) = x$

Given any $x \in G' \exists Ky \in \frac{G}{K}$ s.t. $g(Ky) = f(y)$

Which proves g is onto

Hence, we can say that g and an isomorphism. Therefore, we have :

$$G/K \cong G'$$

Which prove first Theorem on Isomorphism

(c) (i) Suppose ϕ is a homomorphism from $U(30)$ to $U(30)$ and $\text{Ker}(\phi) = \{1, 11\}$.

If $\phi(7) = 7$, find all elements of $U(30)$ that map to 7.

(ii) Let G be a group. Prove that the mapping $\phi(g) = g^{-1}$, for all $g \in G$, is an isomorphism from G onto F if and if G is Abelian.

Sol. (i)

$$\text{Ker}(\phi) = \{1, 11\}$$

$$\phi(7) = 7$$

$$U(30) \rightarrow U(30)$$

All element of $U(30)$ that map of to 7 are calculated i.e.

$$\phi^{-1}(7) = 7 \ker \phi$$

$$= 7 \{1, 11\}$$

$$= \{7, 77\} \pmod{30}$$

$$= \{7, 17\}$$

(ii) Consult text book.

Sr. No. of Q. Paper : 92

Unique Paper Code : 32351303

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Name of the Paper : C-7 Multivariate Calculus

Name of the Course : B.Sc. (Hons.) Mathematics II Year

Semester : III

Duration : 3 Hours

Maximum Marks : 75

All questions carry equal marks.

Section - I

Attempt any six questions from this section

Q. 1. Let f be the function by $f(x, y) = \frac{x^2 + 2y^2}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$.

(a) Find $\lim_{(x, y) \rightarrow (2, 1)} f(x, y)$.

(b) Prove that f has no limit at $(0, 0)$.

Sol. (a) $\lim_{(x, y) \rightarrow (2, 1)} f(x, y) = \lim_{(x, y) \rightarrow (2, 1)} \frac{x^2 + 2y^2}{x^2 + y^2}$

$$= \frac{(2)^2 + 2(1)^2}{(2)^2 + (1)^2} = \frac{4 + 2}{4 + 1} = \frac{6}{5}$$

(b) Let $(x, y) \rightarrow (0, 0)$ along the path $y = mx$, when m is a real number. When $x \rightarrow 0$, then $y \rightarrow 0$ along the path $y = mx$

$$\begin{aligned} \lim_{(x, y) \rightarrow (0, 0)} f(x, y) &= \lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 + 2y^2}{x^2 + y^2} \\ &= \lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 + 2(mx)^2}{x^2 + (mx)^2} \\ &= \lim_{(x, y) \rightarrow (0, 0)} \frac{x^2(1 + 2m^2)}{x^2(1 + m^2)} \\ &= \lim_{(x, y) \rightarrow (0, 0)} \frac{1 + 2m^2}{1 + m^2} = \frac{1 + 2m^2}{1 + m^2} \end{aligned}$$

Which is different for different values of m

$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist.

$\Rightarrow f(x, y)$ has no limit at $(0, 0)$.

Proved.

Q. 2. The temperature at the point (x, y) on a given metal plate in the xy -plane is determined according to the formula $T(x, y) = x^3 + 2xy^2 + y$ degrees. Compute the rate at which the temperature changes with distance if we start at $(2, 1)$ and move:

(a) parallel to the vector j .

(b) parallel to the vector i .

Sol. The temperature at the point (x, y) is given by:

$$T(x, y) = x^3 + 2xy^2 + y \text{ degrees} \quad \dots(1)$$

(a) The rate at which temperature change unit distance and move parallel to vector j is calculated as partial derivation of T w. r. t. y :

$$\begin{aligned} \left. \frac{\partial T}{\partial y} \right|_{(2,1)} &= \left. \frac{\partial}{\partial y} (x^3 + 2xy^2 + y) \right|_{(2,1)} \\ &= (0 + 4xy + 1)|_{(2,1)} = (4xy + 1)|_{(2,1)} \\ &= 4 \times 2 \times 1 + 1 = 8 + 1 \\ &= 9 \text{ degree/unit length} \end{aligned}$$

(b) The rate at which temperature change with distance and move parallel to vector i is calculated as partial derivative of T w. r. t. x :

$$\begin{aligned} \left. \frac{\partial T}{\partial x} \right|_{(2,1)} &= \left. \frac{\partial}{\partial x} (x^3 + 2xy^2 + y) \right|_{(2,1)} \\ &= (3x^2 + 2y)|_{(2,1)} = 3(2)^2 + 4(1) \\ &= 12 + 4 \\ &= 16 \text{ degree/unit length} \end{aligned}$$

Q. 3. The Company sells two brands X and Y of a commercial soap, in thousand-pound units. If x units of brand X and y units of brand Y are sold, the unit price for brand X is $p(x) = 4,000 - 500x$ and for brand Y is $q(y) = 3,000 - 450y$.

(a) Find the total revenue R in terms of p and q .

(b) Suppose the brand X sells for \$ 500 per unit and brand Y sells for \$ 750 per unit. Estimate the change in total revenue if the prices are increased by \$ 20 for brand X and \$ 18 for brand Y.

Sol. The unit price for brand X is:

$$p(x) = 4000 - 500x \quad \dots(1)$$

The unit price for brand y is

$$q(y) = 3000 - 450y \quad \dots(2)$$

Equation (1) can be solved

$$p = 4000 - 500x$$

$$500x = 4000 - p$$

$$x = \frac{4000 - p}{500} \quad \dots(3)$$

Equation (2) can be solved

$$q = 3000 - 450y$$

$$450y = 3000 - q$$

$$y = \frac{3000 - q}{450} \quad \dots(4)$$

(a) The total revenue R is calculated

$$R = (x(p) \cdot p + y(q) \cdot q)$$

$$= xp + yq$$

$$= \left(\frac{4000 - p}{500} \right) p + \left(\frac{3000 - q}{450} \right) q$$

(b)

$$Rp = \frac{4000}{500} - \frac{2p}{500} = 8 - \frac{2p}{500}$$

$$Rq = \frac{3000}{450} - \frac{2q}{450} = \frac{20}{3} - \frac{2q}{450}$$

$$\Delta R = Rp \Delta p + Rq \Delta q$$

$$= \left(8 - \frac{2p}{500} \right) \Delta p + \left(\frac{20}{3} - \frac{2q}{450} \right) \Delta q$$

$$\text{Here, } p = 500 ; \Delta p = 20$$

$$q = 250 ; \Delta q = 18$$

The change in the total scheme i.e

$$= \left(8 - \frac{2 \times 500}{500} \right) (20) + \left(\frac{20}{3} - \frac{2 \times 250}{450} \right) (18)$$

$$= (8 - 2) (20) + \left(\frac{20}{3} - \frac{10}{3} \right) (10)$$

$$= (6) (20) + \frac{10}{3} \times 18$$

$$= 120 + 60 = 180$$

The revenue is increased by \$ 180 (approx)

Q. 4. If $w = f\left(\frac{r-s}{s}\right)$, show that $r \frac{\partial w}{\partial r} + s \frac{\partial w}{\partial s} = 0$.

Sol. If

$$w = f\left(\frac{r-s}{s}\right) \quad \dots(1)$$

Let

$$x = \frac{r-s}{s} = \frac{r}{s} - 1$$

Hence,

$$x = f(x) \quad \forall x \text{ is a function of } r \text{ and } y.$$

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial r}$$

$$= \frac{\partial w}{\partial x} \cdot \frac{\partial}{\partial r} \left(\frac{r}{s} - 1 \right)$$

$$= \frac{\partial w}{\partial x} \cdot \left(\frac{1}{s} - 0 \right) = \frac{1}{s} \frac{\partial w}{\partial x} \quad \dots(2)$$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} = \frac{\partial w}{\partial x} \cdot \frac{\partial}{\partial s} \left(\frac{r}{s} - 1 \right)$$

$$= \frac{\partial w}{\partial x} \cdot \left(-\frac{r}{s^2} - 0 \right) = -\frac{r}{s^2} \frac{\partial w}{\partial x} \quad \dots(3)$$

L.H.S

$$= r \frac{\partial w}{\partial s} + s \frac{\partial w}{\partial r}$$

$$= r \left(\frac{1}{s} \frac{\partial w}{\partial x} \right) + s \left(-\frac{r}{s^2} \frac{\partial w}{\partial x} \right)$$

$$= \frac{r}{s} \frac{\partial w}{\partial x} - \frac{r}{s} \frac{\partial w}{\partial x} = 0 \text{ R.H.S}$$

Proved.

Q. 5. Find the directional derivative of $f(x, y) = e^{x^2 y^2}$ at $P(1, -1)$ in the direction toward $Q(2, 3)$.

Sol.

$$f(x, y) = e^{x^2 y^2} \quad \dots(1)$$

$$P(1, -1) \quad Q(2, 3)$$

A vector in the direction Q is given by:

$$\vec{u} = \vec{PQ}$$

$$= (2-1)\hat{i} + (3-(-1))\hat{j}$$

$$= (1)\hat{i} + (4)\hat{j} = \hat{i} + 4\hat{j}$$

The unit vector $\hat{u}$ in the direction of $\vec{u}$ is

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|} = \frac{(\hat{i} + 4\hat{j})}{\sqrt{1+16}}$$

$$= \frac{(\hat{i} + 4\hat{j})}{\sqrt{17}} = \frac{1}{\sqrt{17}}(\hat{i} + 4\hat{j})$$

$$= \left(\frac{1}{\sqrt{17}} \right) \hat{i} + \left(\frac{4}{\sqrt{17}} \right) \hat{j} = \frac{1}{\sqrt{17}} \hat{i} + \frac{4}{\sqrt{17}} \hat{j}$$

where,

$$u_1 = \frac{1}{\sqrt{17}}$$

$$u_2 = \frac{4}{\sqrt{17}}$$

$$f_x(x, y) = \frac{\partial}{\partial x} (e^{x^2 y^2}) = 2x e^{x^2 y^2}$$

$$f_y(x, y) = \frac{\partial}{\partial y} (e^{x^2 y^2}) = 2y e^{x^2 y^2}$$

$$= 2y e^{x^2 y^2}$$

At $P(1, -1)$

$$f_x(1, -1) = 2(1)e^{(1)^2(-1)^2} = 2e$$

$$f_y(1, -1) = 2(-1)e^{(1)^2(-1)^2} = -2e$$

The directional derivative of the function $f(x, y)$ at the point $P(1, -1)$ in the direction of $\hat{u}$ is

$$D_{\hat{u}} f(x_0, y_0) = f_x(x_0, y_0) u_1 + f_y(x_0, y_0) u_2$$

$$D_{\hat{u}} f(1, -1) = f_x(1, -1) u_1 + f_y(1, -1) u_2$$

$$= (2e) \left(\frac{1}{\sqrt{17}} \right) + (-2e) \left(\frac{4}{\sqrt{17}} \right)$$

$$= \frac{2e}{\sqrt{17}} - \frac{6e}{\sqrt{17}} = -\frac{4e}{\sqrt{17}}$$

$$= -\frac{4e}{\sqrt{17}}$$

Q. 6. Find the absolute extrema of $f(x, y) = 2 \sin x + 5 \cos y$ in the rectangular region with vertices $(0, 0)$, $(2, 0)$, $(2, 5)$ and $(0, 5)$.

Sol.

$$f(x, y) = 2 \sin x + 5 \cos y$$

which gives

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (2 \sin x + 5 \cos y)$$

$$= 2 \cos x$$

and

$$f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (2 \sin x + 5 \cos y)$$

$$= -5 \sin y$$

For critical points of $f(x, y)$, we have

$$f_x = 0$$

and

$$f_y = 0$$

$$2 \cos x = 0$$

and

$$-5 \sin y = 0$$

$$\cos x = 0$$

and

$$\sin y = 0$$

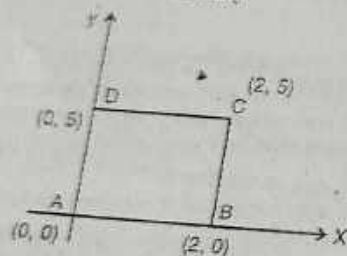
$$x = 90$$

and

$$y = 0$$

Hence, $(90, 0)$ is the critical point.

To find the critical point on boundary



AB:

$$y = 0, x = t; 0 \leq t \leq 2$$

$$G_1(t) = 2 \sin t + 5$$

$$G_1'(t) = 2 \cos t$$

For critical point

$$G_1'(t) = 0$$

$$2 \cos t = 0$$

$$\cos t = 0$$

$$t = \frac{\pi}{2}$$

∴ critical points are $(\frac{\pi}{2}, 0)$, A & B

$$BC: x = 2, y = t; 0 \leq t \leq 5$$

$$G_2(t) = 2 \sin x + 5 \cos t$$

$$G_2'(t) = -5 \sin t$$

For critical points

$$G_2'(t) = 0$$

$$-5 \sin t = 0$$

$$\sin t = 0$$

$$t = 0$$

∴ Critical points are $(2, 0)$, B & C

CD:

$$x = t, y = 5; 0 \leq t \leq 2$$

$$G_3(t) = 2 \sin t + 5 \cos$$

$$G_3'(t) = 2 \cos t$$

For critical point

$$G_3'(t) = 0$$

$$2 \cos t = 0$$

$$\cos t = 0$$

$$t = \frac{\pi}{2}$$

For critical points are $(\frac{\pi}{2}, 5)$, C & D

DA:

$$x = 0, y = t; 0 \leq t \leq 5$$

$$G_4(t) = 2 \sin 0 + 5 \cos t$$

$$= 5 \cos t$$

$$G_4'(t) = -5 \sin t$$

For critical points

$$G_4'(t) = 0$$

$$-5 \sin t = 0$$

$$t = 0$$

∴ Critical points are $(0, 0)$, D & A.

The total critical points are

$$\left(\frac{\pi}{2}, 0\right), (0, 0), (2, 0), (2, 5), \left(\frac{\pi}{2}, 5\right), (0, 5)$$

$$f(x, y) = 2 \sin x + 5 \cos y$$

$$f\left(\frac{\pi}{2}, 0\right) = 2 \sin \frac{\pi}{2} + 5 \cos 0 = 7 = \text{absolute max}$$

$$f(0, 0) = 2 \sin 0 + 5 \cos 0 = 5$$

$$f(2, 0) = 2 \sin 2 + 5 \cos 0 = 6.8186$$

$$f(2, 5) = 2 \sin 2 + 5 \cos 5 = 3.2364$$

$$f\left(\frac{\pi}{2}, 5\right) = 2 \sin \frac{\pi}{2} + 5 \cos 5 = 3.4183$$

$$f(0, 5) = 2 \sin 0 + 5 \cos 5 = 1.4183 = \text{absolute min}$$

Thus, the absolute max value of f is 7 which occurs at $\left(\frac{\pi}{2}, 0\right)$.

Q. 7. Let $\vec{R} = xi + yj + zk$ and $r = \|\vec{R}\|$, evaluate $\text{div} \left(\frac{1}{r^3} \vec{R} \right)$.

Sol. Let $\vec{R} = xi + yj + zk$ and $r = \|\vec{R}\|$

$$\vec{R} = xi + yj + zk$$

$$r = \|\vec{R}\| = \sqrt{x^2 + y^2 + z^2} \quad \text{or } r^2 = x^2 + y^2 + z^2$$

Let

$$\phi = \frac{1}{r^3}$$

$$\vec{A} = \vec{R}$$

$$\frac{\partial}{\partial x} (r^{-3}) = -\frac{3}{r^4} \frac{\partial r}{\partial x} = -\frac{3}{r^4} \left(\frac{x}{r} \right) = -3x r^{-5}$$

Similarly

$$\frac{\partial}{\partial y} (r^{-3}) = -3y r^{-5}$$

$$\frac{\partial}{\partial z} (r^{-3}) = -3z r^{-5}$$

$$\text{grad} \left(\frac{1}{r^3} \right) = \frac{\partial}{\partial x} \left(\frac{1}{r^3} \right) i + \frac{\partial}{\partial y} \left(\frac{1}{r^3} \right) j + \frac{\partial}{\partial z} \left(\frac{1}{r^3} \right) k$$

$$= (-3xr^{-5})i + (-3yr^{-5})j + (-3zr^{-5})k \\ = -3r^{-5}(xi + yj + zk) = -3r^{-5} \vec{R}$$

$$\text{div} \vec{R} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

We know that

$$\text{div} (\phi \vec{A}) = \phi (\text{div} \vec{A}) + \vec{A} \cdot (\text{grad} \phi)$$

$$\text{div} \left(\frac{\vec{R}}{r^3} \right) = \text{div} \left(\frac{1}{r^3} \vec{R} \right)$$

$$= \frac{1}{r^3} (\text{div} \vec{R}) + \vec{R} \cdot \left(\text{grad} \frac{1}{r^3} \right)$$

$$= \frac{1}{r^3} (3) + \vec{R} \cdot (-3r^{-5} \vec{R})$$

$$= \frac{3}{r^3} - \frac{3}{r^5} (\vec{R} \cdot \vec{R}) = \frac{3}{r^3} - \frac{3}{r^5} (r^2) = \frac{3}{r^3} - \frac{3}{r^3} = 0$$

Section II

Attempt any five questions from this section

Q. 8. By using iterated integral, compute

$$\iint_R x \sqrt{1-x^2} e^{3y} dA$$

where R is the rectangle $0 \leq x \leq 1$, $0 \leq y \leq 2$.

$$\text{Ans. } \iint_R x \sqrt{1-x^2} e^{3y} dA$$

$$0 \leq x \leq 1$$

$$0 \leq y \leq 2$$

$$\iint_R x \sqrt{1-x^2} e^{3y} dA = \int_{x=0}^1 \int_{y=0}^2 x \sqrt{1-x^2} e^{3y} dy dx$$

$$= \int_{x=0}^1 x \sqrt{1-x^2} dx \int_{y=0}^2 e^{3y} dy$$

Let

$$\sqrt{1-x^2} = t$$

$$1-x^2 = t^2$$

$$2x dx = 2t dt$$

$$x dx = t dt$$

$$\text{at } x=0, t=1$$

$$\text{at } x=1, t=0$$

$$= \int_{t=1}^0 t(-t dt) \left[\left(\frac{e^{3y}}{3} \right)^2 \right] = \int_{t=0}^1 t^2 dt \left(\frac{1}{3} (e^6 - e^0) \right)$$

$$= \left[\frac{t^3}{3} \right]_0^1 \left(\frac{1}{3} (e^6 - 1) \right) = \frac{(1-0)}{3} \left(\frac{1}{3} (e^6 - 1) \right)$$

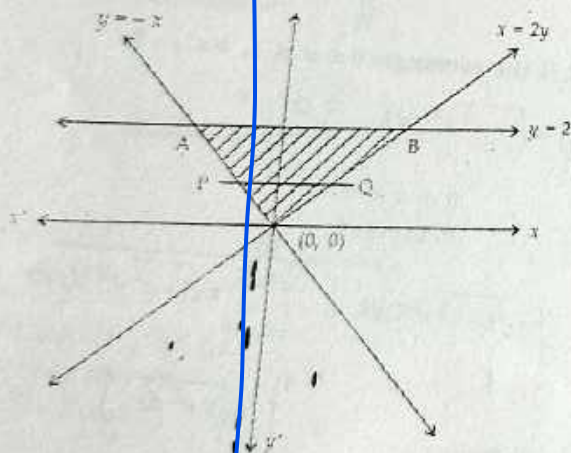
$$= \frac{1}{3} \left(\frac{1}{3} (e^6 - 1) \right) = \frac{1}{9} (e^6 - 1)$$

Q. 9. Evaluate the double integral :

$$\iint_D \frac{dA}{y^2 + 1}$$

where D is the triangular region bounded by $y = -x$ and $y = 2$.

Ans.

The point of intersection of $y = -x$ and $y = 2$ is

$$y = -x \quad \text{and} \quad y = 2$$

$$2 = -x \Rightarrow x = -2$$

$$\Rightarrow A(-2, 2)$$

The point of intersection of $x = zy$ and $y = 2$ i.e.

$$x = zy \quad \text{and} \quad y = 2$$

$$x = 2(2) \quad \text{and} \quad y = 2$$

$$x = 4 \quad \text{and} \quad y = 2$$

$$\Rightarrow B(4, 2)$$

The point of intersection of $x = 2y$ and $y = -x$

$$x = 2y \quad \text{and} \quad y = -x$$

$$x = 2y \Rightarrow x = 2(-x)$$

$$3x = 0 \Rightarrow x = 0$$

$$y = -x = 0 \Rightarrow C(0, 0)$$

Along PQ $\rightarrow y = \text{Constant}$

$$\text{At } P \rightarrow x = -y$$

$$\text{At } Q \rightarrow x = -y$$

$$\iint_D \frac{dA}{y^2 + 1} = \int_{y=0}^2 \int_{x=-y}^{2y} \frac{1}{y^2 + 1} dx dy$$

$$= \int_{y=0}^2 \frac{1}{y^2 + 1} \left[x \right]_{x=-y}^{2y} dy = \int_{y=0}^2 \frac{1}{y^2 + 1} [2y - (-y)] dy$$

$$= \int_{y=0}^2 \frac{1}{y^2 + 1} (2y + y) dy = \int_{y=0}^2 \frac{1}{y^2 + 1} (3y) dy$$

$$= \int_{y=0}^2 \frac{3y dy}{y^2 + 1} = \frac{3}{2} \left[\log(y^2 + 1) \right]_0^2$$

$$= \frac{3}{2} \left[\log(2^2 + 1) - \log(0^2 + 1) \right]$$

$$= \frac{3}{2} [\log 5 - \log 1] = \frac{3}{2} [\log 5 - 0]$$

$$= \frac{3}{2} \log 5$$

Q. 10. Evaluate the double integral

$$\int_0^2 \int_y^{\sqrt{8-y^2}} \frac{1}{\sqrt{1+x^2+y^2}} dx dy$$

by converting to polar co-ordinates.
Ans. Let

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Region :

$$x = y \quad \text{to} \quad x = \sqrt{8-y^2}$$

$$x = y \quad \text{to} \quad x^2 = 8-y^2$$

$$x = y \quad \text{to} \quad x^2 + y^2 = 8$$

Since, y varies from 0 to 2

Hence, we will only consider in 1st quadrant

Point of intersection :

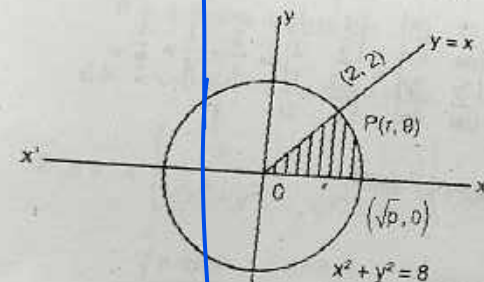
$$\Rightarrow x = y \quad \text{and} \quad x^2 + y^2 = 8$$

$$\Rightarrow y^2 + y^2 = 8 \quad \Rightarrow 2y^2 = 8$$

$$\Rightarrow y^2 = 4 \quad \Rightarrow y = \pm 2$$

$$\Rightarrow x = y = \pm 2$$

Hence, the point of intersection of $x = y$ and $x^2 + y^2 = 8$ are (2, 2) and (-2, -2)



$$I = \int_0^2 \int_y^{\sqrt{8-y^2}} \frac{1}{\sqrt{1+x^2+y^2}} dx dy$$

$$= \int_{\theta=0}^{\frac{\pi}{4}} \int_{r=0}^{\sqrt{8}} \frac{1}{\sqrt{r^2+1}} (r dr d\theta)$$

Let

$$r^2 + 1 = t^2$$

$$2r dr = 2t dt$$

$$r dr = t dt$$

at

$$r = 0, \quad t^2 = r^2 + 1 = 0 + 1 = 1 \quad \Rightarrow \quad t = 1$$

$$r = \sqrt{8}, \quad t^2 = r^2 + 1 = 8 + 1 = 9 \quad \Rightarrow \quad t = 3$$

$$I = \int_{\theta=0}^{\frac{\pi}{4}} \int_{t=1}^3 \frac{1}{t} dt d\theta = \int_{\theta=0}^{\frac{\pi}{4}} \left[\ln t \right]_{t=1}^3 d\theta$$

$$= \int_{\theta=0}^{\frac{\pi}{4}} [\ln 3 - \ln 1] d\theta = \int_{\theta=0}^{\frac{\pi}{4}} \ln 3 d\theta$$

$$= 2 \ln 3 \left[\frac{\pi}{4} - 0 \right] = \frac{\pi}{2} \ln 3$$

Q. 11. Find the volume of the tetrahedron T bounded by the plane $2x + y + 3z = 6$ and the co-ordinates plane $x = 0, y = 0$ and $z = 0$.

Ans.

$$\begin{cases} 2x + y + 3z = 6 \\ x = 0, y = 0, z = 0 \end{cases}$$

from (1)

$$3z = 6 - 2x - y$$

$$z = \frac{1}{3} (6 - 2x - y)$$

Here, T is bounded below $z = 0$ and bounded above by $z = \frac{1}{3} (6 - 2x - y)$

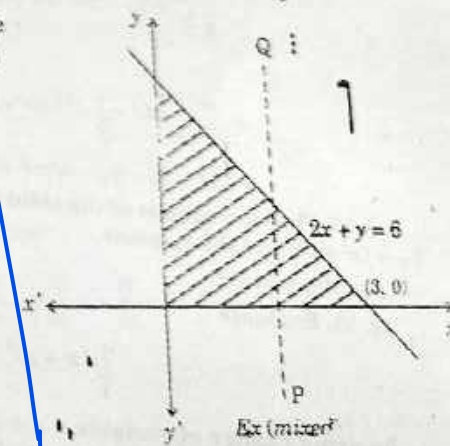
The projection of T on xy plane

$$\text{Put } z = 0, \quad 2x + y + 3z = 6$$

$\Rightarrow$

$$2x + y = 6$$

x	0	3
y	6	0



Along PQ $\rightarrow x = \text{const}$ (from $x = 0$ to $x = 3$)

At P $\rightarrow y = 0$

At Q $\rightarrow y = 6 - 2x$ ($2x + y = 6$)

$$V = \iiint_T dV = \iiint_A \left(\frac{1}{3} (6 - 2x - y) \right) dz dA$$

$$= \int_0^3 \int_0^{6-2x} \left(\frac{1}{3} (6 - 2x - y) \right) dy dx$$

$$= \int_0^3 \int_0^{6-2x} \left(\frac{1}{3} (6 - 2x - y) \right) dy dx$$

$$= \frac{1}{3} \int_0^3 \int_0^{6-2x} (6 - 2x - y) dy dx$$

$$= \frac{1}{3} \int_0^3 \left[6y - 2xy - \frac{y^2}{2} \right]_0^{6-2x} dx$$

$$= \frac{1}{3} \int_0^3 \left[6(6-2x) - 2x(6-2x) - \frac{(6-2x)^2}{2} \right] dx$$

$$= \frac{1}{3} \int_0^3 \left[6(6-2x) - 2(6x-2x^2) - \frac{(6-2x)^2}{2} \right] dx$$

$$= \int_0^3 \left(6 - 4x + \frac{2}{3}x^2 \right) dx = \left[6x - 2x^2 + \frac{2}{9}x^3 \right]_0^3$$

$$= (18 - 18 + 6) - (0 - 0 + 0) = 6$$

Q. 12. Find the volume of the solid D bounded by the paraboloid $z = 1 - 4(x^2 + y^2)$ and the xy -plane.

Ans. Do yourself.

Q. 13. Evaluate

$$\iiint_D (x+y)^5 (x-y)^2 dy dx$$

by using change of variable $u = x + y$ and $v = x - y$, where D is the region in the xy -plane which is bounded by the co-ordinate axes and the line $x + y = 1$.

Ans.

$$u = x + y$$

...(i)

$$v = x - y$$

...(ii)

$$\iiint_D (x+y)^5 (x-y)^2 dy dx$$

Let $I = \iiint_D$

from (i) and (ii):

$$u + v = (x+y) + (x-y)$$

$$u + v = 2x$$

$$\therefore x = \frac{u+v}{2}$$

from (i) (ii) in

$$u - v = (x+y) - (x-y)$$

$$u - v = -y$$

$$y = \frac{-1}{2}(u-v) = \frac{1}{2}(v-u)$$

Let the region D is modified into D^*

$$x + y = 1 \text{ implies } u = 1$$

$$x = 0 \text{ implies } u = -v$$

$$y = 0 \text{ implies } u = v$$

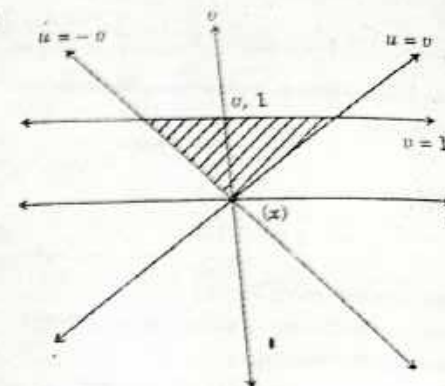
$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$I = \iiint_D (x-y)^5 (x+y)^3 du dv$$

$$= \iiint_{D^*} u^5 v^3 \left| \frac{1}{2} \right| du dv$$

$$= \frac{1}{2} \int_{v=0}^{u=1} \int_{u=-v}^u u^5 v^3 du dv$$

$$= \frac{1}{2} \int_{v=0}^1 v^3 \left[\frac{u^6}{6} \right]_{u=-v}^{u=v} dv$$



Section - III

Attempt any four questions from this section

Q. 14. Evaluate the line integral

$$\int_C \vec{F} \cdot d\vec{R}$$

where

$$\vec{F} = \frac{x}{\sqrt{x^2+y^2}} \vec{i} - \frac{y}{\sqrt{x^2+y^2}} \vec{j}$$

and C is the quarter circle path $x^2 + y^2 = a^2$, traversed from $(a, 0)$ to $(0, a)$.

$$\text{Ans. } \vec{F} = \frac{x}{\sqrt{x^2+y^2}} \vec{i} - \frac{y}{\sqrt{x^2+y^2}} \vec{j}$$

$$I = \int_C \vec{F} \cdot d\vec{R}$$

$$x^2 + y^2 = a^2$$

putting $(a, 0)$ to $(0, 0)$ Let $x = a \cos \theta$

$$y = a \sin \theta$$

$$\vec{R} = a \cos \theta \vec{i} + a \sin \theta \vec{j}$$

$$d\vec{R} = (-a \sin \theta \vec{i} + a \cos \theta \vec{j}) d\theta$$

$$\vec{F} \cdot d\vec{R} = \left(\frac{x}{\sqrt{x^2+y^2}} \vec{i} - \frac{y}{\sqrt{x^2+y^2}} \vec{j} \right) \cdot (-a \sin \theta \vec{i} + a \cos \theta \vec{j}) d\theta$$

$$= \left(\frac{a \cos \theta}{a} \vec{i} - \frac{a \sin \theta}{a} \vec{j} \right) \cdot (-a \sin \theta \vec{i} + a \cos \theta \vec{j}) d\theta$$

$$= (\cos \theta \vec{i} - \sin \theta \vec{j}) \cdot (-a \sin \theta \vec{i} + a \cos \theta \vec{j}) d\theta$$

$$= [(\cos \theta)(-a \sin \theta) + (-\sin \theta)(a \cos \theta)] d\theta$$

$$= (-a \sin \theta \cos \theta - a \sin \theta \cos \theta) d\theta$$

$$= -2a \sin \theta \cos \theta d\theta$$

$$I = \int_C \vec{F} \cdot d\vec{R}$$

$$= \int_0^{\pi/2} (-2a \sin \theta \cos \theta) d\theta$$

$$= -a \int_0^{\pi/2} \sin 2\theta d\theta = -a \left[\frac{-\cos 2\theta}{2} \right]_0^{\pi/2}$$

$$= \frac{a}{2} [\cos 2\theta]_0^{\pi/2} = \frac{a}{2} [\cos \pi - \cos 0]$$

$$= \frac{a}{2} [(-1) - (1)] = \frac{a}{2} (-2) = -a$$

Q. 15. Show that the vector field

$$\vec{F}(x, y, z) = (\sin z, -z \sin y, x \cos z + \cos y)$$

is conservative and evaluate

$$\int_C \vec{F} \cdot d\vec{R}$$

for any piecewise smooth path joining $A(1, 0, -1)$ to $B(0, -1, 1)$.

$$\text{Ans. } \vec{F}(x, y, z) = (\sin z, -z \sin y, x \cos z + \cos y)$$

$$\text{i) } \text{Curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin z & -z \sin y & x \cos z + \cos y \end{vmatrix}$$

$$= \vec{i} [(-\sin y) - (-\sin y)] - \vec{j} [(\cos z) - (\cos z)] + \vec{k} [(0) - (0)]$$

$$= 0 - 0 + 0 = 0$$

$$\Rightarrow \text{Curl } \vec{F} = 0$$

 $\Rightarrow \vec{F}$ is a conservative vector field.(2) let f be a scalar potential of $\vec{F}$ then

$$\vec{F} = \nabla f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

here,

$$\frac{\partial f}{\partial x} = \sin z$$

$$\frac{\partial f}{\partial y} = -z \sin y$$

$$\frac{\partial f}{\partial z} = x \cos z + \cos y$$

From (2.1)

$$\frac{\partial f}{\partial x} = \sin z$$

On integraty w.r.t. x

$$f = x \sin z + C(y, z)$$

Diff. partially w.r.t. y

$$\frac{\partial f}{\partial y} = 0 + \frac{\partial C}{\partial y}$$

From

$$\frac{\partial f}{\partial y} = \frac{\partial C}{\partial y}$$

From (2.2) and (4)

$$-z \sin y = \frac{\partial C}{\partial y}$$

$$\frac{\partial C}{\partial y} = -z \sin y$$

On integrating w.r.t. y

$$C(y) = z \cos y + K$$

Let constant = K = 0

$$C(y) = z \cos y$$

Therefore, from equation (3) we have scalar potential are :

$$f = x \sin z + (1y, z)$$

$$f = x \sin z + z \cos y$$

$$\int_C \vec{F} \cdot d\vec{k}$$

$$= \int_{(1, 0, -1)}^{(0, -1, 1)} \vec{F} \cdot d\vec{k} = f(0, -1, 1) - f(1, 0, -1)$$

$$= [0 + \cos(-1)] - [\sin(-1) - \cos 0]$$

$$= [\cos(-1)] - [\sin(-1) - 1] = [\cos(1)] - [-\sin(1) - 1]$$

$$= \cos 1 + \sin 1 + 1$$

Q. 16. Use Green's theorem, to find the work done by the force field

$$F(x, y) = (3y - 4x)i + (4x - y)j$$

when an object moves once counterclockwise around the ellipse $4x^2 + y^2 = 4$.

$$\text{Ans. } \vec{F}(x, y) = (3y - 4x)i + (4x - y)j \quad \dots (1)$$

$$4x^2 + y^2 = 4 \quad \dots (2)$$

Work done (W) is calculated as :

$$W = \oint_C \vec{F} \cdot d\vec{k} = \oint_C [(3y - 4x)i + (4x - y)j] \cdot [dx i + dy j]$$

$$= \oint_C [(3y - 4x) dx + (4x - y) dy] = \oint_C [M dx - N dy]$$

$$\text{where, } M = 3y - 4x$$

$$N = 4x - 3y$$

Using Green's theorem, we have,

$$W = \iint_D \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

D = region enclosed by ellipse (z)

$$= \iint_D [4 - 3] dA = \iint_D 1 dA$$

$$= \text{area of ellipse} = \pi ab$$

$$\text{Here, } 4x^2 + y^2 = 4$$

$$\frac{x^2}{1} + \frac{y^2}{4} = 1$$

$$\frac{x^2}{1^2} + \frac{y^2}{2^2} = 1$$

$$\Rightarrow a = 1, b = 2$$

$$W = \pi ab = \pi(1)(2) = 2\pi$$

Q. 17. Use Stokes's theorem, to evaluate the line integral

$$\oint_C (3y dx + 2z dy - 5x dz)$$

where C is the intersection of the xy-plane and the hemisphere

$$z = \sqrt{1 - x^2 - y^2}$$

traversed counterclockwise as viewed from above.

Ans. Let $I = \oint_C [3y \, dx + 2z \, dy - 5x \, dz]$

$$= \oint_C \vec{F} \cdot d\vec{R}$$

where, $\vec{F} = 3y \, \vec{i} + 2z \, \vec{j} - 5x \, \vec{k}$
using Stokes' theorem,

$$\oint_C \vec{F} \cdot d\vec{R} = \iint_S (\text{Curl } \vec{F} \cdot \vec{N}) \, ds$$

$$S = xy \text{ plane}; z = 0 = f(x, y) = \sqrt{1-x^2-y^2}$$

$$R = x^2 + y^2 \leq 1$$

R : Projection of given surface on xy -plane

$$f_x = \frac{-x}{\sqrt{1-x^2-y^2}}, \quad f_y = \frac{-y}{\sqrt{1-x^2-y^2}}$$

$$I = \iint_R (\text{Curl } \vec{F}) \cdot (-f_x, -f_y, 1) \, dx \, dy$$

$$\text{Curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3y & 2z & -5x \end{vmatrix}$$

$$= \vec{i}(-2) - \vec{j}(-5-0) + \vec{k}(-2)$$

$$= -2\vec{i} + 5\vec{j} - 3\vec{k} = (-2, 5, -3)$$

$$I = \iint_R (-2, 5, -3) \cdot \left(\frac{x}{\sqrt{1-x^2-y^2}}, \frac{y}{\sqrt{1-x^2-y^2}}, 1 \right) dA$$

$$= \iint_R \left(\frac{-2x}{\sqrt{1-x^2-y^2}} + \frac{5y}{\sqrt{1-x^2-y^2}} - 3 \right) dA$$

Let,

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx \, dy = r \, dr \, d\theta$$

r series from 0 to 1

θ series from 0 to 2π

$$I = -8\pi$$

Q. 18. Evaluate

$$\iint_S (\vec{F} \cdot \vec{N}) \, dS$$

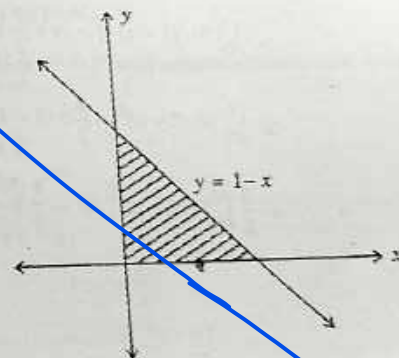
Where $\vec{F} = x^2 \vec{i} + xy \vec{j} + x^2 y^2 \vec{k}$ and S is the surface of the tetrahedron bounded by the plane $x + y + z = 1$ and the coordinate planes, with outward unit normal vector $\vec{N}$.

Ans. $\vec{F} = x^2 \vec{i} + xy \vec{j} + x^2 y^2 \vec{k}$
 $x + y + z = 1$

The projected region in xy plane is

$$\text{div } \vec{F} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial z}(x^2 y^2)$$

$$= 2x + x + 0 = 3x$$



The tetrahedron is the set,

$$R = \{ \text{all } (x, y, z) \text{ s.t. } 0 \leq z \leq 1-x-y \}$$

whenever $0 \leq y \leq 1-x$ for $0 \leq x \leq 1$

This projects onto the triangular region in the xy plane described by

$D : \text{all } (x, y) \text{ s.t. } 0 \leq y \leq 1-x \text{ for } 0 \leq x \leq 1$

Then, by applying the Divergence theorem,

We have,

$$\begin{aligned}
 \iint_S \vec{F} \cdot \vec{N} \, dS &= \iiint_D \operatorname{div} \vec{F} \, dV \\
 &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 3x \, dz \, dy \, dx = \int_0^1 \int_0^{1-x} [3x]_0^{1-x-y} \, dy \, dx \\
 &= \int_0^1 \int_0^{1-x} 3x(1-x-y) \, dy \, dx \\
 &= \int_0^1 \int_0^{1-x} (3x - 3x^2 - 3xy) \, dy \, dx \\
 &= \int_0^1 \left[3x(y) - 3x^2(y) - \frac{3}{2}xy^2 \right]_0^{1-x} dx \\
 &= \int_0^1 \left[3x(1-x) - 3x^2(1-x) - \frac{3}{2}x(1-x)^2 \right] dx \\
 &= \int_0^1 \left[3x(1-x) \left(1-x - \frac{1}{2}(1-x) \right) \right] dx \\
 &= \int_0^1 \left[3x(1-x)^2 - \frac{3}{2}x(1-x)^2 \right] dx \\
 &= \frac{3}{2} \int_0^1 3x(1-x)^2 \, dx = \frac{1}{8}
 \end{aligned}$$

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 Unique Paper Code : 32351301
 Name of the Paper : THEORY OF REAL FUNCTIONS
 Name of Course : B.Sc. (Hons.) Mathematics II Year
 Semester : 3
 Duration : 3 Hours

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Maximum Marks : 75

REVISED SYLLABUS (With effective from Academic year 2019-20)

Attempt any three parts from each question.

All questions are compulsory.

Q. 1. (a) Find the following limit and establish it by using $\epsilon - \delta$ definition of limit :

$$\lim_{x \rightarrow -1} \frac{x+5}{2x+3}$$

$$\begin{aligned}
 \text{Ans. } \lim_{x \rightarrow -1} \frac{x+5}{2x+3} &= \frac{-1+5}{2(-1)+3} \\
 &= 4
 \end{aligned}$$

Let $\epsilon > 0$

To find $\delta > 0$ s.t. if $|x+1| < \delta$ then $\left| \frac{x+5}{2x+3} - 4 \right| < \epsilon$

$$\begin{aligned}
 \text{If } |x+1| &< \frac{1}{4} \\
 \Rightarrow -\frac{1}{4} &< x+1 < \frac{1}{4} \\
 \Rightarrow -\frac{5}{4} &< x < -\frac{3}{4} \\
 \Rightarrow \frac{1}{2} &< 2x+3 < \frac{3}{2} \\
 \Rightarrow \frac{2}{3} &< \frac{1}{2x+3} < 2 \\
 \Rightarrow \frac{1}{|2x+3|} &< 2
 \end{aligned}$$

Consider

$$\begin{aligned}
 \left| \frac{x+5}{2x+3} - 4 \right| &= \left| \frac{x+5-8x-12}{2x+3} \right| \\
 &= \left| \frac{-7x-7}{2x+3} \right| \\
 &= \frac{7|x+1|}{|2x+3|}
 \end{aligned}$$

(277)

$$\Rightarrow \left| \frac{x+5}{2x+3} - 4 \right| < 14(x+1)$$

$$\Rightarrow \left| \frac{x+5}{2x+3} - 4 \right| < \epsilon \text{ if } 14|x+1| < \epsilon$$

$$\Rightarrow |x+1| < \frac{\epsilon}{14}$$

Let $\delta = \min \left\{ \frac{1}{4}, \frac{\epsilon}{14} \right\}$

Since, $|x+1| < \delta$

$$\Rightarrow \left| \frac{x+5}{2x+3} - 4 \right| < \epsilon$$

$$\Rightarrow \lim_{x \rightarrow -1} \frac{x+5}{2x+3} = 4$$

(b) State and prove the sequential criterion for limits of a real valued function.

Ans. Let $f: A \rightarrow \mathbb{R}$ and let c be a cluster point of A . Then the following are equivalent

(i) $\lim_{x \rightarrow c} f(x) = L$

(ii) For every sequence $\langle x_n \rangle$ in A that converges to c s.t. $x_n \neq c$ for all $n \in \mathbb{N}$, the sequence $\langle f(x_n) \rangle$ converges to L .

Proof: Let $\lim_{x \rightarrow c} f(x) = L$

Let $\langle x_n \rangle$ be a sequence of real number in A s.t. $\lim_{n \rightarrow \infty} x_n = c$

To prove $\langle f(x_n) \rangle$ converges to L

Let $\epsilon > 0$ be given then $\exists \delta > 0$ s.t.

$$|f(x_n) - L| < \epsilon \text{ if } |x_n - c| < \delta$$

Since, $\lim_{n \rightarrow \infty} x_n = c$

$\Rightarrow$ For $\delta > 0 \exists$ a positive integer m s.t.

$$|x_n - c| < \delta \quad \forall n \geq m$$

Also $\lim_{x \rightarrow c} f(x) = L$

$$\Rightarrow |f(x) - L| < \epsilon \text{ if } |x - c| < \delta$$

$$\Rightarrow |x - c| < \delta \text{ for } n \geq m$$

$$\Rightarrow |x_n - c| < \delta \quad n \geq m$$

$$\Rightarrow |f(x_n) - L| < \epsilon \text{ for } n \geq m$$

$\Rightarrow$ Sequence $\langle f(x_n) \rangle$ converges to L

Conversely,

Let $\langle x_n \rangle$ be a sequence which converges to c and sequence $\langle f(x_n) \rangle$ converge to L .

We have to show that $\lim_{x \rightarrow c} f(x) = L$

Let if possible $\lim_{x \rightarrow c} f(x) \neq L$

Then there exist an ϵ_0 -neighbourhood $V_{\epsilon_0}(L)$ such that no matter what δ -neighbourhood of c , there will be at least one $x_\delta \in A \cap V_{\epsilon_0}(c)$ with $x_\delta \neq c$ such that $f(x_\delta) \notin V_{\epsilon_0}(L)$

$\Rightarrow$ For $n \in \mathbb{N} \exists x_n \in A$ s.t.

$$|x_n - c| < \frac{1}{n}$$

But $|f(x_n) - L| \geq \epsilon_0 \quad \forall n \in \mathbb{N}$

$\Rightarrow \langle x_n \rangle$ converges to c but $\langle f(x_n) \rangle$ does not converge to L .

It is a contradiction

$\Rightarrow$ Our assumption is wrong

$$\Rightarrow \lim_{x \rightarrow c} f(x) = L$$

(c) Determine whether the following limit exists in $\mathbb{R}$:

$$\lim_{x \rightarrow 0} \operatorname{sgn}(\sin 1/x^2)$$

Ans. Consult your Textbook.

(d) Show that: $\lim_{x \rightarrow 0^+} e^{1/x} = 0$ and establish that $\lim_{x \rightarrow 0^+} e^{1/x}$ does not exist in $\mathbb{R}$.

Ans. Let $x < 0, x \in \mathbb{R}$

To show $\lim_{x \rightarrow 0^-} e^{1/x} = 0$

Since, $0 < t < \epsilon^t$ for $t > 0$

Let $t = -\frac{1}{x}$

Since, $x < 0 \Rightarrow t > 0$

$$\Rightarrow 0 < -\frac{1}{x} < \frac{1}{e^{1/x}}$$

$$\Rightarrow 0 < -\frac{1}{x} < \frac{1}{e^{1/x}}$$

$$\Rightarrow 0 < e^{1/x} < -x$$

Since $\lim_{x \rightarrow 0^-} (-x) = 0$

$\Rightarrow$ By squeeze theorem

$$\lim_{x \rightarrow 0^-} e^{1/x} = 0$$

for $x > 0, x \in \mathbb{R}$

To show $\lim_{x \rightarrow 0^+} e^{1/x}$ does not exist

Since, $0 < t < e^t$ for $t > 0$

Let $t = \frac{1}{x}$

$\Rightarrow 0 < \frac{1}{x} < e^{1/x}$

Since, $x \rightarrow 0^+$

Let $x_n = \frac{1}{n}, n \in \mathbb{N}$

$\Rightarrow n < e^{1/x_n}$

$\Rightarrow \lim_{x \rightarrow 0^+} e^{1/x}$ does not exist in $\mathbb{R}$.

Q. 2. (a) Let $c \in \mathbb{R}$ and f be defined on (c, ∞) and $f(x) > 0$ for all $x \in (c, \infty)$. Show that

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

if and only if

$$\lim_{x \rightarrow \infty} \frac{1}{f(x)} = 0$$

Ans. Let $\lim_{x \rightarrow \infty} f(x) = \infty$

$\Rightarrow$ For $a > 0 \exists k \in \mathbb{R}$ s.t.
 $f(x) > a \forall x > k$

$\Rightarrow \frac{1}{f(x)} < \frac{1}{a} \forall x > k$

$\Rightarrow \frac{1}{f(x)} < \epsilon \forall x > k$

$\Rightarrow \left| \frac{1}{f(x)} - 0 \right| < \epsilon \forall x > k$

$\Rightarrow \lim_{x \rightarrow \infty} \frac{1}{f(x)} = 0$

Conversely, let $\lim_{x \rightarrow \infty} \frac{1}{f(x)} = 0$

$\Rightarrow$ For $\epsilon > 0 \exists$ some $k \in \mathbb{R}$ s.t.

$$\left| \frac{1}{f(x)} - 0 \right| < \epsilon \text{ for } x > k$$

$$\Rightarrow \left| \frac{1}{f(x)} \right| < \epsilon \text{ for } x > k$$

$$\Rightarrow \frac{1}{f(x)} < \epsilon \text{ for } x > 0 (f(x) > 0 \forall x \in (c, \infty))$$

$$f(x) > \frac{1}{\epsilon} \forall x > k$$

$$f(x) > \alpha \forall x > k$$

$$\Rightarrow \lim_{x \rightarrow \infty} f(x) = \infty$$

(b) Evaluate the following limit by using the appropriate definition:

$$\lim_{x \rightarrow 1^+} \frac{x}{x-1}$$

Ans. Since $x \rightarrow 1^+$

Let $\alpha > 1$

$$\text{s.t. } 1 < x < \frac{\alpha}{\alpha-1}$$

$$\Rightarrow \text{For } x < \frac{\alpha}{\alpha-1}$$

$$\Rightarrow x\alpha - x < \alpha$$

$$\Rightarrow x\alpha - \alpha < x$$

$$\Rightarrow \alpha < \frac{x}{x-1}$$

Since, $x \rightarrow 1^+ \Rightarrow |x-1| < \delta$ for $\delta > 0$

and $\alpha > 1, \alpha \in \mathbb{R}$

$$\Rightarrow \text{For } |x-1| < \delta$$

$$\frac{x}{x-1} > \alpha$$

$$\Rightarrow f(x) > \alpha$$

$$\Rightarrow \lim_{x \rightarrow 1^+} \frac{x}{x-1} = \infty$$

(c) Determine the points of continuity of the function $f(x) = x[x]$ where $[.]$ denotes the greatest integer function.

Ans. Let $a, b \in \mathbb{Z}$

Let $c \in \mathbb{R}$ s.t. $c \in]a, b[$

$$\Rightarrow f(x) = x[x]$$

$$\text{and } f(c) = c[c] = ca$$

$$\Rightarrow \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} x[x] = ca$$

$$\Rightarrow \lim_{x \rightarrow c} f(x) = f(c)$$

$\Rightarrow$ Let $f(x) = x[x]$ is continuous for all $c \in]a, b[$.

$$c = a$$

$$\lim_{x \rightarrow a^-} x[x] = \lim_{h \rightarrow 0} (a-h)[a-h]$$

and

$$\lim_{x \rightarrow a^+} x[x] = \lim_{h \rightarrow 0} (a+h)[a+h]$$

$$\lim_{x \rightarrow a^-} f(x) = a \cdot a = a^2$$

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} x[x]$$

$\Rightarrow$ Limit does not exist at all integers
 $\Rightarrow f(x)$ is not continuous for all integers.

But for

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x[x] = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x[x] = 0$$

at

$$x = 0 \quad x[x] = 0$$

$\Rightarrow f(x) = x[x]$ is continuous $\forall c \in \mathbb{R}; c \in \mathbb{Z} - \{0\}$

(d) A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is such that $f(x+y) = f(x) + f(y)$ for all x, y in $\mathbb{R}$. Prove that if f is continuous at some point x_0 , then it is continuous at every point of $\mathbb{R}$.

Ans.

for

$$f(x+y) = f(x) + f(y)$$

$$x = y = 0$$

$$f(0+0) = f(0) + f(0)$$

$$f(0) = 2f(0)$$

$$f(0) = 0$$

for

$$y = -x$$

$$f(x+y) = f(x+(-x))$$

$$f(x-x) = f(x) + f(-x)$$

$$f(0) = f(x) + f(-x)$$

$$f(-x) = -f(x) \quad (\because f(0) = 0 \text{ from (1)})$$

$$f(x-x_0) = f(x) - f(x_0)$$

Since,

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

$$\Rightarrow \lim_{x \rightarrow x_0} f(x-x_0) = \lim_{x \rightarrow x_0} \{f(x) - f(x_0)\}$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x+x_0-x_0) = \lim_{x \rightarrow x_0} f(x) - f(x_0)$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) = f(x_0) - f(x_0)$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) = f(0)$$

$\Rightarrow f(x)$ is continuous at $x = 0$

$\Rightarrow f(x)$ is continuous at every point.

Q. 3. (a) Let $A \subseteq \mathbb{R}$ and $f: A \rightarrow \mathbb{R}$ and let $f(x) \geq 0$, for all $x \in A$. Let $\sqrt{f}$

is defined as $\sqrt{f(x)} = \sqrt{f(x)}$ for $x \in A$. Show that if f is continuous at a point $c \in A$, then $\sqrt{f}$ is continuous at c .

Ans. Since, f is continuous at $x = c \in A$

$\Rightarrow$ for $\epsilon_1 > 0$ be given $\exists \delta > 0$ s.t.

$$|f(x) - f(c)| < \epsilon_1 \quad \text{if } |x - c| < \delta$$

Let,

$$|x - c| < \delta$$

To show $|\sqrt{f(x)} - \sqrt{f(c)}| < \epsilon_2$ for some $\epsilon_2 > 0$.

$$(\sqrt{f(x)} - \sqrt{f(c)})^2 = f(x) + f(c) - 2\sqrt{f(x)} \cdot \sqrt{f(c)}$$

$$< f(x) + f(c)$$

$$= f(x) - f(c) + f(c) + f(c)$$

$$= f(x) - f(c) + 2f(c)$$

$$< \epsilon_1 + 2f(c)$$

$$\Rightarrow |\sqrt{f(x)} - \sqrt{f(c)}| < \sqrt{\epsilon_1 + 2f(c)}$$

$$= \epsilon_2 \text{ (let)}$$

$\Rightarrow$ if $|x - c| < \delta$ then $|\sqrt{f(x)} - \sqrt{f(c)}| < \epsilon_2$

$\Rightarrow \sqrt{f(x)}$ is continuous at $x = c$.

(b) Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $\mathbb{R}$ and that $f(r) = 0$ for every rational number r . Prove that $f(x) = 0$ for all $x \in \mathbb{R}$.

Ans. Let $x \in \mathbb{R}$ be a rational number

$\Rightarrow f(x) = 0$ (given)

Let $x \in \mathbb{R}$ be an irrational number

Since, the irrational number is surrounded by infinite many rational number.

$\Rightarrow$ We get a sequence of rational number s.t. the sequence $\langle x_n \rangle$ converges to x

$$\Rightarrow \lim_{n \rightarrow \infty} x_n = x$$

Since, $x \in \mathbb{R}$ and f is continuous on $\mathbb{R}$.

$\Rightarrow f$ is continuous on x

$$\Rightarrow \lim_{n \rightarrow \infty} f(x_n) = f(x)$$

Since, each x_n is rational

$$\Rightarrow f(x_n) = 0 \quad \forall x_n$$

$$\Rightarrow \lim_{n \rightarrow \infty} f(x_n) = 0$$

$$\Rightarrow f(x) = 0 \quad (\text{Using equation (1)})$$

$$\Rightarrow f(x) = 0 \quad \forall x \in \mathbb{R}$$

(c) Let f be a continuous and real valued function defined on a closed and bounded interval $[a, b]$. Prove that f is bounded. Give an

example to show that the condition of boundedness of the interval cannot be dropped.

Ans. Let if possible f is not bounded on $[a, b]$.

$\Rightarrow$ For each $n \in \mathbb{N}$ we can find points $x_n \in [a, b]$ which satisfy $|f(x_n)| > n$... (1)

Since, each $x_n \in [a, b]$,

$$a \leq x_n \leq b, \forall n$$

$\Rightarrow$ Sequence $\langle x_n \rangle$ is bounded

By Bolzano Weierstrass theorem

The sequence $\langle x_n \rangle$ has a convergent subsequence, say $\langle x_{n_k} \rangle$

Let $\langle x_{n_k} \rangle$ converges to x

Since $x_{n_k} \in [a, b]$

and closed interval contain all its limit point

$\Rightarrow x \in [a, b]$

Since, f is continuous on $[a, b]$

$\Rightarrow f$ is continuous at x

But, from Eq. (1)

$$|f(x_{n_k})| > n_k \geq k$$

$\Rightarrow$

$$|f(x_{n_k})| > k$$

$\Rightarrow \langle f(x_{n_k}) \rangle$ is not bounded and hence not convergent but $x_{n_k} \rightarrow x$ and $f(x_{n_k}) \not\rightarrow f(x)$

$\Rightarrow f$ is not continuous at x .

Which is a contradiction.

Hence our assumption is wrong.

$\Rightarrow f$ is bounded on $[a, b]$

Let $[0, \infty)$ be a semi closed interval

$\Rightarrow [0, \infty)$ is unbounded

Let

$$f(x) = x$$

$\Rightarrow f(x)$ is continuous but not bounded.

$\Rightarrow$ The interval must be bounded.

(d) State the intermediate value theorem. Show that f is continuous on I for some $x \in [0, 1]$.

Ans. Let I be an interval and let $f: I \rightarrow \mathbb{R}$ be continuous on I .

If $a, b \in I$ and if $k \in \mathbb{R}$ satisfies $f(a) < k < f(b)$, then there exist a point $\alpha \in I$ between a and b such that $f(\alpha) = k$.

Q. 4 (a) Show that the function $f(x) = x^2$ is uniformly continuous on $[-2, 2]$, but it is not uniformly continuous on $\mathbb{R}$.

Ans.

$$f(x) = x^2$$

Let $x, c \in [-2, 2]$

$$\Rightarrow -2 \leq x \leq 2 \text{ and } -2 \leq c \leq 2$$

$$\Rightarrow -4 \leq x+c \leq 4$$

Let $|x-c| < \delta$ for $\delta > 0$

Let $\epsilon > 0$ be given s.t. for $\delta > 0$

$$|x-c| < \delta$$

$$\begin{aligned} \text{Consider } |f(x) - f(c)| &= |x^2 - c^2| \\ &= |(x-c)(x+c)| \\ &= |x-c| |x+c| \\ &< \delta \cdot 4 = 4\delta \quad (\text{let}) \end{aligned}$$

$$\Rightarrow |f(x) - f(c)| < \epsilon \text{ if } |x-c| < \delta$$

$\Rightarrow f$ is uniformly continuous on $[-2, 2]$

Let $x, c \in \mathbb{R}$

s.t. $|x-c| < \delta$

$$\begin{aligned} |f(x) - f(c)| &= |x^2 - c^2| \\ &= |(x-c)(x+c)| \\ &= \delta |x+c| \end{aligned}$$

Since, $x, c \in \mathbb{R}$

$\Rightarrow$ There does not exist positive integer k s.t.

$$|x+c| < k \quad \forall x, c \in \mathbb{R}$$

$\Rightarrow$ There does not exist $\epsilon > 0$ s.t.

$$|f(x) - f(c)| < \epsilon \quad \forall |x-c| < \delta$$

$\Rightarrow f(x)$ is not uniformly continuous on $\mathbb{R}$.

(b) Prove that if f and g are uniformly continuous on $A \subseteq \mathbb{R}$ and if they both are bounded on A , then their product fg is uniformly continuous on A .

Ans. Since, f and g are uniformly continuous on $A \subseteq \mathbb{R}$

$\Rightarrow$ For $\epsilon_1, \epsilon_2 > 0$ be given then $\exists \delta > 0$ s.t.

$$|f(x) - f(u)| < \epsilon_1 \text{ if } |x-u| < \delta$$

and

$$|g(x) - g(u)| < \epsilon_2 \text{ if } |x-u| < \delta$$

Also, f and g are bounded on A

$\Rightarrow \exists$ some, $M_1, M_2 \in \mathbb{R}$ s.t.

$$|f(x)| \leq M_1 \text{ and } |g(x)| \leq M_2 \quad \forall x \in A$$

Let

$$M = \max\{M_1, M_2\}$$

$\Rightarrow$

$$|f(x)| \leq M \text{ and } |g(x)| \leq M \quad \forall x \in A$$

$$\begin{aligned} \text{Consider } |f(x)g(x) - f(u)g(u)| &= |f(x)g(x) - f(x)g(u) + f(x)g(u) - f(u)g(u)| \\ &\leq |f(x)| |g(x) - g(u)| + |g(u)| |f(x) - f(u)| \\ &\leq |f(x)| |g(x) - g(u)| + |g(u)| |f(x) - f(u)| \\ &< M \epsilon_2 + M \epsilon_1 \\ &= M(\epsilon_1 + \epsilon_2) \\ &= \epsilon \end{aligned}$$

$$\Rightarrow |f(x)g(x) - f(u)g(u)| < \epsilon \text{ if } |x-u| < \delta$$

$\Rightarrow f(x)g(x)$ is uniformly continuous on A .

(c) Show that the function $f(x) = |x+1| + |x-1|$ is not differentiable at -1 and 1 .

Ans. Ref. Q. 4 (b), Nov./Dec., 2018, p 227.

(d) Prove that if $f: R \rightarrow R$ is an even function and has a derivative at every point, then the derivative f' is an odd function.

Ans. Since, $f: R \rightarrow R$ is an even function
By chain rule,

$$f(-x) = f(x) \quad \forall x \in R$$

$$\Rightarrow [f(-x)]' = f'(-x) \cdot (-x)'$$

$$\Rightarrow [f(x)]' = f'(-x) \cdot (-1)$$

$$\Rightarrow f'(-x) = -f'(x)$$

$\Rightarrow f'(x)$ is an odd function.

Q. 5. (a) State Darboux theorem. Let I be an interval and $f: I \rightarrow R$ be differentiable on I . Show that if the derivative f' is never zero on I , then either $f'(x) > 0$ for all $x \in I$ or $f'(x) < 0$ for all $x \in I$.

Ans. Darboux Theorem

If f is differentiable on $[a, b]$ and if k is a number between $f'(a)$ and $f'(b)$, then there is at least one point c in $[a, b]$ such that $f'(c) = k$.

Since, f is differentiable on I

$\Rightarrow$ For $c \in I$

$$f'(c) \in R$$

Let

Since

$$f'(c) = k$$

$$f'(c) \neq 0$$

$$k \neq 0$$

By law of trichotomy either $k > 0$ or $k < 0$

$\Rightarrow f'(c) > 0$ or $f'(c) < 0$

Since, c is arbitrary

$\Rightarrow f'(x) > 0$ or $f'(x) < 0 \quad \forall x \in I$

(b) Find the Taylor's series for $\cos x$ and indicate why it converges to $\cos x$ for all $x \in R$.

Ans. Let

$$f(x) = \cos x \quad \forall x \in R$$

The Taylor's series of $f(x)$ is

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0) + \dots$$

Since,

$$f(x) = \cos x$$

$$\Rightarrow f(0) = 1$$

$$f'(x) = -\sin x$$

$$\Rightarrow f'(0) = 0$$

$$f''(x) = -\cos x$$

$$\Rightarrow f''(0) = -1$$

$$f^{(n-1)}(x) = \cos\left(x + (n+1)\frac{\pi}{2}\right) \quad \forall x \in R$$

Let

$$R_n(x) = \frac{x^{n-1}}{(n-1)!} \cos\left(x + (n-1)\frac{\pi}{2}\right)$$

Since

$$|R_n(x)| = \left| \frac{x^{n-1}}{(n-1)!} \cos\left(x + (n-1)\frac{\pi}{2}\right) \right|$$

$$\leq \frac{|x|^{n-1}}{(n-1)!}$$

Let

$$a_n = \frac{|x|^{n-1}}{(n-1)!}$$

$$a_{n+1} = \frac{|x|^n}{n!}$$

$\Rightarrow$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{|x|}{n} = 0$$

$\Rightarrow$

$$\lim_{n \rightarrow \infty} |R_n| = 0$$

$\Rightarrow$

$$f(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

This, Taylor's series converges to $\cos x \quad \forall x \in R$ as

$$\lim_{n \rightarrow \infty} R_n(x) = 0$$

(c) Prove that $e^x \geq 1 + x$ for all $x \in R$, with equality occurring if and only if $x = 0$.

Ans. Let

$$f(x) = e^x$$

$$x \in R$$

$\Rightarrow f(x)$ is defined by (1) is continuous and differentiable $\forall x \in R$

Let $x > 0$

Differentiating (1) w.r.t. x

$$f'(x) = e^x > 0$$

$$\Rightarrow f'(x) > 0$$

Consider the interval $[0, x]$

Applying LMVT to the function f on $[0, x]$

We get at least one point $c \in [0, x]$ s.t.

$$f'(c) = \frac{f(x) - f(0)}{x - 0}$$

$\Rightarrow$

$$e^c = \frac{e^x - 1}{x}$$

From Eq. (2) $f'(x) > 0$

$\Rightarrow f$ is an increasing function

i.e., for $x > 0 \quad f(x) > f(0)$

$\Rightarrow e^x > 1$ for $x > 0$

Since, $c \in [0, x]$

$\Rightarrow$

$$c > 0$$

$\Rightarrow$

$$e^c > 1$$

$$= \frac{e^x - 1}{x} > 1 \quad (\text{From Eq. (3)})$$

$$= e^x > x + 1 \quad \forall x > 0$$

$$\text{for } x = 0$$

$$= e^0 = 1 = 1 + 0 = 1 + x$$

$$= e^x = 1 + x$$

$$\text{for } x < 0$$

Applying LMVT to the function f on $[x, 0]$

We get at least one point $d \in [x, 0]$ s.t.

$$f'(d) = \frac{f(0) - f(x)}{0 - x} = \frac{1 - e^x}{-x}$$

$$f'(d) = \frac{e^d - 1}{x} \quad \dots (4)$$

From (2) $f'(x) > 0$ for $x < 0$

$\Rightarrow f$ is an increasing function for $x < 0$

$$= f(x) < f(0)$$

$$= e^x < 1$$

Since $d \in [x, 0]$

$$= d < 0$$

$$= e^d < e^0$$

$$= \frac{e^d - 1}{x} < 1 \quad (\text{From (4)})$$

$$= e^d > 1 + x$$

$$= e^x \geq 1 + x \quad \forall x \in \mathbb{R}$$

(d) Is $f(x) = |x|$, $x \in \mathbb{R}$, a convex function? Is every convex function differentiable? Justify your answer.

Ans. Let $I \subseteq \mathbb{R}$ be an interval.

A function $f: I \rightarrow \mathbb{R}$ is said to be convex on I if for any t satisfying $0 \leq t \leq 1$ and for any points $x_1, x_2 \in I$

$$f((1-t)x_1 + t(x_2)) \leq (1-t)f(x_1) + tf(x_2)$$

To check $f(x) = |x|$ is convex or not $\forall x \in \mathbb{R}$

Let $x_1, x_2 \in \mathbb{R}$

and Let

$$0 \leq t \leq 1$$

$$f((1-t)x_1 + t(x_2)) = |(1-t)x_1 + tx_2|$$

$$\leq |(1-t)x_1| + |tx_2|$$

$$= (1-t)|x_1| + t|x_2|$$

$$= (1-t)f(x_1) + tf(x_2)$$

$f(x) = |x|$ is convex for all $x \in \mathbb{R}$

Since, $f(x) = |x|$ is not differentiable.

$\Rightarrow$ Every convex function may or may not be differentiable.

AGS-S

Nov/Dec. 2019

Sr. No. of Q. Paper : 7464
 Unique Paper Code : 32351302
 Name of the Paper : GROUP THEORY-I
 Name of Course : B.Sc. (Hons.) Mathematics II year
 Semester : 3 (CBCS)
 Duration : 3 Hours
 Maximum Marks : 75

REVISED SYLLABUS (With effective from Academic year 2019-20)

Attempt any two parts from each question.
 All questions carry equal marks.

Q. 1. (a) Let $G = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} : a \in \mathbb{R}, a \neq 0 \right\}$. Show that G is a group under

matrix multiplication.

(b) Let G be a group and H be a subset of G . Prove that H is a subgroup of G if $a, b \in H \Rightarrow ab^{-1} \in H$. Hence prove that $H = \{A \in GL(2, \mathbb{R}) : \det A \text{ is a power of } 3\}$ is a subgroup of $GL(2, \mathbb{R})$.

(c) (i) Suppose G is a group that has exactly eight elements of order 3. How many subgroups of order 3 does G have?

(ii) If $|a| = n$ and k divides n , prove that $|a^{n/k}| = k$. 6×2=12

Sol. (a) $G = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} : a \in \mathbb{R}, a \neq 0 \right\}$

Let $A = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix}$

$$\Rightarrow A, B \in G$$

To show G is closed

$$A \times B = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow A \times B \in G$$

$$\Rightarrow G \text{ is closed}$$

To show G is associative

Let $A, B, C \in G$

and

$$(AB)C = A(BC)$$

$$\Rightarrow G \text{ is associative}$$

For identity

$$\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} e & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} e & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} ae & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ea & 0 \\ 0 & 0 \end{bmatrix}$$

Since,

$$ae = a = ea$$

$$a \neq 0 \text{ } a^{-1} \text{ exist}$$

$$ae = a$$

$$e = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ is identity for } G$$

(Premultiply both sides with a^{-1})

For inverse

$$\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ba & 0 \\ 0 & 0 \end{bmatrix}$$

$$ab = 1 = ba$$

$$a \neq 0$$

$$b = \frac{1}{a}$$

$$\Rightarrow \begin{bmatrix} 1/a & 0 \\ 0 & 0 \end{bmatrix} \text{ is inverse for } \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \in G$$

$\Rightarrow G$ is a group under matrix multiplication.

(b) Since, the operation of H is the same as that of G , it is clear that H is associative.

Let, $x \in H$

Let

$$a = x \text{ and } b = x$$

Since, for $a, b \in H$

$$ab^{-1} \in H$$

$$\Rightarrow xx^{-1} \in H$$

$$\Rightarrow e \in H$$

$\Rightarrow H$ has identity.

To show $x^{-1} \in H \forall x \in H$

Let

$$a = e \text{ and } b = x$$

Since,

$$ab^{-1} \in H$$

$$\Rightarrow e \cdot x^{-1} \in H$$

$$\Rightarrow x^{-1} \in H \forall x \in H$$

Now, to show H is closed

i.e., for $x \in H, y \in H \Rightarrow xy \in H$

Let $x \in H$ and $y \in H$

$$\Rightarrow x \in H \text{ and } y^{-1} \in H^*$$

Let

$$a = x \text{ and } b = y^{-1}$$

$$\Rightarrow ab^{-1} \in H$$

$$\Rightarrow x(y^{-1}) \in H$$

$$\Rightarrow xy \in H$$

$\Rightarrow H$ is closed

$\Rightarrow H$ is subgroup of G .

$$H = \{A \in G : |A| = 3^n, n \in \mathbb{Z}\}$$

Let $A, B \in H$

$$\Rightarrow |A| = 3^n \text{ and } |B| = 3^m, n, m \in \mathbb{Z}$$

$$\text{Since, } |B| = 3^m \neq 0$$

$$\Rightarrow |B^{-1}| = \frac{1}{|B|}$$

$$|B^{-1}| = \frac{1}{3^m} = 3^{-m}$$

$$\Rightarrow |AB^{-1}| = |A| \cdot |B^{-1}| = 3^n \cdot 3^{-m}$$

$$|AB^{-1}| = 3^{n-m}$$

$$\Rightarrow AB^{-1} \in H$$

$\Rightarrow H$ is subgroup of G .

(c) (i) Let G be a group.

Since, G has exactly eight elements of order 3

Let $a \in G$ s.t. $a \neq e$

$$\text{s.t. } |a| = 3$$

$$\Rightarrow a^3 = e$$

$$\Rightarrow a^2 \neq e$$

(Identity of G)

Let H be a subset of G

$$H = \{e, a, a^2\}$$

Since,

$$a \cdot a^2 = a^3 = e$$

$$\Rightarrow a \cdot a^2 = e$$

$$\Rightarrow a^{-1} = a^2$$

$$\text{and } (a^2)^2 = a^4 = a^3 \cdot a = e \text{ and } (a^2)^3 = a^6 = a^3 \cdot a^3 = e \cdot e = e$$

$$\Rightarrow a^2 \in G \text{ s.t. } |a^2| = 3 \text{ and } (a^2)^{-1} = a$$

$\Rightarrow H$ is closed, associative, inverse and has identity.

$\Rightarrow H$ is subgroup of G

Since, H has two elements of order 3

But G has 8 elements of order 3

$\Rightarrow G$ has 4 subgroups of order 3.

$$(ii) \text{ If } |a| = n \quad a^n = e$$

Since, k divides n

$$\Rightarrow (a^{n/k})^k = a^n = e$$

$$\text{To show } |a^{n/k}| = k$$

We have to show that for $t < k$

$$(a^{n/k})^t \neq e$$

$$\text{Since } t < k$$

$$\Rightarrow nt < nk$$

$$\frac{nt}{k} < n$$

Let if possible $(a^{n/k})^t = e$

$$\Rightarrow \frac{nt}{k} = n$$

$$\text{But } \frac{nt}{k} < n \Rightarrow \text{Contradiction}$$

$$\Rightarrow \frac{nt}{k} \neq n$$

$$\Rightarrow |a^{n/k}| \neq t$$

Since, t is arbitrary

$\Rightarrow$ Any number less than k is not an order of $a^{n/k}$

$$\Rightarrow |a^{n/k}| = k$$

Q. 2. (a) Let $G \langle a \rangle$ be a cyclic group of order n . Prove that $G = \langle a^k \rangle$ if and only if $\gcd(k, n) = 1$. List all the generators of Z_{20} .

(b) (i) If a cyclic group has an element of infinite order, how many elements of finite order does it have.

(ii) List all the elements of order 6 and 8 in Z_{30} .

(c) Suppose that a and b are group elements that commute and have orders m and n . If $\langle a \rangle \cap \langle b \rangle = \{e\}$, Prove that the group contains

an element whose order is the least common multiple of m and n . Show that this need not be true if a and b do not commute. 6.5×2=13

Sol. (a) Let $\gcd(k, n) = 1$

$\Rightarrow$ There exist some integers u and v

$$\text{s.t. } ku + nv = 1$$

$$\Rightarrow a = a^{ku + nv}$$

$$\Rightarrow a = a^{ku} a^{nv}$$

$$\Rightarrow a = a^{ku}$$

$$\Rightarrow a \in \langle a^k \rangle$$

$\Rightarrow$ any power of a belongs to $\langle a^k \rangle$

$\Rightarrow a^k$ is generator of G

$$\Rightarrow G = \langle a^k \rangle$$

Let if possible $\gcd(k, n) = d > 1$

$\Rightarrow$ for some integers t and s

$$k = dt \text{ and } n = ds$$

$$\text{Then } (a^k)^s = (a^{dt})^s = (a^{ds})^t = (a^n)^t = e^t = e$$

$$\Rightarrow |a^k| \leq s < n$$

$\Rightarrow a^k$ is not generator of G

Using above result

An integer k in Z_n is a generator of Z_n iff $\gcd(k, n) = 1$

$$Z_{20} = \langle 1 \rangle = \langle 3 \rangle = \langle 7 \rangle = \langle 9 \rangle = \langle 11 \rangle = \langle 13 \rangle = \langle 17 \rangle = \langle 19 \rangle$$

(b) (i) Let G be a cyclic group

Let ' a ' be a generator of G

$$\Rightarrow G = \langle a \rangle$$

Since all the elements of G are of the form $a, a^2, a^3, \dots$

and One element has infinite order

$\Rightarrow$ All the elements of G has infinite order.

$\Rightarrow$ Only element in G which has finite order is identity.

$\Rightarrow$ ' e ' has finite order in G .

(ii) Every subgroup of cyclic group is cyclic. Moreover if $|\langle a \rangle| = n$, then the order of any subgroup is divisor of n ; and for each positive divisor k of n , the group $\langle a \rangle$ has exactly one subgroup of order k - namely $\langle a^{n/k} \rangle$

$$\text{Since } |Z_{30}| = 30$$

and 6 divides 30

$$\langle 5 \rangle = \{0, 5, 10, 15, 20, 25\}$$

is a subgroup of order 6 and each element has order 6.

But 8 does not divide 30

$\Rightarrow$ There does not exist any element of order 8 in Z_{30} .

(c) Let $\text{lcm}(m, n) = t$
and $|ab| = s$

We have to show that $s = t$

Since, a, b are commutative elements.

$$\begin{aligned} \Rightarrow ab &= ba \\ \Rightarrow (ab)^t &= a^t \cdot b^t \\ (ab)^t &= e \cdot e \\ (ab)^t &= e \end{aligned}$$

(m and n divides t)

$\Rightarrow s$ divides t

But

$$\begin{aligned} e &= (ab)^s \\ \Rightarrow e &= a^s \cdot b^s \\ \Rightarrow b^{-s} &= a^s \end{aligned}$$

(a and b are commutative)

$\Rightarrow a^s$ and $b^{-s} \in \langle a \rangle \cap \langle b \rangle$

But $\langle a \rangle \cap \langle b \rangle = e$

$\Rightarrow s$ is divisible by m and n

$\Rightarrow s$ is divisible by $\text{lcm}(m, n)$

$\Rightarrow s$ is divisible by t

$\Rightarrow t$ divides s

From (1) and (2)

$$t = s$$

If a and b does not commute

Then $(ab)^t \neq a^t \cdot b^t$

Result may not be true.

Q. 3. (a) Let G be a group. Is $H = \{x^2 : x \in G\}$ a subgroup of G ? Justify.

(b) Prove that any two left cosets of a subgroup H in a group G are either equal or disjoint.

(c) Show that $(Q, +)$ has no proper subgroup of finite index. $6 \times 2 = 12$

Sol. (a) Let G be a group

Let $H = \{x^2 : x \in G\}$

To show H is subgroup of G

Since, $e \in G$ and $e^2 = e \in H$

$\Rightarrow H$ has identity

Let $a, b \in G$

s.t. $a^2, b^2 \in H$

To show $a^2(b^{-1})^2 \in H$ for $ab^{-1} \in G$

Consider $(ab^{-1})^2 = a^2 \cdot (b^{-1})^2$ (Only if G is abelian)

$\Rightarrow H$ is subgroup of G only if G is abelian.

(b) Let H be a subgroup of G

and Let $a, b \in G$

$\Rightarrow aH$ and bH are left cosets of G

To show $aH = bH$ or $aH \cap bH = \emptyset$

If $aH \cap bH = \emptyset$ there is nothing to prove

Let $aH \cap bH \neq \emptyset$

To show $aH = bH$

Since, $aH \cap bH \neq \emptyset$

Let $x \in aH \cap bH$

$\Rightarrow$ For h_1, h_2 belong to H

$$x = ah_1 \text{ and } x = bh_2$$

$$\Rightarrow a = xh_1^{-1}$$

$$\Rightarrow a = bh_2h_1^{-1}$$

$$\Rightarrow aH = bh_2h_1^{-1}H = bH \quad (\text{For } x \in H, xH = H)$$

$$\Rightarrow aH = bH$$

(c) Let $H \subseteq Q$ be a subgroup of Q

s.t. $|Q : H| = n$

$$\Rightarrow \frac{|Q|}{|H|} = n$$

Then for every $q \in Q$

$$n(q + H) = H$$

$$\Rightarrow nq + nH = H$$

$$\Rightarrow nq \in H$$

$$\Rightarrow nQ \subseteq H$$

$$\text{But } nQ = Q$$

$$\Rightarrow Q \subseteq H$$

$$\text{Also } H \subseteq Q$$

$$\Rightarrow H = Q$$

$\therefore (Q, +)$ has no proper subgroup of finite index.

Q. 4. (a) Prove that every subgroup of index 2 is normal. Show that A_5 is normal subgroup of S_5 .

(b) Let G be a group and H be a normal subgroup of G . Prove that the set of all left cosets of H in G forms a group under the operation $aH \cdot bH = abH$ where $a, b \in G$.

(c) If H is a normal subgroup of G with $|H| = 2$, prove that $H \subseteq Z(G)$. Hence or otherwise show that A_5 cannot have a normal subgroup of order 2. 6.5 × 2 = 13

Sol. (a) Let G be a group

Let H be a subgroup of G

$$\text{s.t. } \frac{|G|}{|H|} = 2$$

To prove H is normal in G

We have to show that $aH = Ha \quad \forall a \in G$

Let $a \in G$

If $a \in H$ then $aH = H$ and $Ha = H$

$$\Rightarrow aH = Ha$$

If $a \notin H$

$\Rightarrow eH$ and aH are two different left coset (If G is a finite group and H is subgroup of G , then number of distinct left coset is $\frac{|G|}{|H|}$)

$$eH + aH = G$$

$$\text{Similarly } He + Ha = G$$

$$\Rightarrow aH = G - eH = G - H = G - He = Ha$$

$$\Rightarrow aH = Ha$$

$$\Rightarrow aH = Ha \quad \forall a \in G$$

$\Rightarrow H$ is normal

$$\text{Since, } |S_5| = 5! = 120$$

$$\text{and } |A_5| = \frac{5!}{2} = 60$$

$$\Rightarrow \frac{|S_5|}{|A_5|} = 2$$

Using above result A_5 is normal subgroup of S_5 .

(b) Let G be a group.

Let H be a normal subgroup of G .

$\frac{G}{H} = \{aH \mid a \in G\}$ is the set of all left cosets of H in G .

To prove $\frac{G}{H}$ is a group under the operation $aH \cdot bH = abH; a, b \in G$.

Let $a, b \in G$ and a', b' also in G .

$$\text{s.t. } aH = a'H \text{ and } bH = b'H$$

$$\Rightarrow a' = ah_1$$

$$\text{and } b' = bh_2 \text{ for } h_1, h_2 \in H$$

$$a'b'H = ah_1bh_2H$$

$$= ah_1bH$$

$$= ah_1Hb$$

$$(H \text{ is normal subgroup of } G \Rightarrow aH = Ha)$$

$$= aHb$$

$$= abH$$

$$\Rightarrow \text{For } a'H, b'H \in aH \in a'b'H \in aH$$

$\Rightarrow$ Operation is well defined

For $e \in G$ be identity

Then $eH = H$ is identity

For $a \in G, a^{-1} \in G$

$$\Rightarrow aH \in \frac{G}{H}$$

$$a^{-1}H \in \frac{G}{H}$$

$\Rightarrow \frac{G}{H}$ has inverse

$$\text{also } (aH)(bH \cdot cH) = aHbcH = abcH$$

$$\text{and } (aHbH)cH = abHcH = abcH$$

$\frac{G}{H}$ is associative.

$\Rightarrow \frac{G}{H}$ is a group.

$$(c) \quad Z(G) = \{a \in G \mid ax = xa \quad \forall x \in G\}$$

Let H be normal subgroup of G .

$$\text{Since, } |H| = 2$$

$$\therefore \text{ Let } H = \{e, a\}$$

also, H is normal subgroup of G

$$\Rightarrow xH = Hx \quad \forall x \in G$$

$\Rightarrow$

and

 $\Rightarrow$ Let H be subgroup of A_5

s.t.

 $\Rightarrow$ To show H is not a normal subgroup of A_5 .

Let

$$(abc) \in A_5$$

$$(ab)(cd)(abc) = (bac)$$

$$(abc)(ab)(cd) = acd$$

$$(ab)(cd)(abc) \neq (abc)(ab)(cd)$$

$$\Rightarrow Ha \neq aH \text{ for some } a \in A_5$$

$$\Rightarrow H \text{ is not normal subgroup of } A_5.$$

Q. 5. (a) Let C be the set of complex numbers and

$$M = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

Prove that C and M are isomorphic under addition and $C^* = C \setminus \{0\}$ and $M^* = M \setminus \{0\}$ are isomorphic under multiplication.

(b) Prove that a finite cyclic group of order n is isomorphic to the group $Z_n = \{0, 1, 2, \dots, n-1\}$ under addition modulo n .

(c) (i) Suppose that ϕ is an isomorphism from a group G onto a group G^* . Prove that G is cyclic if and only if G^* is cyclic.

(ii) Show that Z , the group of integers under addition is not isomorphic to Q , the group of rationals under addition. 6x2=12

Sol. (a) See Q. 5 (a), Nov./Dec. 2018, p 248.

(b) Let $G = \langle a \rangle$ be a finite cyclic group of order n

$$\Rightarrow \langle a \rangle = n$$

$$\Rightarrow a^n = e$$

$$\Rightarrow G = \{e, a, a^2, \dots, a^{n-1}\} \text{ are distinct elements of } G.$$

Let us define a mapping

$$\psi: Z_n \rightarrow G$$

s.t.

$$\psi(x) = a^x \quad \forall a \in G \text{ and } x \in Z_n$$

For well defined

$$x_1 = x_2$$

 $\Rightarrow$

$$x_1 - x_2 = 0$$

 $\Rightarrow$

$$a^{x_1 - x_2} = a^0$$

 $\Rightarrow$

$$a^{x_1} \cdot a^{-x_2} = 1$$

 $\Rightarrow$

$$a^{x_1} = a^{x_2}$$

 $\Rightarrow$

$$\psi(x_1) = \psi(x_2)$$

 $\Rightarrow$

ψ is well defined.

$$\psi(x_1 + x_2) = a^{x_1 + x_2}$$

$$= a^{x_1} \cdot a^{x_2}$$

$$= \psi(x_1) \cdot \psi(x_2)$$

 $\Rightarrow$

ψ preserves operation.

 $\Rightarrow$

ψ is homomorphic.

for 1-1

$$\psi(x_1) = \psi(x_2)$$

 $\Rightarrow$

$$a^{x_1} = a^{x_2}$$

 $\Rightarrow$

$$a^{x_1} \cdot a^{-x_2} = 1$$

 $\Rightarrow$

$$a^{x_1 - x_2} = a^0$$

 $\Rightarrow$

$$x_1 - x_2 = 0$$

 $\Rightarrow$

$$x_1 = x_2$$

 $\Rightarrow$

ψ is one-one

For onto

We have to show that for every $a^i \in G$

There exist some $x \in Z_n$ s.t.

$$\psi(x) = a^i$$

 $\Rightarrow$

$$a^x = a^i$$

Since

$$x = i \pmod{n} \quad \forall i \in Z$$

 $\Rightarrow$

ψ is onto

 $\Rightarrow$

ψ is isomorphic

(c) (i)

Let ϕ be mapping from G to G^*

$$\phi: G \rightarrow G^*$$

s.t.

ϕ is isomorphic

Since, ϕ takes identity from G to G^*

i.e. for $e \in G$ $\phi(e) \in G^*$ is identity in G^*

and

$$\phi(a^2) = \phi(a \cdot a) = \phi(a) \cdot \phi(a) = [\phi(a)]^2$$

⇒ By principle of mathematical induction

$$\phi(a^n) = [\phi(a)]^n$$

$$a^n = e \text{ iff } \phi(a^n) = \phi(e)$$

⇒ G is cyclic iff G^* is cyclic

(ii) Do yourself.

Q. 6 (a) Let ϕ be a group homomorphism from a group G to group G^* then prove that:

(i) $|\phi(x)|$ divides $|x|$, for all x in G .

(ii) ϕ is one-one if and only if $|\phi(x)| = |x|$, for all x in G .

(b) State and prove the third isomorphism Theorem.

(c) (i) Let G be a group. Prove that the mapping $\phi(g) = g^{-1}$, for all $g \in G$, is an isomorphism on G if and only if G is Abelian.

(ii) Determine all homomorphisms from Z_n to itself. 6.5×2=13

Sol. (a) (i) Let $\phi: G \rightarrow G^*$

s.t. ϕ is homomorphic

To show $|\phi(x)|$ divides $|x| \quad \forall x \in G$

Let $|x| = n$

and Let $|\phi(x)| = k$

⇒ $x^n = e$

Since $\phi(e) = \bar{e}$

⇒ $\phi(x^n) = \bar{e}$

⇒ $[\phi(x)]^n = \bar{e}$

(ϕ is homomorphic)

Using the result "let G be a group and let ' a ' be an element of order in G . If $a^n = e$ then k divides n ".

⇒ k divides n

⇒ $|\phi(x)|$ divides $|x|$

(ii) If $|\phi(x)| = |x|$

Using above result ϕ is 1-1

Conversely, Let ϕ is 1-1

⇒ $\forall x \in G \exists \phi(x) \in G^*$

⇒ For $x^n = e$

⇒ $\phi(x^n) = \phi(e)$

⇒ $[\phi(x)]^n = \bar{e}$

⇒ $|\phi(x)| = |x|$

(b) If H and k are normal subgroups of G and $k \leq H$, then $\frac{G/k}{H/k} = \frac{G}{H}$

Let us define $\theta: \frac{G}{k} \rightarrow \frac{G}{H}$

s.t. $\theta(gk) = gH \quad \forall g \in G$

Since $\theta(g_1 g_2 k) = g_1^{-1} g_2 H = g_1^{-1} H g_2 H = \theta(g_1 k) \theta(g_2 k)$

⇒ θ is group homomorphic

Let

$$g_1 k = g_2 k$$

⇒

$$g_1^{-1} g_1 k = k$$

⇒

$$g_2^{-1} g_1 \in k$$

But

$$k \leq H$$

⇒

$$g_2^{-1} g_1 \in H$$

⇒

$$g_2^{-1} g_1 H = H$$

⇒

$$g_1 H = g_2 H$$

⇒

$$\theta(g_1 k) = \theta(g_2 k)$$

⇒ θ is well defined.

for onto

Let $gH \in \frac{G}{H}$ then $\exists gk \in \frac{G}{k}$ s.t. $\theta(gk) = gH$

⇒ θ is onto

Let $gk \in \ker \theta$

Then

$$\theta(gk) = H$$

But

$$\theta(gk) = gH$$

⇒

$$gH = H$$

⇒

$$g \in H$$

⇒

$$gk \in \frac{H}{k}$$

⇒

$$\ker \theta = \frac{H}{k}$$

Using first isomorphism theorem

Let θ be group homomorphism from G to G^* . Then the mapping from $G/\ker \theta$ to $\theta(G)$, given by $g \ker \theta \rightarrow \theta(g)$ is an isomorphism $\frac{G}{\ker \theta} = \theta(G)$

$$\Rightarrow \frac{\frac{G}{k}}{\frac{H}{k}} = \frac{G}{H}$$

(c) Let G be a group

Let $\phi: G \rightarrow G$

s.t.

$$\phi(g) = g^{-1}, \quad \forall g \in G$$

Let G be abelian

⇒

$$ab = ba \quad \forall a, b \in G$$

To show ϕ is isomorphic

Let $g_1, g_2 \in G$

$$\phi(g_1 \cdot g_2) = (g_1 g_2)^{-1}$$

$$\begin{aligned}
 &= g_2^{-1} g_1^{-1} \\
 &= g_1^{-1} g_2^{-1} \\
 \Rightarrow \phi \text{ is homomorphic} &= \phi(g_1) \phi(g_2) \\
 \text{for } 1-1 &
 \end{aligned}$$

(G is abelian)

$$\begin{aligned}
 \Rightarrow \phi(g_1) &= \phi(g_2) \\
 g_1^{-1} &= g_2^{-1}
 \end{aligned}$$

Pre multiply both sides with $g_1 g_2$

$$\Rightarrow g_1 g_2 g_1^{-1} = g_1 g_2 g_2^{-1}$$

$$\Rightarrow g_1 g_1^{-1} g_2 = g_1$$

$$\Rightarrow g_2 = g_1$$

$$\Rightarrow g_1 = g_2$$

$$\Rightarrow \phi \text{ is } 1-1$$

for onto

$$\forall g^{-1} \in G \text{ there exist some } g \in G$$

$$\text{s.t. } \phi(g) = g^{-1}$$

$$\Rightarrow \phi \text{ is onto}$$

$$\Rightarrow \phi \text{ is isomorphic}$$

Conversely,

Let ϕ is isomorphic

To show G is abelian.

Since ϕ is isomorphic

$$\Rightarrow \phi \text{ is homomorphic}$$

$$\Rightarrow \forall g_1, g_2 \in G$$

$$\phi(g_1 \cdot g_2) = (g_1 g_2)^{-1}$$

$$\Rightarrow \phi(g_1) \phi(g_2) = g_2^{-1} g_1^{-1}$$

$$\Rightarrow g_1^{-1} g_2^{-1} = g_2^{-1} g_1^{-1}$$

Pre multiply with $g_1 g_2$

$$g_1 g_2 g_1^{-1} g_2^{-1} = g_1 g_2 g_2^{-1} g_1^{-1}$$

$$g_1 g_2 g_1^{-1} g_2^{-1} = g_1 g_1^{-1}$$

$$g_1 g_2 g_1^{-1} g_2^{-1} = e$$

Post multiply with $g_2 g_1$

$$g_1 g_2 g_1^{-1} g_2^{-1} g_2 g_1 = g_2 g_1$$

$$\Rightarrow g_1 g_2 = g_2 g_1$$

$$\Rightarrow G \text{ is abelian.}$$

Sr. No. of Q. Paper : 7465

Unique Paper Code : 32351303

Name of the Paper : MULTIVARIATE CALCULUS

Name of Course : B.Sc. (Hons.) Mathematics II Year

Semester : 3 (CBCS)

Duration : 3 Hours

Nov/Dec. 2019

Maximum Marks : 75

REVISED SYLLABUS (With effective from Academic year 2019-20)

All Sections are compulsory. Attempt any five questions from each Section. All questions carry equal marks.

SECTION-I

Q. 1. Given that the function

$$f(x, y) = \begin{cases} \frac{3x^3 - 2y^3}{x^2 - y^2} & \text{for } x^2 \neq y^2 \\ B & \text{otherwise} \end{cases}$$

is continuous at the origin, what is B .

$$\text{Sol. } f(x, y) = \begin{cases} \frac{3x^3 - 3y^3}{x^2 - y^2} & \text{for } x^2 \neq y^2 \\ B & \text{otherwise} \end{cases}$$

Since, $f(x, y)$ is continuous at origin.

$$\text{Let } y = mx$$

$$\text{where } m \neq \pm 1$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{3x^3 - 3y^3}{x^2 - y^2}$$

$$= \lim_{x \rightarrow 0} \frac{3x^3 - 3m^3 x^3}{x^2 - m^2 x^2}$$

$$= \lim_{x \rightarrow 0} \frac{3x^3(1 - m^3)}{x^2(1 - m^2)}$$

$$= \lim_{x \rightarrow 0} 3x \frac{(1 - m^3)}{(1 - m^2)}$$

$$= 0$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$$

(303)

But $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$ (as f is continuous at origin)

$\Rightarrow 0 = 0$
Q. 2. In physics, the wave equations is :

$$\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$$

and the heat equations is :

$$\frac{\partial z}{\partial t} = c^2 \frac{\partial^2 z}{\partial x^2}$$

Determine whether $z = \sin 5ct \cos 5x$ satisfies the wave equation, the heat equation, or neither.

Sol. $z = \sin 5ct \cos 5x$

Diff. z w.r.t. t

$$\frac{\partial z}{\partial t} = 5c \cos 5ct \cos 5x \quad \dots(1)$$

Again diff. w.r.t. t

$$\frac{\partial^2 z}{\partial t^2} = -25c^2 \sin 5ct \cos 5x \quad \dots(2)$$

Diff. z w.r.t. x

$$\frac{\partial z}{\partial x} = -5 \sin 5ct \sin 5x \quad \dots(3)$$

Again diff. w.r.t. x

$$\frac{\partial^2 z}{\partial x^2} = -25 \sin 5ct \cos 5x \quad \dots(4)$$

From (2) and (4)

$$\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$$

and from (1) and (3)

$$\frac{\partial z}{\partial t} \neq c^2 \frac{\partial^2 z}{\partial x^2}$$

$\Rightarrow z$ satisfies wave equation but it does not satisfies heat equation.

Q. 3. The radius and height of a right circular cone are measured with errors of at most 3% and 2%, respectively. Use increments to approximate the maximum possible percentage error in computing the volume of the cone using these measurements and the formula $V = \frac{1}{3} \pi R^2 H$.

Sol. Percentage error in measurement of Radius is = 3%

Percentage error in measurement of Height is = 2%

$\Rightarrow$ Let change in radius be ΔR

and Change in height be ΔH

$$\Rightarrow \frac{\Delta R}{R} = 0.03$$

$$\text{and } \frac{\Delta H}{H} = 0.02 \quad \dots(1)$$

Let change in volume be ΔV

$$\Delta V = \frac{dV}{dR} \Delta R + \frac{dV}{dH} \Delta H$$

$$\Rightarrow \frac{\Delta V}{V} = \frac{dV}{dR} \frac{\Delta R}{V} + \frac{dV}{dH} \frac{\Delta H}{V}$$

$$\Rightarrow \frac{\Delta V}{V} = \frac{1}{3} \{ (\pi 2R) H \frac{\Delta R}{3 \pi R^2 H} + \frac{1}{3} \pi R^2 \frac{\Delta H}{\frac{1}{3} \pi R^2 H} \}$$

$$\Rightarrow \frac{\Delta V}{V} = 2 \frac{\Delta R}{R} + \frac{\Delta H}{H}$$

Using Equation (1)

$$\frac{\Delta V}{V} = 2(0.03) + 0.02$$

$$= 0.08$$

$\Rightarrow$ Maximum change in percentage error of measurement in volume is 8%.

Q. 4. If $f(x, y, z) = xy^2 e^{xz}$ and $x = 2 + 3t$, $y = 6 - 4t$, $z = t^2$. Compute $\frac{df}{dt}$

(1).

Sol.

$$f(x, y, z) = xy^2 e^{xz}$$

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$$

$$= (y^2 e^{xz} + xy^2 z e^{xz}) (3) + 2xy e^{xz} (-4) + x^2 y^2 e^{xz} (2t) \\ = e^{xz} \{ (y^2 + xy^2 z) (3) + 2xy (-4) + x^2 y^2 (2t) \} \dots(1)$$

Since

$$\begin{aligned}
 x &= 2 + 3t & \Rightarrow x(1) &= 5 \\
 y &= 6 - 4t & \Rightarrow y(1) &= 2 \\
 z &= t^2 & \Rightarrow z(1) &= 1
 \end{aligned}$$

$$\Rightarrow \frac{df}{dt}(1) = e^{(5+1)} [(2)^2 + 5 \cdot (2)^2 \cdot 1] [3] + [-8] [2 \cdot 5]$$

$$\begin{aligned}
 &+ [2 \times (5)^2 \times (2)^2 \times 1] \\
 &= e^6 [24 \times 3 - 80 + 200] \\
 &= e^6 [72 - 80 + 200] \\
 &= 192e^6
 \end{aligned}$$

Q. 5. Sketch the level curve corresponding to $C = 1$ for the function $f(x, y) = \frac{x^2}{a^2} - \frac{y^2}{b^2}$ and find a unit normal vector at the point $P_0(2\sqrt{3})$.

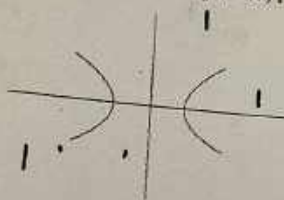
Sol.

$$f(x, y) = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

$$c = 1$$

$$f(x, y) = c$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



$\Rightarrow$ Level curve of $f(x, y)$ at $c = 1$ is hyperbola with vertices $(\pm a, 0)$
 For unit normal vector consult your text book.

Q. 6. Find the point on the plane $2x + y - z = 5$ that is closest to the origin.

Sol. Let $P(x, y, z)$ be the point on plane $2x + y - z = 5$ that is closest to $(0, 0, 0)$.
 It means that the problem is to find out the minimum value of the function.

$$|\overline{OP}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$= \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2}$$

$$= \sqrt{x^2 + y^2 + z^2}$$

Subject to the constraint that

$$2x + y - z = 5$$

Since $|\overline{OP}|$ has the minimum value, whenever the function

$$f(x, y, z) = x^2 + y^2 + z^2 \text{ has minimum value}$$

Let

$$g(x, y, z) = 2x + y - z - 5$$

Then

$$f_x = 2x$$

$$f_y = 2y$$

$$f_z = 2z$$

$$g_x = 2$$

$$g_y = 1$$

$$g_z = -1$$

By Lagrange's multipliers, we have

$$f_x = \lambda g_x$$

$$2x = \lambda(2)$$

 $\Rightarrow$

$$x = \lambda$$

 $\Rightarrow$

$$f_y = \lambda g_y$$

$$2y = \lambda(1)$$

 $\Rightarrow$

$$2y = \lambda$$

$$y = \frac{\lambda}{2}$$

 $\Rightarrow$

$$f_z = \lambda g_z$$

$$2z = \lambda(-1)$$

 $\Rightarrow$

$$2z = -\lambda$$

$$z = -\frac{\lambda}{2}$$

 $\Rightarrow$

On putting above values of (x, y, z) in $g(x, y, z) = 0$

 $\Rightarrow$

$$g(x, y, z) = 0$$

$$2x + y - z - 5 = 0$$

 $\Rightarrow$

$$2\lambda + \frac{\lambda}{2} - \frac{\lambda}{2} = 5$$

$$\lambda = \frac{5}{3}$$

 $\Rightarrow$

$$(x, y, z) = \left(\frac{5}{3}, \frac{5}{6}, -\frac{5}{6}\right) \text{ is the closest point to origin.}$$

SECTION-II

Q. 7. Find the volume of the solid bounded above by the plane $z = y$ and below in the xy -plane by the part of the disk $x^2 + y^2 \leq 1$ in the first quadrant.

Sol.

$$x^2 + y^2 \leq 1$$

 $\Rightarrow$

$$y^2 \leq 1 - x^2$$

In 1st quadrant.

$$0 \leq y \leq \sqrt{1 - x^2}$$

and

$$0 \leq x \leq 1$$

$$V = \int_0^1 \int_0^{1-x} z \, dy \, dx$$

$$V = \int_{x=0}^1 \int_{y=0}^{\sqrt{1-x^2}} y \, dy \, dx$$

$$V = \int_0^1 \left[\frac{y^2}{2} \right]_0^{\sqrt{1-x^2}} dx$$

$$V = \frac{1}{2} \int_0^1 (1-x^2) dx$$

$$V = \frac{1}{2} \left[x - \frac{x^3}{3} \right]_0^1$$

$$V = \frac{1}{2} \left(1 - \frac{1}{3} \right) = \frac{1}{3}$$

Q. 8. Sketch the region of integration and then compute the integral

$$\int_0^{1+2x} \int_0^{2x} e^{y-x} \, dy \, dx \text{ in 2 ways:}$$

(a) with the given order of integration

(b) with the order of integration reversed

Sol. Since

$$0 \leq x \leq 1$$

and

$$x \leq y \leq 2x$$

for

$$x \leq y$$

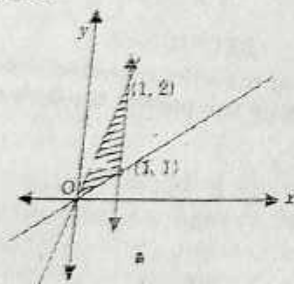
and

$$y \leq 2x$$

x	0	1
y	0	1

x	0	1
y	0	2

Region of integration is



(a)

$$I = \int_{x=0}^1 \int_{y=x}^{2x} e^{y-x} \, dy \, dx$$

$$= \int_0^1 \left[e^{y-x} \right]_x^{2x} dx$$

$$= \int_0^1 (e^x - 1) dx$$

$$= [e^x - x]_0^1$$

$$= e - 1 - 1$$

$$= e - 2$$

(b)

$$I = \int_{y=0}^1 \int_{x=y/2}^y e^{y-x} \, dx \, dy + \int_{y=1}^2 \int_{x=y/2}^1 e^{y-x} \, dx \, dy$$

$$= - \left[\int_0^1 (e^{y-x})_{y/2}^y dy + \int_1^2 (e^{y-x})_{y/2}^1 dy \right]$$

$$= - \left[\int_0^1 (1 - e^{y/2}) dy + \int_1^2 (e^{y-1} - e^{y/2}) dy \right]$$

$$= - \left[\left[y - 2e^{y/2} \right]_0^1 + \left[e^{y-1} - 2e^{y/2} \right]_1^2 \right]$$

$$= - [1 - 2e^{1/2} + 2 + e - 2e - 1 + 2e^{1/2}]$$

$$= - [2 - e]$$

$$= e - 2$$

Q. 9. Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} y \sqrt{x^2+y^2} \, dy \, dx$ by converting to polar coordinates.

Sol.

$$I = \int_0^2 \int_0^{\sqrt{2x-x^2}} y \sqrt{x^2+y^2} \, dy \, dx$$

$$\Rightarrow y \text{ varies from } 0 \text{ to } \sqrt{2x-x^2}$$

and x varies from 0 to 2.

for

$$y = \sqrt{2x-x^2}$$

$\Rightarrow$

$$y^2 = 2x - x^2$$

$\Rightarrow$

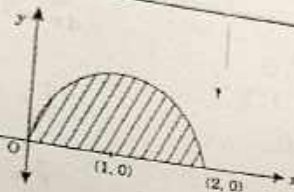
$$x^2 - 2x + 1 + y^2 = 1$$

$\Rightarrow (x-1)^2 + y^2 = 1$
Which is equation of circle with
centre (1, 0)

$\Rightarrow$ Region of integration is

Let

$\Rightarrow \theta$ varies from 0 to π and r varies
from 0 to 1



$$I = \int_0^{\pi} \int_0^1 r \sin \theta \sqrt{r^2 \cos^2 \theta + r^2 \sin^2 \theta} dr d\theta$$

$$I = \int_0^{\pi} \int_0^1 r^3 \sin \theta dr d\theta$$

$$= \int_0^{\pi} \left[\frac{r^4}{4} \right]_0^1 \sin \theta d\theta$$

$$= \int_0^{\pi} \frac{1}{4} \sin \theta d\theta$$

$$= \frac{1}{4} [-\cos \theta]_0^{\pi}$$

$$= \frac{-1}{4} (\cos \pi - \cos 0)$$

$$= -\frac{1}{4} \times (-2)$$

$$I = \frac{1}{2}$$

Q. 10. Find the volume of the tetrahedron bounded by the plane $2x + y + 3z = 6$ and the coordinate planes $x = 0$, $y = 0$ and $z = 0$.

Sol. See Q. 11, Nov/Dec. 2018, Page 267.

Q. 11. Compute $\iiint_D \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}}$ where D is the solid sphere $x^2 +$

$$y^2 + z^2 \leq 3.$$

Sol. Since $x^2 + y^2 + z^2 \leq 3$

Let $x = \rho \sin \phi \cos \theta$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\Rightarrow x^2 + y^2 + z^2 = \rho^2 \leq 3$$

$$\Rightarrow \rho \text{ varies from } 0 \text{ to } \sqrt{3}$$

$$\phi \text{ varies from } 0 \text{ to } \pi$$

$$\theta \text{ varies from } 0 \text{ to } 2\pi$$

$$\Rightarrow \iiint_D \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}} = \int_0^{2\pi} \int_0^{\pi} \int_0^{\sqrt{3}} \frac{1}{\rho} \rho^2 \sin \phi d\rho d\phi d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \rho \sin \phi d\phi d\theta$$

$$= \int_0^{2\pi} d\theta \cdot \int_0^{\pi} \sin \phi d\phi \int_0^{\sqrt{3}} \rho d\rho$$

$$= \left[\theta \right]_0^{2\pi} \left[-\cos \phi \right]_0^{\pi} \left[\frac{\rho^2}{2} \right]_0^{\sqrt{3}}$$

$$= 2\pi \times (2) \times \frac{(\sqrt{3})^2}{2}$$

$$= 6\pi$$

Q. 12. Use the change of variables to compute $\iint_D \frac{(x-y)^4}{(x+y)^4} dy dx$,

where D is the triangular region bounded by the line $x + y = 1$ and the coordinate axes.

Sol. Let

$$x + y = u$$

$$x - y = v$$

$$\Rightarrow x = \frac{u+v}{2}$$

and

$$y = \frac{u-v}{2}$$

Let the region D is modified into 0^*

$$x + y \leq 1$$

$$x = 0$$

$$y = 0$$

$$\Rightarrow \begin{cases} u \leq 1 \\ u = -v \\ u = v \end{cases}$$

$$|J| = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix}$$

$$= \left| -\frac{1}{4} - \frac{1}{4} \right| = \frac{1}{2}$$

$$\iint_D \frac{(x-y)^4}{(x+y)^4} dy dx = \int_{u=0}^1 \int_{v=-u}^u |J| \frac{v^4}{u^4} dv du$$

$$= \frac{1}{2} \int_0^1 \frac{1}{u^4} \left[\frac{v^5}{5} \right]_{-u}^u du$$

$$= \frac{1}{10} \int_0^1 \frac{1}{u^4} u^5 - (-u)^5 du$$

$$= \frac{1}{10} \int_0^1 2 \times \frac{u^5}{u^4} du$$

$$= \frac{1}{5} \left[\frac{u^2}{2} \right]_0^1$$

$$= \frac{1}{10}$$

SECTION-III

Q. 13. Find the work done by the force field $\vec{F} = \frac{x}{\sqrt{x^2+y^2}} \vec{i} - \frac{y}{\sqrt{x^2+y^2}} \vec{j}$

when an object moves from $(a, 0)$ to $(0, a)$ on the path $x^2 + y^2 = a^2$.

Sol.

$$x^2 + y^2 = a^2$$

Let

$$x = t \quad a \leq t \leq 0$$

$\Rightarrow$

$$y = \sqrt{a^2 - t^2}$$

Let

$$\vec{r} = x\vec{i} + y\vec{j}$$

$$\vec{r}(t) = t\vec{i} + \sqrt{a^2 - t^2}\vec{j}$$

$$d\vec{r} = \vec{i} - \frac{t}{\sqrt{a^2 - t^2}} \vec{j}$$

$$\vec{F} = \frac{x}{\sqrt{x^2+y^2}} \vec{i} - \frac{y}{\sqrt{x^2+y^2}} \vec{j}$$

$$\vec{F} = \frac{t}{\sqrt{t^2+a^2-t^2}} \vec{i} - \frac{\sqrt{a^2-t^2}}{\sqrt{t^2+a^2-t^2}} \vec{j}$$

$$\vec{F} = \frac{t}{a} \vec{i} - \frac{\sqrt{a^2-t^2}}{a} \vec{j}$$

$$w = \int \vec{F} \cdot d\vec{r}$$

$$w = \int_a^0 \left(\frac{t}{a} \vec{i} - \frac{\sqrt{a^2-t^2}}{a} \vec{j} \right) \cdot \left(\vec{i} - \frac{t}{\sqrt{a^2-t^2}} \vec{j} \right) dt$$

$$w = - \int_0^a \left(\frac{t}{a} + \frac{t}{a} \right) dt$$

$$w = - \frac{1}{a} \int_0^a 2t dt$$

$$w = - \frac{1}{a} \left[t^2 \right]_0^a$$

$$w = - \frac{a}{a}$$

Q. 14. Verify that the following line integral is independent of the path $\oint (3x^2 + 2x + y^2) dx + (2xy + y^3) dy$ where C is any path from $(0, 0)$ to $(0, 1)$.

Sol. Let $\vec{F}(x, y) = (3x^2 + 2x + y^2)\vec{i} + (2xy + y^3)\vec{j}$

Compare with $\vec{F}(x, y) = f(x, y)\vec{i} + g(x, y)\vec{j}$

$$f(x, y) = 3x^2 + 2x + y^2$$

$$g(x, y) = 2xy + y^3$$

$\oint \vec{F} \cdot d\vec{r}$ is independent of path if $\vec{F}$ is conservative

$\Rightarrow$ If $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$ then $\oint \vec{F} \cdot d\vec{r}$ is independent of path.

Since,

$$f(x, y) = 3x^2 + 2x + y^2$$

$$\Rightarrow \frac{\partial f}{\partial y} = 2y$$

and

$$g(x, y) = 2xy + y^3$$

$$\Rightarrow \frac{\partial g}{\partial x} = 2y$$

From (1) and (2)

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$$

$\Rightarrow$ Line integral is independent of path.

Q. 15. Use Green's theorem to evaluate $\oint_C (x \sin x \, dx - \exp(y^2) \, dy)$ where C is the closed curve joining the points $(1, -1)$, $(2, 5)$ and $(-1, -1)$ in counterclockwise direction.

Sol. Using Green theorem.

$$\oint_C f(x, y) \, dx + g(x, y) \, dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

Compare $\oint_C (x \sin x \, dx - \exp(y^2) \, dy)$ with $\oint_C f(x, y) \, dx + g(x, y) \, dy$

$$\Rightarrow f(x, y) = x \sin x \quad \text{and} \quad g(x, y) = -\exp(y^2)$$

$$\Rightarrow \frac{\partial g}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

$$\Rightarrow \oint_C (x \sin x \, dx - \exp(y^2) \, dy) = 0$$

Q. 16. State Stoke's theorem and use it to evaluate $\iint_S \text{curl } \vec{F} \cdot d\vec{S}$

where $\vec{F} = xz \vec{i} + yz \vec{j} + xy \vec{k}$ and S is the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

Sol. Stoke's Theorem

Let σ be a piecewise smooth oriented surface that is bounded by a simple, closed, piecewise smooth curve C with positive orientation. If the components of the vector field

$$\vec{F}(x, y, z) = f(x, y, z)\vec{i} + g(x, y, z)\vec{j} + h(x, y, z)\vec{k}$$

are continuous and have continuous first partial derivative on some open set containing σ and if T is the unit tangent vector to C , then

$$\oint_C \vec{F} \cdot \vec{T} \, dS = \iint_\sigma \text{curl } \vec{F} \cdot \vec{n} \, dS$$

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$x^2 + y^2 = 1$$

$$x = \cos \theta,$$

$$y = \sin \theta$$

Let

and

$$z = 1$$

$$\iint_S \text{curl } \vec{F} \cdot \vec{n} \, dS = \int_0^{2\pi} \vec{F} \cdot \vec{T} \, d\theta$$

$$= \int_0^{2\pi} (\cos \theta \vec{i} + \sin \theta \vec{j} + \cos \theta \sin \theta \vec{k}) \cdot (-\sin \theta \vec{i} - \cos \theta \vec{j}) \, d\theta$$

$$= \int_0^{2\pi} -\cos \theta \sin \theta + \cos \theta \sin \theta \, d\theta$$

$$= \int_0^{2\pi} 0 \, d\theta$$

$$= 0$$

Q. 17. Use the divergence theorem to evaluate the surface integral $\iint_S \vec{F} \cdot \vec{N} \, dS$, where $\vec{F} = (x^2 + y^2 - z^2)\vec{i} + yx^2\vec{j} + 3z\vec{k}$; S is the surface comprised of the five faces of the unit cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$, missing $z = 0$.

Sol. Using divergence theorem

$$\iint_S \vec{F} \cdot \vec{N} \, dS = \iiint_V \text{div } \vec{F} \, dV$$

Since

$$\vec{F} = (x^2 + y^2 - z^2)\vec{i} + yx^2\vec{j} + 3z\vec{k}$$

$$\text{div } \vec{F} = \frac{\partial}{\partial x}(x^2 + y^2 - z^2) + \frac{\partial}{\partial y}(yx^2) + \frac{\partial}{\partial z}(3z)$$

$$= 2x + x^2 + 3$$

$$\Rightarrow \iint_S \vec{F} \cdot \vec{N} \, dS = \int_{z=0}^1 \int_{y=0}^1 \int_{x=0}^1 (x^2 + 2x + 3) \, dx \, dy \, dz$$

$$= \int_0^1 dz \int_0^1 dy \int_0^1 (x^2 + 2x + 3) \, dx$$

$$= \left[\frac{1}{3} x^3 + x^2 + 3x \right]_0^1$$

$$= \frac{13}{3}$$

Q. 18. Evaluate $\iint_S 2x \, dS$ where S is the portion of the plane $x + y + z = 1$ with $x \geq 0, y \geq 0, z \geq 0$.

Sol. Since, $x + y + z = 1$ ($= g(x, y)$ Let)

$$\Rightarrow z = 1 - x - y$$

$$\Rightarrow \frac{\partial z}{\partial x} = -1$$

$$\Rightarrow \frac{\partial z}{\partial y} = -1$$

Using formula

$$\iint f(x, y, z) \, dS = \iint f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \, dA$$

$$= \int_0^1 \int_0^{1-x} 2x \sqrt{(-1)^2 + (-1)^2 + 1} \, dy \, dx$$

$$= 2\sqrt{3} \int_0^1 x \left[y \right]_0^{1-x} dx$$

$$= 2\sqrt{3} \int_0^1 x(1-x) dx$$

$$= 2\sqrt{3} \int_0^1 (x - x^2) dx$$

$$= 2\sqrt{3} \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$= 2\sqrt{3} \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$= \frac{\sqrt{3}}{3}$$

Paper Code : 32351301
 Name of the Paper : C-5 : Theory of Real Functions
 Name of Course : B.Sc. (Hons.) Mathematics II Year
 Semester : 3
 Duration : 3 Hours
 Maximum Marks : 75

Attempt any four questions. All questions carry equal marks.

1. Use the ϵ - δ definition of the limit to show that

$$\lim_{x \rightarrow 1} \frac{x^3 - 2x + 4}{x^2 + 4x - 3} = \frac{3}{2}$$

Use the sequential criteria for limits, to show that the following does not exist

$$\lim_{x \rightarrow 2} \operatorname{sgn} \left(\cos \left(\frac{1}{(x-2)^3} \right) \right) \text{ where } \operatorname{sgn} \text{ is the signum function.}$$

Sol. $\lim_{x \rightarrow 1} \frac{x^3 - 2x + 4}{x^2 + 4x - 3} = \frac{3}{2}$
 $\epsilon > 0$

Let find $\delta > 0$ s.t. if $|x - 1| < \delta$ then $\left| \frac{x^3 - 2x + 4}{x^2 + 4x - 3} - \frac{3}{2} \right| < \epsilon$

$$|x - 1| < \frac{1}{5}$$

$$1 - \frac{1}{5} < x < 1 + \frac{1}{5}$$

$$\frac{4}{5} < x < \frac{6}{5}$$

$$x^2 + 4x - 3 > \left(\frac{4}{5}\right)^2 + 4\left(\frac{4}{5}\right) - 3$$

$$x^2 + 4x - 3 > \frac{16}{25} + \frac{16}{5} - 3$$

$$x^2 + 4x - 3 > \frac{16 + 80 - 75}{25}$$

$$x^2 + 4x - 3 > \frac{21}{25}$$

$$|x^2 + 4x - 3| > \frac{21}{25}$$

$$\frac{1}{x^2 + 4x - 3} < \frac{25}{21}$$

Consider, $\left| \frac{f(x)-3}{2} \right| = \left| \frac{x^3-2x+4}{x^2+4x-3} - \frac{3}{2} \right|$

$$= \left| \frac{2x^3-4x+8-3x^2-12x+9}{2(x^2+4x-3)} \right|$$

$$= \left| \frac{2x^3-3x^2-16x+17}{2(x^2+4x-3)} \right|$$

$$= \frac{|2x^3-3x^2-16x+17|}{2|x^2+4x-3|}$$

$$< \left(2\left(\frac{6}{5}\right)^2 + \frac{6}{5} + 17 \right) |x-1| \left(\frac{25}{2 \times 21} \right)$$

$$< \left(\frac{72+30+425}{25} \right) \left(\frac{25}{42} \right) |x-1|$$

$$< \frac{527}{42} |x-1|$$

$$< \epsilon \text{ if } |x-1| < \frac{42}{527}$$

Let $\delta = \min \left\{ \frac{1}{5}, \frac{42}{527} \epsilon \right\}$

Since $|x-1| < \delta$

$$\Rightarrow \left| \frac{x^3-2x+4}{x^2+4x-3} - \frac{3}{2} \right| < \epsilon$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^3-2x+4}{x^2+4x-3} = \frac{3}{2}$$

(ii) Sequential criterion for limits

Let $f: A \rightarrow \mathbb{R}$ and let C be a cluster point of A

Then the following are equivalent

(a). $\lim_{x \rightarrow c} f = L$

(b) For every sequence $\langle x_n \rangle$ in A that converges to c such that $x_n \neq c$ for all $n \in \mathbb{N}$, the sequence $\langle f(x_n) \rangle$ converges to L

Let $\langle x_n \rangle$ and $\langle y_n \rangle$ be two sequences defined as

$$x_n = 2 + \frac{1}{[(2n+1)\pi]^{1/3}}, n \in \mathbb{N}$$

$$y_n = 2 + \frac{1}{(2n\pi)^{1/3}}, n \in \mathbb{N}$$

Since, $\lim_{x \rightarrow 2} x_n = \lim_{x \rightarrow 2} y_n = 2$

Let

$$f(x_n) = \operatorname{sgn} \left(\cos \frac{1}{(x_n-2)^3} \right)$$

$$f(y_n) = \operatorname{sgn} \left(\cos \frac{1}{\left[2 + \frac{1}{(2n+1)\pi} - 2 \right]^3} \right)$$

$$f(x_n) = \operatorname{sgn} (\cos (2n+1)\pi)$$

$$f(x_n) = \operatorname{sgn} (-1)$$

$$f(x_n) = -1$$

and let

$$f(y_n) = \operatorname{sgn} \left(\cos \frac{1}{(y_n-2)^3} \right)$$

$$f(y_n) = \operatorname{sgn} \left(\cos \frac{1}{\left[2 + \frac{1}{(2n\pi)^{1/3}} - 2 \right]^3} \right)$$

$$f(y_n) = \operatorname{sgn} (\cos (2n\pi))$$

$$f(y_n) = \operatorname{sgn} (1)$$

$$f(y_n) = 1$$

Thus, for sequence $\langle x_n \rangle$ and $\langle y_n \rangle$ converges to 2 sequences $\langle f(x_n) \rangle$ and $\langle f(y_n) \rangle$ converge to -1 and 1 respectively.

Hence, by sequential criterion for limits given limit does not exist.

Q. 2. Let $r, L \in \mathbb{R}$ and $f: (-\infty, r) \rightarrow \mathbb{R}$ be a function. prove that $\lim_{x \rightarrow r} f(x) = L$ if and only if, for every sequence $\langle x_n \rangle$ in $(-\infty, r)$ such

that $\lim_{n \rightarrow \infty} x_n = r$, then the sequence $\langle f(x_n) \rangle$ converges to L .

Sol. Let $r, L \in \mathbb{R}$

Let $f: (-\infty, r) \rightarrow \mathbb{R}$ be a function

Let $\lim_{x \rightarrow r} f(x) = L$

$\Rightarrow$ for given

$$\epsilon > 0 \exists k < r \text{ such that}$$

$$|f(x) - L| < \epsilon \text{ for } x < k$$

...(1)

Let $\langle x_n \rangle$ be a sequence in $(-\infty, r)$ such that

$$\lim_{n \rightarrow \infty} x_n = r$$

$\Rightarrow \exists m \in \mathbb{N}$ (depending on k) such that

$$x_n < k \text{ for } n \geq m$$

...(2)

From (1) and (2)

$$\begin{aligned} &|f(x_n) - L| < \epsilon \text{ for } x_n > -h \\ &\langle f(x_n) \rangle \text{ converges to } L \end{aligned}$$

Conversely,

Let if possible $\lim_{x \rightarrow -\infty} f(x) \neq L$

$\Rightarrow$ for $\epsilon > 0$ be given $\exists h < r$ such that

$$|f(x) - L| > \epsilon \quad \forall x < h$$

Let $x_1 < h$ such that $|f(x_1) - L| > \epsilon$

Let $x_2 < h$ such that $|f(x_2) - L| > \epsilon$

Continuing in this way we get a sequence

$\langle x_n \rangle \rightarrow -\infty$ such that $\langle f(x_n) \rangle \not\rightarrow L$

But this is contradiction to given

Hence our assumption is wrong

$$\Rightarrow \lim_{x \rightarrow -\infty} f(x) = L$$

Q. 3. Suppose $f: [a, b] \rightarrow \mathbb{R}$ is a monotonically increasing function. If f assumes every value between $f(a)$ and $f(b)$ at least once, then show that f is continuous on $[a, b]$.

Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function on $\mathbb{R}$. Let $a, b \in \mathbb{R}$ be such that $a < b$, then prove that $f^{-1}([a, b])$ is an open interval in $\mathbb{R}$, where $[a, b]$ denotes an open interval in $\mathbb{R}$.

Sol. Let $f: [a, b] \rightarrow \mathbb{R}$ be a monotonically increasing function

$\Rightarrow$ for $x_1, x_2 \in [a, b]$ such that $x_1 < x_2$ then $f(x_1) \leq f(x_2)$

Let $c \in [a, b]$

Since $c^- < c < c^+$

$$\Rightarrow \lim_{x \rightarrow c^-} f(x) \leq f(x) \leq \lim_{x \rightarrow c^+} f(x)$$

$$\text{Let } \lim_{x \rightarrow c^-} f(x) = \sup \{f(x) : x < c\}$$

$$\text{and } \lim_{x \rightarrow c^+} f(x) = \inf \{f(x) : x > c\}$$

Let us assume that f is discontinuous at c

$$\text{Then } \lim_{x \rightarrow c^-} f(x) < \lim_{x \rightarrow c^+} f(x)$$

Let $h \neq f(c)$ be any real number such that

$$\lim_{x \rightarrow c^-} f(x) < h < \lim_{x \rightarrow c^+} f(x)$$

Since, f assume every value between $f(a)$ and $f(b)$ at least once

$$\Rightarrow \exists x_0 \in [a, b] \text{ s.t. } f(x_0) = h$$

$$\text{if } x_0 < c \text{ then } f(x_0) < \sup \{f(x) : x < c\}$$

$$f(x_0) < \lim_{x \rightarrow c^-} f(x)$$

But from (1) $\lim_{x \rightarrow c^-} f(x) < h = f(x_0)$

$$\Rightarrow \begin{aligned} &x_0 < c \\ &x_0 > c \Rightarrow f(x_0) > \inf \{f(x) : x > c\} \\ &\Rightarrow f(x_0) > \lim_{x \rightarrow c^+} f(x) \end{aligned}$$

Now, let

But from (1) $\lim_{x \rightarrow c^-} f(x) > f(x_0) = h$

$$\Rightarrow \begin{aligned} &x_0 > c \\ &x_0 < c \text{ (By law of trichotomy)} \\ &f(c) = h \end{aligned}$$

Our assumption that f is discontinuous at c is wrong

f is continuous at c

Since, $c \in [a, b]$ is arbitrary

f is continuous for all values in $[a, b]$

(ii) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function

Let $a, b \in \mathbb{R}$ s.t. $a < b$ and $[a, b]$ is an open interval.

To show $f^{-1}([a, b])$ is an open interval in $\mathbb{R}$

Let $c \in f^{-1}([a, b])$

$$\Rightarrow f(c) \in [a, b]$$

for $\epsilon > 0$ be given such that $(f(c) - \epsilon, f(c) + \epsilon) \subseteq [a, b]$

Since, f is continuous on $\mathbb{R}$

f is continuous on c

$$\Rightarrow \exists \delta > 0 \text{ s.t.}$$

$$|f(x) - f(c)| < \epsilon \text{ if } |x - c| < \delta$$

$$\Rightarrow f(c) - \epsilon < f(x) < f(c) + \epsilon$$

$$\Rightarrow f(x) \in (f(c) - \epsilon, f(c) + \epsilon) \subseteq [a, b]$$

$$\Rightarrow f(x) \in [a, b]$$

$$\Rightarrow x \in f^{-1}([a, b])$$

$$\Rightarrow \text{For } |x - c| < \delta \Rightarrow x \in f^{-1}([a, b])$$

$$(c - \delta, c + \delta) \subseteq f^{-1}([a, b])$$

$f^{-1}([a, b])$ is a neighbourhood of c

Since c is arbitrary

$f^{-1}([a, b])$ is an open interval in $\mathbb{R}$.

Q. 4. Prove that the function $\sin(x^2)$ is not uniformly continuous on $(0, \infty)$. However, it is uniformly continuous on $[0, a]$, where $a > 0$ is any fixed real number.

Suppose $f: [0, 2\pi] \rightarrow \mathbb{R}$ is continuous and $f(0) = f(2\pi)$. Prove that there exists at least one point $c \in [0, \pi]$ such that $f(c) = f(c + \pi)$.

Sol. Let

Let $\epsilon > 0$ be given such that $\exists \delta > 0$

Let

$$\begin{aligned}
 x_1 &= \sqrt{(n+1)\frac{\pi}{2}} \text{ and } x_2 = \sqrt{\frac{n\pi}{2}} \\
 \Rightarrow |x_1 - x_2| &= \left| \sqrt{(n+1)\frac{\pi}{2}} - \sqrt{\frac{n\pi}{2}} \right| \\
 &= \left| \left(\sqrt{(n+1)\frac{\pi}{2}} - \sqrt{\frac{n\pi}{2}} \right) \left(\sqrt{(n+1)\frac{\pi}{2}} + \sqrt{\frac{n\pi}{2}} \right) \right| \\
 &= \frac{(n+1)\frac{\pi}{2} - \frac{n\pi}{2}}{\sqrt{(n+1)\frac{\pi}{2}} + \sqrt{\frac{n\pi}{2}}} \\
 &= \frac{\frac{\pi}{2}}{\sqrt{\frac{\pi}{2}}(\sqrt{n+1} + \sqrt{n})} \\
 &< \left| \frac{\sqrt{\pi}}{2\sqrt{2n}} \right| \\
 &= \frac{\sqrt{\pi}}{2\sqrt{2n}}
 \end{aligned}$$

$$\Rightarrow |x_1 - x_2| < \frac{\sqrt{\pi}}{2\sqrt{2n}} = \delta \text{ (let)}$$

$$\begin{aligned}
 \text{Consider } |f(x_1) - f(x_2)| &= \left| \sin\left((n+1)\frac{\pi}{2}\right) - \sin\frac{n\pi}{2} \right| \\
 &= \left| 2\cos\left[\frac{(n+1)\frac{\pi}{2} + \frac{n\pi}{2}}{2}\right] \sin\left[\frac{(n+1)\frac{\pi}{2} - \frac{n\pi}{2}}{2}\right] \right| \\
 &= \left| 2\cos\left(\frac{n\pi}{2} + \frac{\pi}{4}\right) \sin\left(\frac{\pi}{4}\right) \right|
 \end{aligned}$$

$$= \left| 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \right|$$

$$= 1$$

$$4 \epsilon$$

For

$$|x_1 - x_2| < \delta \implies |f(x_1) - f(x_2)| < \epsilon$$

But $f(x) = \sin(x^2)$ for $x \in [0, \infty)$ is not continuous on $[0, \infty)$ as $[0, \infty)$ is not a closed and bounded interval and continuous function on closed and bounded interval is uniformly continuous.

(ii) Let $f : [0, 2\pi] \rightarrow \mathbb{R}$ be a continuous function.

Let

$$f(0) = f(2\pi)$$

Define

$$g : [0, \pi] \rightarrow \mathbb{R} \text{ such that}$$

$$g(x) = f(\pi + x) - f(x)$$

Since f is continuous $\Rightarrow g$ is continuous

For

$$x = 0$$

$$g(0) = f(\pi) - f(0)$$

For

$$x = \pi$$

$$g(\pi) = f(2\pi) - f(\pi)$$

$$= f(0) - f(\pi)$$

$$[\text{as } f(2\pi) = f(0)]$$

$$g(\pi) = -(f(\pi) - f(0))$$

$$g(\pi) = -g(0)$$

$$g(0) = 0$$

If

$$f(\pi) - f(0) = 0$$

Then,

$$f(\pi + 0) = f(0)$$

 $\Rightarrow$

$$C = 0 \text{ is the required point}$$

If

$$g(0) \neq 0$$

Consider

$$g(\pi) = -g(0)$$

 $\Rightarrow$

$g(\pi)$ and $g(0)$ are of opposite signs

 $\Rightarrow$

$$\exists C \in (0, \pi)$$

 $\Rightarrow$

Such that

$$g(C) = 0$$

 $\Rightarrow$

$$f(C + \pi) - f(C) = 0$$

 $\Rightarrow$

$$f(C + \pi) = f(C)$$

Since,

$$C \in (0, \pi) \subseteq [0, \pi]$$

$\Rightarrow C$ is the required point.

Q. 5. Let $f(x) = |\cos x|$, $x \in (0, 2\pi)$. Determine that where the function is differentiable, and where it is not differentiable in the interval $(0, 2\pi)$. Also, find the derivative at the points of differentiability.

Let f be a function defined on the real line $\mathbb{R}$ and suppose that it satisfies the condition

$$|f(x) - f(y)| \leq (x - y)^2 \text{ for all } x, y \in \mathbb{R}$$

Prove that f is a constant function.

Sol. Let $f(x) = |\cos x|$; $x \in (0, 2\pi)$

Let $\cos x = 0$; $x \in (0, 2\pi)$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$f(x) = \begin{cases} \cos x & 0 < x < \frac{\pi}{2} \\ -\cos x & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ \cos x & \frac{3\pi}{2} < x < 2\pi \end{cases}$$

At

$$x = \frac{\pi}{2}; f(x) = 0$$

L.H.D

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)}{h} &= \lim_{h \rightarrow 0^-} \frac{\cos(\pi/2 + h) - 0}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-\sin h}{h} \\ &= -1 \end{aligned}$$

$\Rightarrow f$ is not differentiable at $x = \frac{\pi}{2}$

Similarly $f(x)$ is not differentiable at $x = \frac{3\pi}{2}$

Since, $\cos x$ is a differentiable function in $(0, 2\pi)$

$\Rightarrow f(x)$ is differentiable for all $x \in (0, 2\pi) - \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}$

$$f'(x) = \begin{cases} -\sin x & 0 < x < \frac{\pi}{2} \\ \sin x & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ -\sin x & \frac{3\pi}{2} < x < 2\pi \end{cases}$$

(ii) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$|f(x) - f(y)| \leq (x - y)^2 \quad \forall x, y \in \mathbb{R}$$

To prove f is constant

Let $c \in \mathbb{R}$

Let $\varepsilon > 0$ be given $\exists \delta > 0$ s.t. $|x - c| < \delta$

$$\text{Then } \left| \frac{f(x) - f(c)}{x - c} - 0 \right| = \left| \frac{f(x) - f(c)}{x - c} \right|$$

$$\leq \frac{(x - c)^2}{|x - c|}$$

$$= |x - c|$$

$$< \delta = \varepsilon$$

$$\Rightarrow \left| \frac{f(x) - f(c)}{x - c} - 0 \right| < \varepsilon$$

$$f'(c) = 0$$

Since, c is an arbitrary

$$f'(x) = 0 \quad \forall x \in \mathbb{R}$$

$\Rightarrow f$ is a constant function on $\mathbb{R}$.

Q. 6. Using the Taylor's theorem, find the approximate value of $\sin(0.4)$ with the error value being calculated up to less than 10^{-3} .

Let $x \in [0, 1]$ and $n \in \mathbb{N}$, show that the following inequality holds

$$\left| e^x - \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} \right) \right| < \frac{3}{(n+1)!}$$

Using this inequality, approximate the value of $\sqrt{e}$ with an error value determined to be less than 10^{-2} .

Sol. (i) Let $f(x) = \sin x$, $x \in [-1, 1]$; $f(0) = 0$

$$f'(x) = \cos x = \sin\left(\frac{\pi}{2} + x\right); \quad f'(0) = 1$$

$$f''(x) = -\sin x = \sin\left(\frac{2\pi}{2} + x\right); \quad f''(0) = 0$$

$$f'''(x) = -\cos x = \sin\left(\frac{3\pi}{2} + x\right); \quad f'''(0) = -1$$

$$f^{(n)}(x) = \sin\left(\frac{n\pi}{2} + x\right)$$

By Taylor's theorem $\exists c \in (0, 1)$ s.t.

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \frac{e^{n+1}}{(n+1)!} f^{(n+1)}(c)$$

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{e^{n+1}}{(n+1)!} \sin\left((n+1)\frac{\pi}{2} + c\right)$$

Since, error is less than 10^{-3}

$$\text{As } \sin \theta \leq 1 \Rightarrow \sin\left((n+1)\frac{\pi}{2} + c\right) \leq 1$$

$$\text{for } c \in (0, 1) \quad e^{n+1} \leq 1$$

$$\Rightarrow \frac{1}{(n+1)!} < \frac{1}{(10)^3}$$

$$\Rightarrow 10^3 < (n+1)!$$

$$\Rightarrow n = 6$$

Thus taking terms upto $n = 6$

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

$$\sin(0.4) = f(0.4)$$

$$= 0.4 - \frac{(0.4)^3}{6} + \frac{(0.4)^5}{120}$$

$$= 0.38941867$$

(ii) Let

$$f(x) = e^x, \quad x \in [0, 1]; \quad f(0) = 1$$

$$f'(x) = e^x; \quad f'(0) = 1$$

$$f''(x) = e^x; \quad f''(0) = 1$$

$$\vdots$$

$$f^{(n)}(x) = e^x; \quad f^{(n)}(0) = 1$$

By Taylor's theorem $\exists c \in]0, 1[$ s.t.

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \frac{e^{n+1}}{(n+1)!} f^{(n+1)}(c)$$

$$f(x) = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \frac{e^{n+1}}{(n+1)!} e^c$$

Since,

$$0 < c < 1$$

$$e^c < 3$$

$\Rightarrow$

$$e^{n+1} < 1$$

and

$$\Rightarrow \frac{e^{n+1}}{(n+1)!} < \frac{1}{(n+1)!}$$

$$\Rightarrow f(x) < 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{3}{(n+1)!}$$

$$\Rightarrow \left| f(x) - \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}\right) \right| < \frac{3}{(n+1)!}$$

$$\Rightarrow \left| e^x - \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}\right) \right| < \frac{3}{(n+1)!}$$

To determine $e^{\sqrt{x}}$ with error less than 10^{-6}

$$\Rightarrow \frac{3}{(n+1)!} < \frac{1}{100}$$

$$\Rightarrow (n+1)! > 300$$

$$\Rightarrow n = 5$$

$$\text{Substitute } x = \frac{1}{2}, \quad n = 5 \text{ in}$$

$$f(x) = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \frac{1}{2!} + \left(\frac{1}{2}\right)^3 + \frac{1}{3!} + \left(\frac{1}{2}\right)^4 + \frac{1}{4!} + \left(\frac{1}{2}\right)^5 + \frac{1}{5!}$$

$$\sqrt{e} = 1.64869$$

□□□

Unique Paper Code : 32351302
 Name of the Paper : C-6: GROUP THEORY-I
 Name of Course : B.Sc. (Hons.) Mathematics II year
 Semester : 3
 Duration : 3 Hours

O.B. EXAM.
 DEC. 2020

C-6 : Mathematics : Group Theory-I (Dec. 2020, QBE1)

329

Maximum Marks : 75 and

Attempt any four questions. All questions carry equal marks.
 Q. 1. Let A be a non-empty set and $(G, \cdot)$ be a group. Let F be the set of all functions from A to G . Define an operation $*$ on F as follows:
 For $f, g \in F$, let $f * g : A \rightarrow G$ as $(f * g)(x) = f(x) \cdot g(x) \forall x \in A$. Prove that $(F, *)$ is a group.

Find the inverse of $\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ in $GL(2, \mathbb{Z}_5)$, the group of 2×2 non-singular matrices over $\mathbb{Z}_5$. Verify the answer by direct calculation.

Describe the group of symmetries of a non-square rectangle and draw its Cayley's table.

Sol. (i) Since, G is a group

for $f * g : A \rightarrow G$ defined as

$$(f * g)(x) = f(x) \cdot g(x) \forall x \in A$$

$$= f * g \in G \text{ for } f \text{ and } g \in G$$

$$= F \text{ holds closure property.}$$

Since, G hold associative property.

$$= F \text{ holds associative property.}$$

Again, G is a group

Let $e \in G$ be its identity

Let $I \in F$ s.t.

$$I(x) = e \text{ for } x \in A$$

$$(f * I)(x) = f(x) \cdot I(x)$$

$$= f(x) \cdot e$$

$$= f(x)$$

$$(I * f)(x) = I(x) \cdot f(x)$$

$$= e \cdot f(x)$$

$$= f(x)$$

$\Rightarrow I$ is the identity in F .

For $f(x) \in G \exists f^{-1}(x) \in G$ (as G is a group)

$$\text{s.t. } f(x) \cdot f^{-1}(x) = e$$

$$\text{Let } f^{-1}(x) = g(x)$$

$$\Rightarrow f * g(x) = f(x) \cdot g(x)$$

(328)

$$= f(x) \cdot f^{-1}(x)$$

$$= e$$

$$= I(x)$$

$$g * f(x) = g(x) \cdot f(x)$$

$$= f^{-1}(x) \cdot f(x)$$

$$= e$$

$$= I(x)$$

$\Rightarrow$ For every $f \in G \exists f^{-1} \in G$

$\Rightarrow$ Inverse exist

$\Rightarrow (F, *)$ is a group.

(ii) Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be the inverse of $\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ in $GL(2, \mathbb{Z}_5)$

Let

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

$\Rightarrow$

$$A = 6 - 4$$

$\Rightarrow$

$$A = 1 + 1$$

$$A = 2$$

$\Rightarrow A^{-1}$ exist.

$$\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow$

$$2a + c = 1$$

...(1)

$$2b + d = 0$$

...(2)

$$4a + 3c = 0$$

...(3)

$$4b + 3d = 1$$

...(4)

From (1) $\times 2$ and (3)

$$4a + 2c = 2$$

$$4a + 3c = 0$$

$$-1c = 2$$

$$4c = 2$$

$$(4) 4c = (4) 2$$

$$16c = 8$$

$$c = 3$$

$$2a + 3 = 1$$

$$2a = 1 - 3$$

$$2a = -2$$

$$2a = 3$$

$$(3) 2a = (3)(3)$$

$$6a = 9$$

$$a = 4$$

From (2) $\times 2$ and (4)

$$4b + 2d = 0$$

$$4b + 3d = 1$$

$$-1d = 1$$

$$4d = 4$$

$$(4) 4d = 4(4)$$

$$16d = 16$$

$$d = 1$$

$$2b + d = 0$$

$$2b + 1 = 0$$

$$2b = -1$$

$$2b = 4$$

$$(3) 2b = 3(4)$$

$$6b = 12$$

$$b = 2$$

Thus

$$A^{-1} = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$$

Verify

$$A \cdot A^{-1} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8+3 & 4+1 \\ 16+9 & 8+3 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 5 \\ 25 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ Hence verified.}$$

Q. 2. Let a be an element of a group such that $|a| = 3$, prove that $C(a) = C(a^2)$. Give an example to show that the conclusion fails if $|a| = 4$. Find the orders of each of the elements of $U(14)$. Show that it is cyclic and find all its generators.

Sol. Let G be a group

Let $a \in G$ s.t. $|a| = 3$

$$C(a) = \{x \in G : ax = xa\}$$

$$C(a^2) = \{x \in G : a^2x = xa^2\}$$

Let $x \in C(a)$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

From (1) and (2)

$$C(a) = C(a^2)$$

$$G = \{\pm 1, \pm i, \pm j, \pm k\}$$

$$i^2 = j^2 = k^2 = -1, ij = ji = k, jk = -kj = i, ki = -ik = j$$

$$a = i$$

$$|a| = 4$$

$$a^2 = i^2$$

$$= -1$$

$$|a^2| = 2$$

$$C(a) = \{x : ax = xa\}$$

$$C(a) = \{\pm 1, \pm i\}$$

$$C(a) = \{x : a^2x = xa^2\}$$

$$C(a^2) = G$$

$$C(a) \neq C(a^2) \text{ when } |a| = 4$$

$$(ii) \quad U(14) = \{1, 3, 5, 9, 11, 13\}$$

1 is the identity

$$3^1 = 3$$

$$3^2 = 9$$

$$3^3 = 27 = 13$$

$$3^4 = 39 = 11$$

$$3^5 = 33 = 5$$

$$3^6 = 15 = 1$$

$$\Rightarrow \langle 3 \rangle = 6$$

$$\Rightarrow U(14) = \langle 3 \rangle$$

$$5^1 = 5$$

$$5^2 = 25 = 11$$

$$5^3 = 55 = 13$$

$$5^4 = 65 = 9$$

$$5^5 = 45 = 3$$

$$5^6 = 15 = 1$$

$$\Rightarrow U(14) = \langle 5 \rangle$$

$$9^1 = 9$$

$$9^2 = 81 = 11$$

$$9^3 = 99 = 1$$

$$11^1 = 11$$

$$11^2 = 121 = 9$$

$$11^3 = 99 = 1$$

$$13^1 = 13$$

$$13^2 = 169 = 1$$

Thus,

$$|1| = 1; |3| = 6; |5| = 6; |9| = 3;$$

$$|11| = 3; |13| = 2;$$

Since,

$$|3| = 6, \langle 3 \rangle \subseteq U(14)$$

and

$$U(14) \subseteq \langle 3 \rangle$$

$\Rightarrow$

$$U(14) = \langle 3 \rangle$$

$\Rightarrow U(14)$ is a cyclic group with generator 3

Similarly $U(14)$ is a cyclic group with generator 5.

Hence, 3 and 5 are generators of $U(14)$.

Q. 3. Define Centre $Z(G)$ of a group G and prove that $Z(S_4) = \{e\}$.

For $n > 2$, show that every even permutation in S_n is a product of 3-cycles. Let $\sigma = (1, 5, 7)(2, 5, 3)(1, 6)$. Express σ^{17} as a cycle.

Sol. (i) Centre of a group G

$$Z(G) = \{x : xa = ax \forall a \in G\}$$

for the group S_4

$$Z(S_4) = \{x : xa = ax \forall a \in S_4\}$$

$$S_4 = \{(1), (12), (13), (14), (23), (24), (34), (234),$$

$$(123), (132), (124), (142), (134), (143), (243),$$

$$(1234), (1243), (1324), (1342), (1423), (1432),$$

$$(12)(34), (13)(24), (14)(23)\}$$

$$x \in S_4$$

For

$$(12) \in S_4$$

Let

$$x(12) = (12)x \text{ is the only for } x = (1)$$

Check

$$Z(S_4) = \{(1)\}$$

$\Rightarrow$

$$Z(S_4) = \{e\}$$

or

$$n > 2$$

(ii)

Let σ be an even permutation in S_n

$\Rightarrow \sigma$ can be written as even no of transposition

Let $(ab)(cd)$ be a pair of two transposition.

Let a, b, c, d be all distinct

$$(ab)(cd) = (ab)e(cd)$$

$$= (ab)(bc)(bc)(cd)$$

$$= (abc)(bcd)$$

(Product of 3 cycles).

Let

$$b = c$$

$$(ab)(cd) = (ab)(bd)$$

$$= (abd)$$

(a 3 cycle)

Let

$$(ab) = (cd)$$

$\Rightarrow$

$$(ab)(cd) = (ab)(ab)$$

$$= e$$

$$= (abc)(acb)$$

(Product of 3 cycles)

$\Rightarrow$ every pair of transposition is a product of 3 cycles.

$$(iii) \quad \sigma = (157)(253)(16)$$

$$\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 2 & 3 & 4 & 7 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 5 & 2 & 4 & 3 & 6 & 7 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 2 & 3 & 4 & 5 & 1 & 7 \end{bmatrix}$$

$$\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 2 & 3 & 4 & 7 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 5 & 2 & 4 & 3 & 1 & 7 \end{bmatrix}$$

$$\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 7 & 2 & 4 & 3 & 5 & 1 \end{bmatrix}$$

$$\sigma = (165327)$$

Length of cycle = 6

$$\sigma^6 = e$$

$$\sigma^{17} = \sigma^{6 \times 2 + 5}$$

$$= (\sigma^6)^2 + (\sigma^5)$$

$$= \sigma^5$$

$$\sigma^2 = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 7 & 2 & 4 & 3 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 7 & 2 & 4 & 3 & 5 \end{bmatrix}$$

$$\sigma^2 = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 1 & 7 & 4 & 2 & 3 & 6 \end{bmatrix}$$

$$\sigma^4 = \sigma^2 \cdot \sigma^2$$

$$\sigma^4 = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 1 & 7 & 4 & 2 & 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 7 & 4 & 2 & 3 \end{bmatrix}$$

$$\sigma^4 = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 5 & 6 & 4 & 1 & 7 & 3 \end{bmatrix}$$

$$\sigma^5 = \sigma^4 \cdot \sigma$$

$$\sigma^5 = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 5 & 6 & 4 & 1 & 7 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 7 & 2 & 4 & 3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 3 & 5 & 4 & 6 & 1 & 2 \end{bmatrix}$$

Thus

$$\sigma^{17} = \sigma^5$$

$$= (172356)$$

Q. 4. Prove or disprove any six, stating the results used

- (i) $\langle \mathbb{R}, + \rangle \cong \langle \mathbb{Q}, + \rangle$, (ii) $\langle \mathbb{Q}, + \rangle \cong \langle \mathbb{Z}, + \rangle$,
 (iii) $\langle \mathbb{R}, + \rangle \cong \langle \mathbb{R}^+, + \rangle$, (iv) $D_4 = \text{Group } Q \text{ of Quaternions}$,
 (v) $U(20) = D_4$, (vi) $U(8) \cong U(12)$,
 (vii) $U(1) = \mathbb{R}$, (viii) $\frac{GL(2, \mathbb{R})}{SL(2, \mathbb{R})} = \mathbb{R}^*$.

Sol. (i) $\langle \mathbb{R}, + \rangle \cong \langle \mathbb{Q}, + \rangle$

Since, $\mathbb{Q}$ is countable and $\mathbb{R}$ is uncountable.

Thus, cardinality of $\mathbb{Q}$ and $\mathbb{R}$ are not equivalent.

Hence, there does not exist any one-one correspondence between them.

So, given statement is false.

(ii) $\langle \mathbb{Q}, + \rangle \cong \langle \mathbb{Z}, + \rangle$

Since,

$$\mathbb{Z} = \langle 1 \rangle$$

$\Rightarrow \mathbb{Z}$ is cyclic

But $\mathbb{Q} \neq \langle a \rangle$ for any $a \in \mathbb{Q}$

$\Rightarrow \mathbb{Q}$ is non-cyclic

Hence, $\langle \mathbb{Q}, + \rangle$ is not isomorphic to $\langle \mathbb{Z}, + \rangle$

(iii) $\langle \mathbb{R}, + \rangle \cong \langle \mathbb{R}^+, + \rangle$

Let $\phi : \mathbb{R} \rightarrow \mathbb{R}^+$ be defined as

$$\phi(x) = e^x \quad \forall x \in \mathbb{R}$$

Since, ϕ is 1-1 and onto

Also,

$$\phi(x+y) = e^{x+y}$$

$$= e^x \cdot e^y$$

$$= \phi(x) \phi(y)$$

$\Rightarrow \phi$ is group homomorphism

$\Rightarrow \langle \mathbb{R}, + \rangle \cong \langle \mathbb{R}^+, + \rangle$

(iv) $D_4 = \text{group } Q \text{ of quaternions}$

$$D_4 = \langle a, b \rangle$$

$$a^4 = e, b^2 = e$$

$$b^{-1}ab = a^3 = a^{-1}$$

In D_4 , each element a^2, b, ab, a^2b, a^3b are of order 2.

$$Q = \{\pm 1, \pm i, \pm j, \pm k\}$$

$$i^2 = j^2 = k^2 = -1; ij = -ji = k; jk = -kj = i; ki = -ik = j$$

Element with order 2 is (-1) only

$\Rightarrow$ Order is not preserved between D_4 and Q

Hence, D_4 is not isomorphic to Q .

(v) $U(20) = D_4$

$$U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}$$

Let $x, y \in U(20)$

$$\Rightarrow x \cdot y = y \cdot x$$

$\Rightarrow U(20)$ is an abelian group.

But

$$D_4 = \langle a, b \rangle$$

$$a^4 = e = b^2, b^{-1}ab = a^3 = a^{-1}$$

D_4 is not abelian

Hence, $U(20)$ is not isomorphic to Q_4 .

(vi) $U(8) = U(12)$

$$U(8) = \{1, 3, 5, 7\}$$

$$U(12) = \{1, 5, 7, 11\}$$

$\Rightarrow$ Both are Klein-4 groups

Define $\phi : U(8) \rightarrow U(12)$ as

$$\phi(1) = 1; \phi(3) = 5; \phi(5) = 7; \phi(7) = 11$$

$\Rightarrow \phi$ is one-on, onto homomorphism

$$\Rightarrow U(8) = U(12)$$

$$(vii) U(1) = Z_2$$

$$U(10) = \{1, 3, 7, 9\} = \langle 3 \rangle$$

$$Z_4 = \{0, 1, 2, 3\} = \langle 1 \rangle$$

Define $\phi: U(10) \rightarrow Z_4$ s.t. $\phi(3) = 1$ will give required isomorphism

$$(viii) \frac{GL(2, \mathbb{R})}{SL(2, \mathbb{R})} = \mathbb{R}^*$$

Define a mapping $Q: GL(2, \mathbb{R}) \rightarrow \mathbb{R}^*$ as

$$Q(A) = |A|$$

$\Rightarrow Q$ is onto group homomorphism

Also $\text{Ker } Q = SL(2, \mathbb{R})$ (set of 2×2 matrices with determinant 1)

$$\text{By fundamental theorem } \frac{GL(2, \mathbb{R})}{\text{Ker } Q} = \mathbb{R}^*$$

Hence given statement is true.

Q. 5. Let H be a subgroup of a group G . Prove that $aH \rightarrow Ha^{-1}$ is a bijective mapping from the set of all left cosets of H in G to the set of all right cosets of H in G . Can the same be said for $aH \rightarrow Ha$?

If G is a non-abelian group of order 8 with $Z(G) \neq \{e\}$, prove that $|Z(G)| = 2$.

Sol. Let G be a group

Let H be a subgroup of G

Let $a \in G$

$\Rightarrow a^{-1}$ exist.

Let $S = \{aH : a \in G\}$

Let $T = \{Ha^{-1} : a \in G\}$

Define a mapping $\phi: S \rightarrow T$ as

$$\phi(aH) = Ha^{-1}$$

Let $a, b \in G$

Let

$$aH = bH$$

$\Rightarrow$

$$a = bh \text{ for some } h \in H$$

$\Rightarrow$

$$b^{-1}a = h$$

$\Rightarrow$

$$b^{-1}a \in H$$

$\Rightarrow$

$$b^{-1}(a^{-1})^{-1} \in H$$

or

$$Hb^{-1} = Ha^{-1}$$

$$Ha^{-1} = Hb^{-1}$$

$\Rightarrow \phi$ is well defined

Since, $\forall Ha^{-1} \in T \exists a \in G$ s.t.

$$\phi(aH) = H(a^{-1}) = Ha^{-1}$$

$\Rightarrow \phi$ is onto

Also

$$\phi(aH) = \phi(bH)$$

$$Ha^{-1} = Hb^{-1}$$

$$a^{-1}(b^{-1})^{-1} \in H$$

$$a^{-1}b \in H$$

$$aH = bH$$

$\Rightarrow \phi$ is 1-1

$\Rightarrow \phi: S \rightarrow T$ is a bijective

But $\phi: S \rightarrow T$ defined as

$\phi(aH) = Ha$ may not be well defined

Let

$$G = S_3$$

$H = \{e, (12)\}$ be a subgroup of G

$a = (23)$ and $b = (132)$

Let

$$aH = \{(23), (132)\}$$

$\Rightarrow$

$$bH = \{(23), (132)\}$$

$\Rightarrow$

$$aH = bH$$

$\Rightarrow$

$$Ha = \{(23), (123)\}$$

and

$$Hb = \{(13), (132)\}$$

$\Rightarrow$

$$Ha \neq Hb$$

$\Rightarrow \phi$ is not well defined

(ii) Let G be a non-abelian group

Let

$$|G| = 8$$

Let

$$Z(G) \neq \{e\}$$

By Lagrange theorem, $|Z(G)|$ divides $|G|$

$\Rightarrow |Z(G)|$ divides 8

$\Rightarrow |Z(G)|$ is one from 1, 2, 4, 8

Since $Z(G) \neq \{e\}$ (given)

$\Rightarrow$

$$|Z(G)| \neq 1$$

If

$$|Z(G)| = 8$$

Then,

$$Z(G) = G$$

as

$$|G| = 8$$

$\Rightarrow$

G is an abelian group

But G is non-abelian (given)

$\Rightarrow$

$$|Z(G)| \neq 8$$

If

$$|Z(G)| = 4$$

Then

$$|Z(G)| = \frac{|G|}{|Z(G)|} = 2$$

$\Rightarrow Z(G)$ is normal in G

$\Rightarrow \frac{G}{Z(G)}$ is cyclic

$\Rightarrow G$ is abelian

But G is non abelian (given)

Hence,

$$|Z(G)| = 2$$

Q. 6. Let N be a normal subgroup of G and M be a normal subgroup of N . If N is cyclic, prove that M is a normal subgroup of G . Show by an example that the conclusion fails to hold if N is not cyclic.

If ϕ is a homomorphism from a finite group G to a finite group G' prove that $|\phi(G)|$ divides the gcd of $|G|$ and $|G'|$.

Sol. Let G be a group

Let N be a normal subgroup of G

Let M be a normal subgroup of N

Let N be a cyclic group.

$\Rightarrow \exists a \in N$ s.t.

$$N = \langle a \rangle$$

Since, subgroup of cyclic group is cyclic

$\Rightarrow M$ is cyclic

$\Rightarrow M = \langle a^k \rangle$ where $a \in N$ and k is an integer.

To show $g^{-1} a^k g \in M \forall g \in G$.

For $g \in G$

$$g^{-1} a g \in N = \langle a \rangle$$

$\Rightarrow g^{-1} a g = a^t$ for some integer t .

$$\begin{aligned} \Rightarrow g^{-1} a^k g &= (g^{-1} a g)^k \\ &= (a^t)^k \\ &= (a^k)^t \\ &\in M \end{aligned}$$

$\Rightarrow M$ is normal subgroup in G .

Let

$$G = A_4$$

$$= \{e, (12)(34), (13)(24), (14)(23), (123), (132), (124), (142), (134), (143), (234), (243)\}$$

Let

$$N = \{e, (12)(34), (13)(24), (14)(23)\}$$

Since, N is normal subgroup of G

and N is not cyclic

Let

$$M = \{e, (12)(34)\}$$

Since,

$$i_N(M) = \frac{O(N)}{O(M)} = \frac{4}{2} = 2$$

M is normal in N

$$(123)(12)(34)(123)^{-1} = (123)(12)(34)(132)$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

$$= (14)(23)$$

$$\notin M$$

$\Rightarrow M$ is not normal in G .

(ii) Let ϕ be a homomorphism from finite group G to finite group G'

$$\phi: G \rightarrow G' \text{ s.t.}$$

$$\phi(G) = \{\phi(a) : a \in G\}$$

$$\phi(G) \neq \phi$$

Also

$$\phi(G) \subseteq G'$$

Let

$$\phi(a), \phi(b) \in \phi(G)$$

$\Rightarrow$

$$\phi(a)\{\phi(b)\}^{-1} = \phi(a)\phi(b^{-1})$$

$$= \phi(ab^{-1})$$

$$\in \phi(G)$$

$\Rightarrow \phi(G)$ is a subgroup of finite group G'

$\Rightarrow$ By Lagrange's theorem

$$|\phi(G)| \text{ divides } |G'|$$

...(1)

By 1st isomorphism theorem

$$\frac{G}{\text{Ker } \phi} \cong \phi(G)$$

$\Rightarrow$

$$|\phi(G)| = \frac{|G|}{|\text{Ker } \phi|}$$

$\Rightarrow$

$$|G| = |\phi(G)| |\text{Ker } \phi|$$

...(2)

$\Rightarrow$

$$|\phi(G)| \text{ divides } |G|$$

From (1) and (2)

$$|Q(G)| \text{ divides gcd of } |G| \text{ and } |G'|.$$

ooo

Unique Paper Code : 32351303
 Name of the Paper : C-7: MULTIVARIATE CALCULUS
 Name of Course : B.Sc. (Hons.) Mathematics II Year
 Semester : 3 (CBCS)
 Duration : 3 Hours

O.B. EXAM
 DEC. 2020

C-7: Mathematics: Multivariate Calculus (Dec. 2020, OBE)

341

Maximum Marks: 70

Attempt any four questions. All questions carry equal marks.

Q. 1. Let $f(x, y) = \begin{cases} \frac{y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

Is the function f continuous at $(0, 0)$? Justify your answer.

Find an equation for the tangent plane to the surface $z = f(x, y)$ defined above at the point $P_0(1, 2, \frac{8}{5})$.

Also find the directional derivative of $f(x, y)$ at $P_0(1, 2)$ in the direction of $\vec{v} = 3\vec{i} - 2\vec{j}$.

Sol. (a) Since, $f(x, y) = \begin{cases} \frac{y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

Let

$$y = mx$$

$$\begin{aligned} \lim_{(x, y) \rightarrow (0, 0)} f(x, y) &= \lim_{x \rightarrow 0} \frac{m^3 x^3}{x^2 + m^2 x^2} \\ &= \lim_{x \rightarrow 0} \frac{m^3 x^3}{x^2(1 + m^2)} \\ &= \lim_{x \rightarrow 0} \frac{m^3 x}{1 + m^2} \\ &= 0 \\ &= f(0, 0) \end{aligned}$$

$\Rightarrow$ Function f is continuous at $(0, 0)$.

(b)

$$f(x, y) = \frac{y^3}{x^2 + y^2}$$

$$f_x = -\frac{y^3}{(x^2 + y^2)^2} (2x)$$

$$f_x = \frac{-2xy^3}{(x^2 + y^2)^2}$$

(340)

$$f_x = \frac{(x^2 + y^2)(2y^2) - y^3(2y)}{(x^2 + y^2)^2}$$

$$f_y = \frac{3x^2 y^2 + y^4}{(x^2 + y^2)^2}$$

Equation of tangent plane to function $f(x, y)$ at point (x_0, y_0, z_0) is
 $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

Equation of tangent plane at the point $P_0(1, 2, \frac{8}{5})$

$$z - \frac{8}{5} = \frac{-2(1)(2)^3}{((1)^2 + (2)^2)^2} (x - 1) + \frac{3(1)^2(2)^2 + (2)^4}{((1)^2 + (2)^2)^2} (y - 2)$$

$$z - \frac{8}{5} = -\frac{16}{25} (x - 1) + \frac{28}{25} (y - 2)$$

$$25z - 40 = -16x + 16 + 28y - 56$$

$$16x - 28y + 25z = 0$$

(c) Unit vector $\hat{v}$ in direction of $\vec{v}$ is

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|}$$

$$= \frac{3\vec{i} - 2\vec{j}}{\sqrt{(3)^2 + (-2)^2}}$$

$$= \frac{3}{\sqrt{13}} \vec{i} - \frac{2}{\sqrt{13}} \vec{j}$$

$$f(x, y) = \frac{y^3}{x^2 + y^2}$$

$$f_x = \frac{-2xy^3}{(x^2 + y^2)^2} ; f_y = \frac{3x^2 y^2 + y^4}{(x^2 + y^2)^2}$$

$$f_x(1, 2) = -\frac{16}{25} ; f_y(1, 2) = \frac{28}{25}$$

$$\Delta f = f_x \vec{i} + f_y \vec{j}$$

$$\Delta f(1, 2) = -\frac{16}{25} \vec{i} + \frac{28}{25} \vec{j}$$

Directional derivative of $f(x, y)$ at the point $P_0(1, 2)$ in the direction
 $\vec{v} = 3\vec{i} - 2\vec{j}$ is

$$\begin{aligned}\Delta f(1, 2) &= \Delta f(1, 2) \cdot \vec{v} \\ &= \left(-\frac{16}{25} \hat{i} + \frac{28}{25} \hat{j} \right) \cdot \left(\frac{3}{\sqrt{13}} \hat{i} - \frac{2}{\sqrt{13}} \hat{j} \right) \\ &= -\frac{104}{25\sqrt{13}}\end{aligned}$$

Q. 2. Find the critical point and classify each point as a relative minimum, relative maximum, or a saddle point of $f(x, y) = xye^{-8(x^2+y^2)}$. Find the maximum and minimum values of $f(x, y, z) = xyz$ subject to the constraint $x^2 + 2y^2 + 4z^2 = 24$.

Where is the function $f(x, y) = \sqrt{x^2 + y^2}$ differentiable?

Sol. (i)

Differentiating $f(x, y) = xye^{-8(x^2+y^2)}$ with respect to x

$$\begin{aligned}f_x &= ye^{-8(x^2+y^2)} - 8(2x)xye^{-8(x^2+y^2)} \\ &= (1 - 16x^2)ye^{-8(x^2+y^2)}\end{aligned}$$

Differentiating $f(x, y)$ with respect to y

$$\begin{aligned}f_y &= xe^{-8(x^2+y^2)} - 8(2y)xye^{-8(x^2+y^2)} \\ &= (1 - 16y^2)xe^{-8(x^2+y^2)}\end{aligned}$$

For critical points

$$(1 - 16x^2)ye^{-8(x^2+y^2)} = 0 = f_x$$

$$\Rightarrow x = \pm \frac{1}{4}, y = 0$$

$$(1 - 16y^2)xe^{-8(x^2+y^2)} = 0$$

$$\Rightarrow x = 0, y = \pm \frac{1}{4}$$

Thus, critical points are

$$(0, 0), \left(\frac{1}{4}, \frac{1}{4}\right), \left(-\frac{1}{4}, -\frac{1}{4}\right), \left(-\frac{1}{4}, \frac{1}{4}\right), \left(\frac{1}{4}, -\frac{1}{4}\right)$$

Now,

$$f_{xx} = (1 - 16x^2)(-8)(2x)ye^{-8(x^2+y^2)} - 32xye^{-8(x^2+y^2)}$$

$$f_{xx} = -16xye^{-8(x^2+y^2)}[1 - 16x^2 + 2]$$

$$f_{xx} = 16xye^{-8(x^2+y^2)}(16x^2 - 3)$$

$$f_{yy} = 16xye^{-8(x^2+y^2)}(16y^2 - 3)$$

Similarly

$$\text{and } f_{xy} = e^{-8(x^2+y^2)}[(1 - 16x^2) + (1 - 16y^2)y(-8(2y))]e^{-8(x^2+y^2)}$$

$$f_{xy} = e^{-8(x^2+y^2)}(1 - 16x^2)[1 - 16y^2]$$

Let

$$D = f_{xx}f_{yy} - (f_{xy})^2$$

For

$$(x, y) = (0, 0)$$

$$D = -1$$

$\Rightarrow (0, 0)$ is a saddle point

For

$$(x, y) = \left(\frac{1}{4}, \frac{1}{4}\right)$$

$$\Rightarrow \left(\frac{1}{4}, \frac{1}{4}\right) \text{ is a point of relative maxima.}$$

$$\text{At } (x, y) = \left(-\frac{1}{4}, -\frac{1}{4}\right)$$

$$D > 0 \text{ and } f_{xx} < 0$$

$$\Rightarrow \left(-\frac{1}{4}, -\frac{1}{4}\right) \text{ is a point of relative maxima.}$$

$$\text{For } (x, y) = \left(-\frac{1}{4}, \frac{1}{4}\right)$$

$$D > 0 \text{ and } f_{xx} > 0$$

$$\left(-\frac{1}{4}, \frac{1}{4}\right) \text{ is a point of relative minima}$$

$$\text{For } (x, y) = \left(\frac{1}{4}, -\frac{1}{4}\right)$$

$$D > 0 \text{ and } f_{xx} > 0$$

$$\Rightarrow \left(\frac{1}{4}, -\frac{1}{4}\right) \text{ is a point of relative minima.}$$

$$(ii) f(x, y, z) = xyz \quad \dots(1)$$

s.t.c

$$x^2 + 2y^2 + 4z^2 = 24 \quad \dots(2)$$

$$\text{Let } g(x, y, z) = x^2 + 2y^2 + 4z^2 - 24$$

$$\text{Let } f_x = \lambda g_x$$

$$\Rightarrow yz = 2\lambda x \quad \dots(3)$$

$$\text{Let } f_y = \lambda g_y$$

$$\Rightarrow xy = 4\lambda y \quad \dots(4)$$

$$\text{Let } f_z = \lambda g_z$$

$$\Rightarrow xy = 8\lambda z \quad \dots(5)$$

From Equation (3) and (4)

$$xz = 4\left(\frac{zy}{2x}\right)y$$

$$x^2 = 2y^2$$

From (4) and (5)

$$xy = 8\left(\frac{xz}{4y}\right)z$$

$$y^2 = 2z^2$$

$$x^2 = 2y^2 = 4y^2$$

$\Rightarrow$ From Equation (2)

$$x^2 + x^2 + x^2 = 24$$

$$3x^2 = 24$$

$$\begin{aligned}x^2 &= 8 \\x &= \pm 2\sqrt{2} \\y &= \pm 2 \\z &= \pm \sqrt{2}\end{aligned}$$

$\Rightarrow$ Extreme points are

$$\begin{aligned}&(2\sqrt{2}, 2, \sqrt{2}), (2\sqrt{2}, 2, -\sqrt{2}), (2\sqrt{2}, -2, \sqrt{2}), (2\sqrt{2}, -2, -\sqrt{2}) \\&(-2\sqrt{2}, 2, \sqrt{2}), (-2\sqrt{2}, 2, -\sqrt{2}), (-2\sqrt{2}, -2, \sqrt{2}), (-2\sqrt{2}, -2, -\sqrt{2})\end{aligned}$$

Since

$$\begin{aligned}f(2\sqrt{2}, 2, \sqrt{2}) &= f(2\sqrt{2}, -2, -\sqrt{2}) = f(-2\sqrt{2}, -2, \sqrt{2}) \\&= f(-2\sqrt{2}, 2, -\sqrt{2}) = 8\end{aligned}$$

$$\begin{aligned}\text{and } f(2\sqrt{2}, 2, -\sqrt{2}) &= f(2\sqrt{2}, -2, \sqrt{2}) = f(-2\sqrt{2}, 2, \sqrt{2}) \\&= f(-2\sqrt{2}, -2, -\sqrt{2}) = -8\end{aligned}$$

$\Rightarrow$ Maximum value of f is 8,
Minimum value of f is -8

$$\begin{aligned}\text{(iii) } f(x, y) &= \sqrt{x^2 + y^2} \\ \text{Since } x^2 &> 0, \forall x \in \mathbb{R} \\ y^2 &> 0, \forall y \in \mathbb{R} \\ \Rightarrow x^2 + y^2 &> 0, \forall x, y \in \mathbb{R} \\ \Rightarrow f(x, y) &\text{ is continuous } \forall (x, y) \in \mathbb{R}^2\end{aligned}$$

Differentiating f with respect to x

$$f_x = \frac{x}{\sqrt{x^2 + y^2}}$$

Differentiating f with respect to y

$$f_y = \frac{y}{\sqrt{x^2 + y^2}}$$

$\Rightarrow f_x$ and f_y are also continuous for every $(x, y) \in \mathbb{R}^2$ except the point $(x, y) = (0, 0)$.

At $(0, 0)$

$$\begin{aligned}f_x(0, 0) &= \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} \\&= \lim_{h \rightarrow 0} \frac{\sqrt{h^2}}{h} \\&= \begin{cases} \lim_{h \rightarrow 0} \frac{h}{h} & \text{if } h > 0 \\ \lim_{h \rightarrow 0} \frac{-h}{h} & \text{if } h < 0 \end{cases}\end{aligned}$$

$$= \begin{cases} 1 & \text{if } h > 0 \\ -1 & \text{if } h < 0 \end{cases}$$

$\Rightarrow$ Limit does not exist at $(0, 0)$

Hence $f(x, y)$ is not differentiable at $(0, 0)$.

Q. 3. Compute $\iint_R x e^{xy} dA$, where R is the rectangle $0 \leq x \leq 1, 1 \leq y \leq 2$

using iterated integrals in both orders.

Evaluate $\iint_R 6x^2 y dA$ if R is the region bounded between the curves

$$y = x, y = 1 \text{ and } 4y = x^2.$$

Find the area of the region bounded between the curves $r_1(\theta) = 2 + \sin 3\theta$ and $r_2(\theta) = 4 - \cos 3\theta$.

Sol. $I = \iint_R x e^{xy} dA, R = \{0 \leq x \leq 1, 1 \leq y \leq 2\}$

$$\begin{aligned}I &= \int_0^1 \int_1^2 x e^{xy} dy dx \\&= \int_0^1 \left[x \frac{e^{xy}}{x} \right]_1^2 dx \\&= \int_0^1 (e^{2x} - e^x) dx \\&= \left[\frac{e^{2x}}{2} - e^x \right]_0^1 \\&= \left(\frac{e^2}{2} - e^1 \right) - \left(\frac{1}{2} - 1 \right) \\&= \frac{e^2}{2} - e + \frac{1}{2}\end{aligned}$$

On reversing order

$$\begin{aligned}I &= \int_1^2 \int_0^1 x e^{xy} dx dy \\&= \int_1^2 \left[\frac{x e^{xy}}{y} - \int \frac{e^{xy}}{y^2} dy \right]_0^1 dy \\&= \int_1^2 \left[\frac{x}{y} e^{xy} - \frac{e^{xy}}{y^2} \right]_0^1 dy\end{aligned}$$

$$I = \int_1^2 \left(\frac{e^y}{y} - \frac{e^y}{y^2} + \frac{1}{y^2} \right) dy$$

$$I = \left[\frac{1}{y} e^y \right]_1^2 + \int_1^2 \frac{e^y}{y^2} dy - \int_1^2 \frac{e^y}{y^2} dy - \left[\frac{1}{y} \right]_1^2$$

$$I = \frac{1}{2} e^2 - e - \left[\frac{1}{2} - 1 \right]$$

$$I = \frac{1}{2} e^2 - e + \frac{1}{2}$$

$$I = \iint_R 6x^2 y \, dA$$

(iii) Let

R is the region bounded between the curves

From $y = x$, $y = 1$ and $4y = x^2$
 Point of intersection $(1, 1)$
 From $y = x$ and $y = 1$

$\Rightarrow y = x$ and $4y = x^2$
 $4x = x^2$
 $x^2 - 4x = 0$

$$x = 0, 4$$

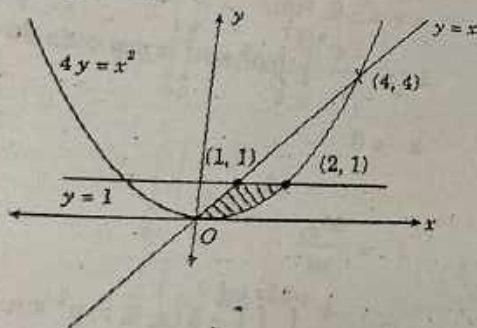
Point of intersection $(0, 0)$ and $(4, 4)$

From $y = 1$ and $4y = x^2$
 $x^2 = 4$

$$x = \pm 2$$

Point of intersection $(-2, 1)$; $(2, 1)$

Common shaded region is.



Thus, integral is

$$I = \int_{y=0}^1 \int_{x=y}^{2\sqrt{y}} (6x^2 y) \, dx \, dy$$

$$I = \int_0^1 2y (x^2)^2 \sqrt{y} \, dy$$

$$I = 2 \int_0^1 y (8y^{3/2} - y^3) \, dy$$

$$I = 2 \int_0^1 (8y^{5/2} - y^4) \, dy$$

$$I = 2 \left[8 \times \frac{2}{7} y^{7/2} - \frac{y^5}{5} \right]_0^1$$

$$I = 2 \left[\frac{16}{7} - \frac{1}{5} \right]$$

$$I = \frac{146}{35}$$

(iii)

$$A = \int_0^{2\pi} \int_{2+\sin 3\theta}^{4-\cos 3\theta} r \, dr \, d\theta$$

$$A = \int_0^{2\pi} \left[\frac{r^2}{2} \right]_{2+\sin 3\theta}^{4-\cos 3\theta} d\theta$$

$$A = \frac{1}{2} \int_0^{2\pi} [(4 - \cos 3\theta)^2 - (2 + \sin 3\theta)^2] d\theta$$

$$A = \frac{1}{2} \int_0^{2\pi} [16 + \cos^2 3\theta - 8 \cos 3\theta - 4 - \sin^2 3\theta - 4 \sin 3\theta] d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} [12 - 8 \cos 3\theta - 4 \sin 3\theta + \cos 6\theta] d\theta$$

$$= \frac{1}{2} \left[12\theta - \frac{8}{3} \sin 3\theta + \frac{4}{3} \cos 3\theta + \frac{1}{6} \sin 6\theta \right]_0^{2\pi}$$

$$= 12\pi$$

Q. 4. Find the mass of the ellipsoid $4x^2 + 4y^2 + z^2 = 16$ lying above the xy -plane if the density is given by $\delta(x, y, z) = z$.

Determine the centroid of the solid bounded above by the sphere $x^2 + y^2 + z^2 = 1$ below by the xy -plane where the density is given by $\delta(x, y, z) = z$.

Compute $\int_0^1 \int_0^1 x^2 y \, dx \, dy$ by changing $u = x$ and $v = xy$.

Sol. (i) $4x^2 + 4y^2 + z^2 = 16$

Let $x = r \cos \theta$

$y = r \sin \theta$

$z = z$

$4r^2 + z^2 = 16$

$z^2 = 4(4 - r^2)$

$z = 2\sqrt{4 - r^2}$

Also, for

$z = 0$

$4r^2 = 16$

$r^2 = 4$

$r = 2$

Thus, limits are

$0 \leq r \leq 2$

$0 \leq \theta \leq 2\pi$

$0 \leq z \leq 2\sqrt{4 - r^2}$

$$\text{Mass} = \int_0^{2\pi} \int_0^2 \int_0^{2\sqrt{4-r^2}} \delta(x, y, z) r dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^2 \int_0^{2\sqrt{4-r^2}} r z dz dr d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \int_0^2 [z^2]_0^{2\sqrt{4-r^2}} r dr d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \int_0^2 4(4 - r^2) r dr d\theta$$

$$= 2 \int_0^{2\pi} \int_0^2 (4r - r^3) dr d\theta$$

$$= 2 \int_0^{2\pi} \left[2r^2 - \frac{r^4}{4} \right]_0^2 d\theta$$

$$= 2(8 - 4) (\theta)_0^{2\pi}$$

$$= 16\pi$$

(ii) Solid bounded above by the sphere $x^2 + y^2 + z^2 = 1$ and below by the xy plane

Changing to spherical coordinates

$x = \rho \cos \theta \sin \phi$

$y = \rho \sin \theta \sin \phi$

$z = \rho \cos \phi$

$\rho^2 = 1$

$0 \leq \rho \leq 1$

$0 \leq \theta \leq 2\pi$

$0 \leq \phi \leq \frac{\pi}{2}$

$$m = \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \delta(x, y, z) \rho^2 \sin \phi d\rho d\theta d\phi$$

$$m = \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^3 \sin \phi \cos \phi d\rho d\theta d\phi$$

$$m = \left[\frac{\rho^4}{4} \right]_0^1 (\theta)_0^{2\pi} \int_0^{\pi/2} \frac{\sin 2\phi}{2} d\phi$$

$$m = \frac{1}{8} (2\pi) \left[-\frac{\cos 2\phi}{2} \right]_0^{\pi/2}$$

$$m = \frac{\pi}{8} [-(-1 - 1)]$$

$$m = \frac{\pi}{4}$$

$$\bar{x} = \frac{M_{yz}}{m}$$

$$\bar{x} = \frac{4}{\pi} \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 x \delta(x, y, z) \rho^2 \sin \phi d\rho d\theta d\phi$$

$$\bar{x} = \frac{4}{\pi} \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho \cos \theta \sin \phi \rho \cos \phi \rho^2 \sin \phi d\rho d\theta d\phi$$

$$\bar{x} = \frac{-4}{\pi} \int_0^{2\pi} \int_0^{\pi/2} \rho^4 \sin^2 \phi \cos \phi d\phi d\theta (-\sin \theta)_0^{2\pi}$$

$$\bar{x} = 0$$

Similarity $y = 0$

$$\bar{z} = \frac{M_{xy}}{m}$$

$$\bar{z} = \frac{4}{\pi} \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 z \delta(x, y, z) \rho^2 \sin \phi d\rho d\theta d\phi$$

$$\bar{z} = \frac{4}{\pi} \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 (\rho \cos \phi)^2 \rho \sin \phi d\rho d\theta d\phi$$

$$\begin{aligned}\bar{z} &= \frac{4}{\pi} \int_0^{\pi/2} \cos^2 \phi \sin \phi d\phi \int_0^{2\pi} d\theta \int_0^1 \rho^3 d\rho \\ \bar{z} &= \frac{4}{\pi} \left(\frac{-\cos^3 \phi}{3} \right)_0^{\pi/2} (2\pi) \left(\frac{\rho^4}{4} \right)_0^1 \\ \bar{z} &= \frac{4}{5\pi} (2\pi) \left(0 + \frac{1}{3} \right) \\ \bar{z} &= \frac{8}{15}\end{aligned}$$

Thus, coordinate of centroid is $\left(0, 0, \frac{8}{15} \right)$

(iii)

$$I = \int_0^1 \int_0^1 x^2 y dx dy$$

Let

Jacobian is

$$\begin{aligned}u &= x & v &= xy \\ J &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \\ &= \begin{vmatrix} 1 & 0 \\ y & x \end{vmatrix} \\ &= x\end{aligned}$$

 $\Rightarrow$ $\Rightarrow$

Also for

 $\Rightarrow$

Thus,

$$\begin{aligned}x dx dy &= du dv \\ x^2 y dx dy &= xy du dv \\ &= v du dv \\ 0 \leq x \leq 1 &\text{ and } 0 \leq y \leq 1 \\ 0 \leq u \leq 1 &\text{ and } 0 \leq v \leq u \\ I &= \int_0^1 \int_0^u v dv du \\ &= \int_0^1 \left[\frac{v^2}{2} \right]_0^u du \\ &= \frac{1}{2} \int_0^1 u^2 du \\ &= \frac{1}{2} \left[\frac{u^3}{3} \right]_0^1 \\ &= \frac{1}{6}\end{aligned}$$

$$= \frac{1}{6}$$

Q. 5, Evaluate $\oint_C (x^2 z dx - yx^2 dy + 3 dz)$ where C is the boundary of the triangle with vertices $(0, 0, 0)$, $(1, 1, 0)$ and $(1, 1, 1)$.

Find a non-zero function h for which

$F(x, y) = h(x)(x \sin y + y \cos y) i + h(x)(x \cos y - y \sin y) j$ is conservative.

Using line integral, find the area of the region enclosed by the steroid.

$$x = a \cos^2 t, y = a \sin^2 t \quad (0 \leq t \leq 2\pi)$$

Sol. (i) Let C_1 be the curve from $(0, 0, 0)$ to $(1, 1, 0)$ parametric equation will be

$$x = t; \quad y = t; \quad z = 0 \quad \text{for } 0 \leq t \leq 1$$

 $\Rightarrow$

$$\int_{C_1} x^2 z dx - yx^2 dy + 3 dz = \int_0^1 (t^2 \cdot 0) dt - t(t)^2 dt + 3(0)$$

$$= - \int_0^1 t^3 dt$$

$$= - \left[\frac{t^4}{4} \right]_0^1$$

$$= - \frac{1}{4}$$

Let C_2 be the curve from $(1, 1, 0)$ to $(1, 1, 1)$ parametric equation will be

$$x = 1; \quad y = 1; \quad z = t \quad \text{for } 0 \leq t \leq 1$$

 $\Rightarrow$

$$\begin{aligned}\int_{C_2} x^2 z dx - yx^2 dy + 3 dz &= \int_0^1 3 dt \\ &= 3 \left[t \right]_0^1 \\ &= 3\end{aligned}$$

Let C_3 be curve from $(1, 1, 1)$ to $(0, 0, 0)$ parametric equation will be

$$x = -t; \quad y = -t; \quad z = -t \quad \text{for } 1 \leq t \leq 0$$

$$dx = -dt \quad dy = -dt \quad dz = -dt$$

$$\begin{aligned}\int_{C_3} x^2 z dx - yx^2 dy + 3 dz &= \int_1^0 t^3 dt - t^3 dt - 3 dt \\ &= - \int_0^1 3 dt\end{aligned}$$

$$= -3 \left[\frac{1}{t} \right]_0^1$$

$$= -3$$

Thus,

$$\oint_C x^2 z dx - yx^2 dy + 3 dz = \frac{-1}{4} + 3 - 3 = \frac{-1}{4}$$

$$(iii) \vec{F}(x, y) = h(x)(x \sin y + y \cos y) \hat{i} + h(x)(x \cos y - y \sin y) \hat{j}$$

Compare with $\vec{F}(x, y) = M(x, y) \hat{i} + N(x, y) \hat{j}$ $\vec{F}(x, y)$ is conservative iff

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\begin{aligned} &= h(x)(x \cos y + \cos y - y \sin y) = h'(x)(x \cos y - y \sin y) + h(x) \cos y \\ &+ h(x) \cos y - h(x) \cos y - h'(x) y \sin y = h'(x) \cos y - h'(x) y \sin y + h(x) \cos y \\ &= h(x)(x \cos y - y \sin y) - h'(x)(x \cos y - y \sin y) = 0 \\ &= [h(x) - h'(x)](x \cos y - y \sin y) = 0 \\ &= h(x) - h'(x) = 0 \\ &= h'(x) = h(x) \\ &= \frac{h'(x)}{h(x)} = 1 \\ &= \frac{1}{h(x)} \end{aligned}$$

Integrating both sides

$$\ln h(x) = x + \ln C$$

$$h(x) = C e^x$$

(iii)

$$x = a \cos^3 t$$

$$dx = -3a \cos^2 t \sin t dt$$

$$y = a \sin^3 t$$

$$dy = 3a \sin^2 t \cos t dt$$

$$\text{Area} = \frac{1}{2} \oint_C -y dx + x dy$$

$$= \frac{1}{2} \int_0^{2\pi} [3a^2 \cos^2 t \sin^4 t + 3a^2 \sin^3 t \cos^4 t] dt$$

$$= \frac{3a^2}{2} \int_0^{2\pi} \sin^2 t \cos^2 t (\sin^2 t + \cos^2 t) dt$$

$$= \frac{3a^2}{2} \int_0^{2\pi} \left[\frac{\sin 2t}{2} \right]^2 dt$$

$$= \frac{3a^2}{8} \int_0^{2\pi} \frac{1 - \cos 4t}{2} dt$$

$$= \frac{3a^2}{16} \left[t - \frac{\sin 4t}{4} \right]_0^{2\pi}$$

$$= \frac{3a^2}{16} (2\pi)$$

$$= \frac{3a^2}{8} \pi$$

Q. 6. Find the mass of the lamina that is the portion of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 2$ with constant density δ_0 .Verify Stokes' Theorem if $F(x, y, z) = (x - y) \hat{i} + (y - z) \hat{j} + (z - x) \hat{k}$ and S be the portion of the plane $x + y + z = 1$ in the first octant assuming that the surface has an upward orientation.Using the Divergence Theorem, evaluate $\iint_S F \cdot N dS$, where $F(x, y, z) = (z^3 \hat{i} - x^3 \hat{j} + y^3 \hat{k})$ and S is the sphere $x^2 + y^2 + z^2 = a^2$, with outward unit normal vector N .

Sol. (i)

$$m = \iint_R \delta_0 \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA$$

For

$$z = x^2 + y^2$$

$$\frac{\partial z}{\partial x} = 2x \quad \text{and} \quad \frac{\partial z}{\partial y} = 2y$$

 $\Rightarrow$

$$m = \iint_R \delta_0 \sqrt{4x^2 + 4y^2 + 1} dA$$

For

$$z = x^2 + y^2 \quad \text{and} \quad z = 2$$

 $\Rightarrow$

$$x = r \cos \theta, \quad y = r \sin \theta$$

 $\Rightarrow$

$$0 \leq r \leq \sqrt{2}$$

$$0 \leq \theta \leq 2\pi$$

$$m = \delta_0 \int_0^{2\pi} \int_0^{\sqrt{2}} \sqrt{4r^2 + 1} r dr d\theta$$

For

$$4r^2 + 1 = t$$

$$8r dr = dt$$

$$r dr = \frac{1}{8} dt$$

Also

$$0 \leq r \leq \sqrt{2} \Rightarrow 1 \leq t \leq 9$$

$$m = \frac{\delta_0}{8} \int_0^{2\pi} \int_1^9 \sqrt{t} dt d\theta$$

$$= \frac{\delta_0}{8} \int_0^{2\pi} d\theta \cdot \int_1^9 \sqrt{t} dt$$

$$\begin{aligned}
 &= \frac{\delta_0}{8} (\theta)^{2\pi} \frac{2(t^{3/2})_0^1}{3} \\
 &= \frac{\delta_0}{12} (2\pi) (27-1) \\
 &= \frac{13\pi}{12} \delta_0
 \end{aligned}$$

(ii) Stoke's Theorem is

$$\oint_C \vec{F} \cdot d\vec{R} = \iint_S (\text{curl } \vec{F} \cdot \vec{N}) dS$$

$$\vec{F} = (x-y)\vec{i} + (y-z)\vec{j} + (z-x)\vec{k}$$

$$\begin{aligned}
 \text{Curl } \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x-y & y-z & z-x \end{vmatrix} \\
 &= \vec{i} + \vec{j} + \vec{k}
 \end{aligned}$$

Unit normal vector outward from $x+y+z=1$ is

$$\vec{N} = \frac{1}{\sqrt{3}} (\vec{i} + \vec{j} + \vec{k})$$

$$\begin{aligned}
 \Rightarrow \text{Curl } \vec{F} \cdot \vec{N} &= (\vec{i} + \vec{j} + \vec{k}) \cdot \frac{1}{\sqrt{3}} (\vec{i} + \vec{j} + \vec{k}) \\
 &= \frac{3}{\sqrt{3}} \\
 &= \sqrt{3}
 \end{aligned}$$

Also,

$$\begin{aligned}
 \Rightarrow x+y+z &= 1 \\
 \Rightarrow z &= 1-x-y \\
 \Rightarrow \frac{\partial z}{\partial x} &= -1 \quad \text{and} \quad \frac{\partial z}{\partial y} = -1
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow dS &= \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA \\
 &= \sqrt{3} dA
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus, } \iint_S \text{Curl } \vec{F} \cdot \vec{N} dA &= \iint_S \sqrt{3} \cdot \sqrt{3} dA \\
 &= 3 \int_0^1 \int_0^{1-x} dy dx
 \end{aligned}$$

$$\begin{aligned}
 &= 3 \int_0^1 (1-x) dx \\
 &= 3 \left[x - \frac{x^2}{2} \right]_0^1 \\
 &= \frac{3}{2}
 \end{aligned}$$

Now, S is the portion of plane $x+y+z=1$ in the first octant $\Rightarrow S$ is a triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ Curve passing through $(1, 0, 0)$ to $(0, 1, 0)$ parametric equation is

$$\begin{aligned}
 x &= 1-t; \quad y=t; \quad z=0 \quad \text{for } 0 \leq t \leq 1 \\
 dx &= -dt \quad dy=dt \quad dz=0
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \int_{C_1} \vec{F} \cdot d\vec{R} &= \int_0^1 [(x-y)\vec{i} + (y-z)\vec{j} + (z-x)\vec{k}] \cdot d(x\vec{i} + y\vec{j} + z\vec{k}) \\
 &= \int_{C_1} (x-y) dx + (y-z) dy + (z-x) dz \\
 &= \int_0^1 (1-t-t)(-dt) + (t-0) dt + (0-(1-t))(0) \\
 &= \int_0^1 (2t-1+t) dt \\
 &= \int_0^1 3t-1 dt \\
 &= \left[\frac{3t^2}{2} - t \right]_0^1 \\
 &= \frac{1}{2}
 \end{aligned}$$

Curve passing through $(0, 1, 0)$ to $(0, 0, 1)$ parametric equation is

$$\begin{aligned}
 x &= 0; \quad y=1-t; \quad z=t \quad \text{for } 0 \leq t \leq 1 \\
 dx &= 0; \quad dy=-dt; \quad dz=dt
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \int_{C_2} \vec{F} \cdot d\vec{R} &= \int_0^1 (x-y) dx + (y-z) dy + (z-x) dz \\
 &= \int_0^1 (0-(1-t))(0) + (1-t-t)(-dt) + (t-0) dt \\
 &= \int_0^1 2t-1+t dt
 \end{aligned}$$

$$= \int_0^1 (3t-1) dt$$

$$= \frac{1}{2}$$

Curve passing through $(0, 0, 1)$ to $(1, 0, 0)$ parametric equation is
 $x=t; y=0; z=1-t$ for $0 \leq t \leq 1$
 $dx=dt; dy=0; dz=-dt$

$$\int_C \vec{F} \cdot d\vec{R} = \int_0^1 (x-y) dx + (y-z) dy + (z-x) dz$$

$$= \int_0^1 (t-0) dt + (0-(1-t))(0) + (1-t-t)(-dt)$$

$$= \int_0^1 (t+2t-1) dt$$

$$= \int_0^1 (3t-1) dt$$

$$= \frac{1}{2}$$

Thus, $\oint_C \vec{F} \cdot d\vec{R} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2}$

$$= \frac{3}{2}$$

$$= \iiint (\text{Curl } \vec{F} \cdot \hat{N}) dS$$

Hence, Stokes theorem verified.

(iii) Since $\iiint_S \vec{F} \cdot \hat{N} dS = \iiint_V \text{div } \vec{F} dV$

$$\vec{F} = z^2 \hat{i} - x^3 \hat{j} + y^3 \hat{k}$$

$$\text{div } \vec{F} = \frac{\partial}{\partial x} z^2 + \frac{\partial}{\partial y} (-x^3) + \frac{\partial}{\partial z} (y^3) = 0$$

$$= \iiint_S \vec{F} \cdot \hat{N} dS = 0$$

AGS-S

Unique Paper Code : 32351301

Name of the Paper : C-5 : Theory of Real Functions.

Name of the Course : B.Sc. (Hons.) Mathematics II Year

Semester : 3

Duration : 3 Hours

Maximum Marks : 75

Attempt any four questions.
 All questions carry equal marks.

Q. 1. Prove that $\lim_{x \rightarrow -1} \frac{x^2-5}{x^2+7} = \frac{-5}{7}$.

Use ε - δ definition of limit to prove that $\lim_{x \rightarrow -1} \frac{x^3-3}{x^2+1} = -1$

Also prove that $\lim_{x \rightarrow 0} x^2 \text{sgn}(x)$ exists. Here sgn denotes the signum function.

Aps. $\lim_{x \rightarrow -1} \frac{x^2-5}{x^2+7} = \frac{(-1)^2-5}{(-1)^2+7}$

$$= \frac{1-5}{1+7} = \frac{-4}{8} = -\frac{1}{2} \neq \frac{-5}{7}$$

Second part,

To show $\lim_{x \rightarrow 1} \frac{x^3-3}{x^2+1} = -1$

$\Rightarrow$ for given $\varepsilon > 0$, there exist $\delta > 0$ such that

$$\left| \frac{x^3-3}{x^2+1} + 1 \right| < \varepsilon \text{ if } |x-1| < \delta$$

Let $|x-1| < \frac{1}{5}$

$$\Rightarrow -\frac{1}{5} < x-1 < \frac{1}{5}$$

$$\Rightarrow \frac{4}{5} < x < \frac{6}{5}$$

$$\Rightarrow \frac{16}{25} < x^2 < \frac{36}{25}$$

$$\Rightarrow x^2 + 1 > \frac{16}{25} + 1$$

$$\Rightarrow \frac{1}{x^2 + 1} < \frac{25}{41}$$

Consider $\left| \frac{x^3 - 3}{x^2 + 1} + 1 \right| = \left| \frac{x^3 - 3 + x^2 + 1}{x^2 + 1} \right| = \left| \frac{x^3 + x^2 - 2}{x^2 + 1} \right|$

$$= \frac{|x^2 + 2x + 2|}{x^2 + 1} |x - 1| < \left(\frac{36}{25} + 2 \left(\frac{6}{5} \right) + 2 \right) \times \frac{25}{41} |x - 1|$$

$$= \frac{36 + 60 + 50}{25} \times \frac{25}{41} |x - 1| = \frac{146}{41} |x - 1| < \epsilon$$

$$\Rightarrow \frac{146}{41} |x - 1| < \epsilon$$

$$\Rightarrow |x - 1| < \frac{41}{146} \epsilon$$

Let $\delta = \min \left\{ \frac{41\epsilon}{146}, \frac{1}{5} \right\}$

$$\Rightarrow \text{for } |x - 1| < \delta$$

$$\Rightarrow \left| \frac{x^3 - 3}{x^2 + 1} + 1 \right| < \epsilon$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^3 - 3}{x^2 + 1} = -1$$

Third part

$$\operatorname{sgn}(x) = \begin{cases} \frac{|x|}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$\Rightarrow x^2 \operatorname{sgn}(x) = x|x|$$

$$\lim_{x \rightarrow 0^-} x^2 \operatorname{sgn}(x) = \lim_{x \rightarrow 0^-} x|x|$$

$$= \lim_{h \rightarrow 0} (0 - h)(0 - h) = \lim_{h \rightarrow 0} -h^2 = 0$$

also, $\lim_{x \rightarrow 0^+} x^2 \operatorname{sgn}(x) = \lim_{x \rightarrow 0^+} x|x| = \lim_{h \rightarrow 0} (0 - h)(0 + h) = \lim_{h \rightarrow 0} -h^2 = 0$

Thus, LHC = RHL

$$\Rightarrow \lim_{x \rightarrow 0} x^2 \operatorname{sgn}(x) \text{ exist.}$$

Q. 2. Let $A \subset \mathbb{R}$, function $f, g: A \rightarrow \mathbb{R}$, c be a cluster point of A and $L, L \neq 0$.

If $\lim_{x \rightarrow c} f = L$ and $\lim_{x \rightarrow c} g = \infty$, then find $\lim_{x \rightarrow c} fg$.

If $\lim_{x \rightarrow c} f = 0$ and $\lim_{x \rightarrow c} g = \infty$ then justify by an example that $\lim_{x \rightarrow c} fg$ need not be infinity.

Ans. given $\lim_{x \rightarrow c} f = L \neq 0$

and $\lim_{x \rightarrow c} g = \infty$

since $\lim_{x \rightarrow c} fg = \lim_{x \rightarrow c} f \cdot \lim_{x \rightarrow c} g$

$$\Rightarrow \lim_{x \rightarrow c} fg = L \cdot \infty = \infty$$

Thus $\lim_{x \rightarrow c} fg = \infty$

second part $\lim_{x \rightarrow c} f = 0$ and $\lim_{x \rightarrow c} g = \infty$

let $f(x) = x - c$ and $g(x) = \frac{1}{x - c}$

$$\Rightarrow \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x - c) = 0$$

and $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} \frac{1}{x - c} = \infty$

Now, $\lim_{x \rightarrow c} fg = \lim_{x \rightarrow c} (x - c) \times \frac{1}{(x - c)} = \lim_{x \rightarrow c} (1) = 1$

$\Rightarrow$ If $L = 0$ then $\lim_{x \rightarrow c} fg$ need not be infinity.

Q. 3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 5x - 6 & \text{if } x \text{ is irrational} \end{cases}$$

Find all the points at which f is continuous.

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be an additive function, that is, $f(a+b) = f(a) + f(b)$ for all $a, b \in \mathbb{R}$. Prove that if f is continuous at some point x_0 , then f is continuous at every point of $\mathbb{R}$.

Ans. let: $\mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 5x - 6 & \text{if } x \text{ is irrational} \end{cases}$$

f is said to be continuous if for $c \in \mathbb{R}$

$$\lim_{x \rightarrow c} f(x) = f(c)$$

Let c be rational number such that c' and c'' are irrational.

Let f be continuous at c

$$\Rightarrow \lim_{x \rightarrow c} f(x) = f(c)$$

$$\Rightarrow \lim_{x \rightarrow c} (5x - 6) = c^2$$

$$\Rightarrow \lim_{x \rightarrow 0} (5(-x) - 6) = c^2$$

$$\Rightarrow 5c - 6 = c^2$$

$$\Rightarrow c^2 - 5c + 6 = 0$$

$$\Rightarrow c^2 - 3c - 2c + 6 = 0$$

$$\Rightarrow c(c-3) - 2(c-3) = 0$$

$$(c-3)(c-2) = 0$$

$$c = 2, 3$$

Thus, f is continuous at $x = 2$ and 3 .

For second parts

Refer Q. 2, (d) Nov/Dec. 2019, Page-282

Q. 4. Show that $f(x) = \frac{x-1}{x+1}$ is uniformly continuous on $[0, \infty)$ and $g(x) =$

$\cos\left(\frac{1}{x}\right)$ is not uniformly continuous on $(0, \infty)$.

If f is continuous on $[0, 2]$ and $f(0) = f(2)$, then prove that there exists $x, y \in [0, 2]$ such that $|y - x| = 1$ and $f(x) = f(y)$.

$$\text{Ans. } f(x) = \frac{x-1}{x+1}$$

$f(x)$ is uniformly continuous on $[0, \infty)$ if for $x_1, x_2 \in [0, \infty)$

and for $\epsilon > 0 \exists \delta > 0$ such that

$$|f(x_1) - f(x_2)| < \epsilon \text{ if } |x_1 - x_2| < \delta.$$

let $x_1, x_2 \in [0, \infty)$ such that $|x_1 - x_2| < \delta$.

Consider

$$|f(x_1) - f(x_2)| = \left| \frac{x_1 - 1}{x_1 + 1} - \frac{x_2 - 1}{x_2 + 1} \right|$$

$$= \left| \frac{(x_1 - 1)(x_2 + 1) - (x_2 - 1)(x_1 + 1)}{(x_1 + 1)(x_2 + 1)} \right|$$

$$= \left| \frac{(x_1 x_2 + x_1 - x_2 - 1) - (x_1 x_2 + x_2 - x_1 - 1)}{(x_1 + 1)(x_2 + 1)} \right|$$

$$= \left| \frac{2x_1 - 2x_2}{(x_1 + 1)(x_2 + 1)} \right|$$

$$= \frac{2}{(x_1 + 1)(x_2 + 1)} |x_1 - x_2| < \frac{2}{(x_1 + 1)(x_2 + 1)} \delta = \epsilon$$

$$\Rightarrow |f(x_1) - f(x_2)| < \epsilon$$

$\Rightarrow f(x)$ is uniformly continuous on $[0, \infty)$

$$g(x) = \cos\left(\frac{1}{x}\right)$$

let $x_1, x_2 \in (0, \infty)$ such that $|x_1 - x_2| < \delta$

$$|g(x_1) - g(x_2)| = \left| \cos\left(\frac{1}{x_1}\right) - \cos\left(\frac{1}{x_2}\right) \right|$$

$$= \left| -2 \sin\left(\frac{\frac{1}{x_1} + \frac{1}{x_2}}{2}\right) \sin\left(\frac{\frac{1}{x_1} - \frac{1}{x_2}}{2}\right) \right|$$

$$= \left| -2 \sin \left(\frac{x_1 + x_2}{2x_1 x_2} \right) \sin \left(\frac{x_2 - x_1}{2x_1 x_2} \right) \right|$$

$$= \left| 2 \sin \left(\frac{x_1 x_2}{2x_1 x_2} \right) \sin \left(\frac{x_1 - x_2}{2x_1 x_2} \right) \right|$$

since, $-1 \leq \sin x \leq 1 \quad \forall x \in (0, \infty)$

$$\Rightarrow |g(x_1) - g(x_2)| < 2 < 4 \in$$

$\Rightarrow g(x)$ is not uniformly continuous.

Second part,

f is continuous on $[0, 2]$ and $f(0) = f(2)$

let $x, y \in [0, 2]$ such that $|y - x| = 1$

To show $f(x) = f(y)$.

let $y = 1 + x$

let $g(x) = f(1 + x) - f(x)$

as $f(x)$ is continuous on $[0, 2]$

$\Rightarrow g(x)$ is continuous on $[0, 1]$

$$g(0) = f(1) - f(0)$$

$$g(1) = f(2) - f(1)$$

$$\text{as } f(2) = f(0)$$

$$g(1) = -g(0)$$

$$g(0) = 0$$

$$f(1) - f(0) = 0$$

$$f(1) = f(0)$$

$$\text{for } x = 0 \text{ and } y = 1$$

$$f(y) = f(x)$$

$$\text{If } g(0) \neq 0$$

$$\text{But } g(1) = -g(0)$$

$$\exists c \in (0, 1) \text{ such that}$$

$$g(c) = 0$$

$$f(1 + c) - f(c) = 0$$

$$f(1 + c) = f(c)$$

$$x = c \text{ and } y = 1 + c$$

$$f(y) = f(x)$$

Q. 5. Find the points of relative extrema of the following function on the specified domain $f(x) = |x^2 - 25|$, $-7 \leq x \leq 7$.

Prove that $ex \leq e^x$, for all $x \in \mathbb{R}$.

Use Mean Value Theorem to find an approximate value of $\sqrt{51}$.

$$\text{Ans. } f(x) = |x^2 - 25|, -7 \leq x \leq 7$$

$$\text{for } -7 \leq x \leq 7$$

$$\Rightarrow 0 \leq x^2 \leq 49$$

$$\Rightarrow -25 \leq x^2 - 25 \leq 49 - 25$$

$$\Rightarrow -25 \leq x^2 - 25 \leq 24$$

$$\Rightarrow 0 \leq |x^2 - 25| \leq 25$$

$$\Rightarrow 0 \leq f(x) \leq 25$$

$$\Rightarrow \text{relative minimum of } f(x) = 0$$

$$\Rightarrow \text{and relative maximum of } f(x) = 25$$

Second Part,

$$\text{Let, } f(x) = e^{x-1} - x$$

$$\Rightarrow f(x) = e^{x-1} - 1$$

$f(x)$ is continuous and differentiable on $\mathbb{R}$

By LMVT, for $f(x)$ continuous on $[a, b]$ and differentiable on $(a, b) \exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

For $x > 1$

Let $c \in (1, x)$

$$f'(c) = \frac{f(x) - f(1)}{x - 1}$$

$$e^{c-1} - 1 = \frac{(e^{x-1} - x) - (e^{1-1} - 1)}{x - 1}$$

$$e^{c-1} - 1 = \frac{e^{x-1} - x}{x - 1}$$

for $x > 1$ and $c > 1$

$$e^{c-1} > 1$$

$$e^{c-1} - 1 > 0$$

$$\frac{e^{x-1} - x}{x - 1} > 0$$

$$e^{x-1} - x > 0$$

$$e^{x-1} > x$$

$$e^x > ex$$

Similarly for $x < 1$ let $c \in (x, 1)$

$$f'(c) = \frac{f(1) - f(x)}{1 - x}$$

$$e^c - 1 = \frac{e^{x-1} - x}{1 - x}$$

$$e^c - 1 = \frac{(e^{x-1} - x)}{x - 1}$$

for $c < 1$

$$e^{c-1} < 1$$

$$e^{c-1} - 1 < 0$$

$$\frac{(e^{x-1} - x)}{x - 1} < 0$$

$$e^{x-1} - x > 0 \text{ (as } x < 1 \Rightarrow x - 1 < 0)$$

$$e^{x-1} > x$$

$$e^x > ex \dots (ii)$$

at $x = 1$

$$e^{1/1} = e$$

and $e(1) = e$

$$e^x = ex \dots (iii)$$

from 1, 2 and 3 $ex \leq e^x \forall x \in \mathbb{R}$

Third part, Mean value theorem

let f be continuous on $[a, b]$

f be differentiable on (a, b)

then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Let $f(x) = \sqrt{x} \Rightarrow f'(x) = \frac{1}{2\sqrt{x}}$

Since, $\sqrt{49} = 7$

So, let $x \in [49, 51]$

$\exists c \in (49, 51)$ such that

$$f'(c) = \frac{f(51) - f(49)}{51 - 49}$$

$$\frac{1}{2\sqrt{c}} = \frac{\sqrt{51} - \sqrt{49}}{2}$$

$$\frac{1}{\sqrt{c}} = \sqrt{51} - 7$$

$$\sqrt{51} = 7 + \frac{1}{\sqrt{c}}$$

since $\sqrt{x}$ is an increasing function for $x \in (49, 51)$

$\frac{1}{\sqrt{x}}$ is decreasing function for $x \in (49, 51)$

$$7 < \sqrt{51} = 7 + \frac{1}{\sqrt{c}} < 7 + \frac{1}{\sqrt{49}}$$

$$7 < \sqrt{51} < 7 + \frac{1}{7}$$

$$7 < \sqrt{51} < 7.143$$

Q. 6. If $x \in [0, 1]$ and $n \in \mathbb{N}$, show that

$$\ln(1+x) - \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{n-1} \frac{x^n}{n} \right) < \frac{x^{n+1}}{n+1}$$

Use this to approximate $\ln(1.5)$ with an error less than 0.01.

Use Taylor's theorem to prove that for all $x > 0$

$$1 + x + \frac{x^2}{2} + \frac{x^3}{3!} < e^x < 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} e^x$$

Ans. Let $f(x) = \ln(1+x)$, $x \in [0, 1]$ and $n \in \mathbb{N}$

$$\Rightarrow f(x) = \ln(1+x) \Rightarrow f(0) = 0$$

$$f'(x) = \frac{1}{1+x} \Rightarrow f'(0) = 1$$

$$f''(x) = \frac{-1}{(1+x)^2} \Rightarrow f''(0) = -1$$

$$f'''(x) = \frac{2}{(1+x)^3} \Rightarrow f'''(0) = 2$$

$$f^{(n)}(x) = \frac{(-1)^{n-1} (n-1)!}{(1+x)^n} \Rightarrow f^{(n)}(0) = (-1)^{n-1} (n-1)!$$

and

$$f^{(n+1)}(x) = \frac{(-1)^n n!}{(1+x)^{n+1}}$$

By Taylor's theorem

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \frac{x^{n+1}}{(n+1)!} f^{(n+1)}(c)$$

For $c \in (0,1)$

$$\frac{c^{n+1} (-1)^n}{(n+1) (1+c)^{n+1}} < \frac{c^{n+1} (-1)^n}{n+1}$$

$$\Rightarrow \left| \frac{c^{n+1} (-1)^n}{(n+1) (1+c)^{n+1}} \right| < \frac{c^{n+1}}{n+1}$$

$$\Rightarrow \left| f(x) - \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{n-1} \frac{x^n}{n} \right) \right| < \frac{x^{n+1}}{n+1}$$

$$\Rightarrow \left| \ln(1+x) - \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{(-1)^{n-1} x^n}{n} \right) \right| < \frac{x^{n+1}}{n+1}$$

To approximate value of $\ln(1.5)$ with error less than 0.01

$$\Rightarrow x = 0.5 \text{ and } \frac{x^{n+1}}{n+1} < 0.01$$

$$\Rightarrow x = 0.5 \text{ and } n+1 = 5$$

$$\Rightarrow x = 0.5 \text{ and } n = 4$$

$$\ln(1.5) = 0.5 - \left(\frac{0.5}{2} \right)^2 + \frac{(0.5)^3}{3} - \frac{(0.5)^4}{4} = 0.401041667$$

Second part

$$\text{Let } f(x) = e^x \Rightarrow f(0) = 1$$

$$f'(x) = e^x \Rightarrow f'(0) = 1$$

$$f''(x) = e^x \Rightarrow f''(0) = 1$$

$$f^{(n)}(x) = e^x \Rightarrow f^{(n)}(0) = 1$$

By Taylor's theorem $\exists c > 0$ s.t.

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \frac{x^{n+1}}{(n+1)!} f^{(n+1)}(c)$$

$$\Rightarrow f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!} e^c$$

Since $x > 0$

$$\Rightarrow 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} < 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!} e^c$$

$$\Rightarrow 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} < e^x \dots (i)$$

also

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!} e^c = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \left(\frac{3!}{4!} x + \frac{3! x^2}{5!} + \dots \right)$$

$$< 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right)$$

$$< 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} e^x$$

$$\Rightarrow e^c < 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} e^x$$

From (i) and (ii)

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} e^x < e^c < 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} e^x$$

Unique Paper Code: 2351302

Name of the Paper: T-6: Group Theory-1

Dec. 2021

Name of the Course: B.Sc. (Hons.) Mathematics II Year

Semester: 3

Duration: 3 Hours

Maximum Marks: 75

Attempt any four questions.
All questions carry equal marks.

Q. 1. Show that the set S of all ordered pairs (a, b) of non-zero real numbers is an abelian group under the multiplication defined by

$$(a, b)(c, d) = (ac, bd) \quad \forall a, b, c, d \in S$$

Consider the group $G = GL(2, \mathbb{R})$ under multiplication. Then find the centralizer of

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}. \text{ Also, find the center of } G.$$

Let $A = \begin{pmatrix} 3 & 4 \\ 4 & 4 \end{pmatrix}$. Find A^{-1} in $SL(2, \mathbb{Z})$. Verify the answer by direct calculation.

Ans. $\forall a, b, c, d \in S$

ac and $bd \in S$

$$\Rightarrow \forall (a, b) \text{ and } (c, d) \in S$$

$$\Rightarrow (ac, bd) \in S$$

S is closed

Since, S is associative

Now, for identity of S

Consider $(1, 1) \in S$

$$\text{as } (a, b)(1, 1) = (a \cdot 1, b \cdot 1)$$

$$= (a, b)$$

$$= (1, 1)(a, b)$$

$$= (1, 1)(a, b)$$

$$\Rightarrow (1, 1) \text{ is identity for } S.$$

Now, for inverse

$$\text{let } (a, b)(c, d) = (1, 1)$$

$$\Rightarrow (ac, bd) = (1, 1)$$

$$ac = 1 \text{ and } bd = 1$$

$$c = \frac{1}{a}, d = \frac{1}{b}$$

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$$\Rightarrow \text{for } (a, b) \neq (0, 0) \text{ inverse of } (a, b) \text{ is } \left(\frac{1}{a}, \frac{1}{b}\right)$$

S has inverse

Now

$$(a, b)(c, d) = (ac, bd)$$

$$= (ca, db)$$

$$= (c, d)(a, b)$$

S is an abelian group

(ii) $G = GL(2, \mathbb{R})$ is a group under multiplication

Centralizer of $a \in G$ is $C(a) = \{g \mid ga = ag\}$

$$\Rightarrow C(a) = \left\{ g \mid g \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} g \right\}$$

let

$$g = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a+b & a \\ c+d & c \end{bmatrix} = \begin{bmatrix} a+b & b+d \\ a & b \end{bmatrix}$$

$$\Rightarrow a+b = a+c \Rightarrow b=c$$

$$\Rightarrow a = b+d \Rightarrow d = a-b$$

$$\Rightarrow g = \begin{bmatrix} a & b \\ b & a-b \end{bmatrix}$$

and

$$C(b) = \left\{ g \mid g \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} g \right\}$$

$$\text{let } g = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} b & a \\ d & c \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

$$\Rightarrow c = b \text{ and } d = a$$

$$\Rightarrow g = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

Thus $c(a) = \left\{ \begin{bmatrix} a & b \\ b & a-b \end{bmatrix} \mid \forall a, b \in \mathbb{R} \right\}$

and $(b) = \left\{ \begin{bmatrix} a & b \\ b & a \end{bmatrix} \mid \forall a, b \in \mathbb{R} \right\}$

Center of g is $z(g) = \{a \in g \mid ax = xa \forall x \in G\}$

as $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is identity of G

and $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\Rightarrow z(g) = \left\{ \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \mid \forall a \in \mathbb{R} \right\}$$

(iii) Let $n = \begin{bmatrix} 3 & 4 \\ 4 & 4 \end{bmatrix} \in sl(2, \mathbb{Z}_5)$

let $n^{-1} = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$

s.t. $an^{-1} = 1$

$$\Rightarrow \begin{bmatrix} 3 & 4 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 3x + 4z &= 1 \pmod{5} \\ 4x + 4z &= 0 \pmod{5} \end{aligned}$$

$$\begin{aligned} -x &= 1 \pmod{5} \\ x &= -1 \pmod{5} \\ x &= 4 \pmod{5} \end{aligned}$$

$$\Rightarrow \begin{aligned} 3x + 4z &= 1 \pmod{5} \\ 3(4) + 4z &= 1 \pmod{5} \end{aligned}$$

$$4z = 1 - 12 \pmod{5}$$

$$4z = -11 \pmod{5}$$

$$4z = 4 \pmod{5}$$

$$z = 1 \pmod{5}$$

Also $3y + 4w = 0 \pmod{5}$

$$4y + 4w = 1 \pmod{5}$$

$$-y = -1 \pmod{5}$$

$$y = 1 \pmod{5}$$

and $4y + 4w = 1 \pmod{5}$

$$4(1) + 4w = 1 \pmod{5}$$

$$4w = 1 - 4 \pmod{5}$$

$$4w = -3 \pmod{5}$$

$$4w = 2 \pmod{5}$$

$$w = 3 \pmod{5}$$

Thus $n^{-1} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$

Q. 2. Find all the subgroups of z :

(a) containing $20z$

(b) contained in $20z$.

Prove that an abelian group which contains two distinct elements which are their own inverses must have a subgroup of order 4.

Suppose a group contains elements a and b such that $|a| = 4$ and $|b| = 5$ and that $a^3b = ba$. Find $|ab|$.

Ans. (i) Do your self.

(ii) let g be an abelian group

let $a, b \in g$ s.t. $a^{-1} = a$ and $b^{-1} = b$

let e be the identity of g

$$\begin{aligned} a^2 &= a.a \\ &= a.a^{-1} \\ &= e \end{aligned}$$

$$\text{Similarly } b^2 = e$$

let h be a subset of g

let $h = \{e, a, b, ab\}$

$$\begin{aligned} a.ab &= a^2b = eb = b \\ ab.a &= a.ab = b \end{aligned}$$

and

$\Rightarrow h$ is closed.

Also h is associative and e is its identity.

since $a^{-1} = a$ and $b^{-1} = b$

Now consider

$$(a b) (ab)^{-1} = (a b) (b^{-1} a^{-1}) = (a (b b^{-1}) a^{-1}) = (a e a^{-1}) = (a a^{-1}) = e$$

$$(a b)^{-1} = ab$$

inverse of all elements of h exist

h is a subgroup with atleast 4 elements.

$$(ii) \quad |a| = 4, |b| = 5 \text{ and } a^4 b = ba$$

To find

$$(ab)^2$$

find

$$(ab)^2 = (a b) (a b) = a (ba) b = a (a^4 b) b = a^5 b^2 = e b^2 = b^2$$

Since,

$$(b)^2 = 5$$

$$(b^2)^5 = 5$$

$$(ab)^{10} = \{(a b)^2\}^5 = \{b^2\}^5 = e$$

$$|ab| = 10$$

Q. 3. State Cayley's theorem and verify theorem for $U(10)$.

Let a and b be elements of a group G . If $o(a) = 12$, $o(b) = 22$ and $\langle a \rangle \cap \langle b \rangle \neq \{e\}$. Prove that $a^6 = b^{11}$.

Find a non-cyclic group of order 4 in $U(40)$.

Ans. (i) Every group G is isomorphic to a group of permutation.

$$U(10) = \{1, 3, 7, 9\}$$

To show $U(10)$ is isomorphic to $S_4 = \{1, 2, 3, 4\}$

For $g \in U(10)$ let $T_g(x) = gx$.

$$T_1(x) = gx \quad 1 \quad 3 \quad 7 \quad 9$$

$$T_3(x) = 3x \quad 1 \quad 3 \quad 7 \quad 9$$

$$T_7(x) = 7x \quad 3 \quad 9 \quad 1 \quad 7$$

$$T_9(x) = 9x \quad 7 \quad 1 \quad 9 \quad 3$$

$$T_5(x) = 5x \quad 5 \quad 7 \quad 3 \quad 1$$

$$T_1(x) = \begin{bmatrix} 1 & 3 & 7 & 9 \\ 1 & 3 & 7 & 9 \end{bmatrix} = (1379) \cong S_4$$

$$T_3(x) = \begin{bmatrix} 1 & 3 & 7 & 9 \\ 3 & 9 & 1 & 7 \end{bmatrix} = (1397) \cong S_4$$

$$T_7(x) = \begin{bmatrix} 1 & 3 & 7 & 9 \\ 7 & 1 & 9 & 3 \end{bmatrix} = (1793) \cong S_4$$

$$T_9(x) = \begin{bmatrix} 1 & 3 & 7 & 9 \\ 9 & 7 & 3 & 1 \end{bmatrix} = (19)(37) \cong S_4$$

(ii) let $a, b \in G$

$$O(a) = 12 \text{ and } O(b) = 22$$

$$\text{also, } \langle a \rangle \cap \langle b \rangle \neq \{e\}$$

$$\text{let } x \in \langle a \rangle \cap \langle b \rangle$$

$$\text{and } x \in \langle a \rangle \text{ and } x \in \langle b \rangle$$

$$\Rightarrow x^{12} = e \text{ and } x^{22} = e$$

$$\text{since, } \gcd(12, 22) = 2$$

$$\Rightarrow (x^2)^6 = e \text{ and } (x^2)^{11} = e$$

$$\text{since, } x \in \langle a \rangle$$

$$\text{let } x^2 = a$$

$$\Rightarrow a^6 = (x^2)^6 = e \dots (i)$$

$$\text{and } x \in \langle b \rangle$$

$$\text{let } x^2 = b$$

$$\Rightarrow b^{11} = (x^2)^{11} = e \dots (ii)$$

from 1 and 2

$$a^6 = b^{11}$$

$$(iii) \quad U(40) = \{1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39\}$$

$$\langle 3 \rangle = \{1, 3, 9, 27\} = \langle 27 \rangle$$

$$\langle 7 \rangle = \{1, 7, 9, 23\} = \langle 23 \rangle$$

$$\langle 9 \rangle = \{1, 9\}$$

$$\langle 11 \rangle = \{1, 11\}$$

$$\langle 13 \rangle = \{1, 9, 13, 37\} = \langle 37 \rangle$$

$$\langle 17 \rangle = \{1, 9, 17, 33\} = \langle 33 \rangle$$

$$\langle 19 \rangle = \{1, 19\}$$

$$\langle 21 \rangle = \{1, 21\}$$

$$\langle 29 \rangle = \{1, 29\}$$

$$\langle 31 \rangle = \{1, 31\}$$

$$\langle 39 \rangle = \{1, 39\}$$

$$\text{let } h = \{1, 9, 11, 19\}$$

for each $a \in h$ $a^4 = 1$

$\Rightarrow h$ is non cyclic subgroup of order 4.

Q. 4. Let p be a prime. If a group has more than $(p-1)$ elements of order p . then prove that the group cannot be cyclic.

Let $\beta = (1 \ 2 \ 3) (1 \ 4 \ 5)$. Write β^{10} as a cycle.

Given a permutation $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 3 & 8 & 7 & 6 & 5 & 2 & 4 \end{pmatrix}$

(a) Write as product of disjoint cycle.

(b) Find $|\alpha|$

(c) Find α^{-1} and verify by calculation.

Ans. (i) Let g be a group

If g is infinite then g cannot be cyclic

let g be a finite group.

also let g be a cyclic group such that p divides $|g|$.

By Fundamental theorem of cyclic group, g has a unique subgroup of order p .

Since, $\phi(p) = p - 1$

$\Rightarrow g$ has $p - 1$ element of order p .

But it is given that g has more than $p - 1$ elements of order p

$\Rightarrow g$ has another group of order p also

But this contradicts Fundamental theorem of cyclic group

$\Rightarrow g$ can not be cyclic.

(ii)

$$\beta = (1\ 2\ 3)(1\ 4\ 5)$$

$$\beta^2 = (1\ 2\ 3)(1\ 4\ 5)(1\ 2\ 3)(1\ 4\ 5)$$

$$= (1\ 2\ 3)(1\ 2\ 3)(1\ 4\ 5)(1\ 4\ 5) = (1\ 3\ 2)(1\ 5\ 4)$$

$$\beta^3 = \beta \cdot \beta^2$$

$$= (1\ 2\ 3)(1\ 4\ 5)(1\ 3\ 2)(1\ 5\ 4)$$

$$= (1\ 2\ 3)(1\ 3\ 2)(1\ 4\ 5)(1\ 5\ 4) = (1)(2)(3)(4)(5)$$

$$\beta^{99} = (\beta^3)^{33} = (1)(2)(3)(4)(5)$$

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 3 & 8 & 7 & 6 & 5 & 2 & 4 \end{bmatrix}$$

$$\alpha = (2\ 3\ 8\ 4\ 7)(5\ 6)$$

$$|\alpha| = \text{LCM}(5, 2) = 10$$

$$\alpha^2 = (2\ 3\ 8\ 4\ 7)(5\ 6)(2\ 3\ 8\ 4\ 7)(5\ 6)$$

$$= (2\ 3\ 8\ 4\ 7)(2\ 3\ 8\ 4\ 7)(5\ 6)(5\ 6) = (2\ 8\ 7\ 3\ 4)$$

$$\alpha^4 = (2\ 8\ 7\ 3\ 4)(2\ 8\ 7\ 3\ 4) = (2\ 7\ 4\ 8\ 3)$$

$$\alpha^6 = (2\ 7\ 4\ 8\ 3)(2\ 7\ 4\ 8\ 3) = (2\ 4\ 3\ 7\ 8)$$

$$\alpha^8 = \alpha^6 \cdot \alpha^2$$

$$= (2\ 4\ 3\ 7\ 8)(2\ 3\ 8\ 4\ 7)(5\ 6) = (2\ 7\ 4\ 8\ 3)(5\ 6)$$

$$\alpha^{-1} = \alpha^9 = (2\ 7\ 4\ 8\ 3)(5\ 6)$$

Verify

$$\alpha \cdot \alpha^{-1} = (2\ 3\ 8\ 4\ 7)(5\ 6)(2\ 7\ 4\ 8\ 3)(5\ 6) \\ = (2\ 3\ 8\ 4\ 7)(1\ 7\ 4\ 8\ 3)(5\ 6)(5\ 6) = (1)$$

Hence, verified.

Q. 5. Let G be the additive group $\mathbb{R} \times \mathbb{R}$ and $H = \{(x, x) : x \in \mathbb{R}\}$ be a subgroup of G . Give a geometric description of cosets of H .

If N is a normal subgroup of order 2 of a group g then show that $N \subseteq Z(g)$.

If H is a subgroup of a group G such that $(aH)(Hb)$ for any $a, b \in g$ is either a left or a right coset of H in g , prove that H is normal.

Ans. (i) Do your self

(ii) Given, n is a normal subgroup of order 2 of a group G .

To show $n \subseteq Z(G)$

let $n = \{e, a\}$

let $x \in G$

$$s.t. \quad xN = Nx$$

$$\Rightarrow xa = ax$$

$$\Rightarrow a \in Z(G)$$

$$\Rightarrow \text{also } e \in Z(G)$$

$$\Rightarrow N \subseteq Z(G)$$

(iii) Do your self

Q. 6. If N be a normal subgroup of a group G . If N is cyclic subgroup of G then prove that every subgroup of N is normal in G .

Prove that the mapping from $x \mapsto x^a$ from C^* to C^* where C^* denotes the set of non-zero complex numbers is a homomorphism. What is the kernel?

Ans. (i) given : ϕ be a homomorphism from Z_n onto a group of order 5.

let G be the required group

$$\Rightarrow |G| = 5$$

By first isomorphism theorem $\frac{Z_n}{\text{Ker } \phi}$ is of order 5

By Lagrange's theorem $\text{Ker } \phi = 6$

Since, Z_n is a cyclic group and $\text{ker } \phi$ is subgroup of Z_n of order 6

$$\Rightarrow \text{ker } \phi = \{0, 5, 10, 15, 20, 25\} \\ = \langle 5 \rangle$$

(ii) let N be normal subgroup of group G

let N be cyclic subgroup of G

$$\Rightarrow N = \langle a \rangle$$

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let H be a subgroup of N

$$H = \langle a \rangle$$

let $g \in G$ and $n \in H$

$$g n g^{-1} = g a^n g^{-1} = (g a g^{-1})^n$$

Since, $a \in N$

$$\Rightarrow g a g^{-1} \in N$$

$$\Rightarrow g a g^{-1} = a^m \text{ for some integer } m$$

$$\Rightarrow (g a g^{-1})^n = (a^m)^n$$

$$= (a^n)^m = (a^n)^m$$

$$\in H$$

$$\Rightarrow g H g^{-1} = H$$

$\Rightarrow H$ is normal in G

(ii) let the mapping f from C^* to C^* such that

$$\phi : x \rightarrow x^6$$

$$\phi(x) = x^6$$

let $x, y \in C^*$

$$\begin{aligned} \text{Since, } \phi(xy) &= (xy)^6 \\ &= x^6 y^6 \\ &= \phi(x) \phi(y) \end{aligned}$$

$\Rightarrow \phi$ is homomorphism.

$$\ker \phi = \{x \mid x^6 = 1\}$$

$$\text{let } x = \cos \phi + i \sin \phi$$

$$\Rightarrow x^6 = 1$$

$$\Rightarrow (\cos \phi + i \sin \phi)^6 = 1$$

$$\Rightarrow \cos 6\phi + i \sin 6\phi = \cos 0^\circ + i \sin 0^\circ \text{ (By de Moivre's Theorem)}$$

$$\Rightarrow \cos 6\phi = \cos 0^\circ$$

$$\Rightarrow 6\phi = 360^\circ + 0^\circ$$

$$6\phi = 360^\circ$$

$$\phi = 60^\circ$$

$$x = \cos 60^\circ + i \sin 60^\circ$$

$$\ker \phi = \langle \cos 60^\circ + i \sin 60^\circ \rangle$$

Unique Paper Code : 32351303

Name of the Paper : C : 7 Multivariate Calculus

Name of the Course : B.Sc. (Hons.) Mathematics II Year

Semester : 3

Duration : 3 Hours

Dec. 2021

Maximum Marks : 75

Attempt any four questions.
All questions carry equal marks.

$$Q. 1. \text{ Let } f(x, y) = \begin{cases} \frac{xy + y^3}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

(a) Show that $f_x(0, 0)$ and $f_y(0, 0)$ both exist but f is not differentiable at $(0, 0)$.

(b) Find the rate of change of f at the point $P(2, 0)$ in the direction from P to

$$Q\left(\frac{1}{2}, 2\right).$$

(c) Find the direction in which f decreases most rapidly and increases most rapidly at the point $(2, 1)$.

(d) Find the tangent plane to the given surface $z = f(x, y)$ at the point $(0, 2, 2)$.

$$\text{Ans. (a) } f(x, y) = \begin{cases} \frac{xy + y^3}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^3}{h^2}}{h} = 1$$

$$\Rightarrow f_x(0, 0) \text{ and } f_y(0, 0) \text{ both exist}$$

check continuity of $f(x, y)$ at $(0, 0)$

let $y = mx$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = \lim_{x \rightarrow 0} \frac{x(mx) + (mx)^3}{x^2 + (mx)^2}$$

$$\lim_{x \rightarrow 0} \frac{x^2 (m + m^3 x)}{x^2 (1 + m^2)} = \frac{m}{1 + m^2}$$

$$\lim_{x \rightarrow 0, y \rightarrow 0} f(x, y) \neq f(0, 0)$$

$f(x, y)$ is not continuous at $(0, 0)$
 (x, y) is not differentiable at $(0, 0)$

(b)

$$f(x, y) = \frac{xy + y^3}{x^2 + y^2}$$

$$f_x = \frac{(x^2 + y^2)y - (xy + y^3)(2x)}{(x^2 + y^2)^2}$$

$$f_y = \frac{x^2 y + y^3 - 2x^2 y - 2x y^3}{(x^2 + y^2)^2}$$

$$f_x = \frac{y^3 - x^2 y - 2x y^3}{(x^2 + y^2)^2}$$

and f

$$f_y = \frac{(x^2 + y^2)(x + 3y^2) - (xy + y^3)(2y)}{(x^2 + y^2)^2}$$

$$f_x = \frac{x^3 + 3x^2 y^2 + xy^2 + 3y^4 - 2xy^2 - 2y^4}{(x^2 + y^2)^2}$$

$$= \frac{y^4 + 3x^2 y^2 - xy^2 + x^3}{(x^2 + y^2)^2}$$

rate of change of f at the point $P(2, 0)$

$$\nabla f = f_x \hat{i} + f_y \hat{j}$$

$$\nabla f(2, 0) = 0 \hat{i} + \frac{1}{2} \hat{j} = \frac{1}{2} \hat{j}$$

let $\vec{u}$ be a vector from point P to $Q(\frac{1}{2}, 2)$

$$\vec{u} = Q - P = \left(\frac{1}{2}, 2\right) - (2, 0) = \left(-\frac{3}{2}, 2\right)$$

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|} = \frac{\left(-\frac{3}{2}, 2\right)}{\sqrt{\left(-\frac{3}{2}\right)^2 + (2)^2}} = \frac{\left(-\frac{3}{2}, 2\right)}{\frac{5}{2}} = \left(-\frac{3}{5}, \frac{4}{5}\right)$$

rate of change of f at the point $P(2, 0)$ in the direction from P to Q is

$$D_{\vec{u}} f(2, 0) = \left(\frac{1}{2}\right) \left(-\frac{3}{5} \hat{i} + \frac{4}{5} \hat{j}\right) = \frac{2}{5}$$

(c) from (b) part

$$\nabla f = f_x \hat{i} + f_y \hat{j} = \frac{y^3 - x^2 y - 2x y^3}{(x^2 + y^2)^2} \hat{i} + \frac{y^4 + 3x^2 y^2 - xy^2 + x^3}{(x^2 + y^2)^2} \hat{j}$$

$$\nabla f(2, 1) = \frac{-7}{25} \hat{i} + \frac{19}{25} \hat{j}$$

$$|\nabla f| = \sqrt{\left(\frac{-7}{25}\right)^2 + \left(\frac{19}{25}\right)^2} = \frac{\sqrt{410}}{25}$$

Direction in which f increases most rapidly is

$$\frac{-\nabla f}{|\nabla f|} = \frac{+7}{\sqrt{410}} \hat{i} - \frac{19}{\sqrt{410}} \hat{j}$$

(d) tangent plane to the given surface $z = f(x, y)$ at the point $(0, 2, 2)$ is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$z - 2 = f_x(0, 2)(x - 0) + f_y(0, 2)(y - 2)$$

$$z - 2 = \frac{x}{2} + 1(y - 2)$$

$$2z - 4 = x + 2y - 4$$

$$x + 2y - 2z = 0$$

Q. 2. Let $f(x, y) = x^2 - 4xy + y^2 + 4y$.

(a) Find the absolute extrema of $f(x, y)$ on the closed bounded rectangular region $0 \leq x \leq 3, 0 \leq y \leq 2$

(b) Use the method of constrained optimization to find the largest and the smallest value of $f(x, y)$ subject to the constraint $x + y = 2$

(c) Compare the results obtained in parts (a) and (b) and interpret it.

Ans. (a) $f(x, y) = x^2 - 4xy + y^2 + 4y$

$$f_x = 0$$

$$2x - 4y = 0$$

$$x = 2y$$

From (1) and (2)

$$-4(2y) + 3y^2 + 4 = 0$$

$$3y^2 - 8y + 4 = 0$$

$$3y^2 - 6y - 2y + 4 = 0$$

$$3y(y - 2) - 2(y - 2) = 0$$

$$(3y - 2)(y - 2) = 0$$

$$y = 2, \frac{2}{3}$$

$$y = 2, x = 4$$

$$y = \frac{2}{3}, x = \frac{4}{3}$$

Critical point $(4, 2), (\frac{4}{3}, \frac{2}{3}), (0, 0), (3, 0), (0, 2)$ and $(3, 2)$.

$$f(4, 2) = 0$$

$$f\left(\frac{4}{3}, \frac{2}{3}\right) = \frac{16}{9}$$

$$f(0, 0) = 0$$

$$f(3, 0) = 9$$

$$f(0, 2) = 16$$

$$f(3, 2) = 1$$

$\Rightarrow$ absolute maxima = 16 at $(0, 2)$
and absolute minima = 0 at $(0, 0)$ and $(4, 2)$

$$f_y = 0$$

$$-4x + 3y^2 + 4 = 0$$

$$(ii) \quad \dots(ii)$$

$$(b) \quad f(x, y) = x^2 - 4xy + y^2 + 4y$$

$$f_x = 2x - 4y$$

$$f_y = -4x + 3y^2 + 4$$

$$\text{Let, } g(x, y) = x + y$$

$$g_x = 1$$

$$g_y = 1$$

$$\text{Let } f_x = \lambda g_x$$

$$\Rightarrow 2x - 4y = \lambda$$

$$\text{and } f_y = \lambda g_y$$

$$\Rightarrow -4x + 3y^2 + 4 = \lambda$$

from equation (1) and (2)

$$2x - 4y = \lambda = -4x + 3y^2 + 4$$

$$\Rightarrow 3y^2 - 6x + 4y + 4 = 0$$

$$\Rightarrow 3y^2 - 6(2 - y) + 4y + 4 = 0$$

$$\Rightarrow 3y^2 + 10y - 8 = 0$$

$$3y^2 + 12y - 2y - 8 = 0$$

$$3y(y + 4) - 2(y + 4) = 0$$

$$(3y - 2)(y + 4) = 0$$

$$y = \frac{2}{3}, -4$$

$$x = \frac{4}{3}, 6$$

(Using $x + y = 2$)

Critical points are $\left(\frac{4}{3}, \frac{2}{3}\right)$ and $(6, -4)$

$$f\left(\frac{4}{3}, \frac{2}{3}\right) = \frac{16}{9}$$

$$f(6, -4) = 52$$

$$\Rightarrow \text{maxima} = 52 \text{ and minima} = \frac{16}{9}$$

(c) Do yourself.

Q. 3. (a) Let R be the region which lies between two squares of sides 2 and 4 with center at the origin and sides parallel to the coordinate axes. Compute the double integral $\iint_R e^{x+y} dA$

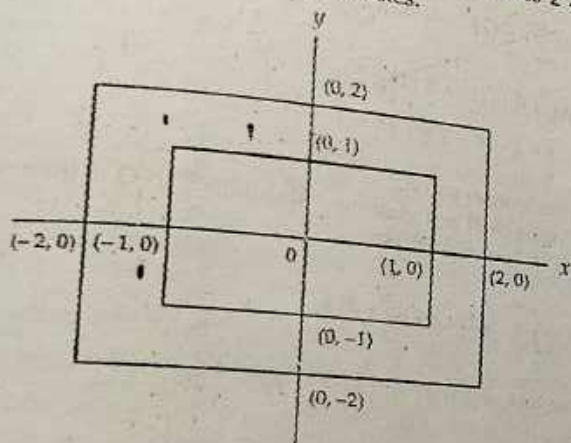
$$\iint_R e^{x+y} dA$$

(b) Evaluate $\iint_D \sin^2(x+y) dy dx$

where D is the parallelogram with vertices $(0, \pi)$, $(\pi, 0)$, $(\pi, 2\pi)$, $(2\pi, \pi)$

(c) Compute $\iiint_D \frac{z}{(x-1)^2 + (y-1)^2} dV$, where D is the solid bounded by the surface $(x-1)^2 + (y-1)^2 = 2z$ and the plane $z = 2$.

Ans. (a) R is the region between two squares of sides 2 and 4, centered at origin and sides parallel to the coordinate axes.



$$\iint_R e^{x+y} dA = \int_{x=-2}^2 \int_{y=-2}^2 e^{x+y} dy dx - \int_{x=-1}^1 \int_{y=-1}^1 e^{x+y} dy dx$$

$$= \int_{-2}^2 \left[e^{x+y} \right]_{y=-2}^2 dx - \int_{-1}^1 \left[e^{x+y} \right]_{y=-1}^1 dx$$

$$= \int_{-2}^2 \left[e^{x+2} - e^{x-2} \right] dx - \int_{-1}^1 \left[e^{x+1} - e^{x-1} \right] dx$$

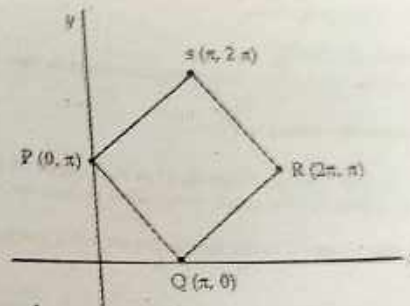
$$= \left[e^{x+2} - e^{x-2} \right]_{-2}^2 - \left[e^{x+1} - e^{x-1} \right]_{-1}^1$$

$$= (e^4 - e^0 - e^0 + e^{-4}) - (e^2 - e^0 - e^0 + e^{-2}) = e^4 + e^{-4} - e^2 - e^{-2}$$

(b) D is the parallelogram with vertices $(0, \pi)$, $(\pi, 0)$, $(\pi, 2\pi)$ and $(2\pi, \pi)$
equation of line PQ $y = \pi - x$

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equation of line QR $y = x - \pi$
equation of line RS $y = -x + 3\pi$
equation of line SP $y = x + \pi$



$$\iint_D \sin^2(x+y) dy dx = \int_{x=0}^{\pi} \int_{y=\pi-x}^{\pi} \sin^2(x+y) dy dx + \int_{x=\pi}^{2\pi} \int_{y=x-\pi}^{2\pi-x} \sin^2(x+y) dy dx$$

$$= \int_{x=0}^{\pi} \int_{y=\pi-x}^{\pi} \frac{1 - \cos(2x+2y)}{2} dy dx + \int_{x=\pi}^{2\pi} \int_{y=x-\pi}^{2\pi-x} \frac{1 - \cos(2x+2y)}{2} dy dx$$

$$= \frac{1}{2} \left[\int_0^{\pi} \left(y - \frac{\sin(2x+2y)}{2} \right)_{y=\pi-x}^{\pi} dx + \int_{\pi}^{2\pi} \left(y - \frac{\sin(2x+2y)}{2} \right)_{y=x-\pi}^{2\pi-x} dx \right]$$

$$= \frac{1}{2} \left[\int_0^{\pi} \left(2x - \frac{\sin 4x}{2} \right) dx + \int_{\pi}^{2\pi} \left(4\pi - 2x + \frac{\sin 4x}{2} \right) dx \right]$$

$$= \frac{1}{2} \left[\left(x^2 + \frac{\cos 4x}{8} \right)_{\pi}^{\pi} + \left(4\pi x - x^2 - \frac{\cos 4x}{8} \right)_{\pi}^{2\pi} \right]$$

$$= \frac{1}{2} \left[\pi^2 + 8\pi^2 - 4\pi^2 - 4\pi^2 + \pi^2 \right] = \pi^2$$

(c) Consult your textbook.

Q. 4. Let the vector field $\vec{F}$ be given as

$$\vec{F}(x, y, z) = (y^2 \cos x + z^3) \hat{i} + (2y \sin x - 4) \hat{j} + (2y \sin x - 4) \hat{j} + (3xz^2 + 2) \hat{k}$$

- (a) Show that $\vec{F}$ is conservative.
 (b) Find the scalar potential for $\vec{F}$.
 (c) What will be the work done in moving an object in this force field from the point $P(0, 1, -1)$ to the point $Q\left(\frac{\pi}{2}, -1, 2\right)$
 (d) If the path P to Q is traversed through ten additional in between points then what will be the effect on the work done. Give reason for your answer.

(e) Evaluate $\int_C \vec{F} \cdot d\vec{R}$, where C is the line segment $x=1, y=1, 0 \leq z \leq 1$.

Ans. (a)

$$\vec{F}(x, y, z) = (y^2 \cos x + z^3)\vec{i} + (2y \sin x - 4)\vec{j} + (3xz^2 + 2)\vec{k}$$

$\vec{F}$ is conservative if $\text{curl } \vec{F} = 0$

$$\text{Curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 + 2 \end{vmatrix}$$

$$= 0\vec{i} - (3z^3 - 3z^3)\vec{j} + (2y \cos x - 2y \cos x)\vec{k}$$

$$= 0\vec{i} - 0\vec{j} + 0\vec{k} = 0$$

$\Rightarrow (x, y, z)$ is conservative.

(b) Let f be scalar potential of $\vec{F}$

$$\Rightarrow f_x = y^2 \cos x + z^3, f_y = 2y \sin x - 4, f_z = 3xz^2 + 2$$

$$\Rightarrow f = \int (y^2 \cos x + z^3) dx$$

$$\Rightarrow f = y^2 \sin x + xz^3 + \phi(y) + \psi(z)$$

$$\Rightarrow f_y = 2y \sin x + \phi'(y)$$

$$\text{But } f_y = 2y \sin x - 4$$

$$\Rightarrow \phi'(y) = -4$$

$$\Rightarrow \phi(y) = -4y$$

$$\text{also, } f_z = 3xz^2 + \psi'(z)$$

$$\text{But, } f_z = 3xz^2 + 2$$

$$\Rightarrow \psi'(z) = 2$$

$$\Rightarrow \psi(z) = 2z$$

$$\text{Thus, } f = y^2 \sin x + xz^3 - 4y + 2z + c$$

(c) Since, $\vec{F}$ is conservative

$\Rightarrow$ Work done in moving an object from $P(0, 1, -1)$ to $Q\left(\frac{\pi}{2}, -1, 2\right)$ is

$$W = f\left(\frac{\pi}{2}, -1, 2\right) - f(0, 1, -1) \text{ where } f \text{ is scalar potential}$$

$$W = \left[(-1)^2 \sin \frac{\pi}{2} + \frac{\pi}{2} (2)^3 - 4(-1) + 2(2) + c \right]$$

$$= \left[(1)^2 \sin 90^\circ + 0(-1)^3 - 4(1) + 2(-1) + c \right]$$

$$= 1 + 4\pi + 4 - 4 - 2 = 4\pi + 3$$

(d) Since, $\vec{F}$ is conservative.

Thus, work done will not change if the path P to Q is traversed through ten additional points in between them.

$$(e) \int_C \vec{F} \cdot d\vec{R} = \int_C \vec{F} \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k})$$

since, C is the line segment $x=1, y=1, 0 \leq z \leq 1$

$$\Rightarrow dx=0, dy=0$$

$$\int_C \vec{F} \cdot d\vec{R} = \int_0^1 \vec{F} \cdot dz\vec{k} = \int_0^1 (3(1)z^2 + 2) dz = \left[z^3 + 2z \right]_0^1$$

$$= 1^3 - 0 + 2(1 - 0) = 3$$

Q. 5. Let D be a ball of radius 3 with center at the origin and $\vec{F}$ be the vector field given by

$$\vec{F}(x, y, z) = 2xy\vec{i} + yz^2\vec{j} + xz\vec{k}$$

(a) Use spherical coordinates to compute $\iiint_D \Delta \cdot \vec{F} \, dV$

(b) If S denotes the upper half of the ball D , then verify Stokes' Theorem for $\vec{F}$.

$$\text{Ans. (a)} \quad \vec{F}(x, y, z) = 2xy\vec{i} + yz^2\vec{j} + xz\vec{k}$$

D is the ball of radius 3 centered at origin

$$\Rightarrow D = x^2 + y^2 + z^2 = (3)^2$$

$$\Delta \cdot \vec{F} = 2y + z^2 + x$$

To compute $\iiint_D \nabla \cdot \vec{F} dV$

Let

$$x = r \sin \phi$$

$$y = r \sin \theta \cos \phi$$

$$z = r \cos \phi$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

$$\iiint_D \nabla \cdot \vec{F} dV = \int_{r=0}^3 \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} (zr \sin \theta \sin \phi + r^2 \cos^2 \theta + r \sin \theta \cos \phi) \cdot r^2 \sin \theta d\theta d\phi dr$$

$$= \int_0^3 \int_0^{2\pi} \int_0^{\pi} (2r^3 \sin^2 \theta \sin \phi + r^4 \cos^2 \theta \sin \theta + r^3 \sin^2 \theta \cos \phi) d\theta d\phi dr$$

$$= \int_0^3 \int_0^{2\pi} \int_0^{\pi} \left[r^3 (1 - \cos 2\theta) \sin \phi + r^4 \cos^2 \theta \sin \theta + r^3 \left(\frac{1 - \cos 2\theta}{2} \right) \cos \phi \right] d\theta d\phi dr$$

$$= \int_0^3 \int_0^{2\pi} \left[r^3 \sin \phi \left[\theta - \frac{\sin 2\theta}{2} \right] - \frac{r^4 \cos^3 \theta}{3} + r^3 \cos \phi \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right] \right]_{\theta=0}^{\pi} d\phi dr$$

$$= \int_0^3 \int_0^{2\pi} \left[\pi r^3 \sin \phi + \frac{2}{3} r^4 + \frac{\pi}{2} r^3 \cos \phi \right] d\phi dr$$

$$= \pi \left[\frac{r^4}{4} \right]_0^3 \left[-\cos \phi \right]_0^{2\pi} + \frac{2}{15} \left[r^5 \right]_0^3 \left[\phi \right]_0^{2\pi} + \frac{\pi}{2} \left[\frac{r^4}{4} \right]_0^3 \left[\sin \phi \right]_0^{2\pi}$$

$$= 0 + \frac{2}{15} (243 - 0)(2\pi - 0) + 0 = \frac{324\pi}{5}$$

(b) Consult your text book.

Q. 6. (a) Let the force field $\vec{F}$ be given by

$$\vec{F}(x, y, z) = xyz(\hat{i} + \hat{j} + \hat{k}).$$

If $\vec{N}$ is the outward drawn normal, then use Divergence theorem to evaluate $\iiint_S \vec{F} \cdot \vec{N} dS$, where S is the surface of the box: $0 \leq x \leq 2$, $0 \leq y \leq 1$, $0 \leq z \leq 4$.

(b) Evaluate: $\int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} dz dy dx$ using

(i) Cylindrical coordinates

(ii) Spherical coordinates.

Ans. $\vec{F} = xyz(\hat{i} + \hat{j} + \hat{k}) = xyz\hat{i} + xyz\hat{j} + xyz\hat{k}$
Using divergence theorem,

$$\iiint_V \vec{F} \cdot \vec{N} dV = \iiint_V \text{div } \vec{F} dV$$

$$\text{div } \vec{F} = yz + xz + xy$$

$$\iiint_V \vec{F} \cdot \vec{N} dV = \int_{x=0}^3 \int_{y=0}^{\sqrt{9-x^2}} \int_{z=0}^{\sqrt{9-x^2-y^2}} (yz + xz + xy) dz dy dx$$

$$= \int_0^3 \int_0^{\sqrt{9-x^2}} \left[y \left[\frac{z^2}{2} \right]_0^{\sqrt{9-x^2-y^2}} + x \left[\frac{z^2}{2} \right]_0^{\sqrt{9-x^2-y^2}} \right] dy dx = \int_0^3 \int_0^{\sqrt{9-x^2}} [8y + 8x + 4xy] dy dx$$

$$= \int_0^3 \left[4y^2 + 8xy + 2xy^2 \right]_0^{\sqrt{9-x^2}} dx = \int_0^3 (4 + 8x + 2x) dx$$

$$= \int_0^3 (4 + 10x) dx = (4x + 5x^2) \Big|_0^3 = 8 + 20 = 28$$

$$(b) I = \int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} dz dy dx$$

Cylindrical Coordinates

Let, $x = r \cos \theta$

$$y = r \sin \theta$$

$$z = z$$

$$dz dy dx = r dz d\theta dr$$

$$\text{For } 0 < x < 3, 0 < y < \sqrt{9-x^2}, 0 < z < \sqrt{9-x^2-y^2}$$

$$0 < r < 3, 0 < \theta < \frac{\pi}{2}, 0 < \phi < \sqrt{9-r^2}$$

$$I = \int_0^3 \int_0^{\pi/2} \int_0^{\sqrt{9-r^2}} r^2 \sin \theta \, d\phi \, d\theta \, dr$$

$$= \int_0^3 \left[\int_0^{\pi/2} \int_0^{\sqrt{9-r^2}} r^2 \sin \theta \, d\phi \, d\theta \right] dr = \frac{\pi}{2} \int_0^3 r \sqrt{9-r^2} \, dr$$

Let $9-r^2 = t$

$$-2r \, dr = \frac{-1}{2} \times \frac{\pi}{2} \int_9^0 \sqrt{t} \, dt$$

$$= -\frac{\pi}{4} \times \frac{2}{3} \left[t^{3/2} \right]_9^0 = \frac{-\pi}{2 \times 3} \times (0 - (9)^{3/2})$$

$$= \frac{\pi}{2 \times 3} \times 27 = \frac{9\pi}{2}$$

Spherical Coordinates

Let $x = r \sin \theta \cos \phi$

$y = r \sin \theta \sin \phi$

$z = r \cos \theta$

$dx \, dy \, dz = r^2 \sin \theta \, d\phi \, d\theta \, dr$

and $0 < r < 3, 0 < \theta < \frac{\pi}{2}, 0 < \phi < \frac{\pi}{2}$

$$I = \int_0^3 \int_0^{\pi/2} \int_0^{\pi/2} r^2 \sin \theta \, d\phi \, d\theta \, dr$$

$$= \left[\frac{r^3}{3} \right]_0^3 \left[\theta \right]_0^{\pi/2} \left[-\cos \phi \right]_0^{\pi/2}$$

$$= 9 \times \frac{\pi}{2} \times \left[-\left(\cos \frac{\pi}{2} - \cos 0 \right) \right] = \frac{9\pi}{2}$$